

# Level 4

Watsonx Orchestrate

Hands-on lab guide – Build an AI Assistant



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# Introduction

## Before you start

This lab is the first part of a series of 2 independent labs.

- Lab I: Create an AI Assistant
  - How to create an assistant and preview an assistant
  - How to define custom actions
  - How to define user interactions with the assistant
  - How to specify an end-to-end discussion
  - How to preview / publish an assistant
- Lab II: Integrate your AI Assistant in your web interface
  - How to leverage the disambiguation and digression features
  - How to debug and trace an assistant before its deployment
  - How to work with sub-actions
  - How to configure and deploy your assistant an existing web site

**IMPORTANT NOTE:** Then number of assistants being limited on each tenant, when you are finished with all the assistant labs, or should you need to suspend the learning for a moment, please DELETE YOUR ASSISTANT for the environment. Instructions to export your work and delete an assistant are provided at the end of this guide. **Assistants aged more than 3 weeks will be deleted by administrators.**

## About this lab

This Assistant lab 1 is designed to teach you how to rapidly and easily create and deploy an intelligent assistant using the low-code no-code editor provided in watsonx Orchestrate.

This lab is composed of different sections that can be performed independently. You will create only one assistant to covers all the activities related to the L4 Assistant labs.

**Note:** It is strongly recommended to export your work at the end of each lab, should your assistant be deleted by the administrator for inactivity or delay expiration reasons. Instructions to export/import your assistant are given in this guide.

## Environment

This lab has been created and tested on watsonx Orchestrate Standard running on IBM Cloud. To avoid issues please use a watsonx Standard environment running on IBM Cloud as there are minor

differences between AWS, IBM Cloud, and Cloud Pak for Data tenants.

Instructions for requesting access to an enablement tenant to complete these labs are provided in the Preparation guide included in this learning plan.

## Lab use-case

In this lab, you'll explore the capabilities of the watsonx Orchestrate AI Assistant Builder to create and deploy an intelligent assistant. The lab is set within a banking scenario focused on enhancing customer service. Specifically, you'll learn how to build an assistant that guides customers through the process of opening a new account. While the scenario is intentionally simple, it showcases a broad range of features and tools that will help you design and customize your own AI assistant, including:

- Custom actions creation
- Conditional action routing
- Assistant testing and deployment

And many more implementation details.

## How to get support

If you encounter an issue with this lab,

- IBM employees can open an IBM Support ticket at <http://ibm.biz/support>.
- Business Partners can request help at the [Partner Plus Support](#) website.

## Prerequisites

Please note the following prerequisites for this lab:

- You have Completed the wxO L4 Preparation guide
  - You have access to a wxO SaaS instance, as specified in wxO L4 Preparation guide.

# Create your L4\_assistant

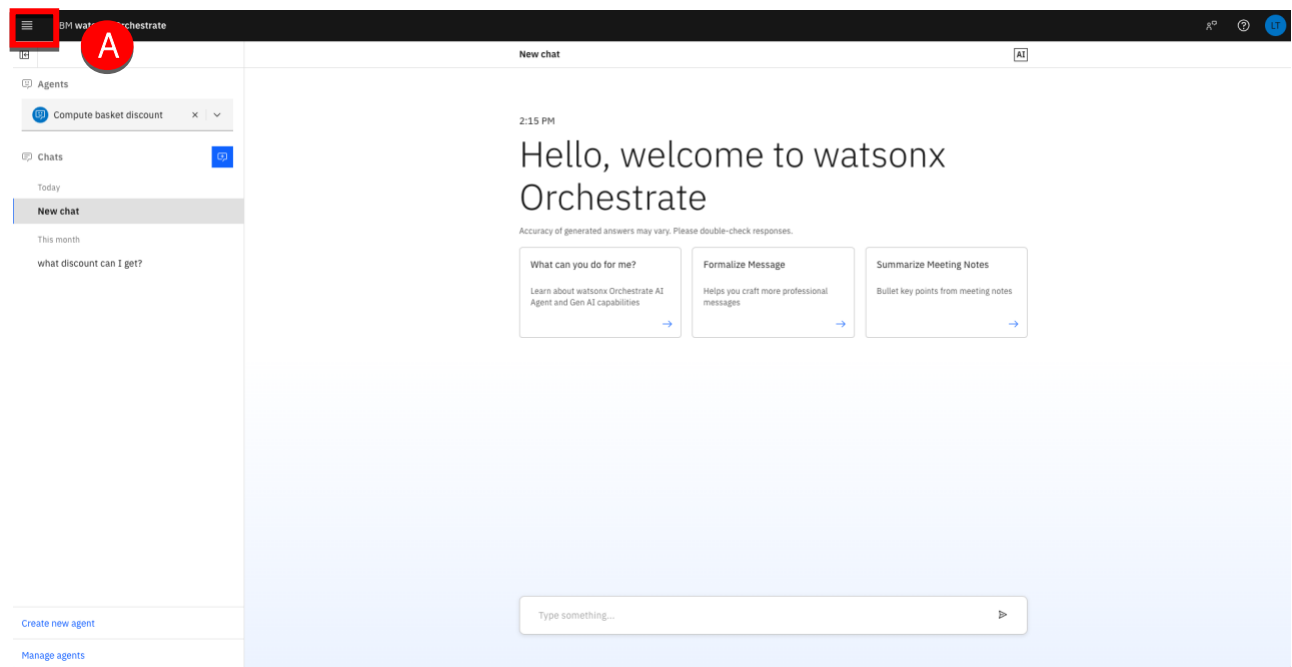
## Introduction

The assistant you create in this lab will be used throughout all the watsonx Orchestrate Assistant labs in this Level 4 learning plan. If you need to pause your progress, be sure to export your assistant as an archive. This will allow you to easily resume the course later without losing your work.

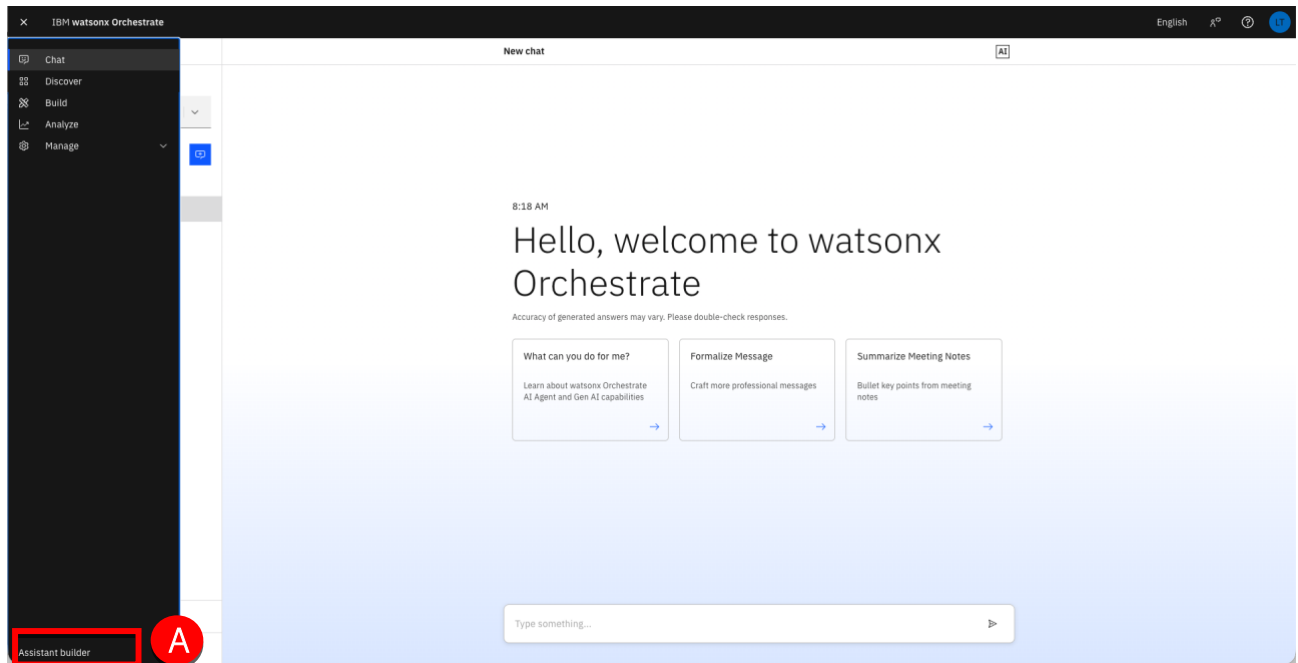
The assistant you will create must have a unique name. We suggest naming your assistant L4\_labs\_XXXYYMM to rapidly identify it among the other assistants created on your watsonx Orchestrate instance.

Let's create the assistant first,

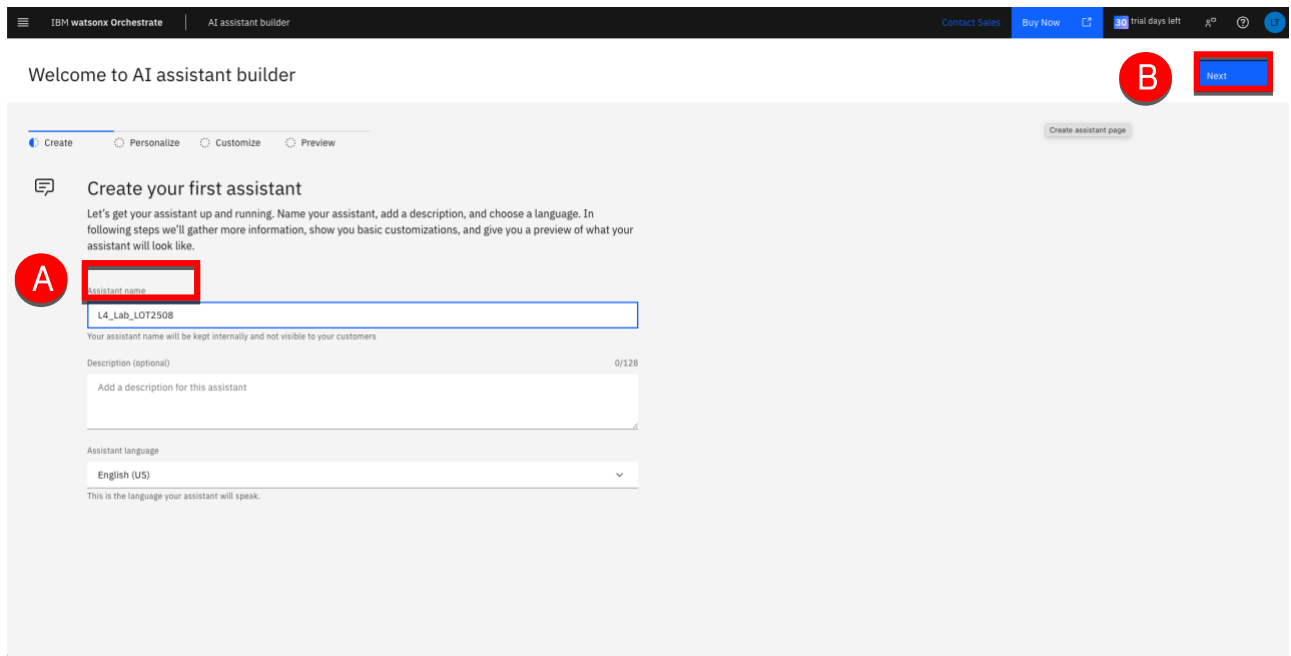
1. Login to your SaaS instance of watsonx Orchestrate
2. Click the main menu icon (A).



3. Click Assistant Builder (A).



4. Enter L4\_lab\_XXXXYYMM in the Assistant name field (A). Click Next (B).



5. Select **Web** in the **Where do you plan...** drop down menu (A). Select any value in the other ones (B). Click **Next** (C).

IBM watsonx Orchestrate | AI assistant builder

Welcome to AI assistant builder

Create Personalize Customize Preview

### Personalize your assistant

**Tell us where your assistant will live**  
Where do you plan to use your assistant?  
Web

**Tell us about yourself**  
This information will be used to personalize your onboarding experience.

Which industry do you work in?  
Banking and financial services

What is your role on the team building the assistant?  
Lead strategist

Which statement describes your needs best?  
I want to provide confident answers to common questions

This is what your customers will experience

watsonx Assistant

Do you have the Speed Demons in stock?

The Speed Demons are in stock at our Downtown and Northgate locations, which are both within 5 miles of you.

What size and color do you need?

I'm looking for a size 9 in white

Great news! The Speed Demons are available in white in a size 9.

You can purchase them for curbside pickup or we can ship them to you. Which would you prefer?

I'll pick them up! Ship them to me!

Type something...

6. Click **Next** (A).

IBM watsonx Orchestrate | AI assistant builder

Welcome to AI assistant builder

Create Personalize Customize Preview

### Customize your chat UI

Update the style to match your brand and your website. You can change these settings later. A developer can also add more advanced styling changes with code. [Learn more](#)

Assistant's name as known by customers  
Assistant

Intended purpose  
☒ Standard: For virtual agents and customer support experiences.  
☐ Carbon for AI: For use in internal IBM products.

Choose a theme  
Light Dark

Primary color  
#FFFFFF

Secondary color  
#3D3D3D

Chat header  
Accent color  
#0354E9

Significant and interactive objects

Size  
The size of the web chat on this page will not change by updating these fields.

Width: 380px Height: 640px

Add an avatar image

Restart conversation

Assistant

Hi! I'm a virtual assistant. How can I help you today?

Example: Find nearby location

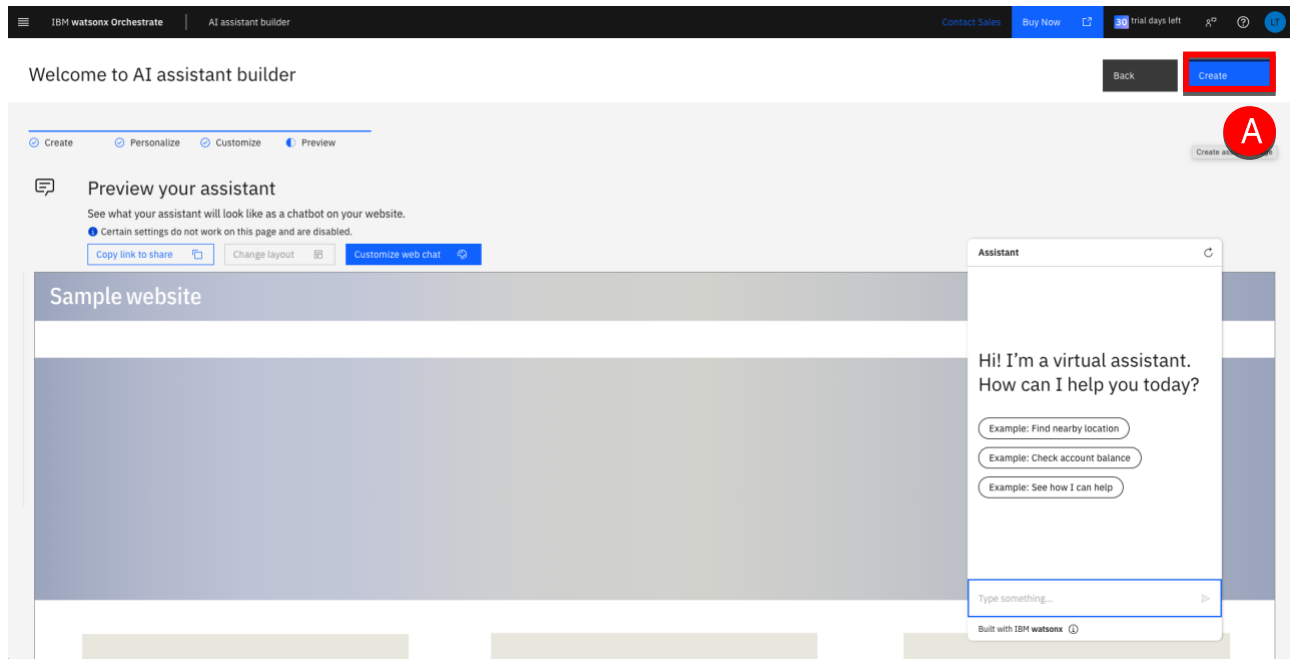
Example: Check account balance

Example: See how I can help

Type something...

Built with IBM watsonx

7. Click **Create** (A).



You are now ready to perform the L4 Assistant labs.

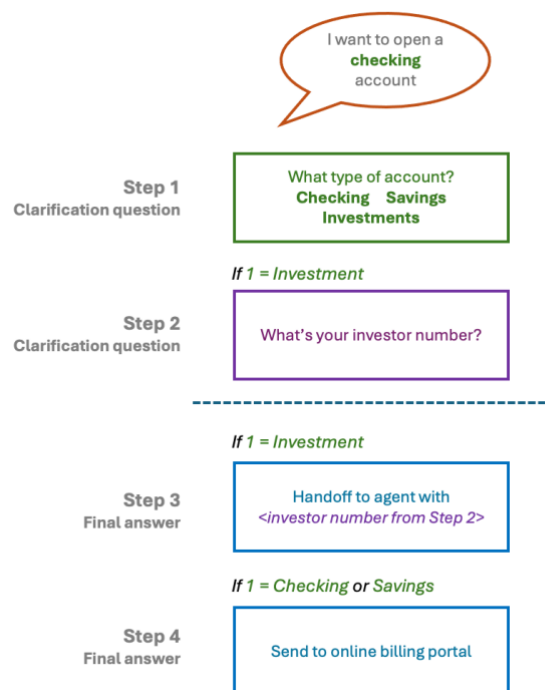
## Action fundamentals

### What is an action?

In AI assistant builder, an *action* represents a discrete outcome that you want your AI assistant to accomplish in response to a user's request. An action consists of steps, which are the predefined prompts for the conversation between a customer and the AI assistant. This conversation begins with the user input that starts the action (for example, I want to withdraw money). It might then include more exchanges as the AI assistant gathers more information, and it ends when the AI assistant completes the requested tasks or answers the customer's question.

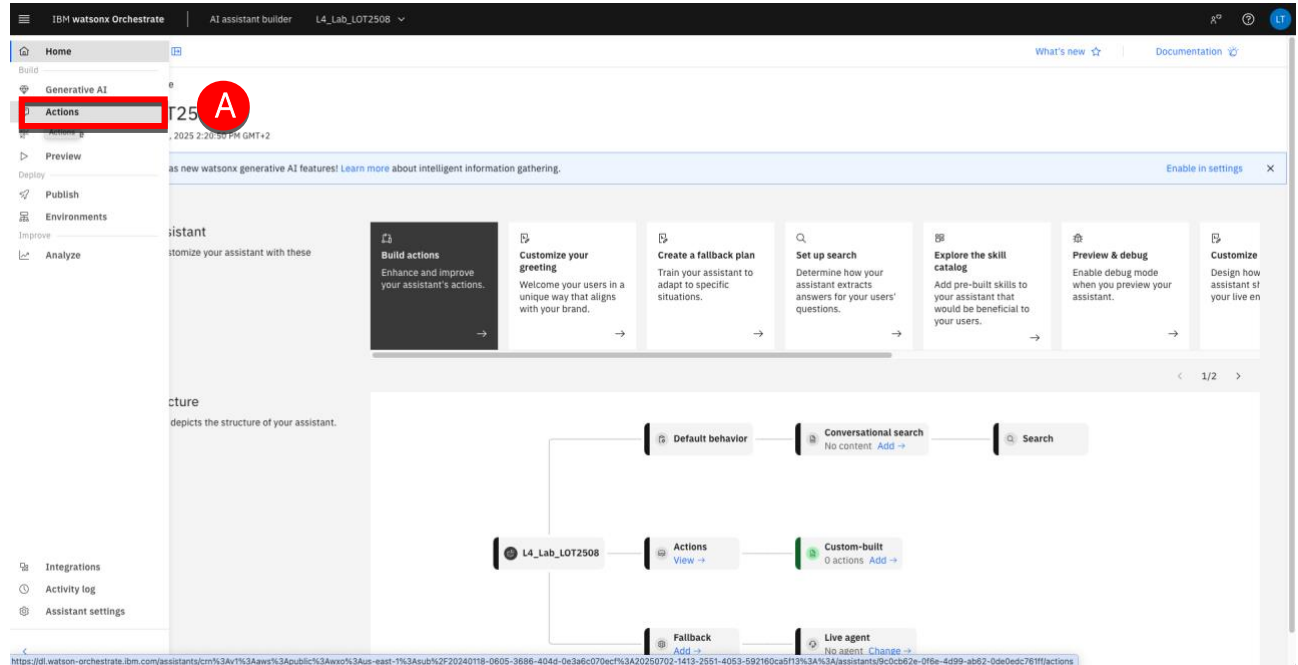
You can use AI assistant builder to build AI assistant that act like a human personal assistant. You can build your AI assistant by creating [AI-guided actions](#), [skill-based actions](#), or [custom-built actions](#). (sourced from documentation: <https://www.ibm.com/docs/en/watsonx/watson-orchestrate/current?topic=builder-building-your-ai-assistant-actions>)

In this Exercise, you begin by building a simple custom action in a *Banking* business context. The diagram below shows a breakdown of this example **Open a new Account** action, broken down into discrete steps. Each color represents the links between the information and the assistant step. Thus, the word checking in the initial request will trigger the execution of step 1. The condition '1=' refers to the customer response in step 1.

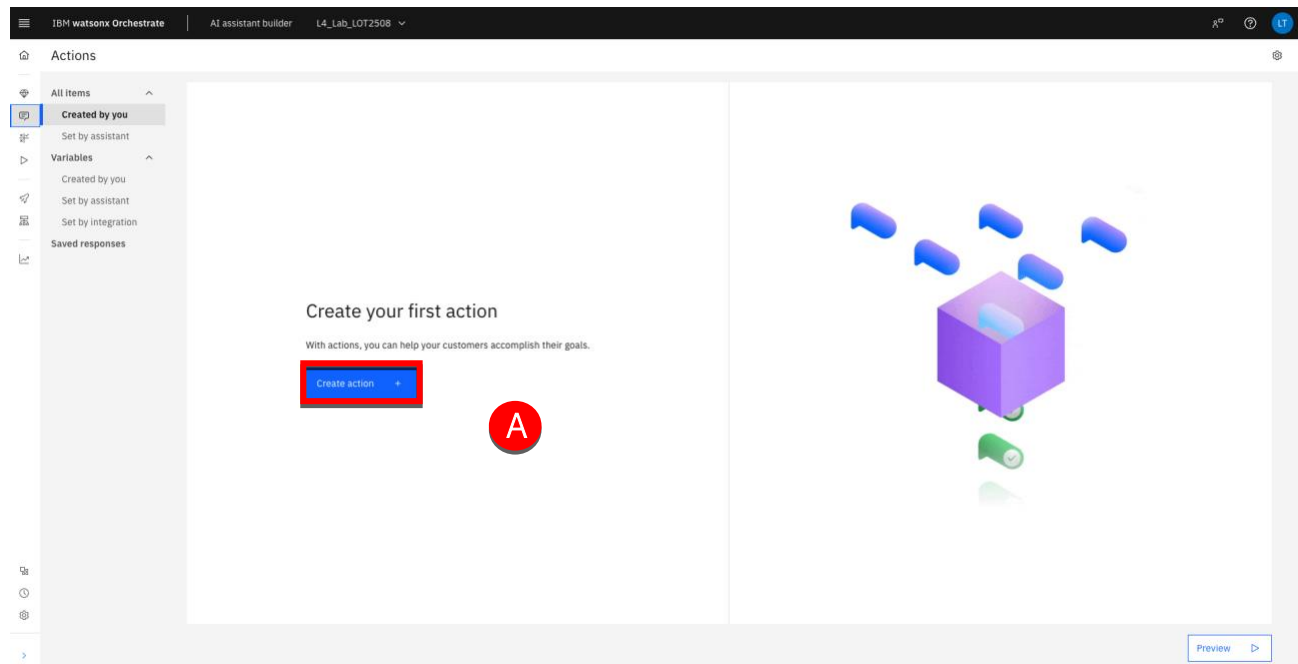


## Building your first action

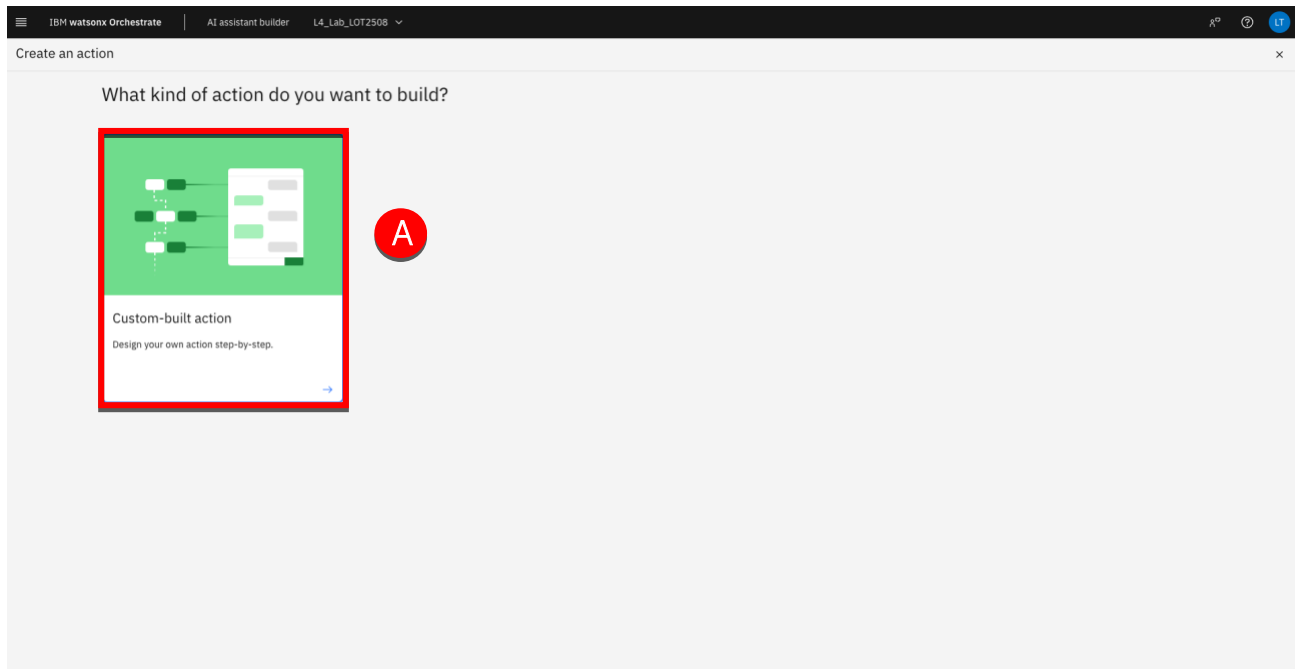
1. In the AI assistant builder homepage, navigate to the lefthand navigation pane and click **Actions** (A).



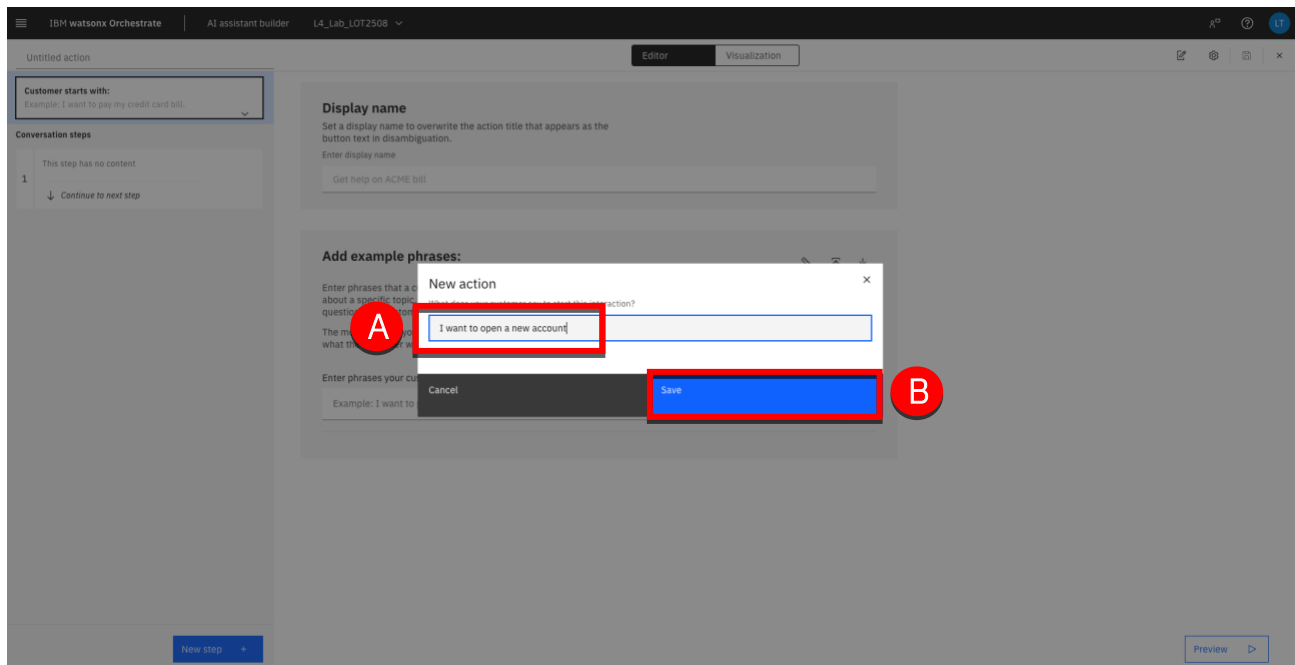
2. Click **Create action** (A).



- Click **Custom-built action** (A).



- Type I want to open a new account in the **What does your customer say to start this interaction?** field (A), click **Save** (B).



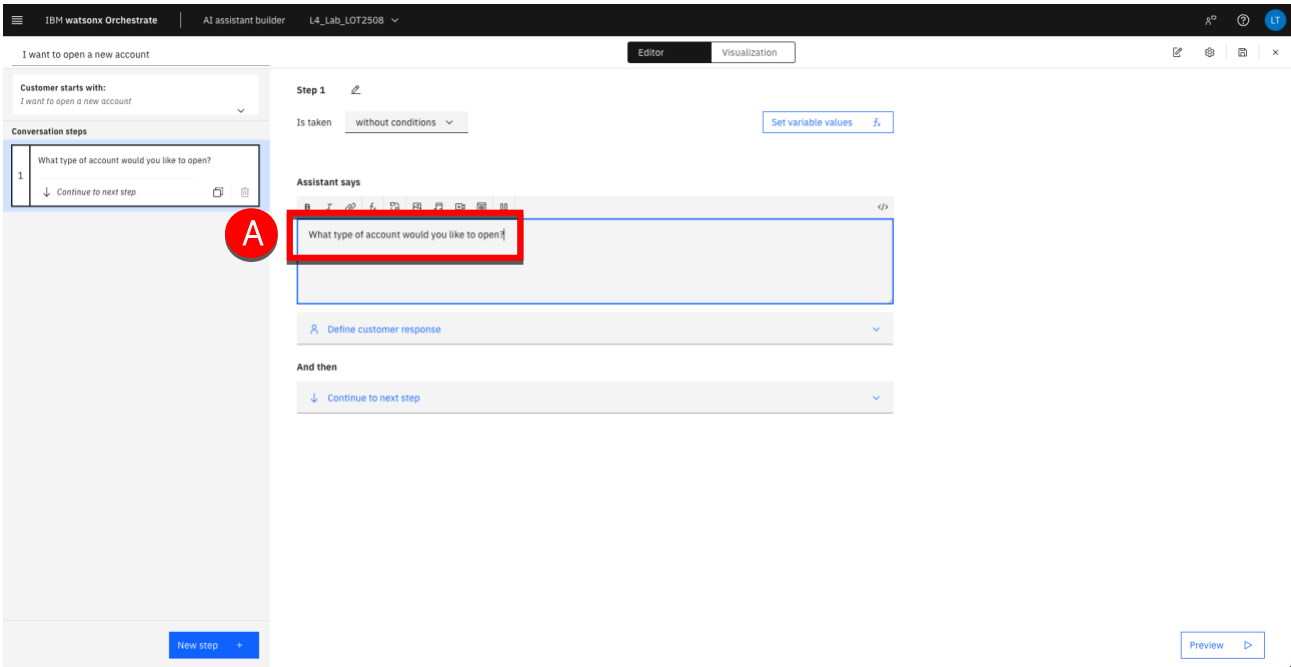
This names the newly created action. Also, it trains your assistant to start this action when a user enters a phrase that is similar the starting phrase you provided (“I want to open a new account”).

Now it's time to create the first step for this action. You start with the clarification question about the type of account the user wants to open.

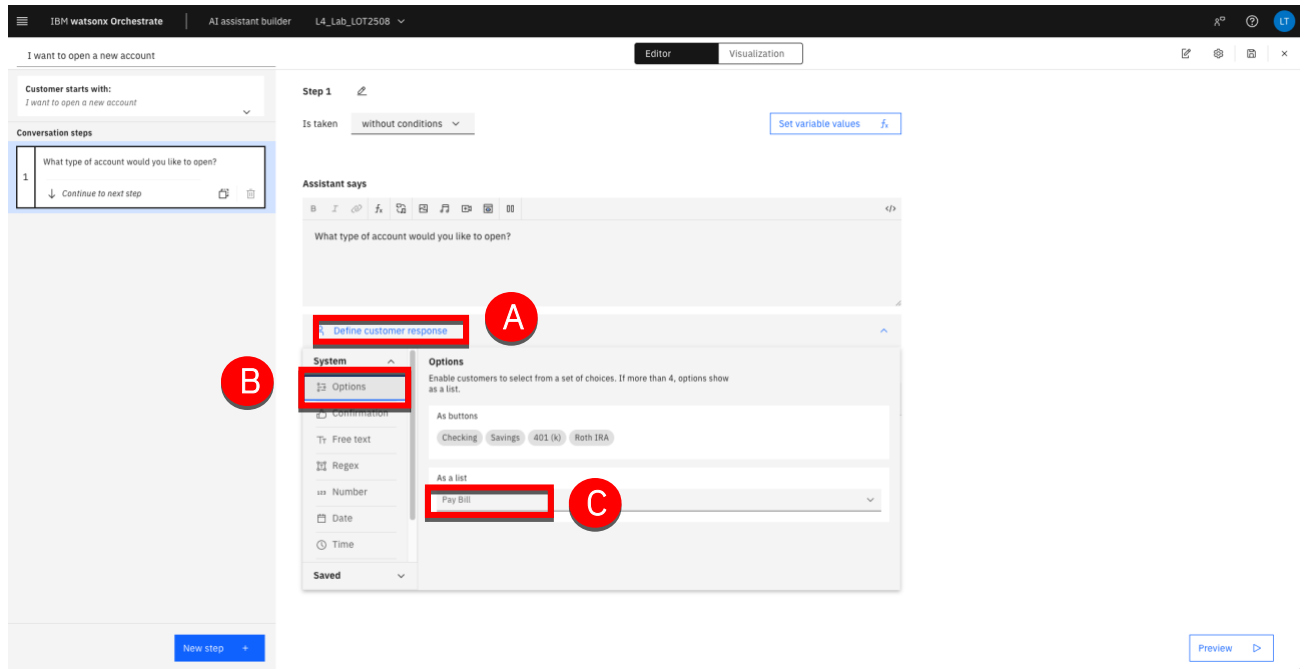
**Step 1**  
**Clarification question**

What type of account?  
Checking Savings  
Investments

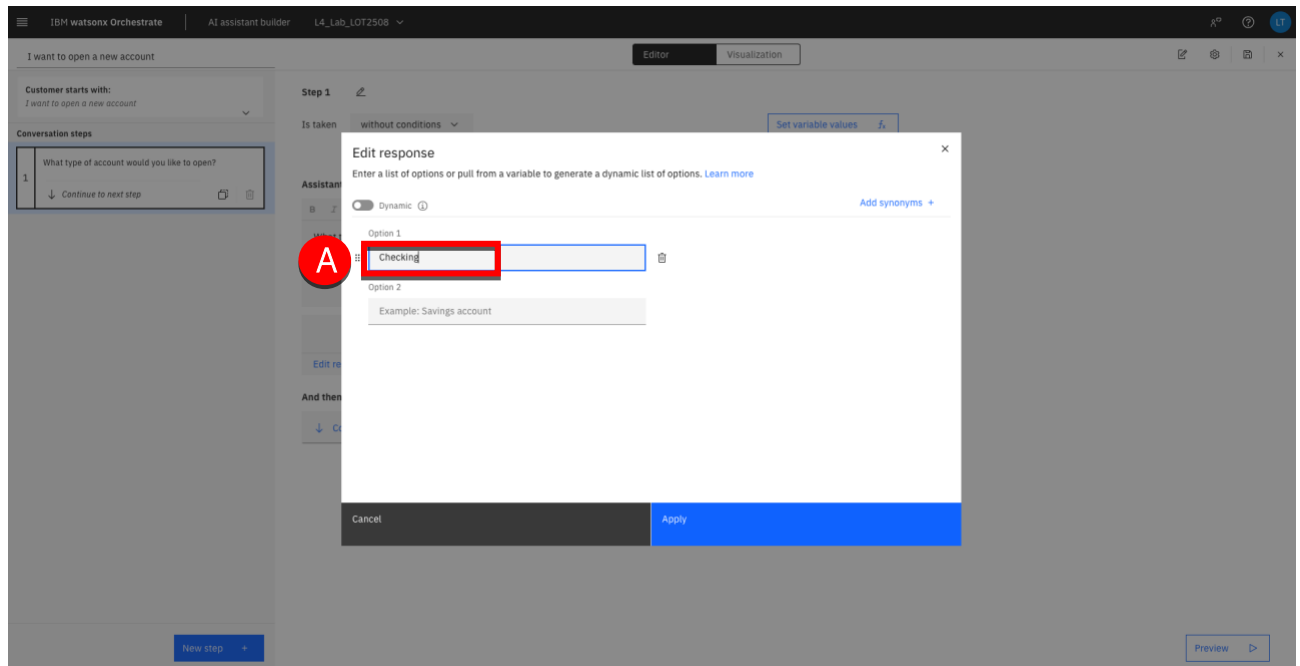
5. Type What type of account would you like to open? in the **Assistant says** text box (A).



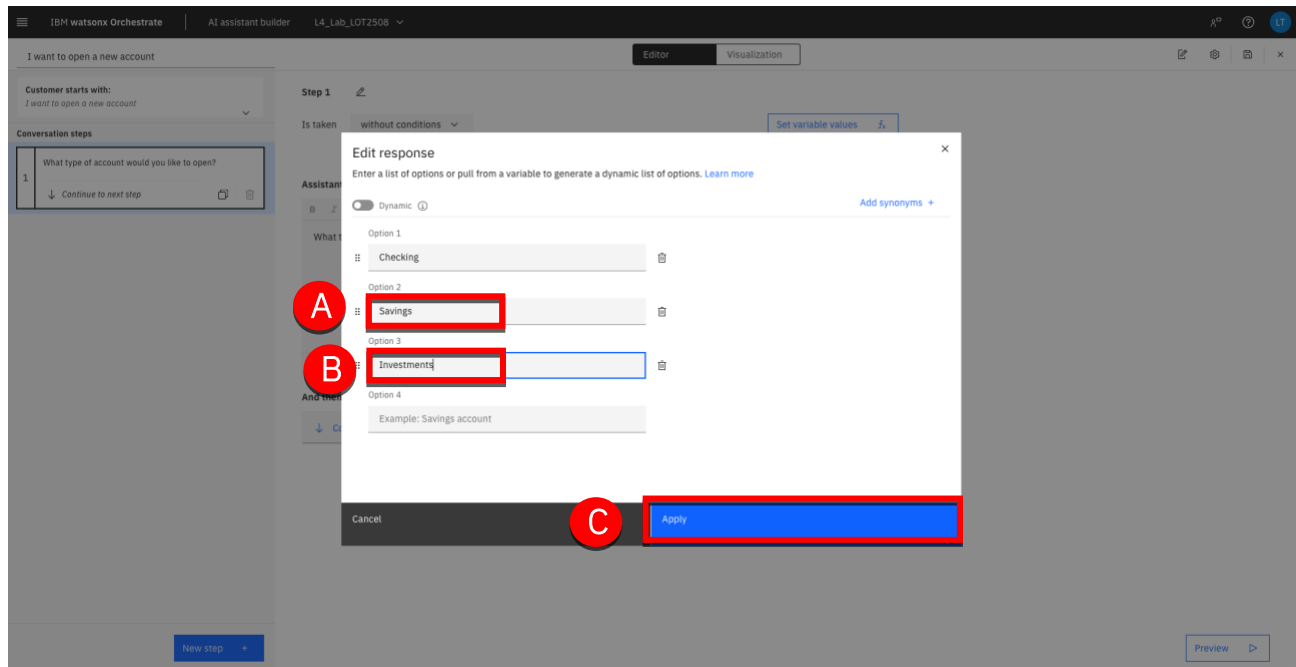
6. Click **Define customer response** (A) then hover over **Options** (B), then click **As a list** (C). This setting tells the assistant what type of response is needed from the customer (Free text, number, date, etc.).



7. Type Checking (A) in the **Option 1** field.



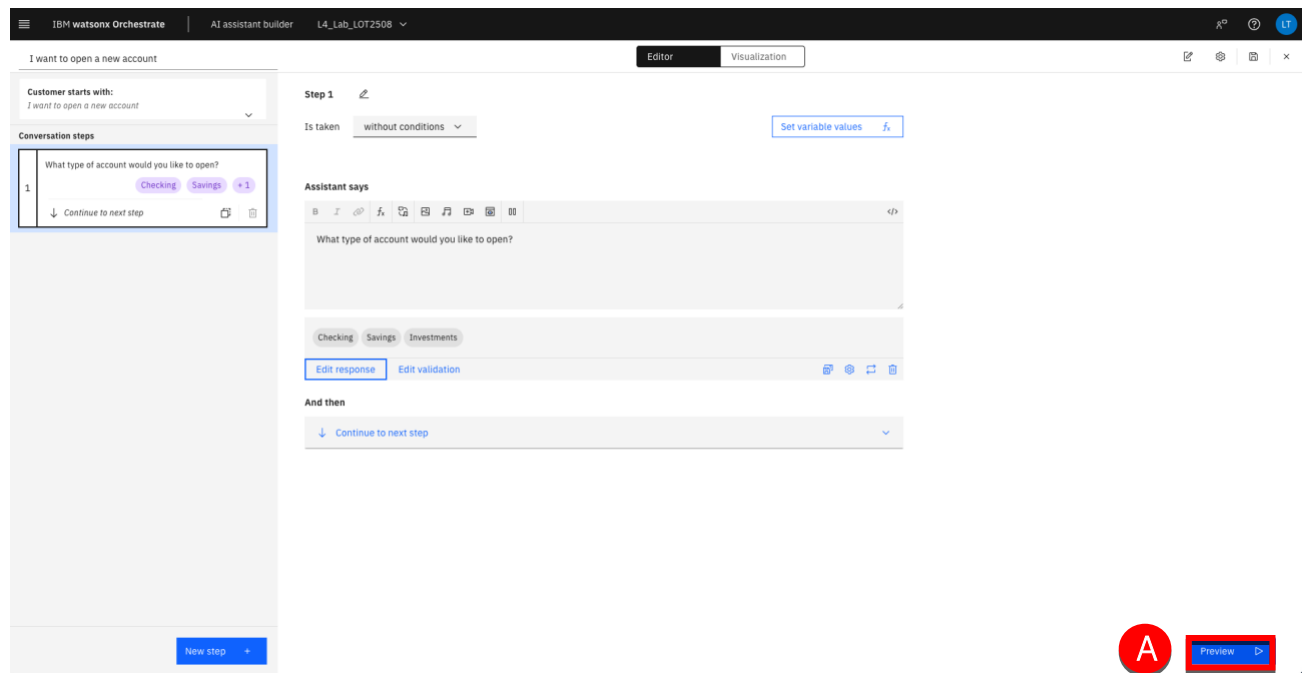
8. Repeat step 7 adding Savings (A) and then Investment (B) to the list. Click **Apply** (C) to save the changes.



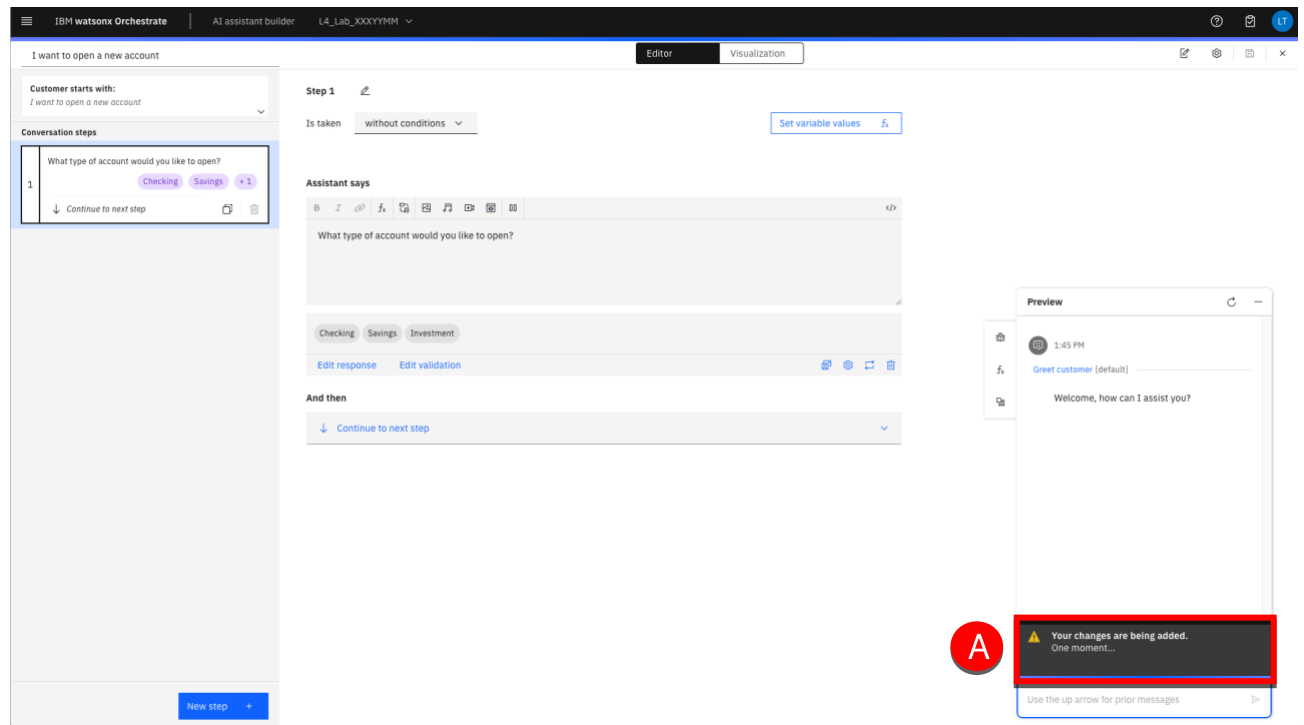
## Preview your action

You can test the action at any time to see whether the resulting interaction works as intended. The **Preview** button in the action page shows you what customers see when they use web chat to interact with this assistant.

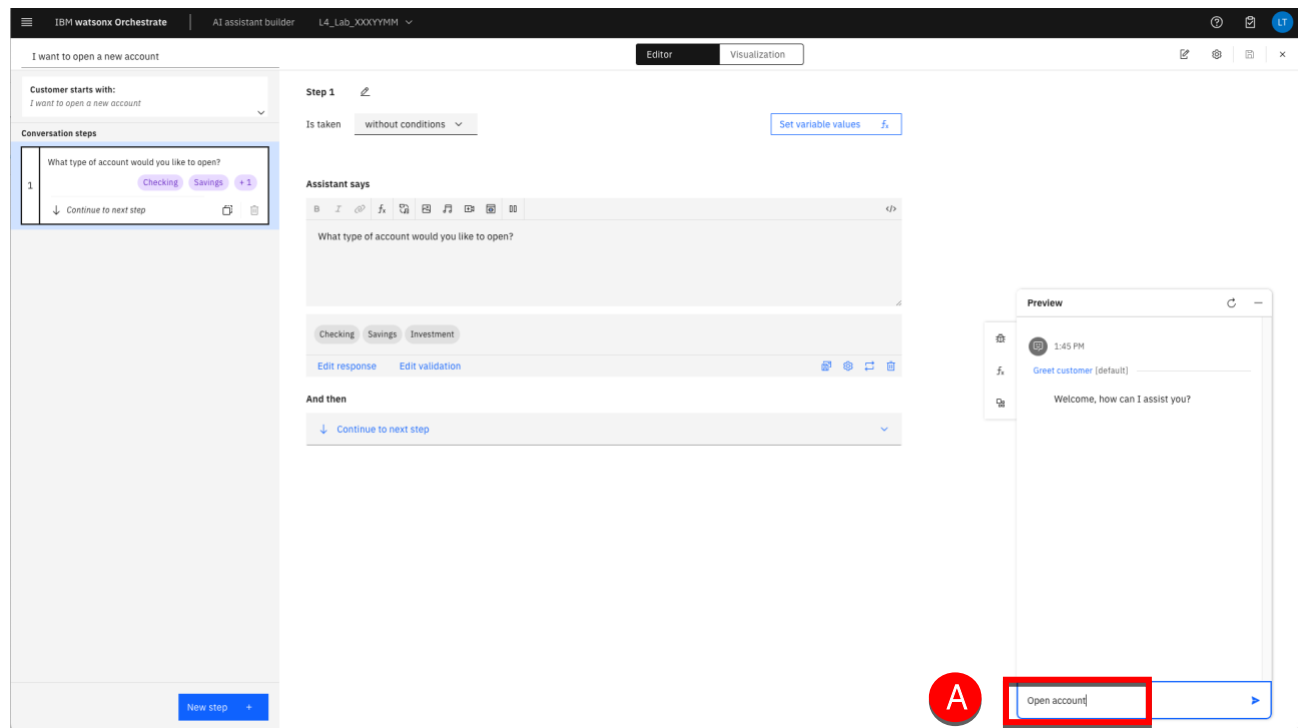
1. Click the **Preview** (A) button on the lower right-hand side of the action page.



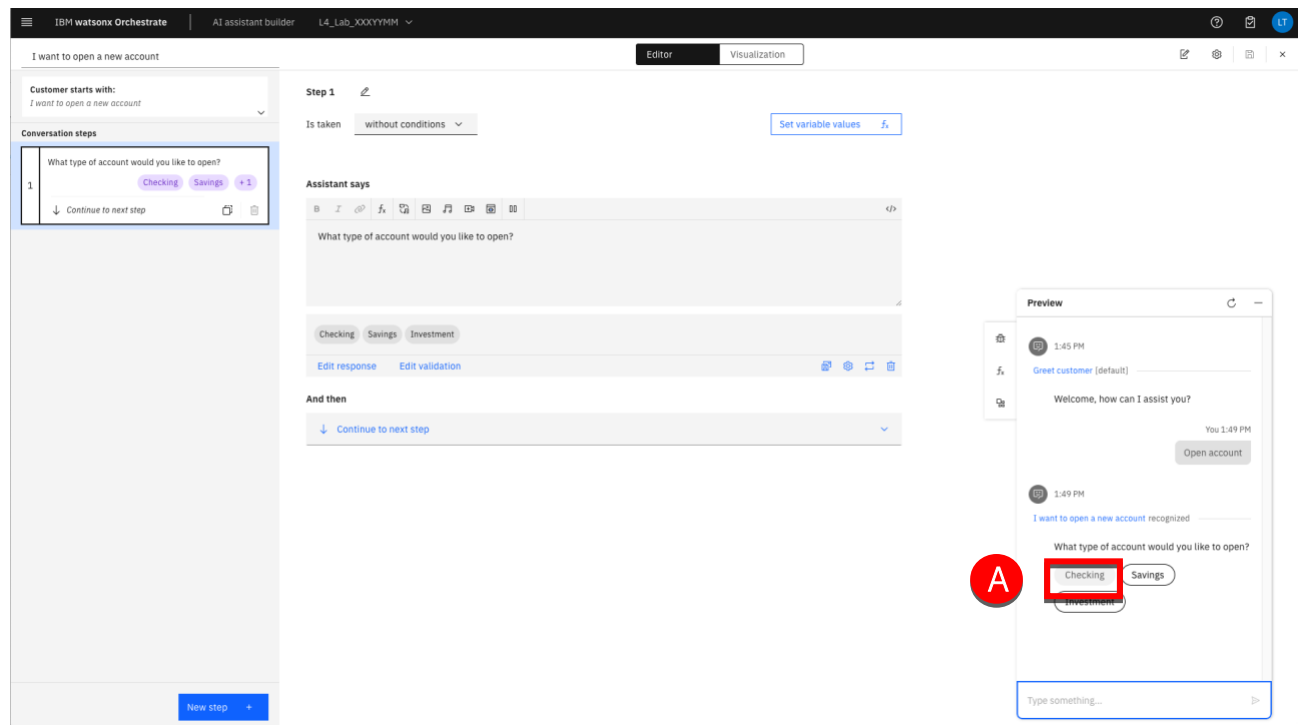
This opens the **Preview** panel. If the preview panel states, **Your changes are being added** (A), wait a few moments for that message to clear.

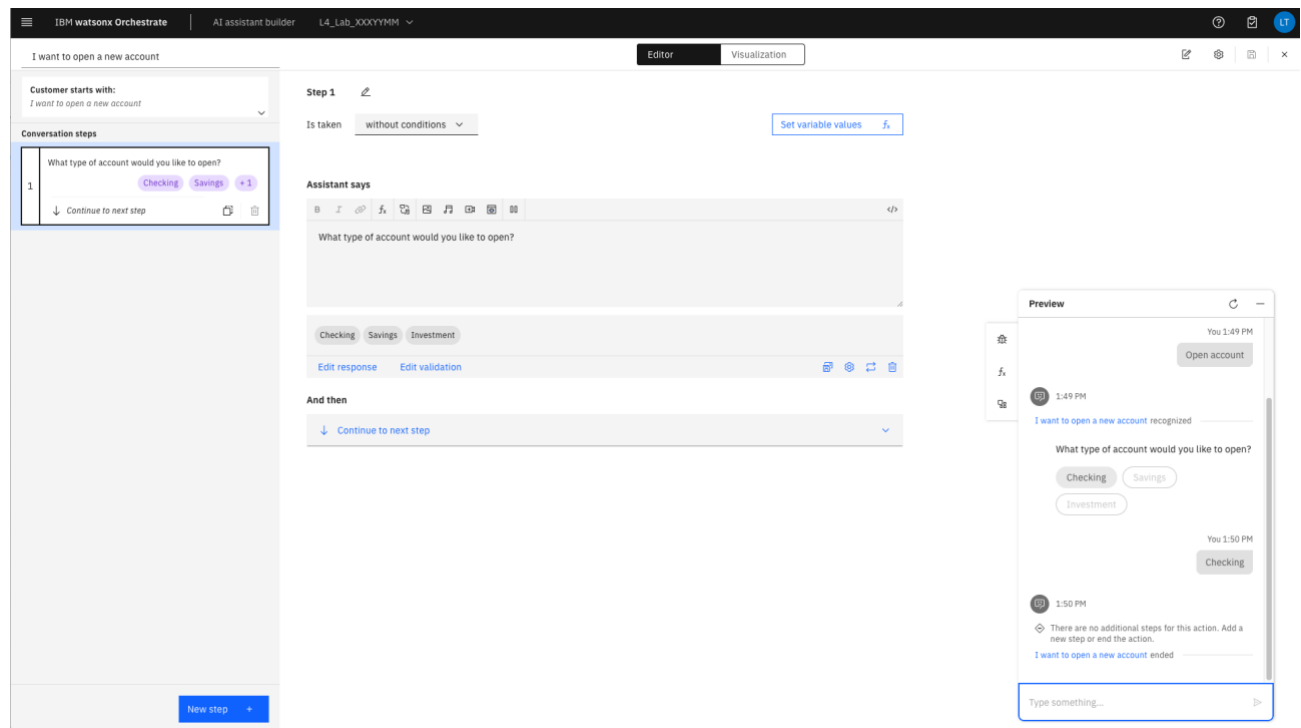


2. Type a phrase such as Open account (A).



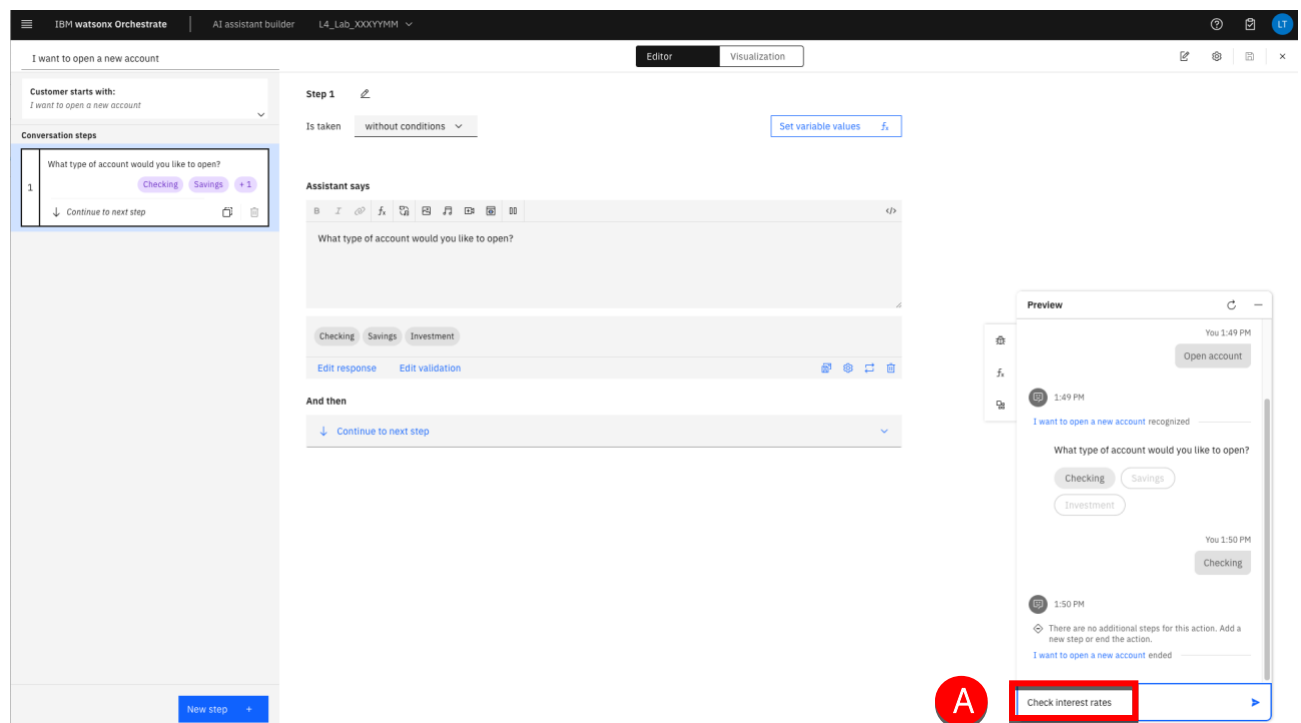
3. Select **Checking** (A).



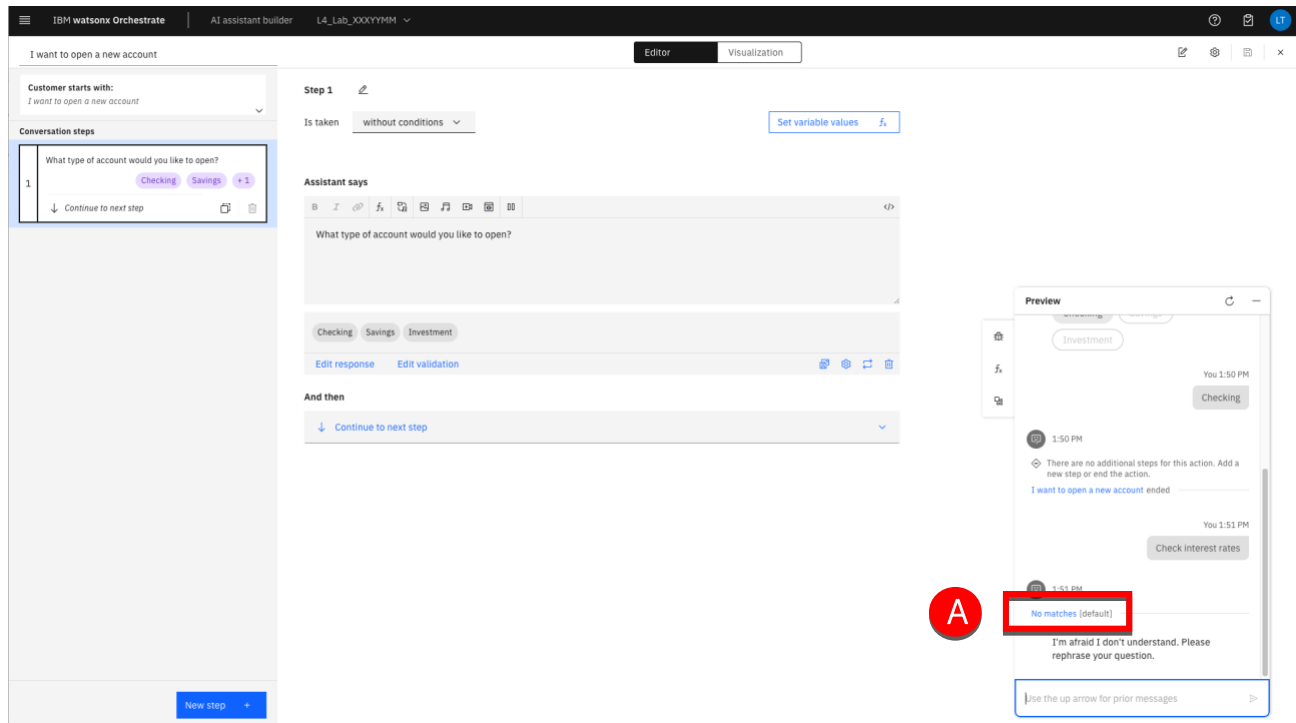


Notice that the user input “Open account” does not exactly match the provided example phrase for starting the action (I want to open a new account), but the user input still starts the action. This demonstrates the assistant’s NLP capabilities – the assistant can detect underlying intent of a user request.

4. Enter another phrase such as Check interest rates (A) or Apply for a loan.

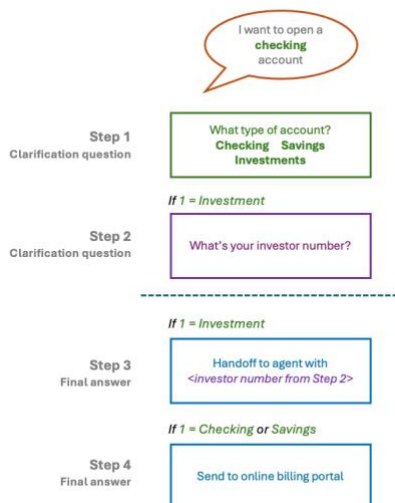


5. These inputs are not recognized by the assistant, so the assistant routes to the built in [No matches action](#) (A).



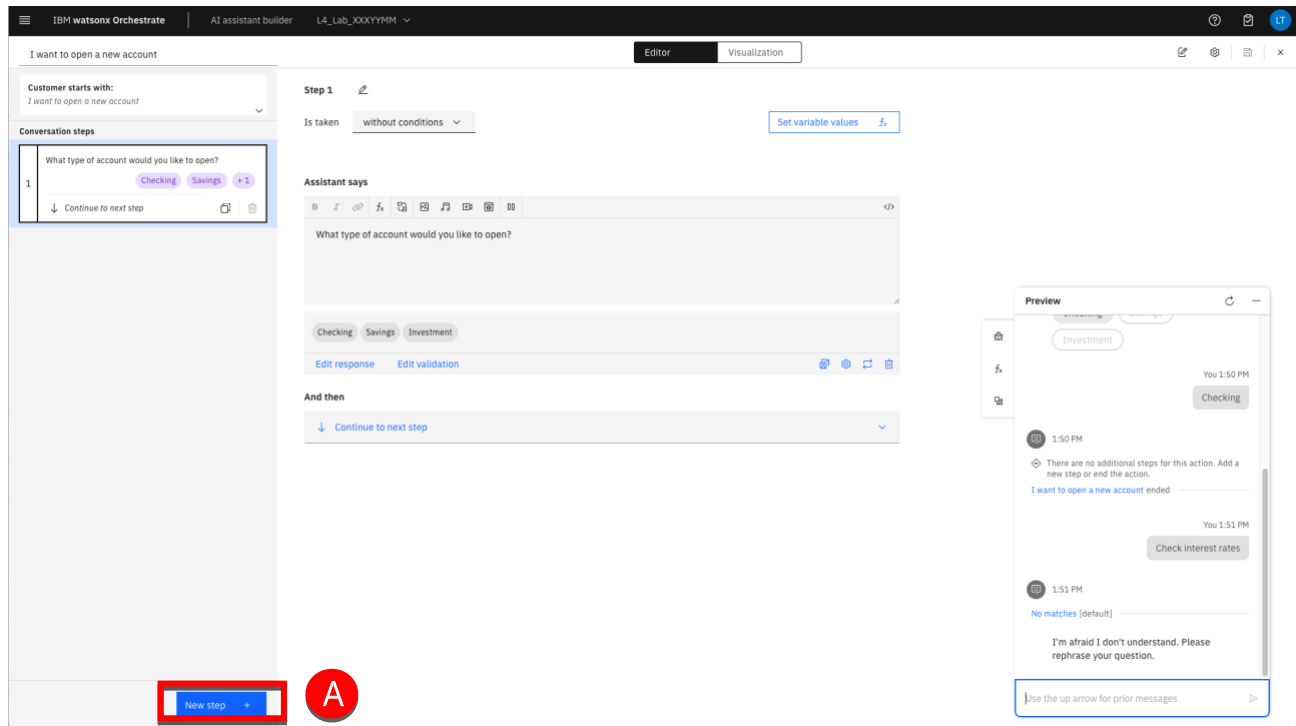
### Create another clarification step

With your first step (Step 1) created and tested, you continue to build this Action by creating another step (Step 2). The diagram below shows the logic you will build into your action.

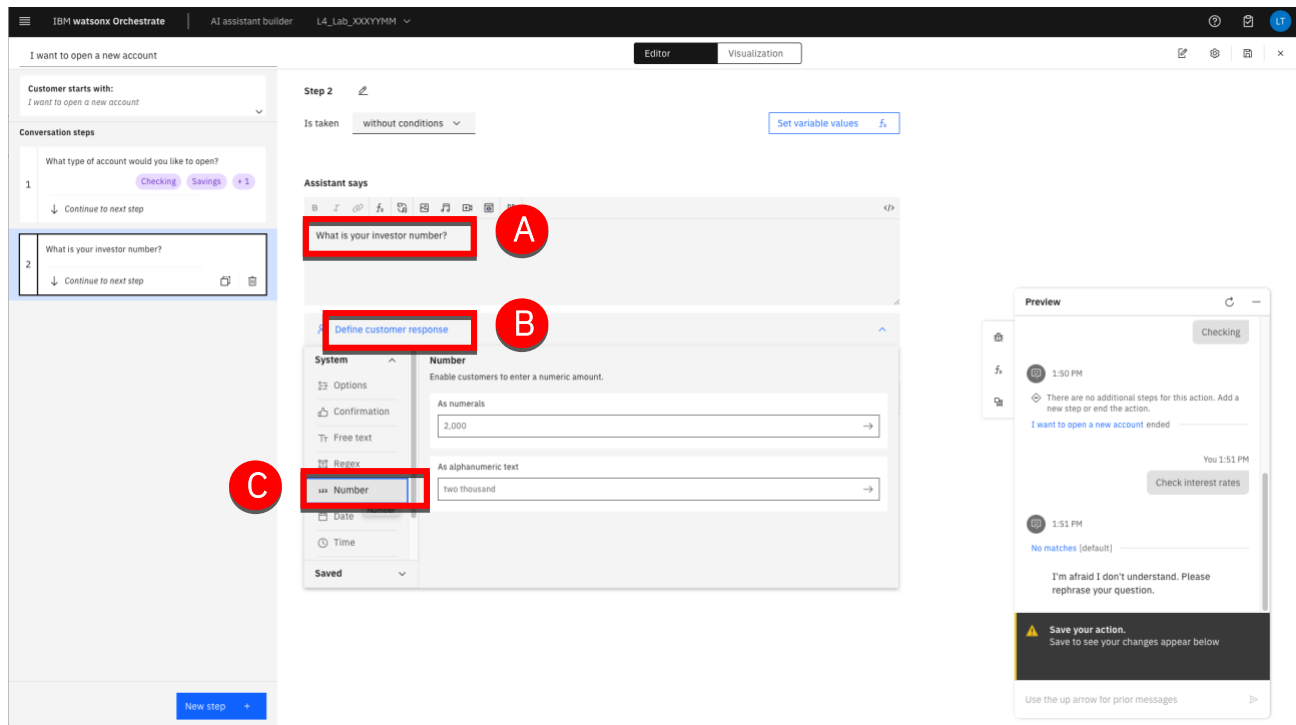


Let's create the corresponding Step 2.

1. Click **New Step +** (A).



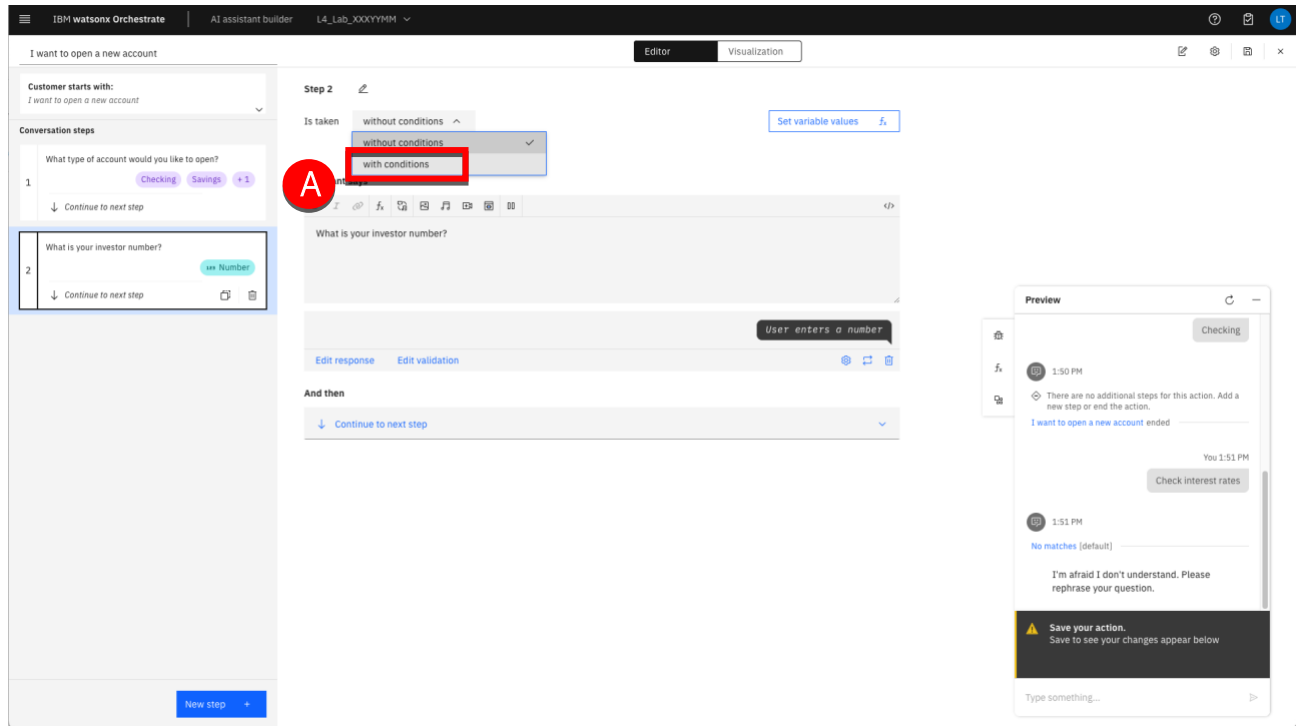
2. Type **What is your investor number?** in the **Assistant Says** box (A), click **Define customer response** (B) then select **Number** (C).



Next, you need to add some flow logic. Currently, this action will always ask for an investor number, even if the user wants to open a checking or savings account.

An investor number should only be gathered if the customer wants to open an investment account. To implement this logic, you need to add a condition to this step.

3. Change the **Is taken** value from **without conditions** to **with conditions** (A).



An action condition specifies requirements which must be met for a step to be triggered.

More formally, an action condition is a Boolean test, based on some runtime value; the action runs only if the test evaluates as true.

A basic action condition is expressed in the following form:

If {variable} {operator} {value}

Let's configure the conditions of Step 2.

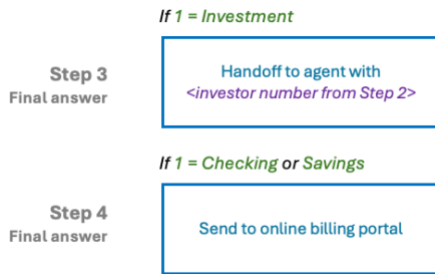
4. Verify that the Variable of the condition is set to the Action Step variable: **1. What type of account would you like to open?** (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, the 'Conversation steps' panel shows Step 1: 'What type of account would you like to open?' with a 'Checking' button and a '+1' icon. Step 2: 'What is your investor number?' with a 'Use Number' button. The main editor area shows Step 2 configuration. Under 'Is taken', 'with conditions' is selected. A red box labeled 'A' highlights the condition variable '1. What type of account would you like to open?'. The condition operator is 'is'. The condition value is 'Checking'. The 'Assistant says' panel shows the prompt 'What is your investor number?'. The 'Preview' panel on the right shows a chat history with a user message 'I want to open a new account' and an assistant response 'Check interest rates'.

5. Verify that the operator of the condition is set to **is** (A) **then** set the value of the condition to **Investment** (B).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, the 'Conversation steps' panel shows Step 1: 'What type of account would you like to open?' with a 'Checking' button and a '+1' icon. Step 2: 'What is your investor number?' with a 'Use Number' button. The main editor area shows Step 2 configuration. Under 'Is taken', 'with conditions' is selected. A red box labeled 'A' highlights the condition variable '1. What type of account would you like to open?'. The condition operator is 'is'. A dropdown menu is open, showing options: 'defined', 'Checking', 'Savings', and 'Investment'. A red box labeled 'B' highlights the 'Investment' option. The 'Assistant says' panel shows the prompt 'What is your investor number?'. The 'Preview' panel on the right shows a chat history with a user message 'I want to open a new account' and an assistant response 'Check interest rates'.

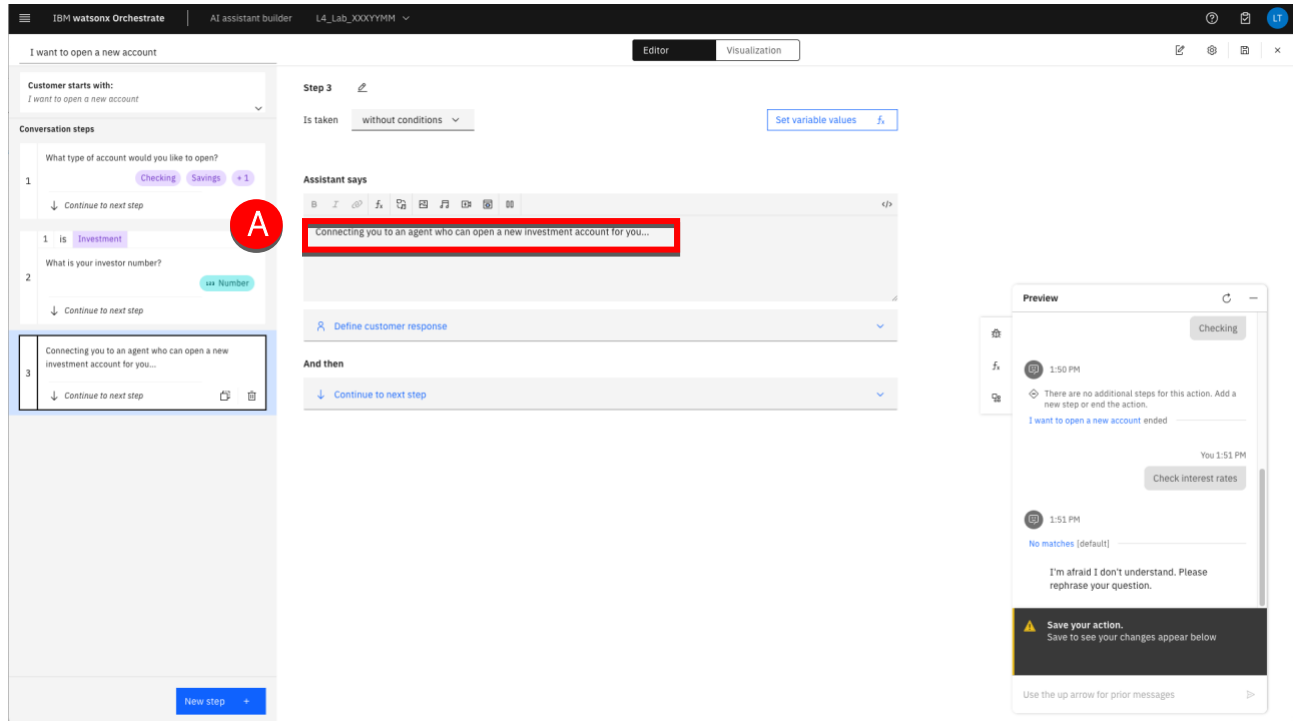
Now, you will add steps 3 and 4, each of which provide a final outcome for the user:



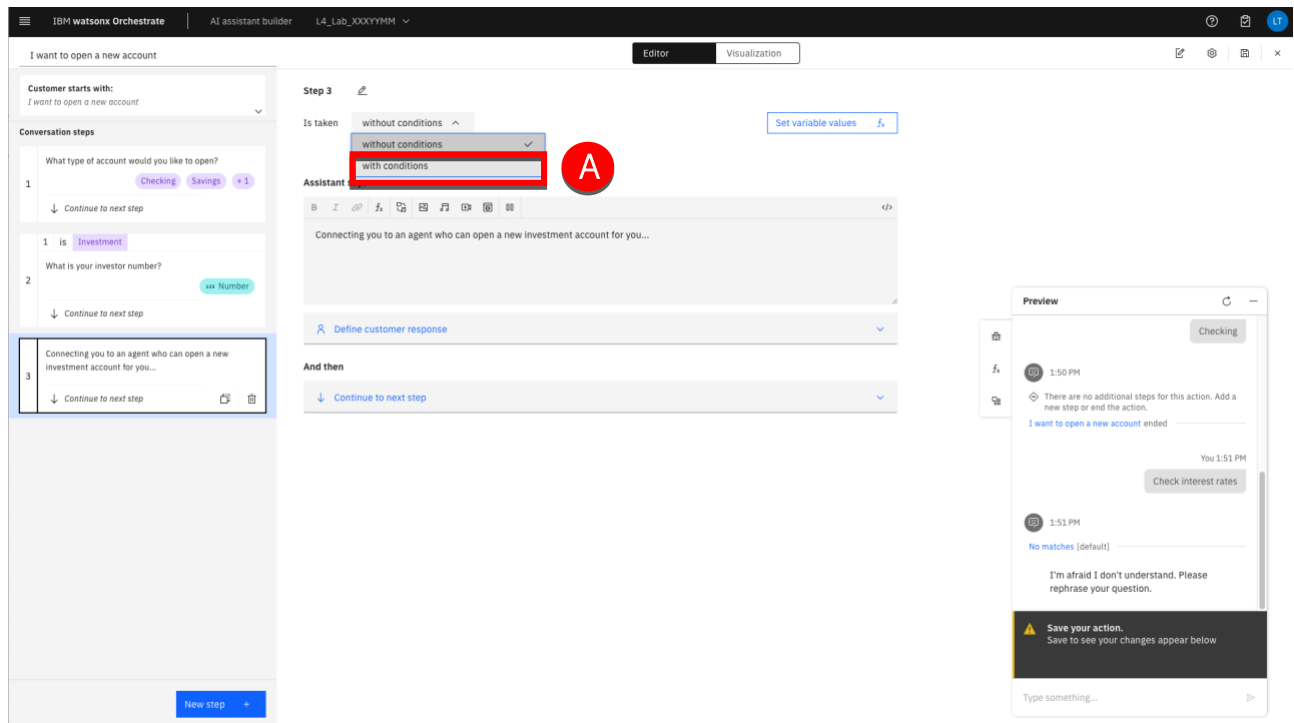
6. Click New Step + (A).

The screenshot displays the IBM Watsonx Orchestrate AI assistant builder interface. The main workspace shows Step 2 configuration. The 'Conditions' section is set to 'with conditions' and includes a condition: 'If All of this is true: 1. What type of account would you ... is Investment'. The 'Assistant says' section contains the prompt 'What is your investor number?'. The 'And then' section is set to 'Continue to next step'. A red circle labeled 'A' highlights the 'New step +' button at the bottom left. On the right, a 'Preview' panel shows a chat conversation: 'Checking', '1:50 PM', 'There are no additional steps for this action. Add a new step or end the action.', 'I want to open a new account ended', 'You 1:51 PM', 'Check interest rates', '1:51 PM', 'No matches [default]', 'I'm afraid I don't understand. Please rephrase your question.', and a 'Save your action.' warning at the bottom.

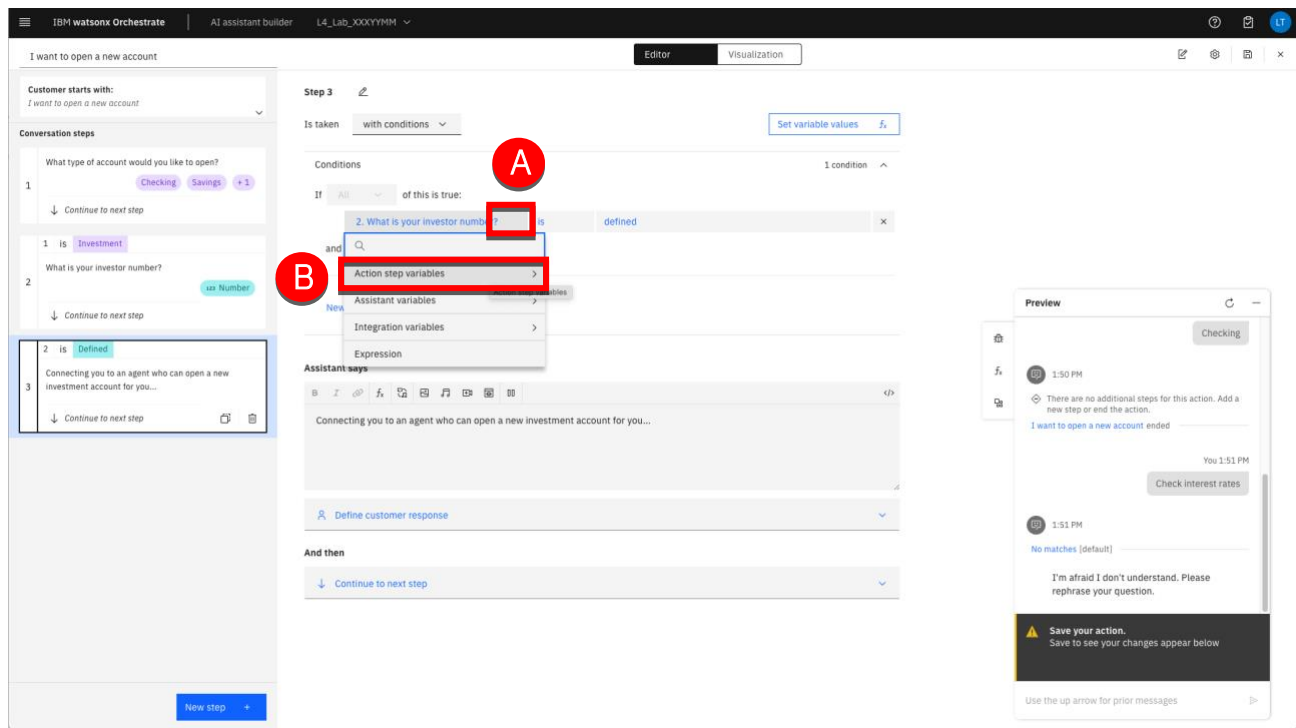
7. Enter **Connecting you to an agent who can open a new investment account for you...** in the **Assistant says** field (A).



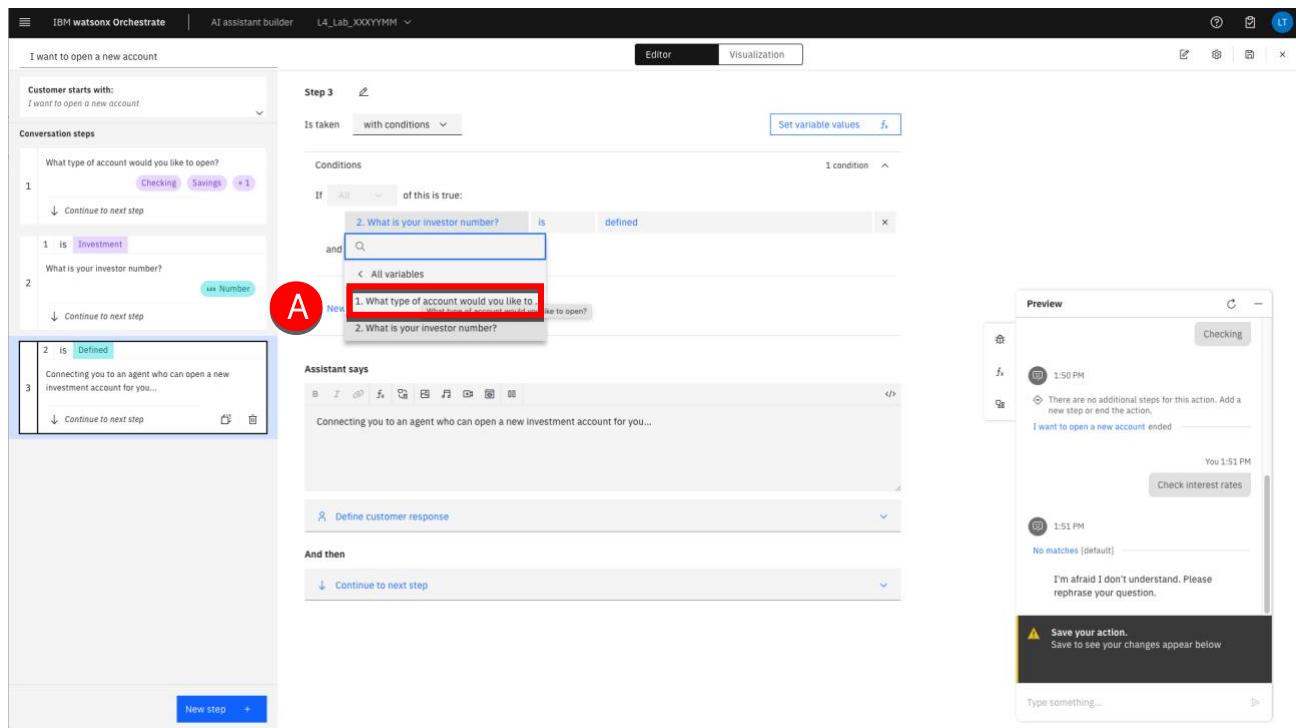
8. Select **with conditions** to make this step conditional (A).



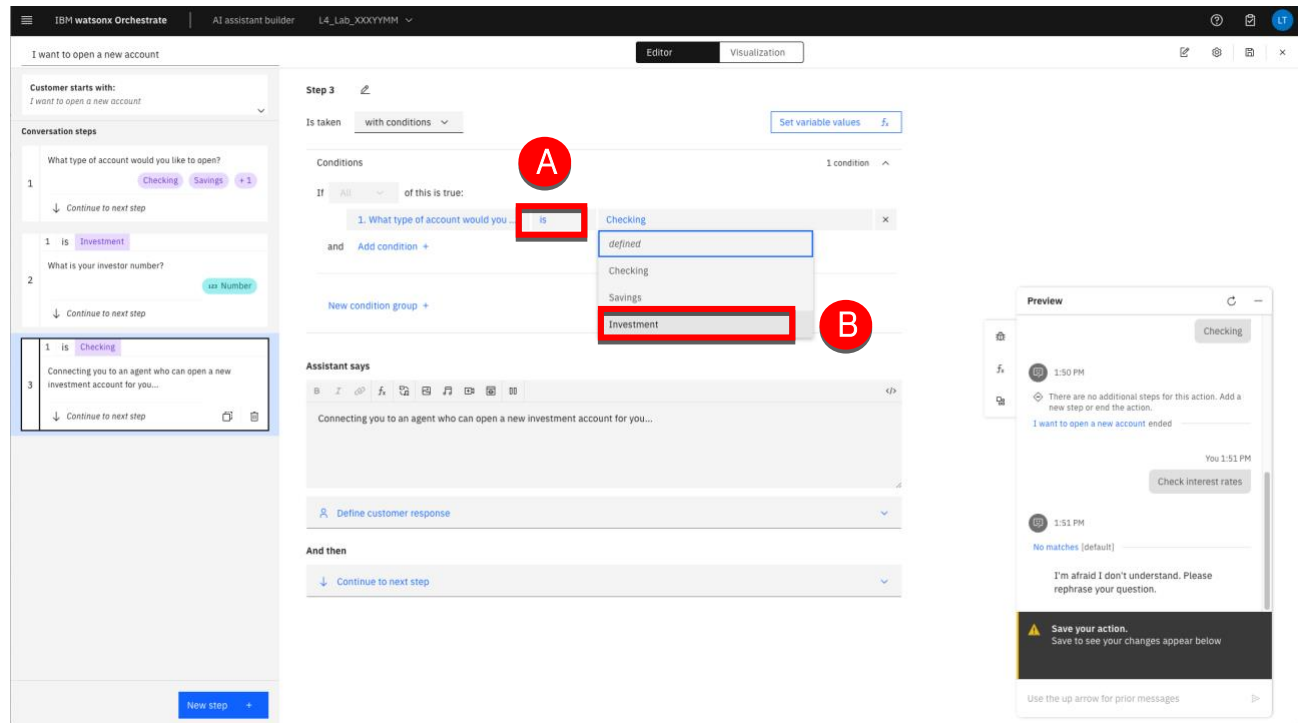
9. Click the **Variable** box (A) then select **Action step variables** in the dropdown (B).



10. Select **1. What type of account would you like...** in the dependent dropdown (A).



11. Verify that the **operator** is set to **is** (A) then select **Investment** as the variable value for the condition (B).



For this step, you don't need to gather any information from the user, so you can leave the **Define customer response** section empty. However, you do need to set the assistant to route this conversation to a human agent.

12. Expand the **And then** setting (A).

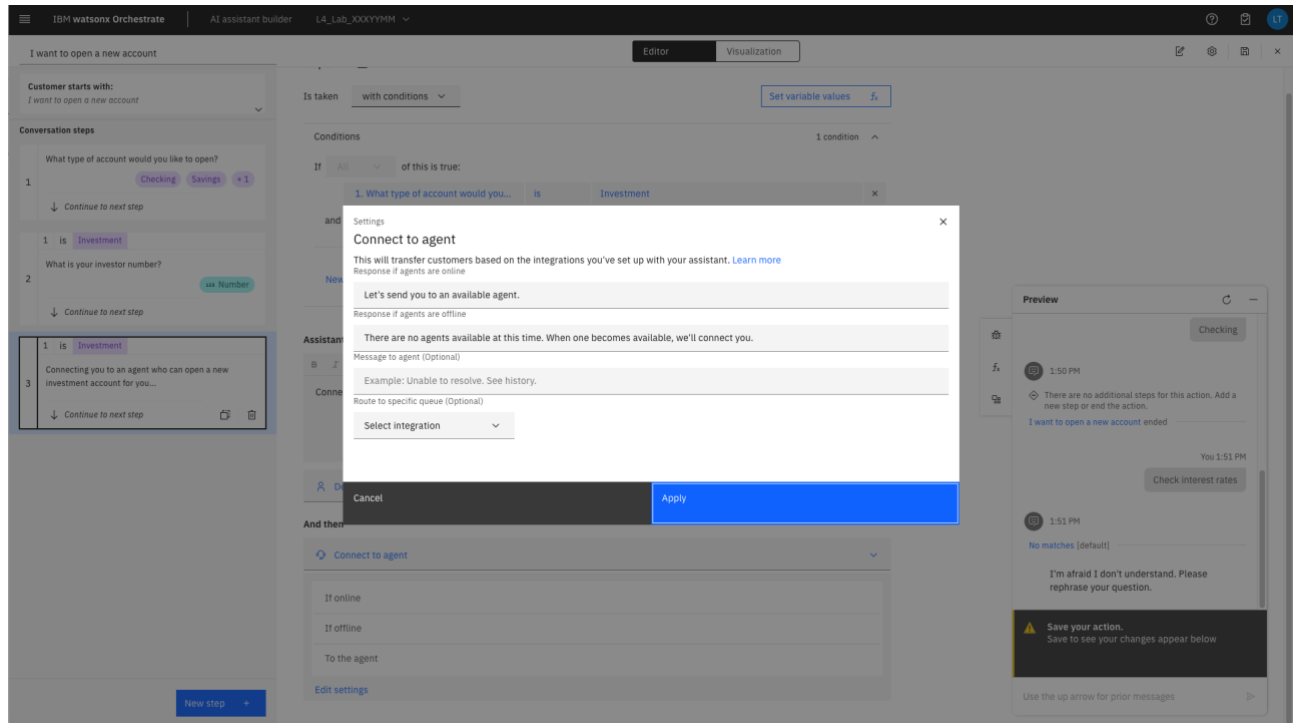
The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. On the left, the 'Conversation steps' panel shows a flowchart with three steps. Step 3 is highlighted with a blue border. The main editor area shows the configuration for Step 3. The 'Is taken' dropdown is set to 'with conditions'. The 'Conditions' section shows a single condition: '1. What type of account would you ... is Investment'. The 'Assistant says' section contains the text 'Connecting you to an agent who can open a new investment account for you...'. The 'And then' section is expanded, showing a dropdown menu with the option 'Connect to agent' highlighted. A red box and a red circle with the letter 'A' are placed over the 'Connect to agent' option. On the right, the 'Preview' panel shows a chat conversation with a customer and an assistant. The assistant's response is 'I'm afraid I don't understand. Please rephrase your question.' A warning message at the bottom of the preview panel says 'Save your action. Save to see your changes appear below'.

13. Select **Connect to agent** (A). (Human agent in this context).

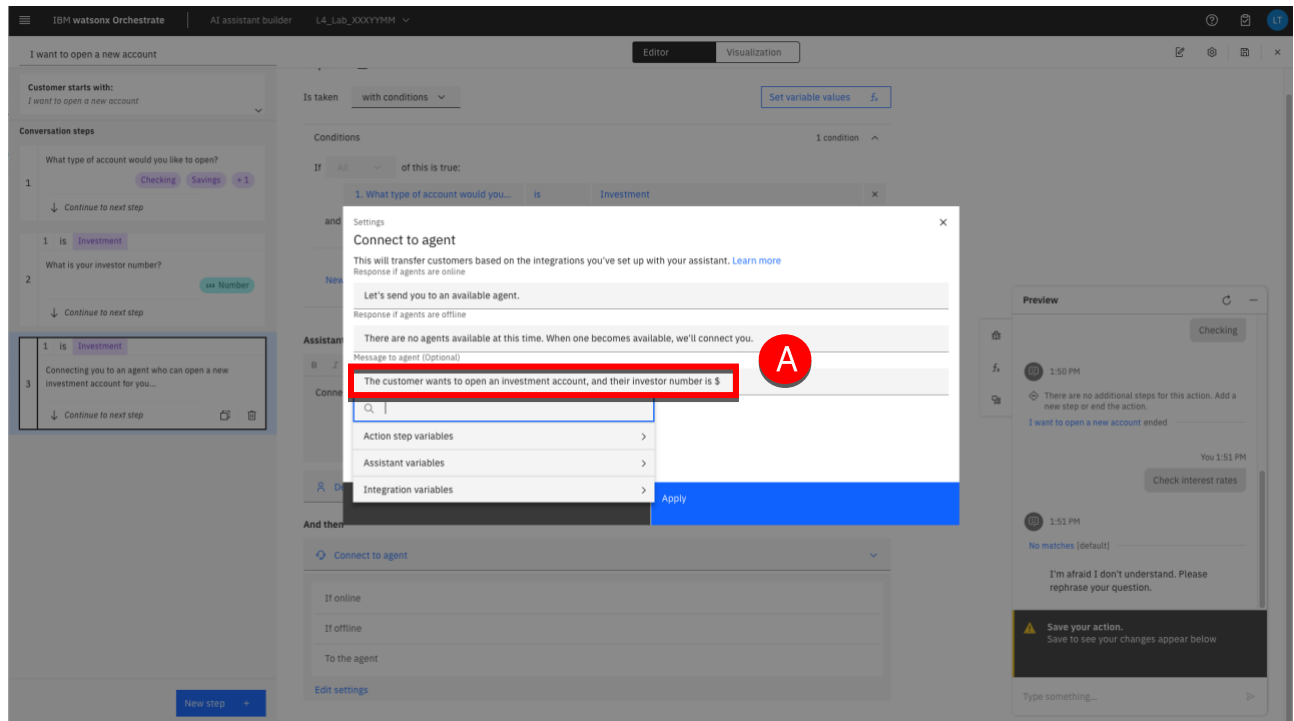
This screenshot is identical to the one above, showing the IBM watsonx Orchestrate AI assistant builder interface. The 'And then' dropdown menu is expanded, and the 'Connect to agent' option is highlighted with a red box and a red circle with the letter 'A'. The 'Preview' panel on the right shows the same chat conversation, with the assistant's response 'I'm afraid I don't understand. Please rephrase your question.' and the warning message 'Save your action. Save to see your changes appear below'.

In the **Connect to agent** popup shown below, note that the first two inputs are pre-populated – **Response if agents are online** and **Response if agents are offline**. You may rephrase these two pre-populated responses if you wish.

In the third input, **Message to agent (Optional)**, you can insert the context that you gathered for the human agent to review: the customer’s intent to open an investment account, and the customer’s investor number.



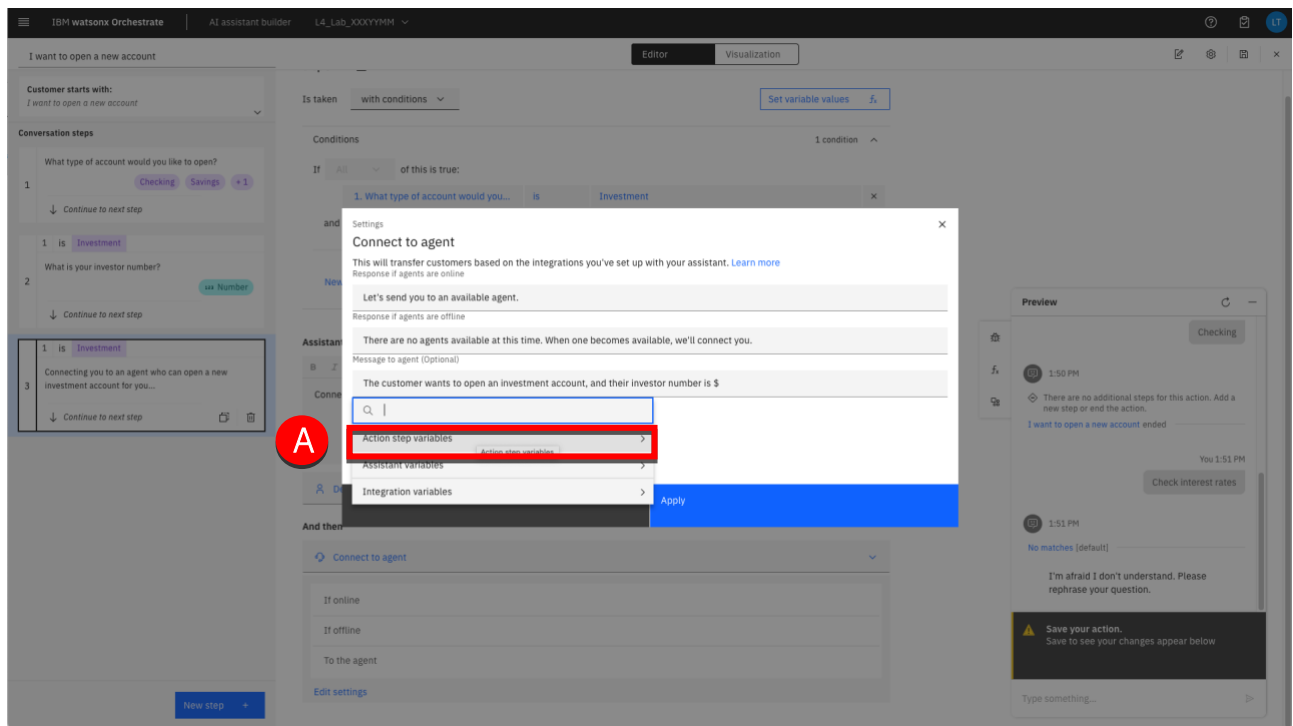
14. Type The customer wants to open an investment account, and their investor number is \$ in the **Message to Agent** field (A).



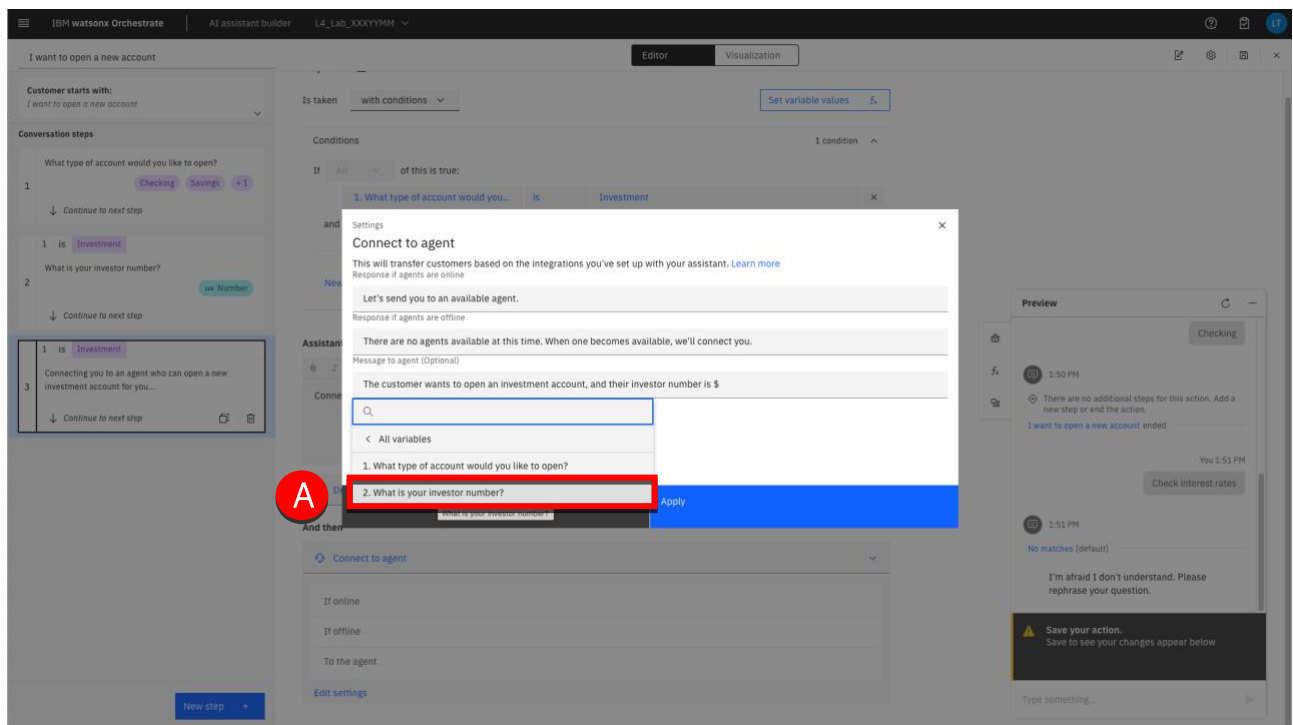
15.

**Note:** as you type the “\$” sign, a quick select box appears.

16. Select Action step variables (A).

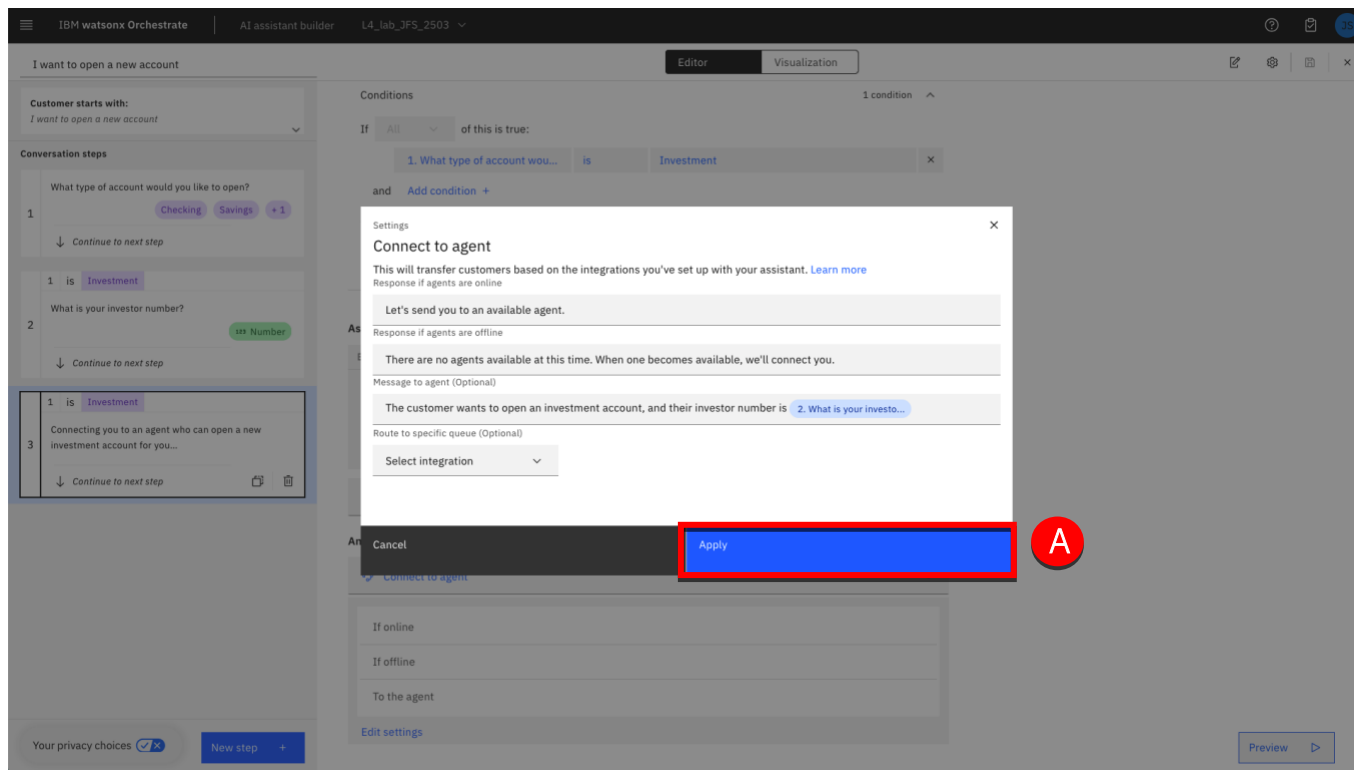


17. Select 2. What is your investor number? (A).



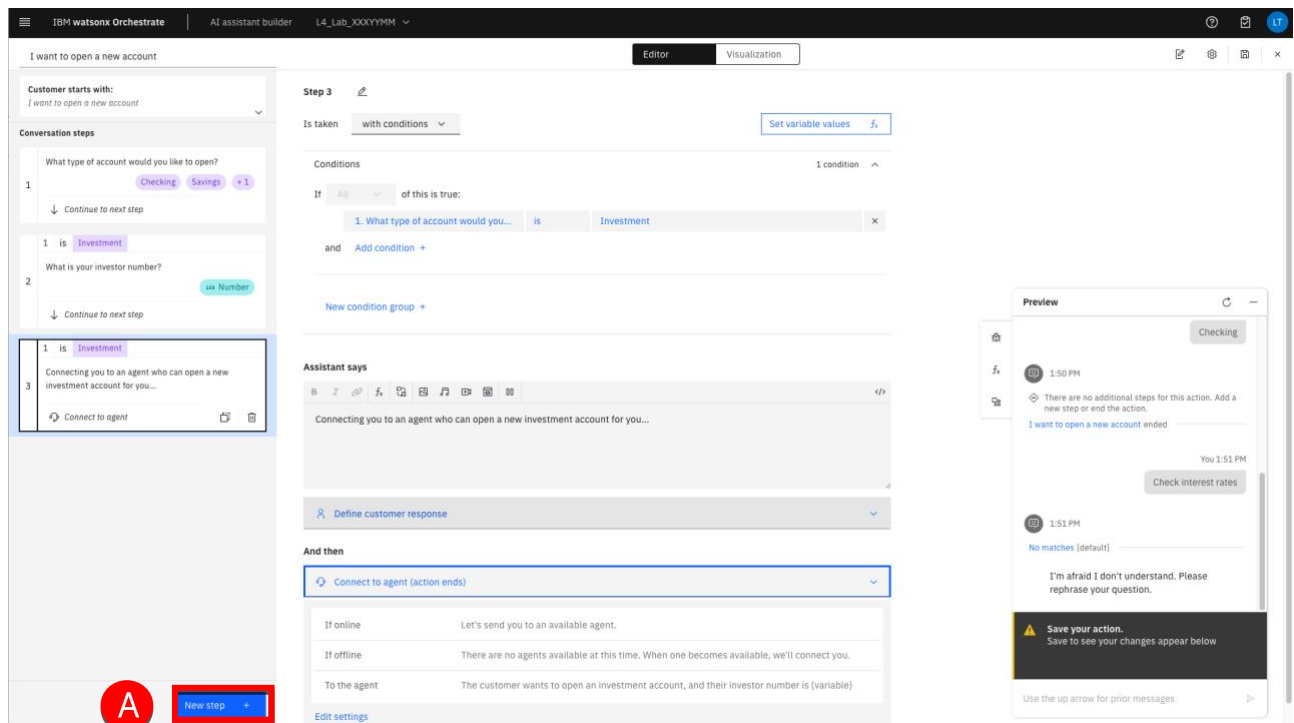
This will pass on the customer's response with their investor number to the agent.

18. The popup should now look like the image below; click **Apply** (5).

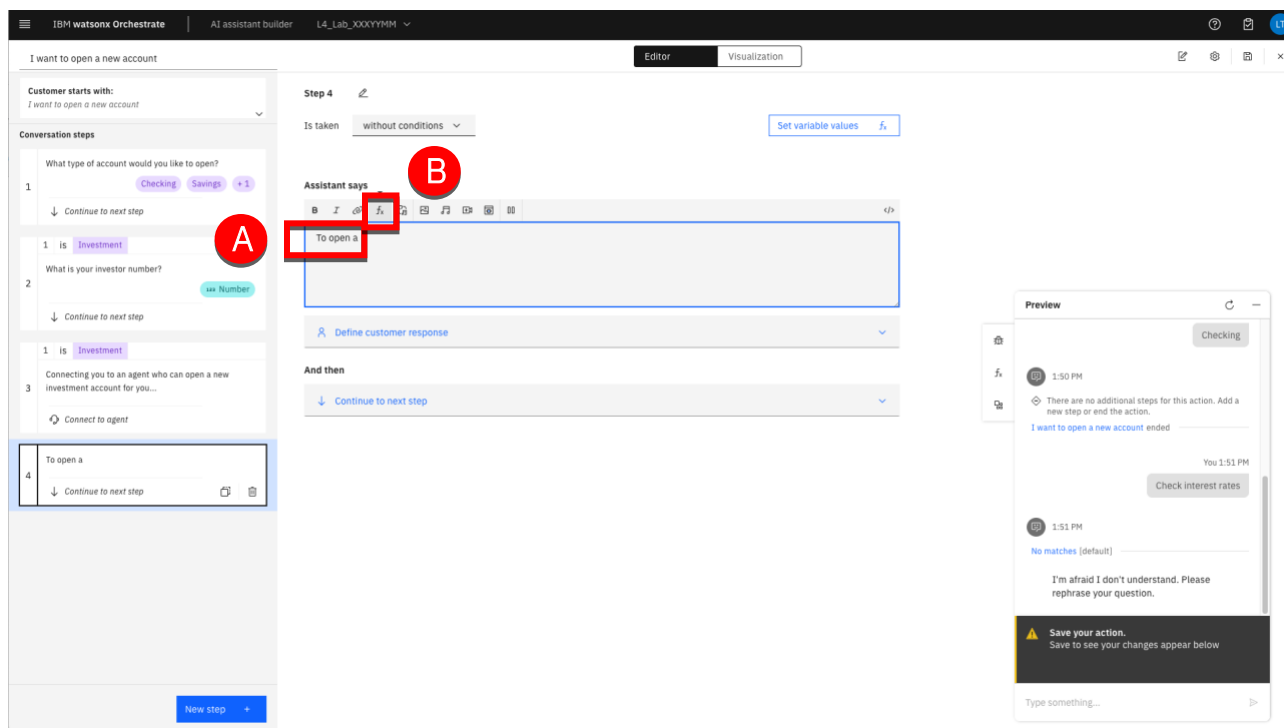


Let's now create step 4 – the final response step.

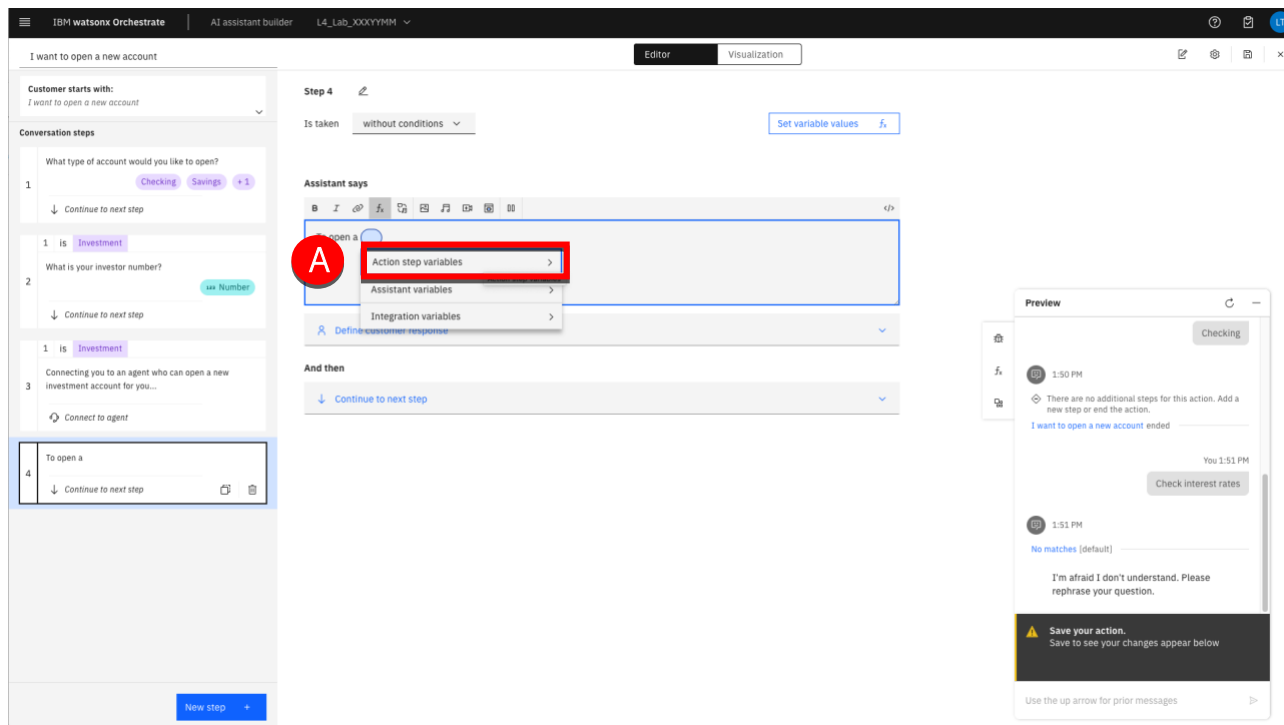
19. Click on **New step +** (A).



20. In **Assistant says**, enter To open a (A) then click the **Variable** icon (B) (or type “\$”).



21. Select **Action step variable** (A).



22. Select 1. What type of account would you like to open? (A).

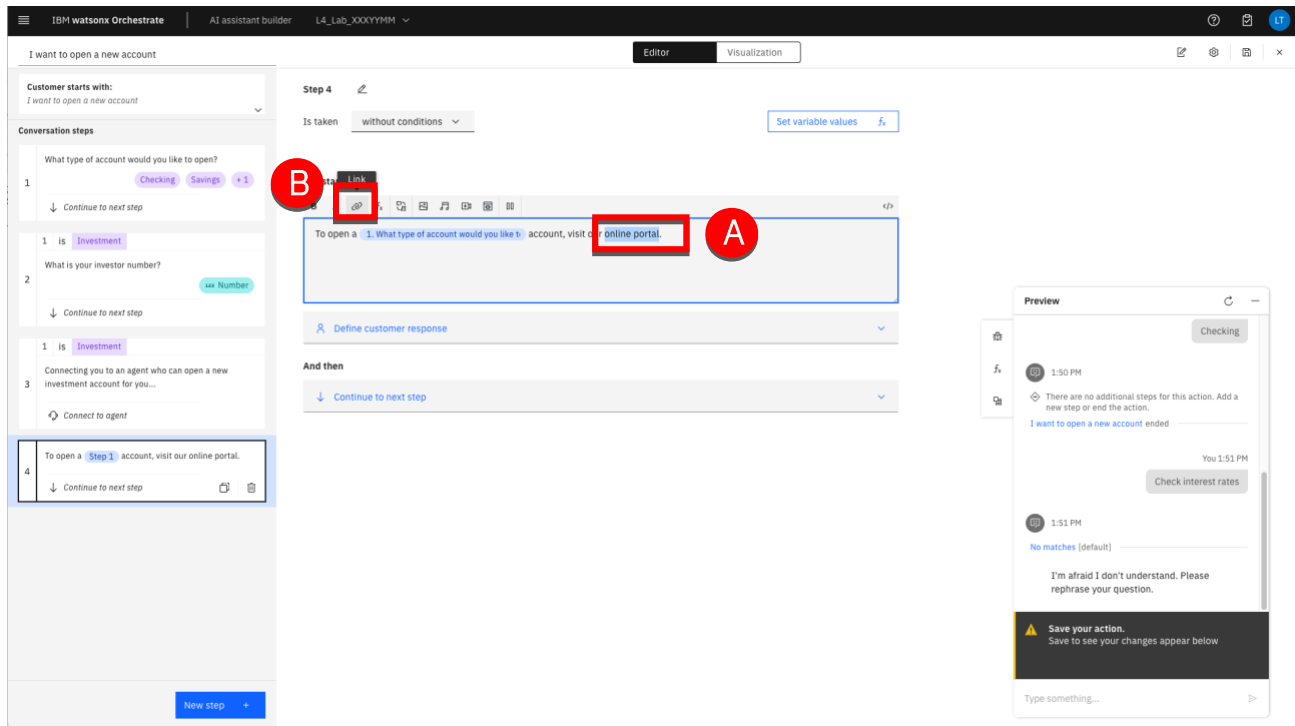
The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The top bar includes the logo, 'AI assistant builder', and a dropdown menu showing 'L4\_lab\_3FS\_2503'. The main workspace is divided into two panels: 'Editor' and 'Visualization'. The 'Editor' panel shows the configuration for Step 4. The 'Assistant says' section contains a text input field with a red box around the text '1. What type of account would you like to open?'. A red circle with the letter 'A' is next to this text. The 'And then' section shows a dropdown menu with 'Continue to next step' selected. The 'Visualization' panel shows a preview of the assistant's response, with the same text '1. What type of account would you like to open?' highlighted in a red box. A red circle with the letter 'A' is next to this text. The left sidebar shows the conversation steps, with Step 4 highlighted. The bottom of the interface includes a 'Your privacy choices' link and a 'New step' button.

23. Complete the **Assistant says** phrase by typing account, visit our online portal. (A).

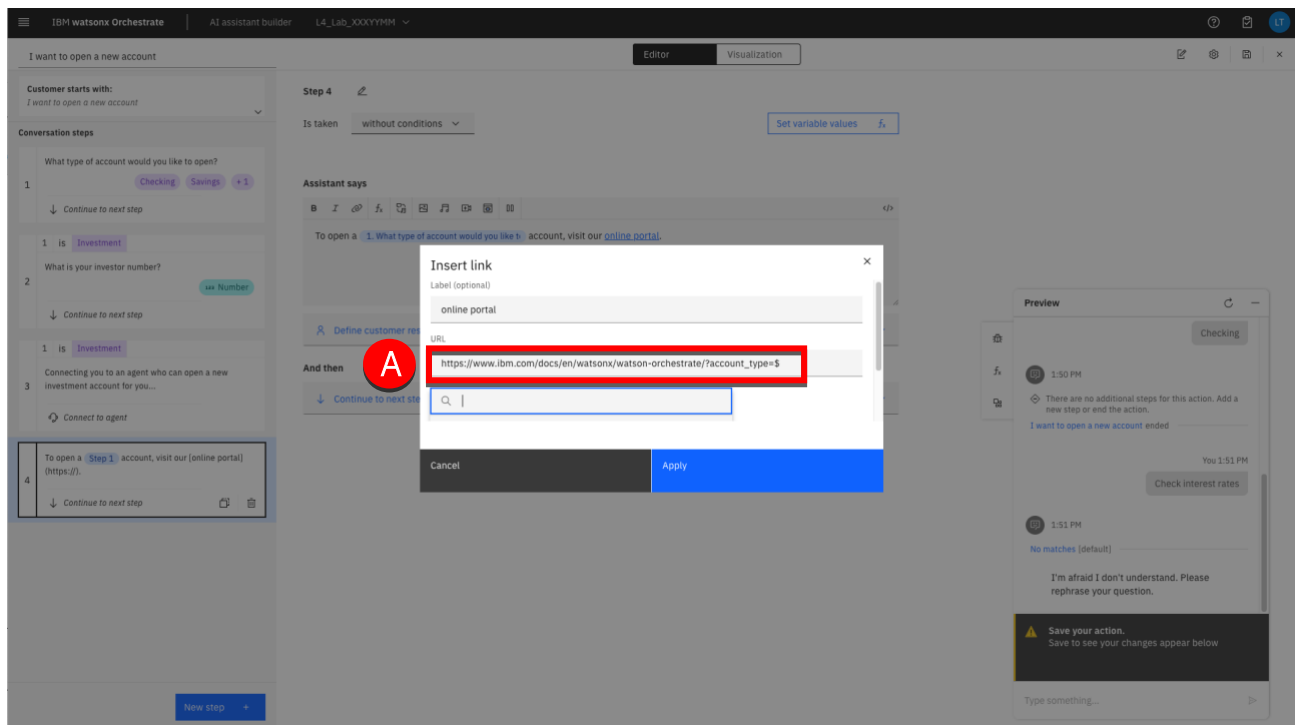
The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The top bar includes the logo, 'AI assistant builder', and a dropdown menu showing 'L4\_lab\_3FS\_2503'. The main workspace is divided into two panels: 'Editor' and 'Visualization'. The 'Editor' panel shows the configuration for Step 4. The 'Assistant says' section contains a text input field with a red box around the text '1. What type of account would you like to open? account, visit our online portal.'. A red circle with the letter 'A' is next to this text. The 'And then' section shows a dropdown menu with 'Continue to next step' selected. The 'Visualization' panel shows a preview of the assistant's response, with the same text '1. What type of account would you like to open? account, visit our online portal.' highlighted in a red box. A red circle with the letter 'A' is next to this text. The left sidebar shows the conversation steps, with Step 4 highlighted. The bottom of the interface includes a 'Your privacy choices' link and a 'New step' button.

Let's now add a hyperlink to the **Assistant Says** section:

24. Highlight the **online portal** text (A) then click the URL link button (B).

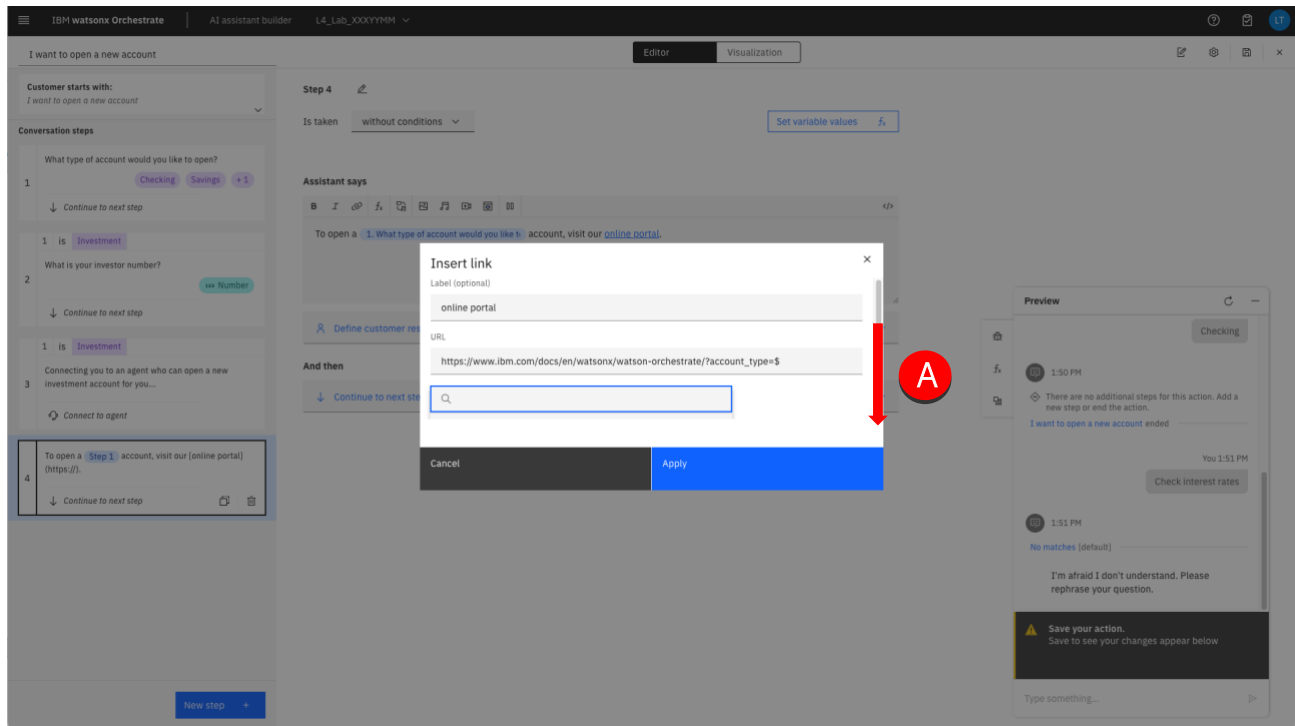


25. Type the url: [https://www.ibm.com/docs/en/watsonx/watson-orchestrate/?account\\_type=\\$](https://www.ibm.com/docs/en/watsonx/watson-orchestrate/?account_type=$) in the **URL** (A).

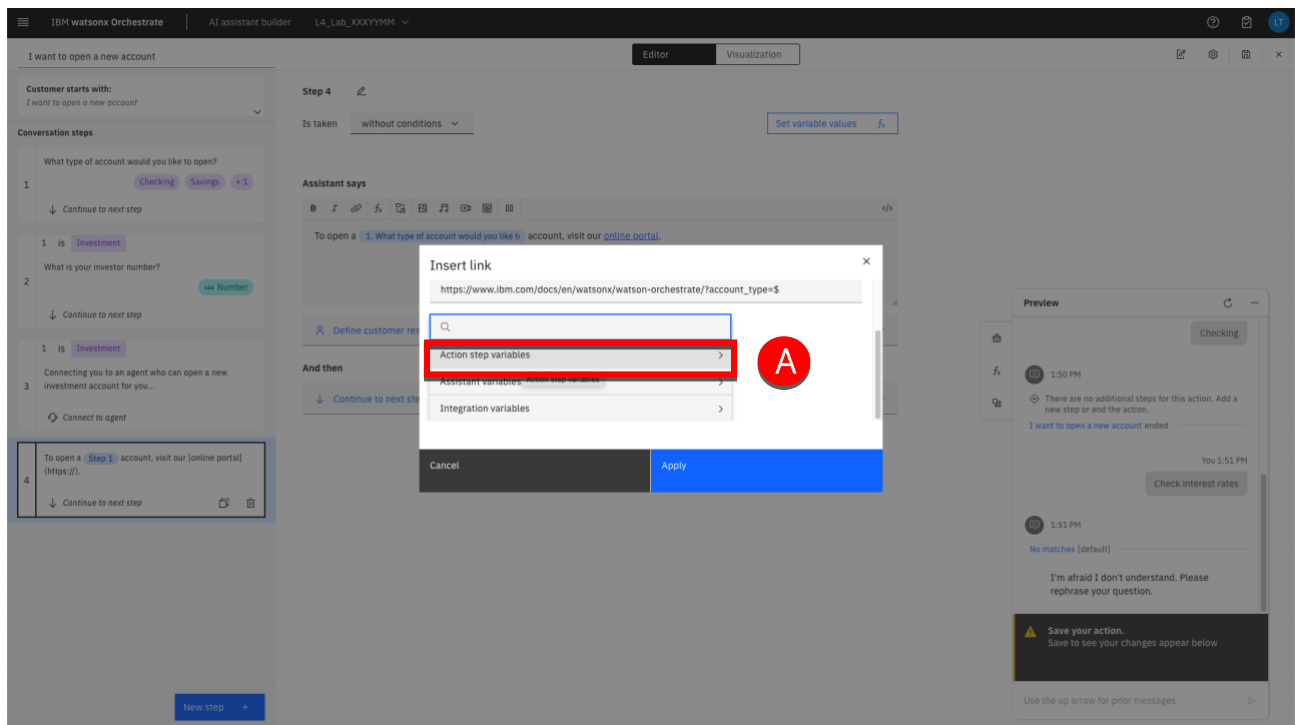


The “\$” sign will once again invoke our Action variables, and you can select the type of account.

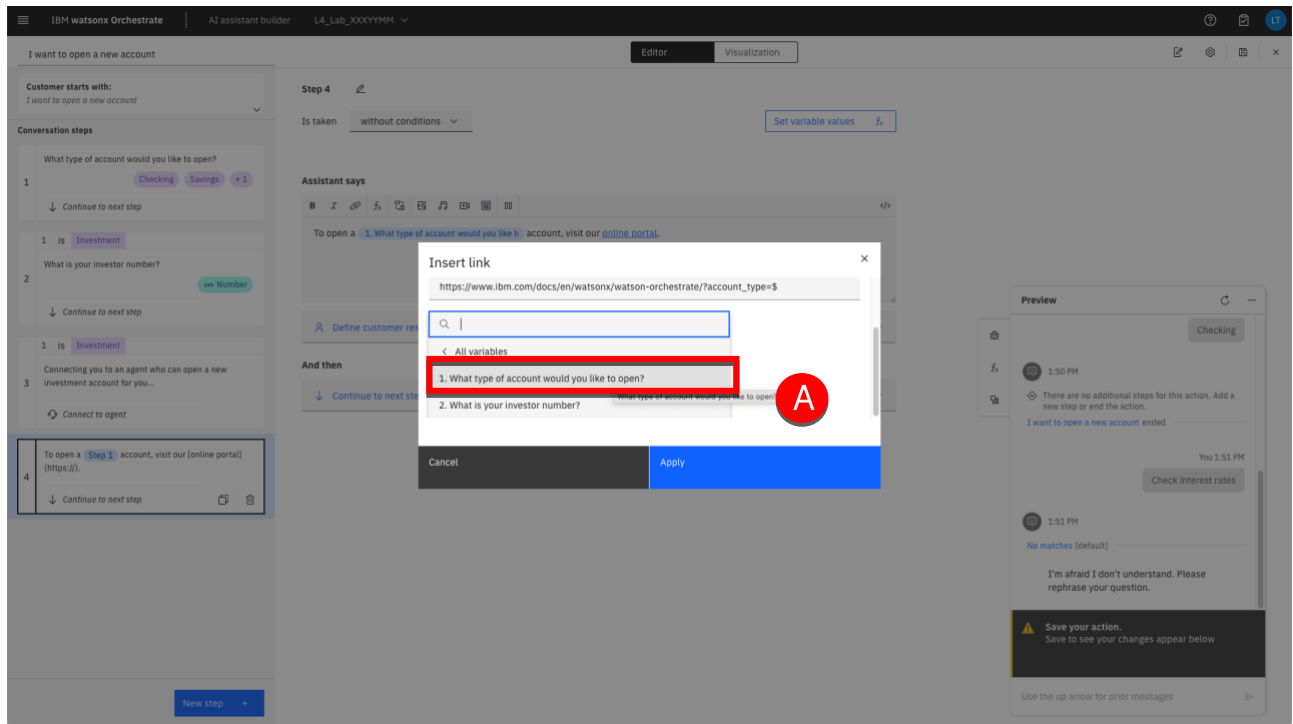
26. Scroll down (A).



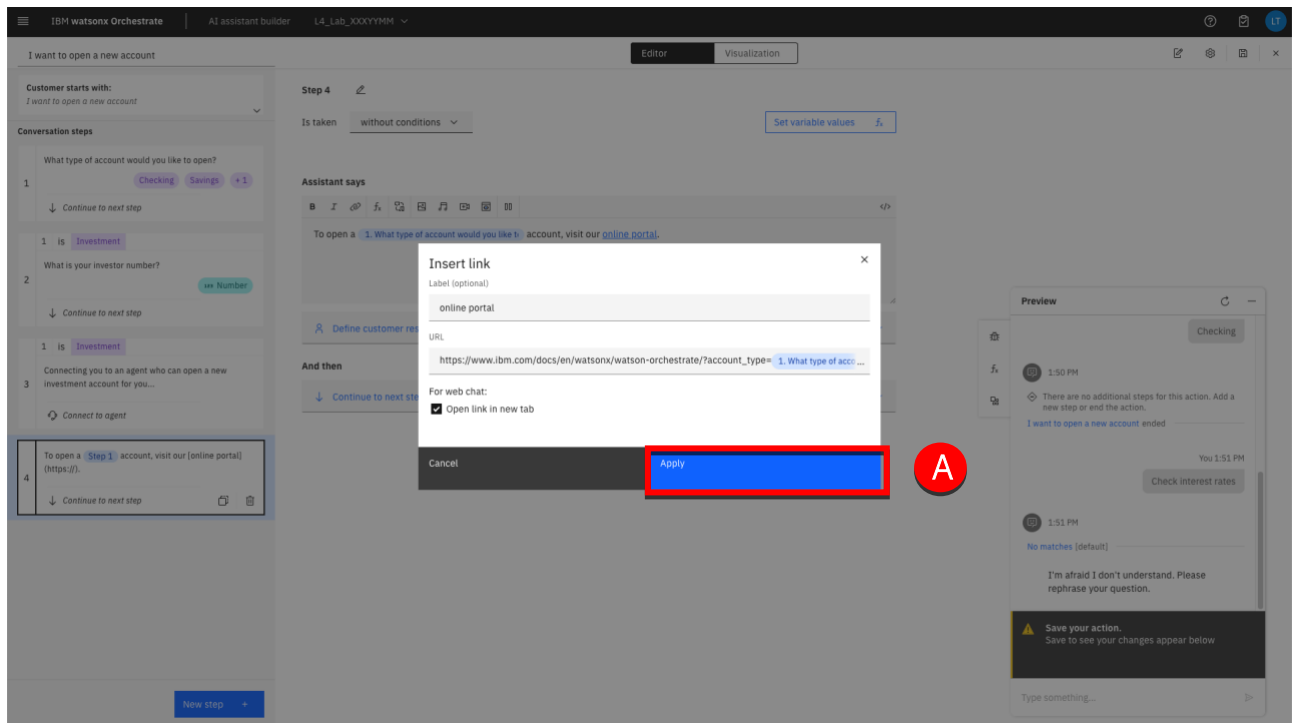
27. Select the Action Step Variable (A).



28. Select 1. What type of account would you like to open? (A) .



29. Click Apply (A).



Note that this URL does not link to a savings or checking account. It is an illustration of how a builder would pass a parameter through a link. Either link will redirect to the watsonx Orchestrate documentation.

Now that the action is completed, you will end the action.

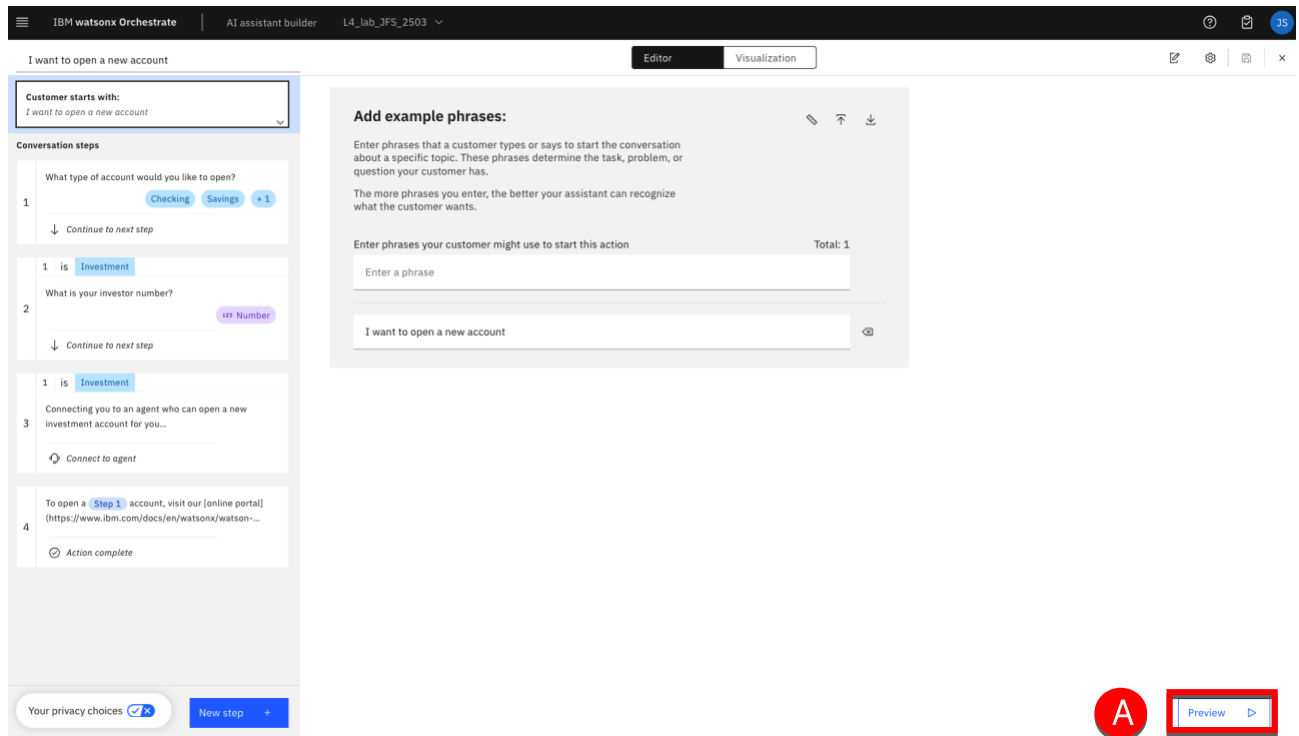
30. Expand the **And then** menu (A) then select **End the action** (B).

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar shows a conversation flow with steps 1 through 4. Step 4 is highlighted, showing a customer request to open a new account. The main panel shows the configuration for Step 4, which is 'without conditions'. The 'Assistant says' section contains a message: 'To open a [1. What type of account would you like to open?] account, visit our [online portal]'. The 'And then' menu is expanded, showing options like 'Continue to next step', 'Re-ask previous step(s)', 'Go to a subaction', 'Use an extension', 'Search for the answer', 'Connect to agent', and 'End the action'. A red box labeled 'A' highlights the 'And then' menu, and a red box labeled 'B' highlights the 'End the action' option.

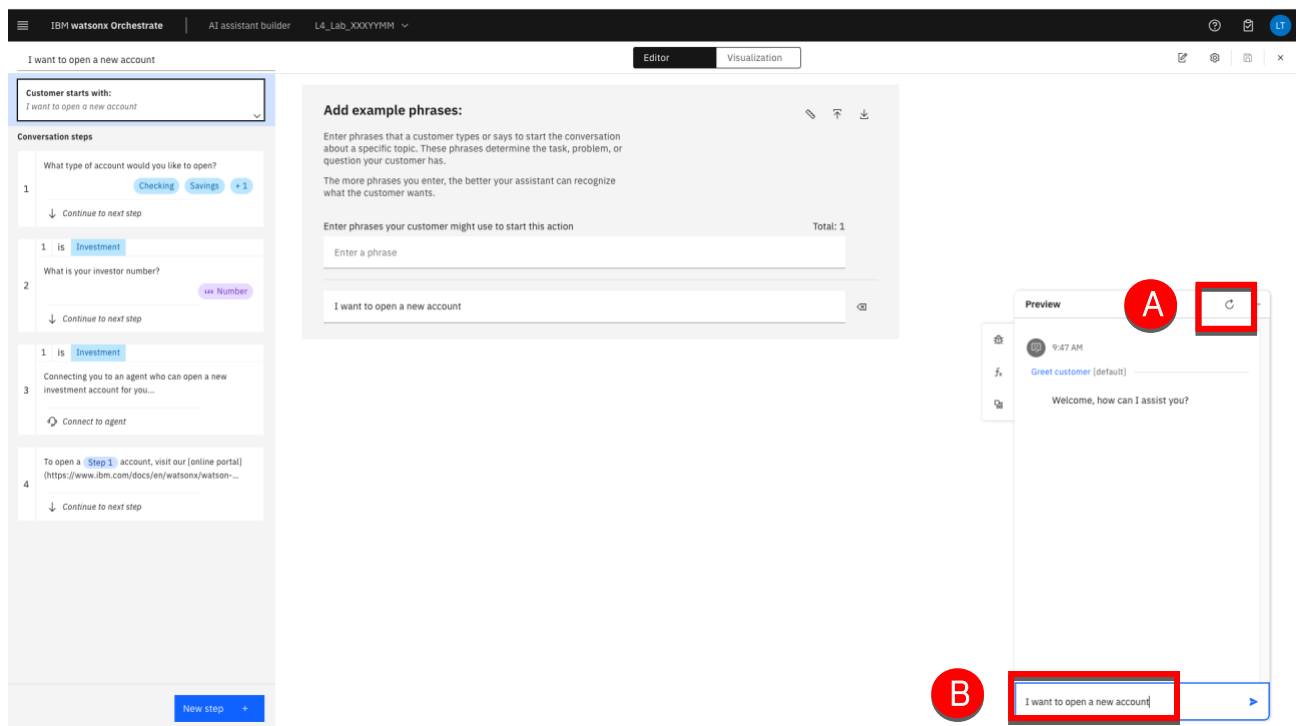
## Test the action

Now that you have configured this action, test out the assistant's behavior.

1. Click **Preview** (A).



2. Click the refresh button in the preview panel (A), then type the phrase: I want to open an account (B).



3. Click the **Investment** button (A) or type the word investment.

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The top navigation bar includes the IBM watsonx Orchestrate logo, the AI assistant builder title, and a dropdown menu showing 'L4\_lab\_3FS\_2503'. The main interface is divided into three sections:

- Customer starts with:** A text box containing 'I want to open a new account'.
- Conversation steps:** A list of steps for the conversation flow. Step 1 asks 'What type of account would you like to open?' with buttons for 'Checking', 'Savings', and '+ 1'. Step 2 asks 'What is your investor number?' with a '+ Number' button. Step 3 connects to an agent. Step 4 provides a link to the online portal and marks the action as complete.
- Add example phrases:** A section for entering phrases that a customer might use to start the conversation. It includes a text box for 'Enter a phrase' and a 'Total: 1' indicator. Below this, a list of phrases is shown, including 'I want to open a new account'.

On the right side, there is a **Preview** window showing a chat flow. The chat flow starts with a greeting from the customer, followed by a response from the assistant asking 'Welcome, how can I assist you?'. The customer then says 'I want to open an account'. The assistant responds with 'I want to open a new account recognized' and asks 'What type of account would you like to open?'. The preview shows three buttons: 'Checking', 'Savings', and 'Investment'. The 'Investment' button is highlighted with a red box and labeled with a red circle containing the letter 'A'.

**Note:** Slight misspellings or abbreviations of the phrase investment will still work. For instance, you can try typing the phrase “invest” and the assistant will correctly recognize the input. [This behavior](#) can be tuned by clicking **Edit Validation** in the **Define Customer Response** section of a step.

4. Enter a numeric value, such as 87654321 (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The main workspace is titled "I want to open a new account" and is in the "Editor" tab. On the left, the "Conversation steps" panel shows a sequence of steps: 1. "What type of account would you like to open?" with buttons for "Checking", "Savings", and "+1"; 2. "What is your investor number?" with a "+ Number" button; 3. "Connecting you to an agent who can open a new investment account for you..."; 4. "To open a Step 1 account, visit our [online portal] (https://www.ibm.com/docs/en/watsonx/watson...)"; and 5. "Action complete". The "Add example phrases" section on the right allows entering phrases that a customer might use to start this action. A "Preview" chat window on the right shows a simulated conversation: "Welcome, how can I assist you?" (You 9:56 AM), "I want to open a new account" (9:56 AM), "I want to open a new account recognized" (9:56 AM), "What type of account would you like to open?" (You 9:56 AM), "Investment" (9:56 AM), "What is your investor number?" (9:56 AM), and "87654321" (9:56 AM). A red circle with the letter "A" highlights the input field for the investor number in the preview window.

You have tested a full progression through your action.

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface, specifically the "Step 1" configuration. The main workspace is titled "I want to open a new account" and is in the "Editor" tab. The "Conversation steps" panel on the left shows the same sequence of steps as the previous screenshot. The "Step 1" configuration panel on the right shows the "Is taken" condition set to "without conditions" and a "Set variable values" button. The "Assistant says" section shows the prompt "What type of account would you like to open?" with buttons for "Checking", "Savings", and "Investment". The "And then" section shows the action "Continue to next step". A "Preview" chat window on the right shows a simulated conversation: "Investment" (You 11:01 AM), "What is your investor number?" (11:01 AM), "87654321" (You 11:02 AM), "Connecting you to an agent who can open a new investment account for you..." (11:02 AM), "Connect to an agent" (11:02 AM), "Live agent support" (11:02 AM), "Let's send you to an available agent." (11:02 AM), "Connect to agent" (11:02 AM), and "Use the up arrow for prior messages" (11:02 AM). A red circle with the letter "A" highlights the input field for the investor number in the preview window.

5. Refresh the Preview window (A), then type the phrase: Open account (B).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The main editor displays a conversation flow for opening a new account. The 'Add example phrases' section is active, showing a list of phrases including 'I want to open a new account'. The 'Preview' window on the right is labeled with a red circle 'A' and a refresh icon. Below the preview, a text input field is labeled with a red circle 'B' and contains the text 'Open account'.

6. Select Investment (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The main editor displays the conversation flow for opening a new account. The 'Assistant says' section is active, showing the text 'To open a 1. What type of account would you like to account, visit our online portal.' The 'Preview' window on the right is labeled with a red circle 'A' and shows the 'Investment' option selected in the 'What type of account would you like to open?' dialog.

## 7. Type actually I meant checking (A).

The screenshot shows the IBM Watsonx Orchestrate AI Assistant Builder interface. The top bar includes the IBM Watsonx Orchestrate logo, the AI assistant builder title, and a dropdown menu showing 'L4\_Lab\_XXXXYMM'. The main workspace is divided into two tabs: 'Editor' and 'Visualization'. The 'Editor' tab is active, showing a conversation flow for the task 'I want to open a new account'. The flow starts with a customer input 'I want to open a new account'. The first step asks 'What type of account would you like to open?' with buttons for 'Checking', 'Savings', and '+1'. The second step asks 'What is your investor number?' with a '+1 Number' button. The third step connects to an agent. The fourth step provides a link to the online portal. A 'Preview' window on the right shows a simulated conversation with a customer. A red circle with the letter 'A' highlights the input 'actually I meant checking' in the preview window.

The screenshot shows the IBM Watsonx Orchestrate AI Assistant Builder interface, specifically the configuration for Step 4. The top bar includes the IBM Watsonx Orchestrate logo, the AI assistant builder title, and a dropdown menu showing 'L4\_Lab\_JFS\_2503'. The main workspace is divided into two tabs: 'Editor' and 'Visualization'. The 'Editor' tab is active, showing the configuration for Step 4. The step is titled 'Step 4' and is configured to be taken 'without conditions'. The 'Assistant says' section contains a message: 'To open a 1. What type of account would you like to open account, visit our online portal.' The 'And then' section contains an action: 'End the action'. A 'Preview' window on the right shows a simulated conversation with a customer. The customer input 'actually I meant checking' is highlighted in the preview window.

The Assistant was able to recognize that the user changed their intent while in step 2. Then, it reassigned the Action Step Variable **1: What type of account would you like to open** to **checking**, then continued the flow .

8. Refresh the Preview window (A), then type the phrase Open a new Checking account (B).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The main editor displays a conversation flow for the intent "I want to open a new account". The flow includes steps for selecting account type (Checking, Savings, +1), investment type, and investor number. The "Add example phrases" section on the right contains the phrase "I want to open a new account". On the right side, there is a "Preview" window (A) and a text input field (B) containing "Open a new Checking account".

Note that the assistant skipped past step 1.

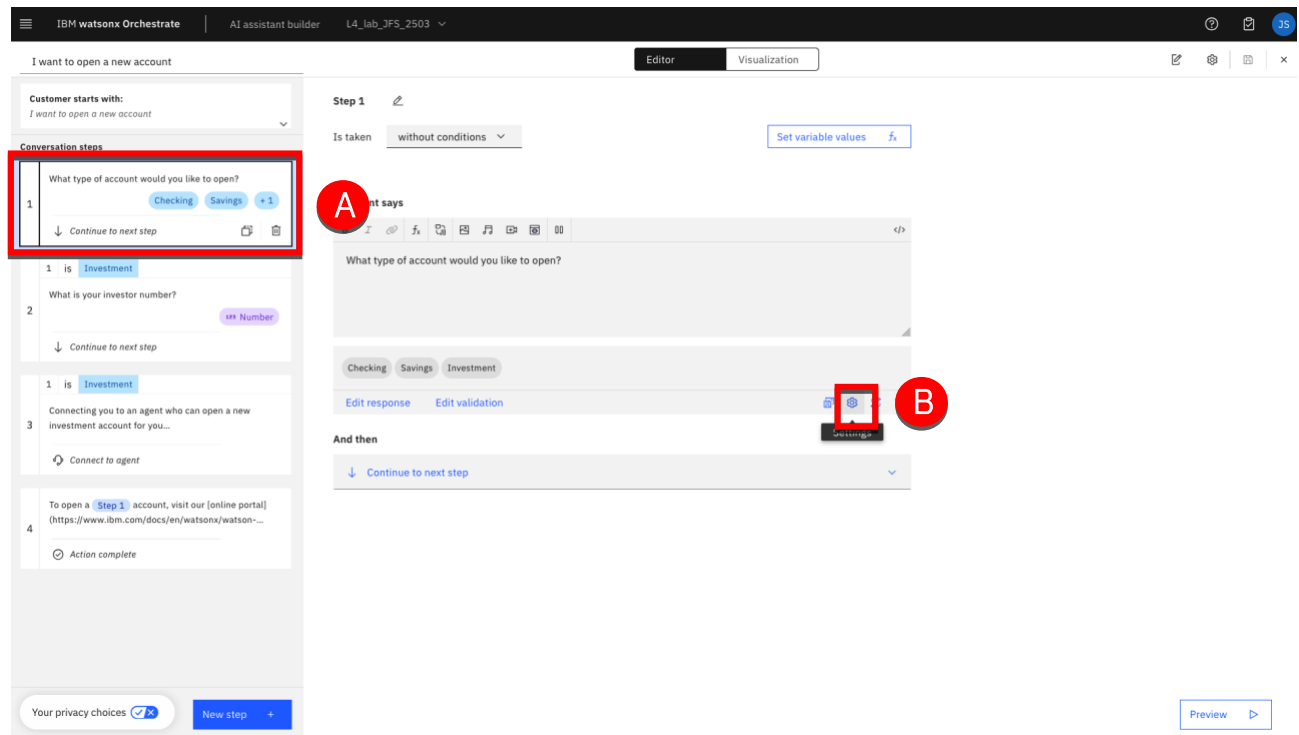
The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface, specifically the "Step 4" configuration. The "Assistant says" section contains the text "To open a 1. What type of account would you like to open account, visit our online portal." The "And then" section contains the action "End the action". The "Preview" window (A) shows the conversation history, including the user's input "Open a new Checking account" and the assistant's response "To open a Checking account, visit our online portal." (B).

The phrase “Open a new checking account” triggered the action to start, and the assistant automatically recognized the intent for a checking account. The assistant did not need to ask which type of account to open, so Step 1 was skipped.

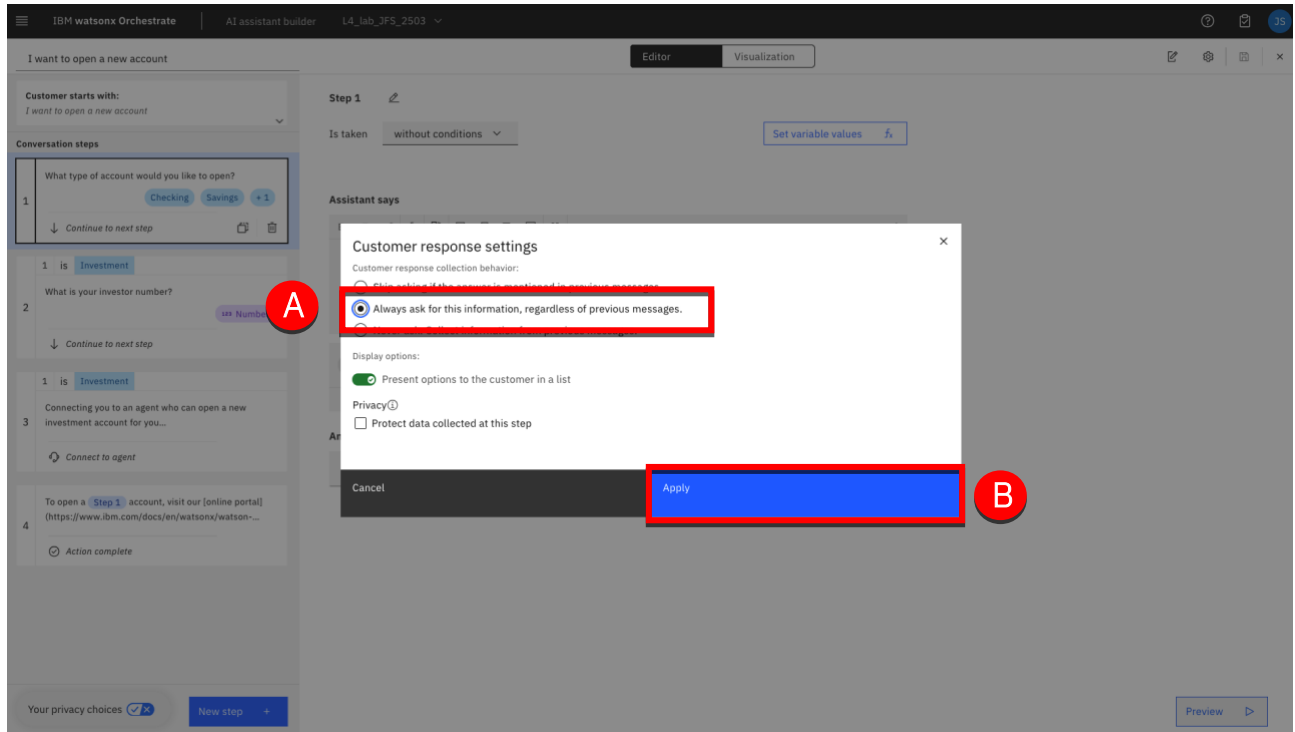
This is a simple example of the AI assistant’s [slot filling](#) feature.

This is useful behavior, but you may want to disable this behavior in certain circumstances.

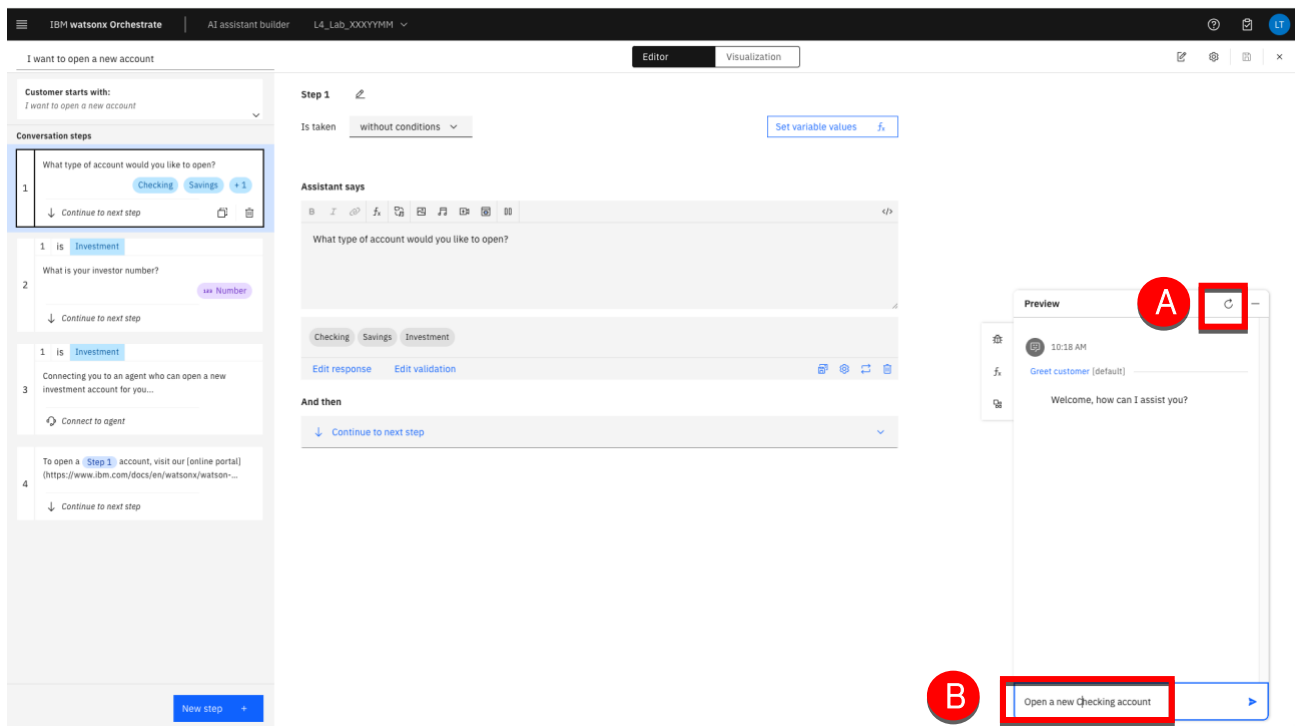
9. Click on **Step 1** (A), then click the **Settings** icon (B) in the define customer response section.



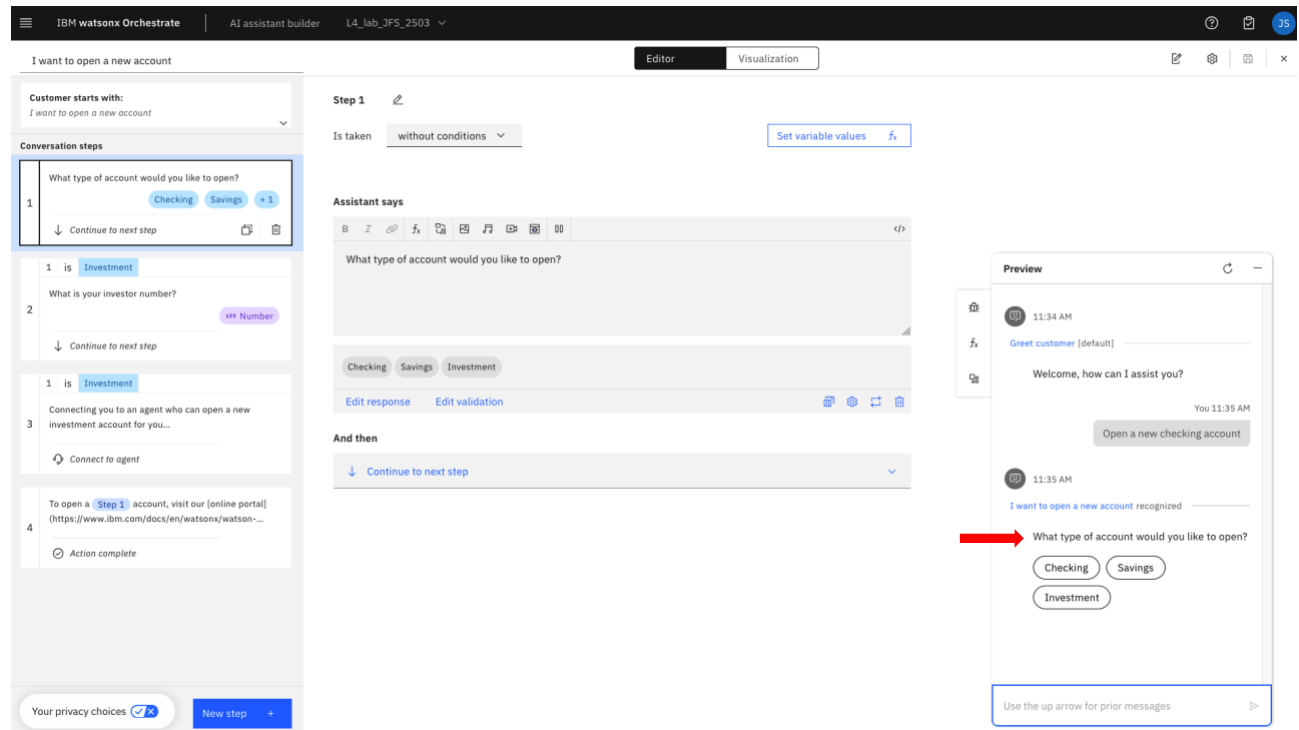
10. In the customer response window, select the option **Always ask for this information, regardless of previous messages** (A), then click **Apply** (B).



11. Refresh the **Preview** window (A), then type **Open a new Checking account** (B).



Notice that the assistant no longer skips step 1. It always stops to verify the customer response for this step.



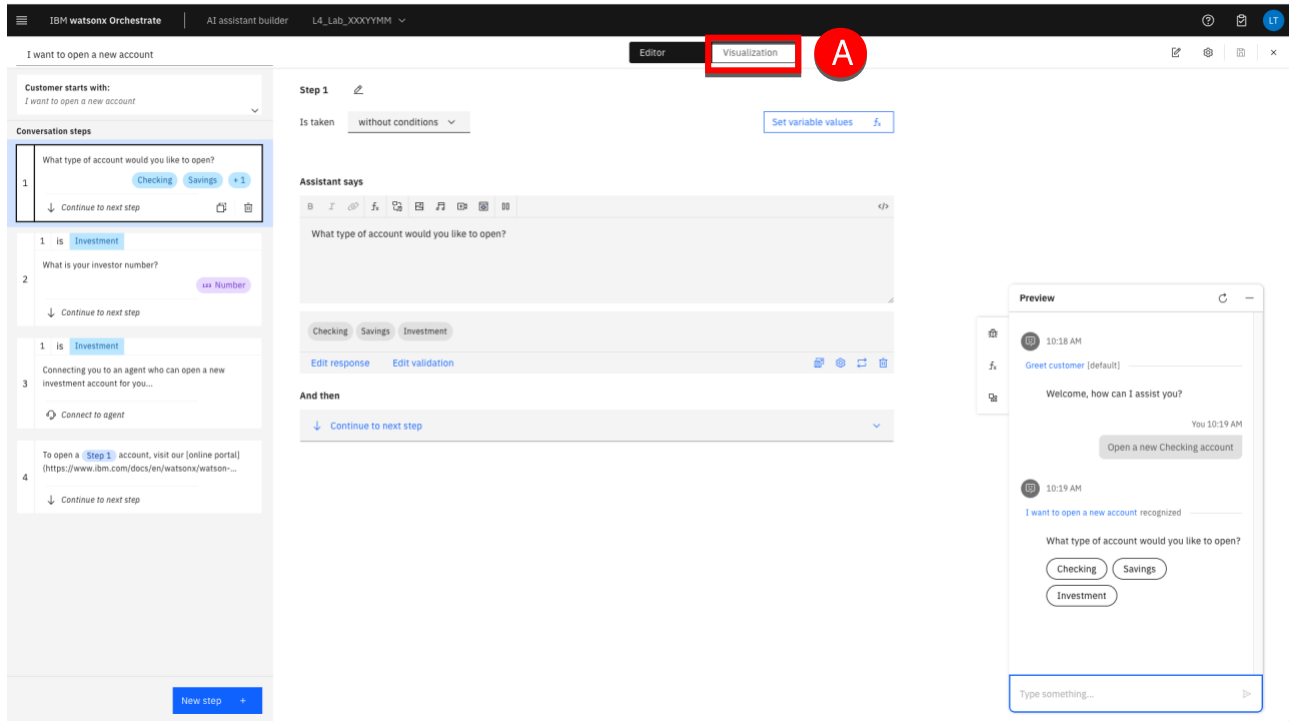
You may test more scenarios in the assistant.

## Additional action configurations

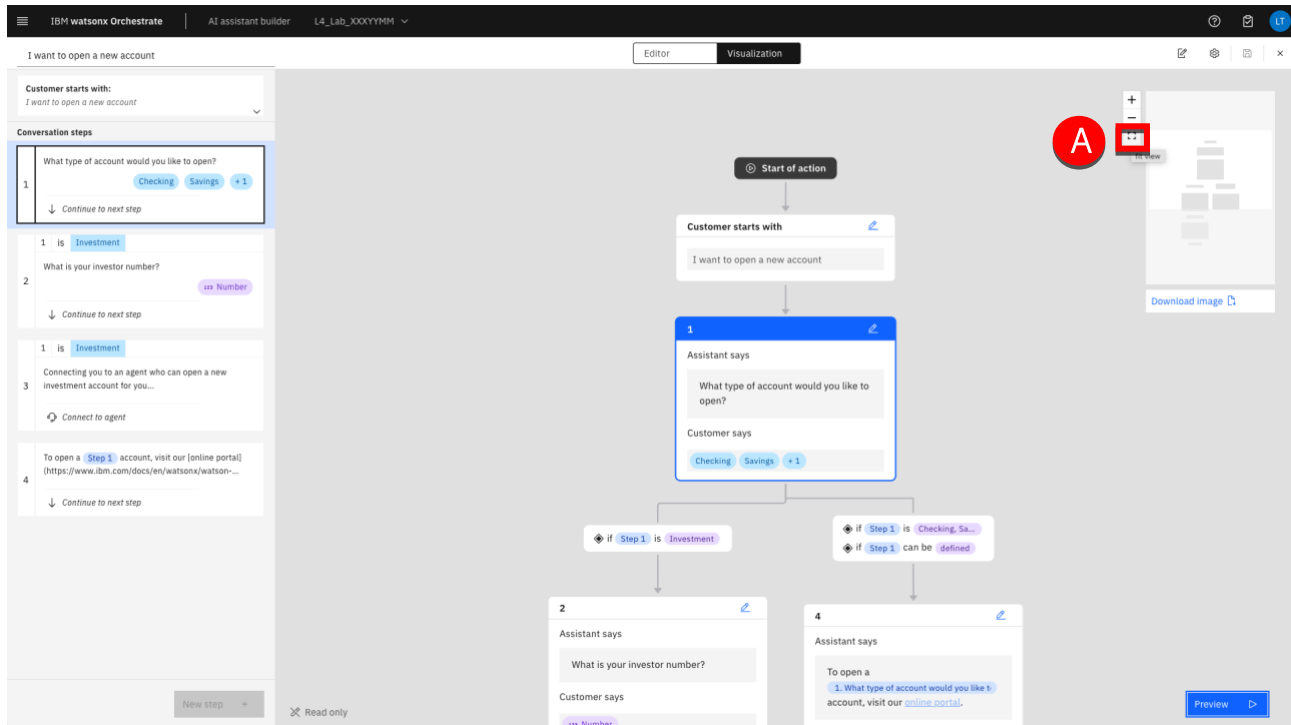
After completing and testing an action, you continue to iterate and build out more features for the action.

First, you study the **visualize** view.

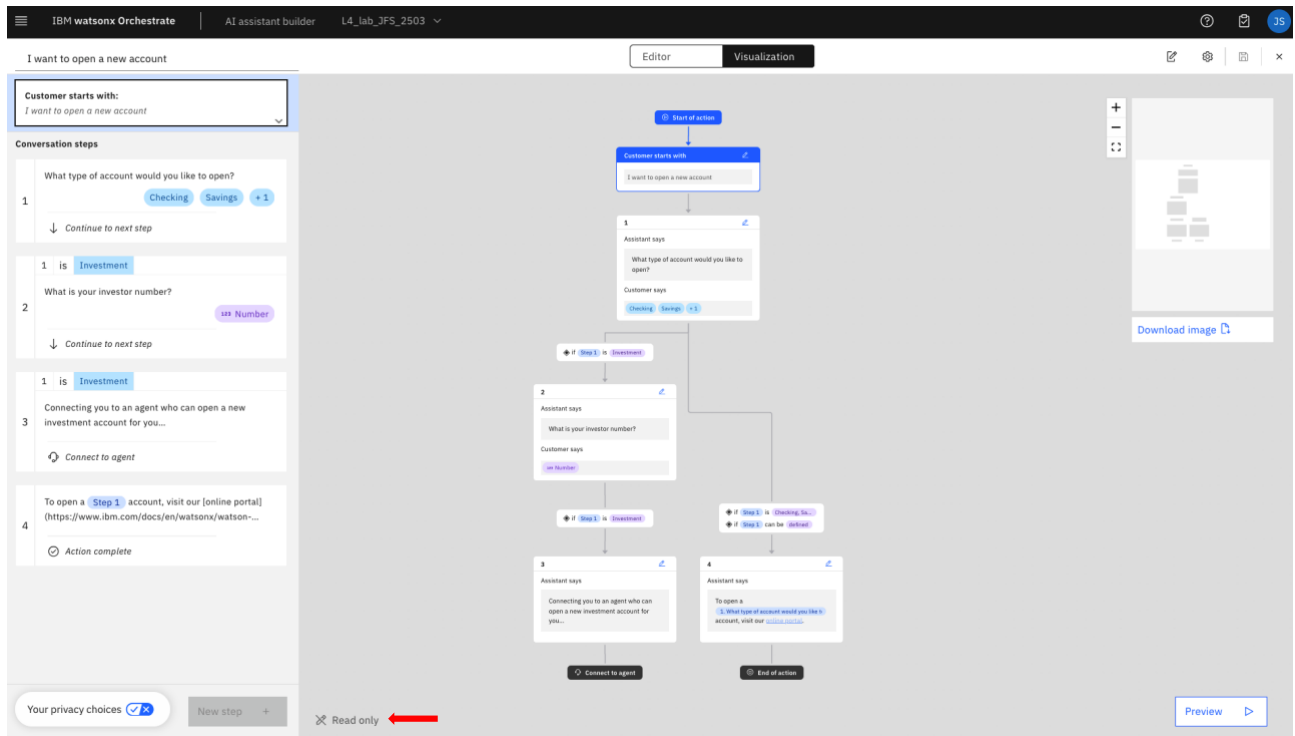
1. Select the **Visualization** tab (A).



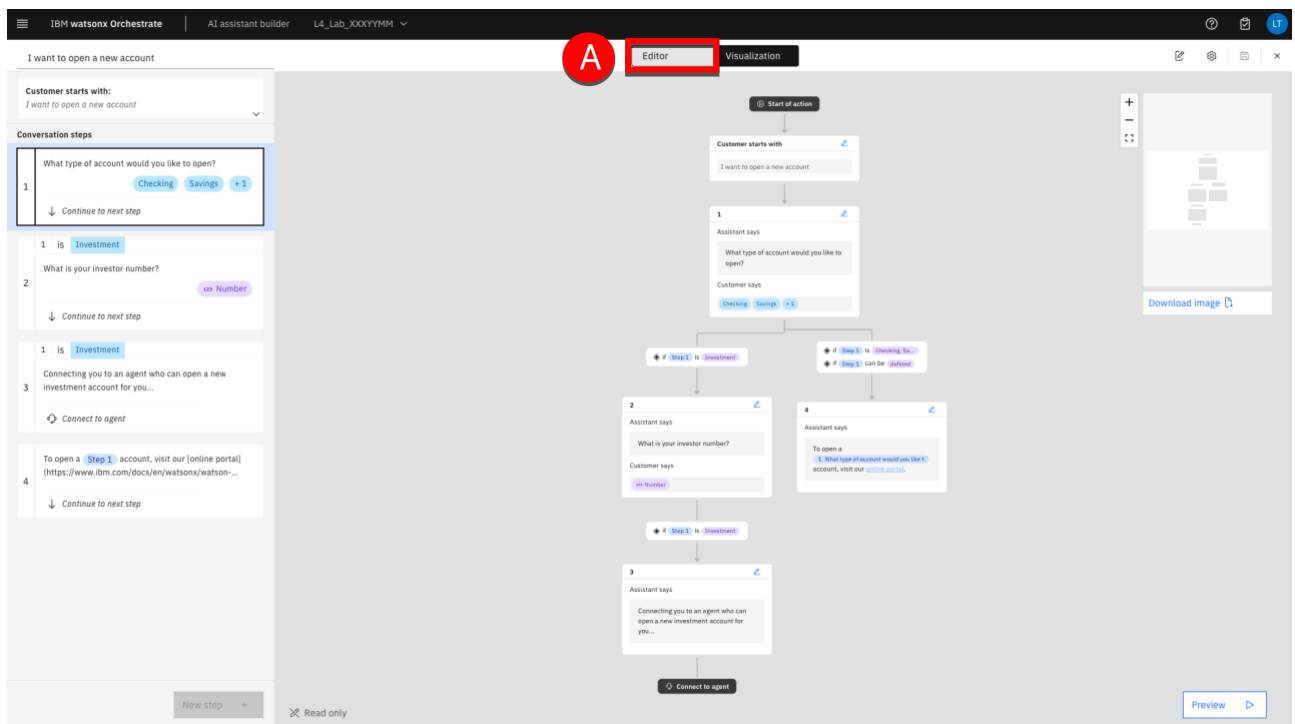
2. Click the **fit view** icon (A).



Study the diagram. The Visualization view is helpful when trying to understand the logic of actions. Note that you cannot edit any steps, values, or conditions when in this view (Read-only).

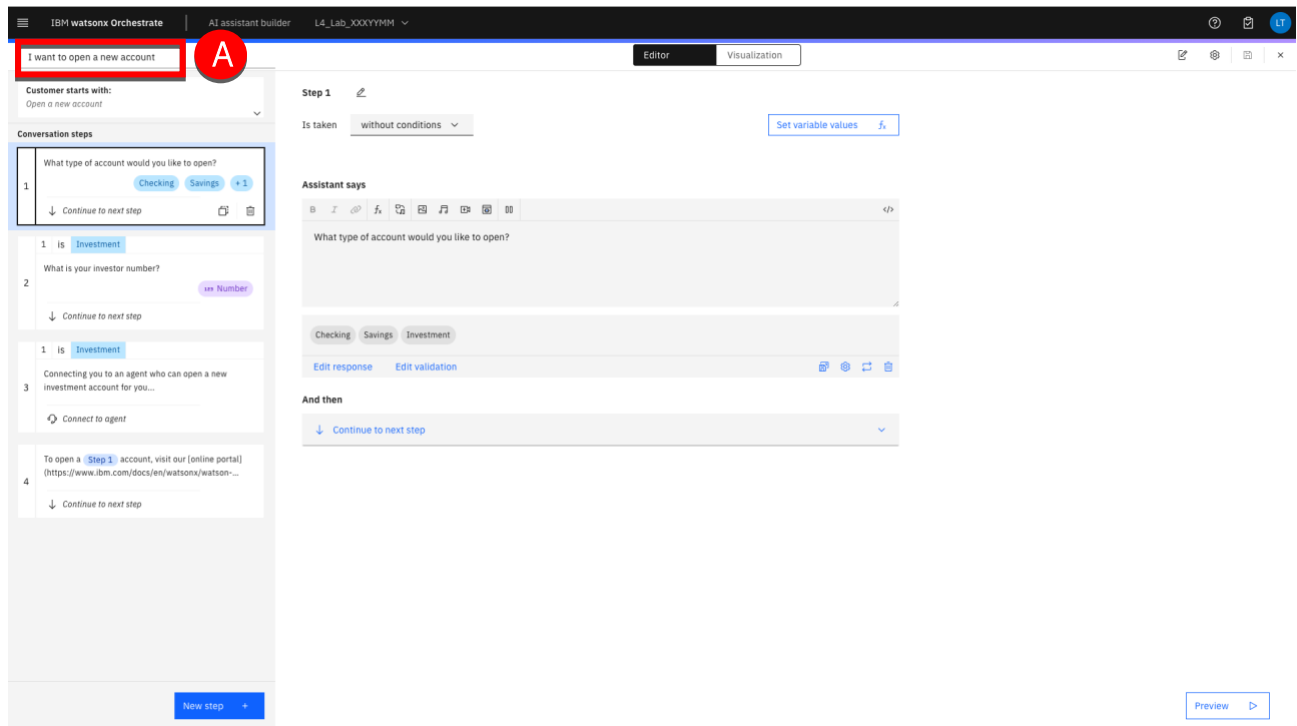


3. Select the Editor tab (A).

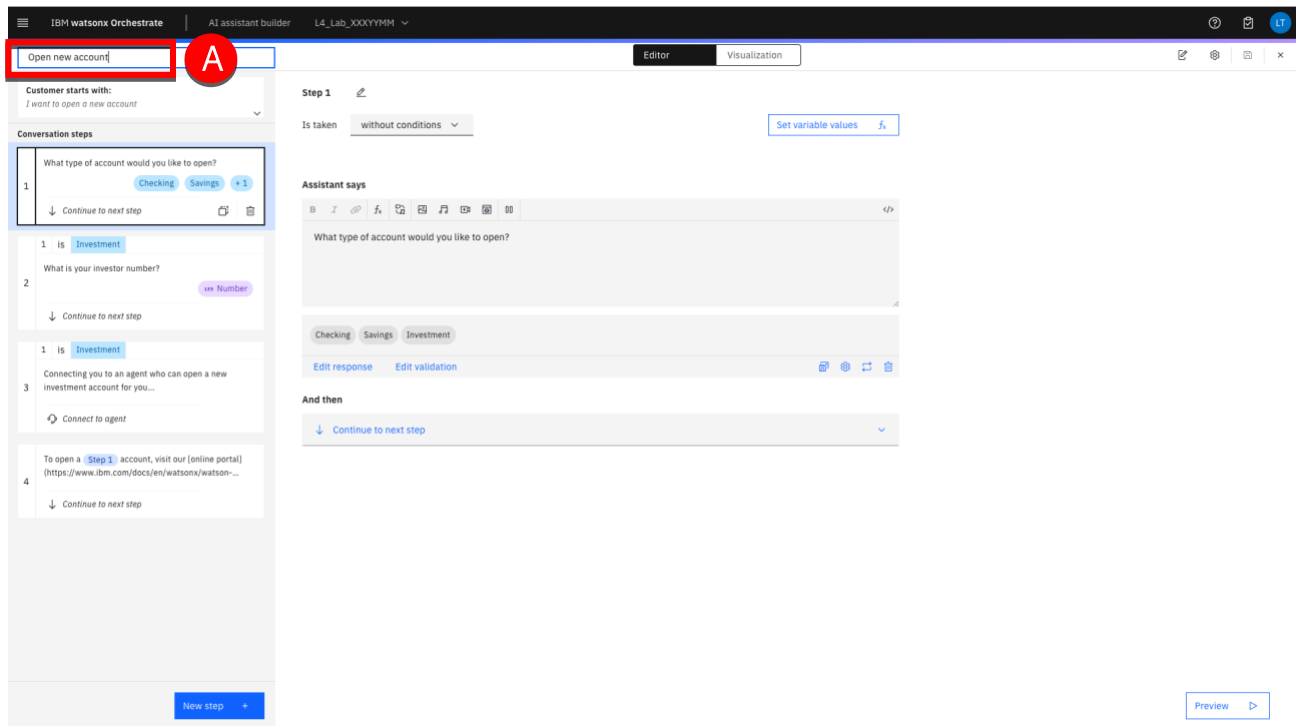


Next, you will rename the action.

4. Click the Customer starts with action (A), which currently says I want to open a new account (A).



5. Replace the text I want to open a new account, with Open a new account (A).



You have renamed the action. The new name is more concise– which is especially relevant if you are managing many actions in one assistant.

Now, you will add more example phrases to trigger this action.

6. Ensure you have the **Customer starts with** tile selected (A), then type the example phrases (B):

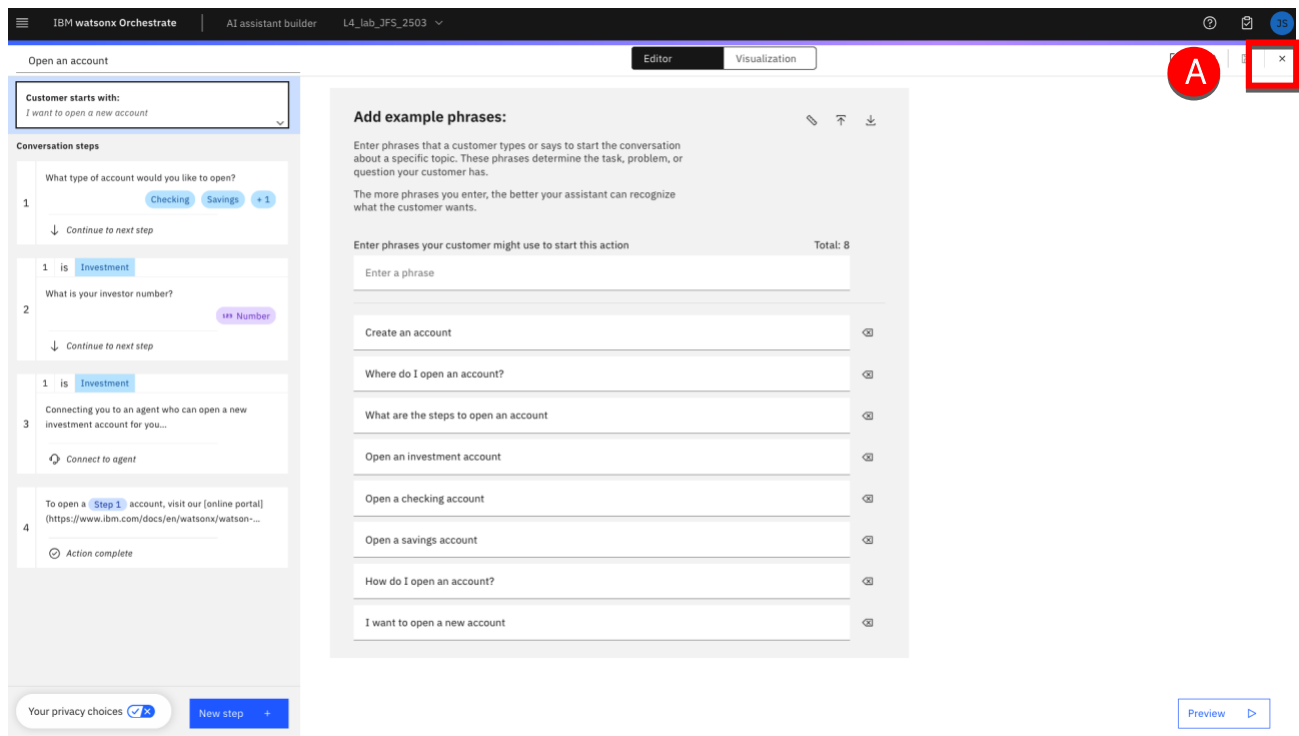
- How do I open an account?
- Open a savings account
- Open a checking account
- Open an investment account
- What are the steps to open an account?
- Where do I open an account?
- Create an account

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The top navigation bar shows 'IBM watsonx Orchestrate' and 'AI assistant builder'. The main area is divided into two tabs: 'Editor' and 'Visualization'. On the left, the 'Conversation steps' section shows a flowchart with steps for opening an account. On the right, the 'Add example phrases' section is active, showing a list of phrases that trigger the 'Create an account' action. A red box labeled 'A' highlights the 'Customer starts with' dropdown menu, which currently shows 'I want to open a new account'. Another red box labeled 'B' highlights a list of example phrases: 'Create an account', 'Where do I open an account?', 'What are the steps to open an account?', 'Open an investment account', 'Open a checking account', 'Open a savings account', and 'How do I open an account?'. The interface also includes a 'Conversation steps' section on the left and a 'Preview' button at the bottom right.

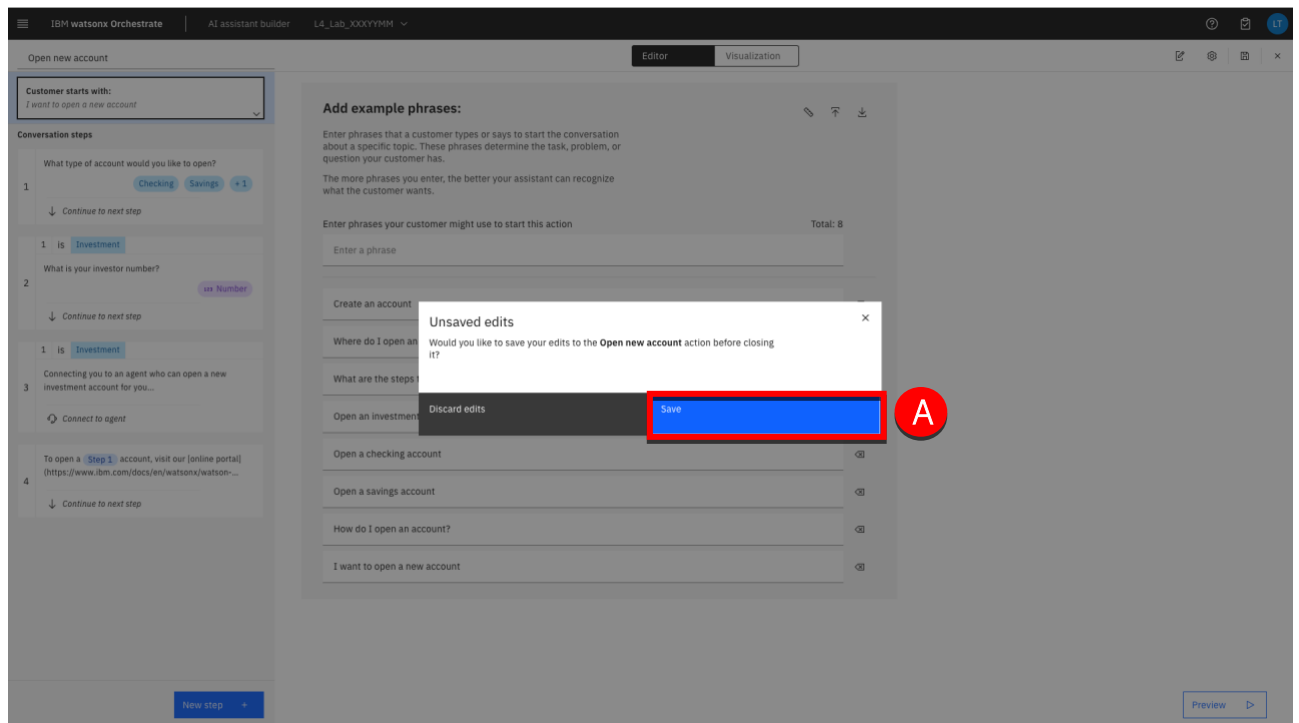
By adding these phrases, your AI assistant learns what is the right action for what a customer wants. The additional examples build the training data that the machine learning engine of AI assistant uses to create a natural language processing model. You can learn about this process by referencing the watsonx [Orchestrate Documentation](#).

Next, you customize your assistant's greeting.

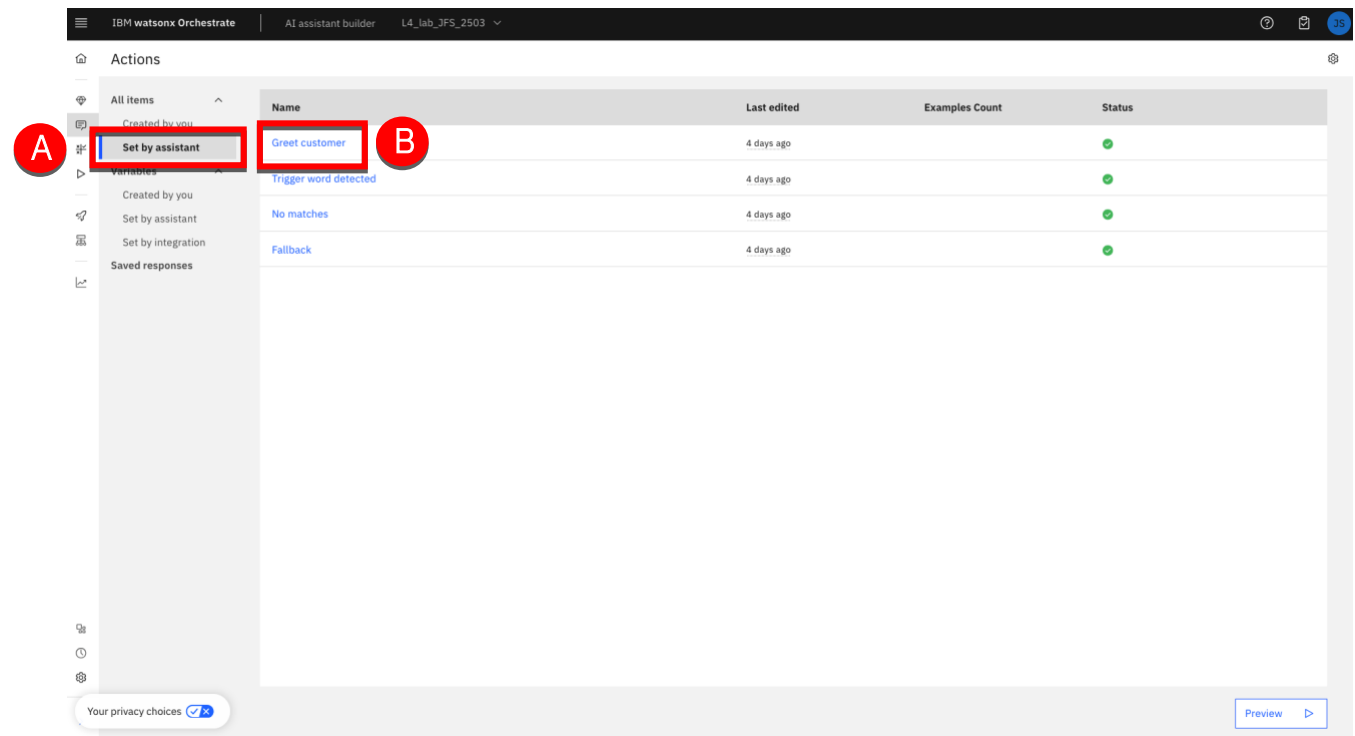
7. Click the X button (A) in the upper right corner to exit the current action editor.



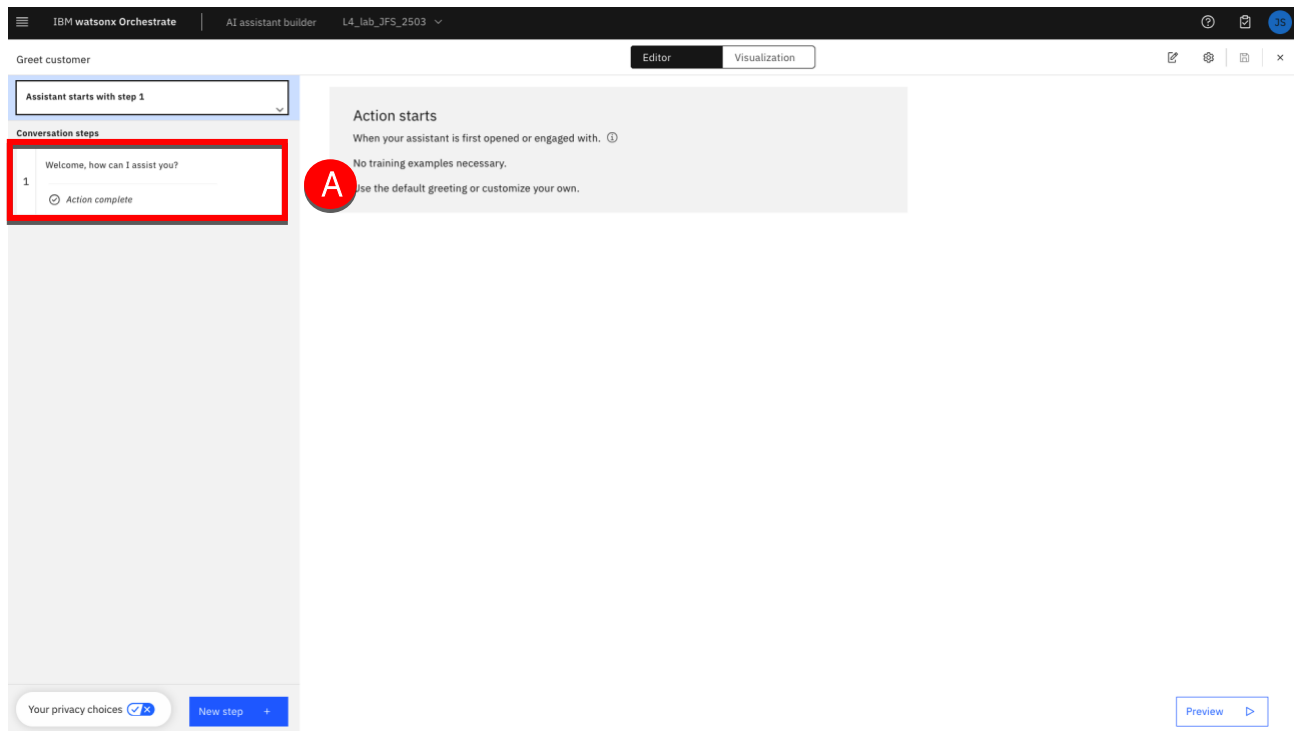
8. Click Save (A).



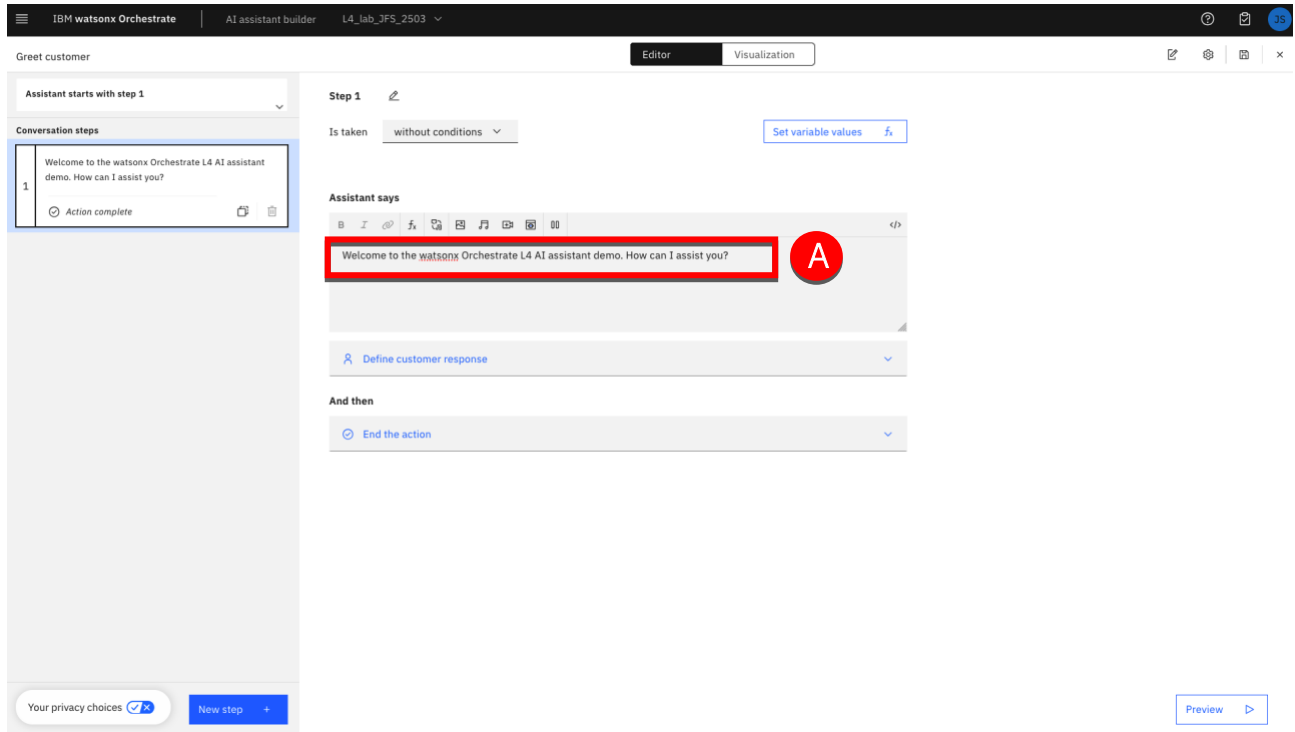
9. Select **Set by assistant** (A), then the **Greet customer** action (B).



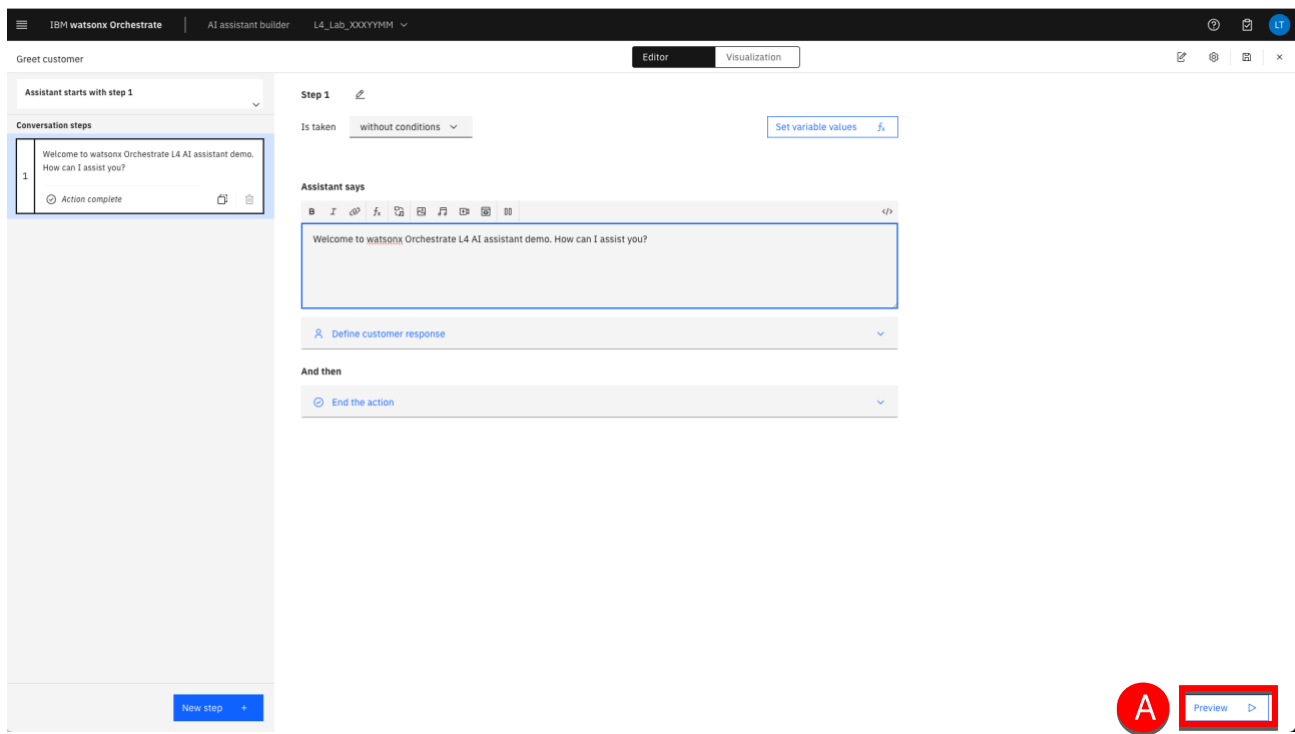
10. Click **Step 1** (A).



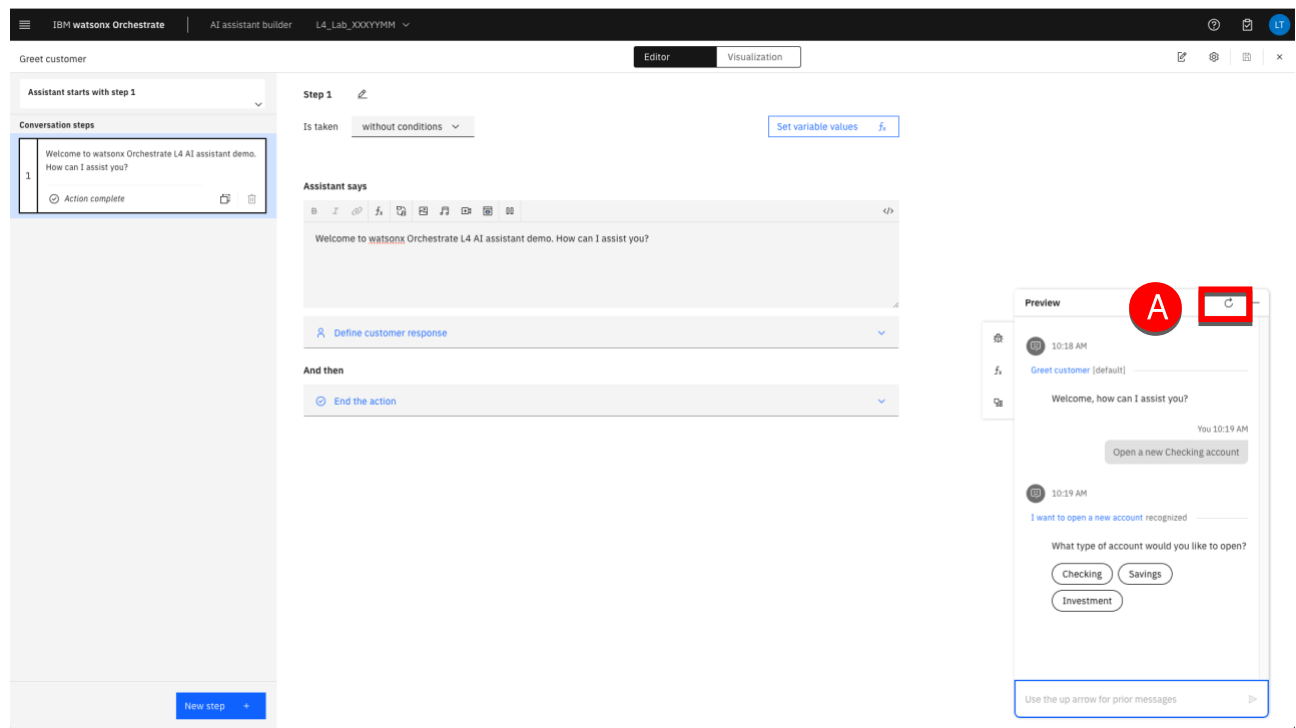
11. Change the **Assistant says** text to Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you? (A).



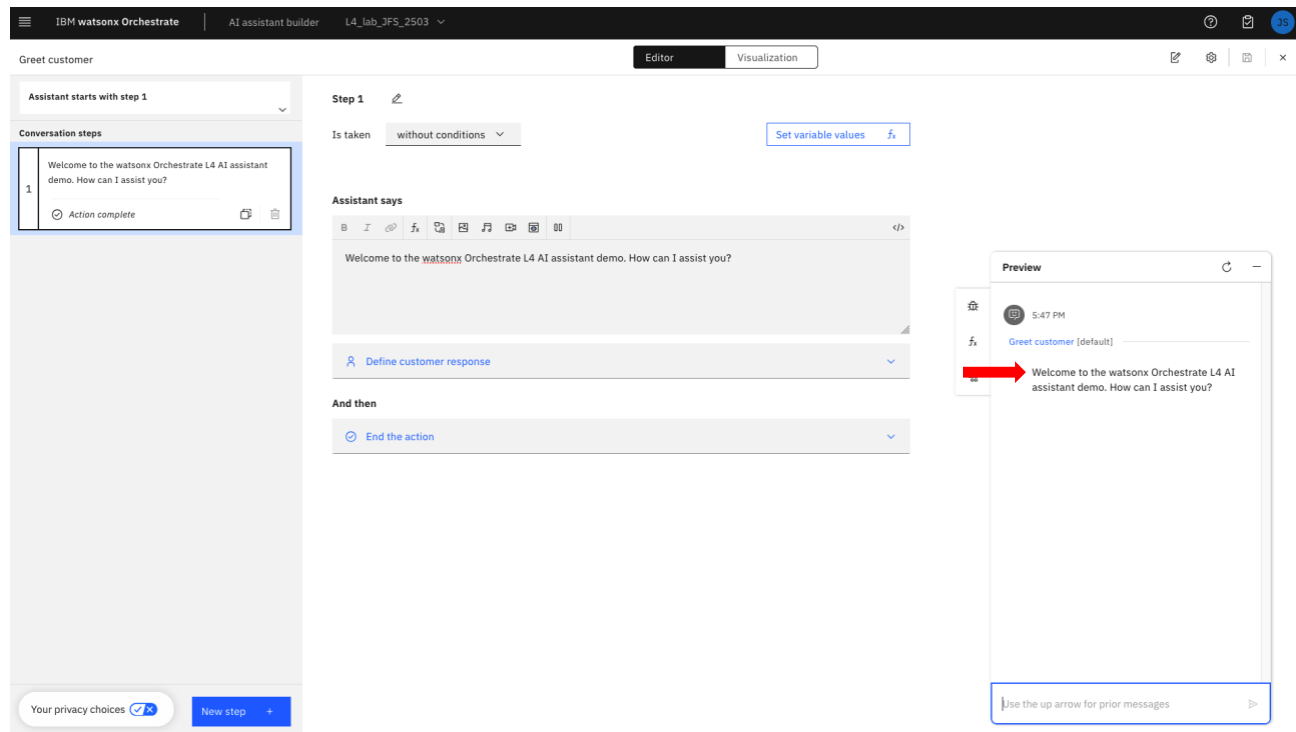
12. Click Preview (A).



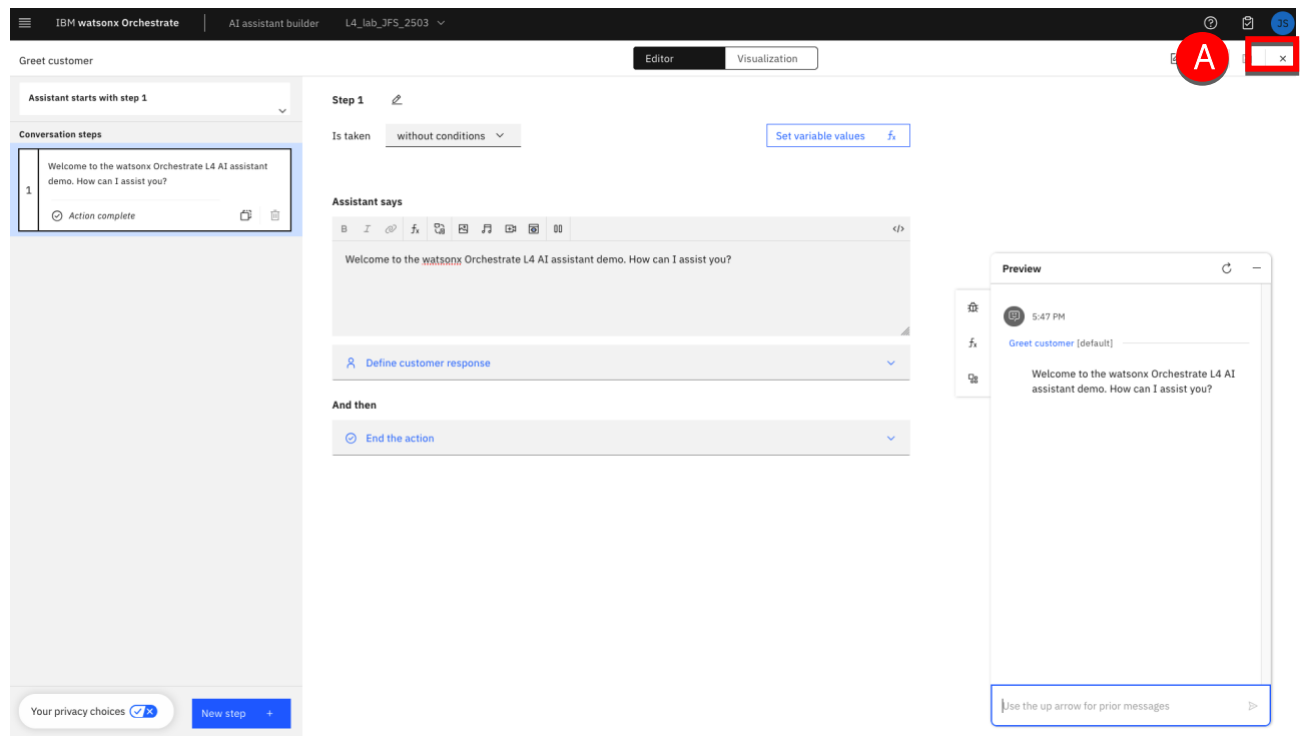
13. Click the **Preview** refresh Icon (A).



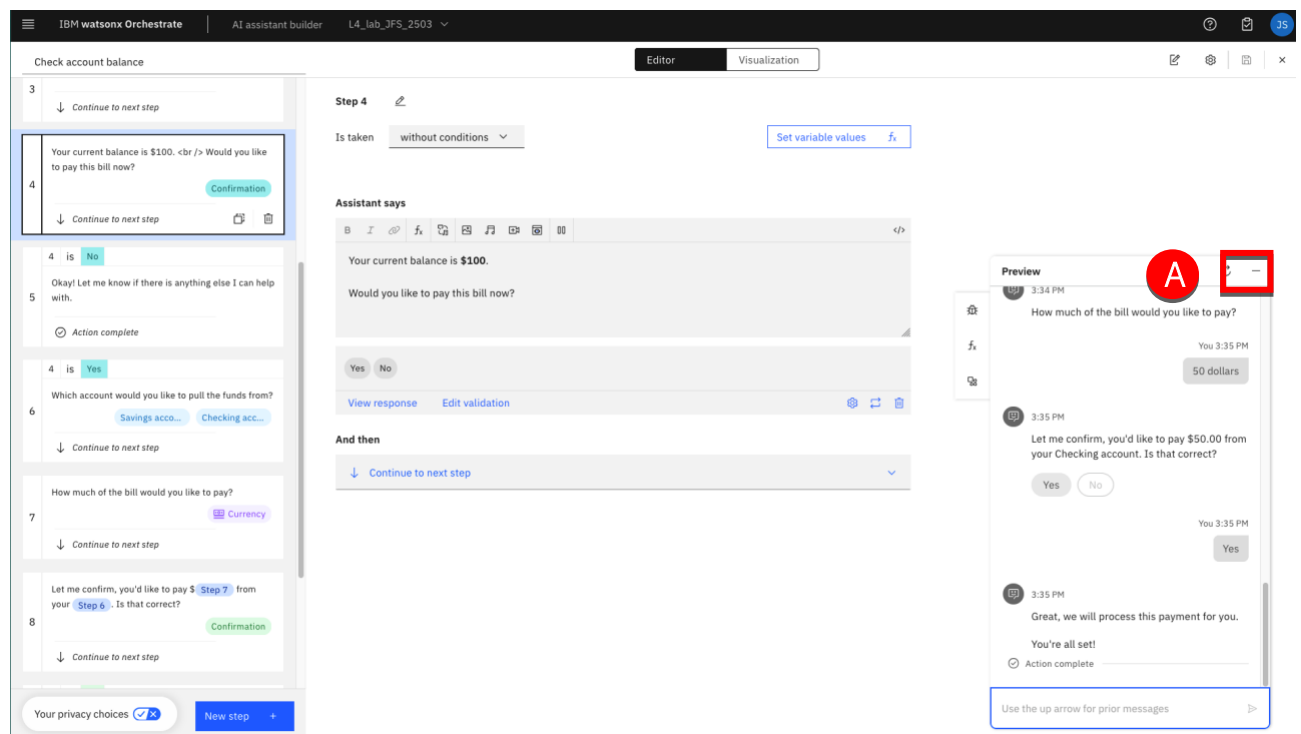
Notice that the greeting has changed.



14. Click the close button (A).



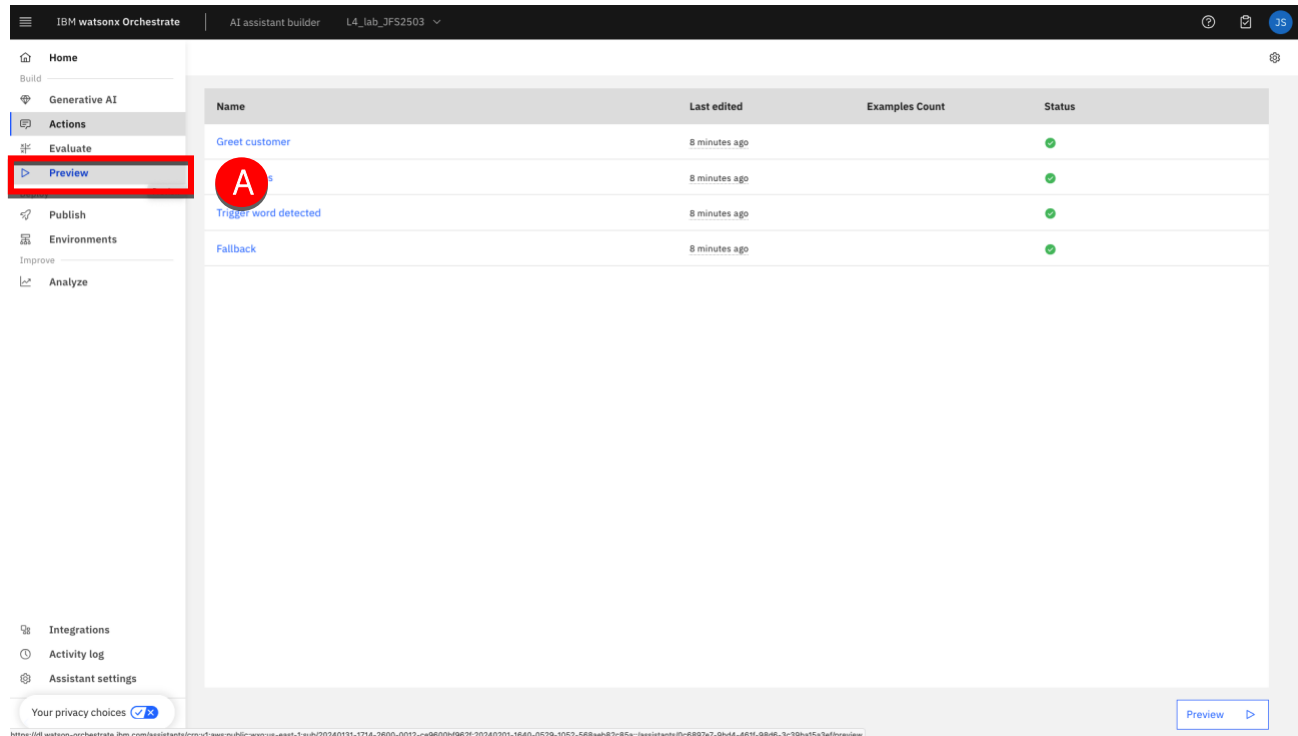
15. Close the Preview panel (A).



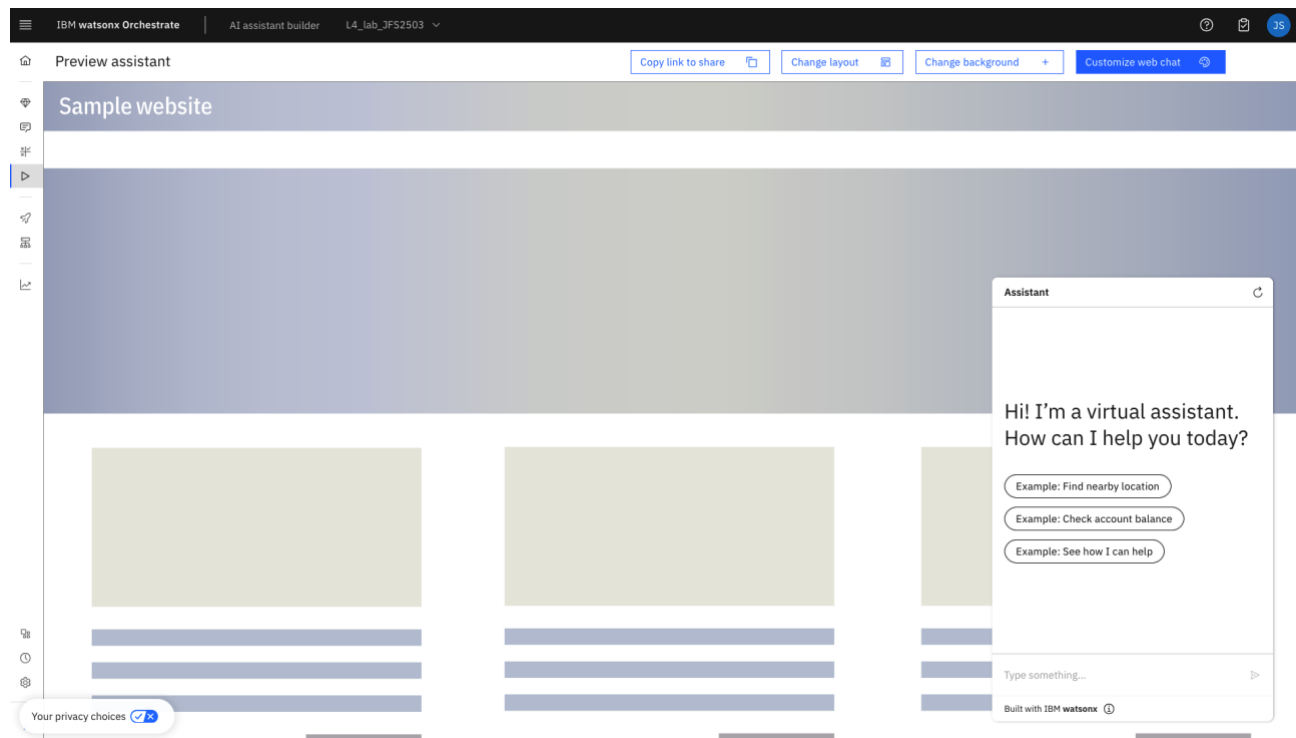
## Preview your assistant

So far, you have been previewing your assistant in the Actions editor. This is a useful way to test and iterate action behavior. Now, you will preview your assistant in the dedicated [Preview page](#).

1. In the left navigation pane, click **Preview** (A).



Here, you will see a preview of your assistant using the [Web Chat integration](#).

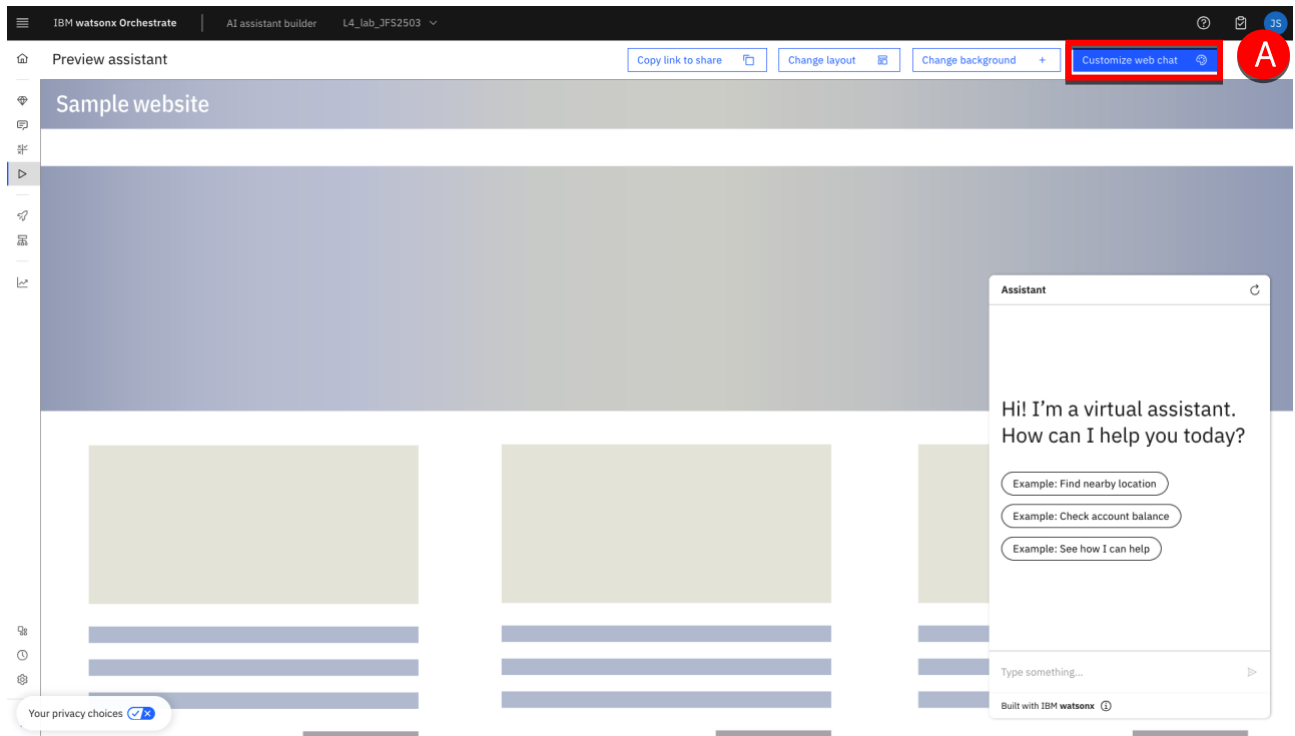


The Web Chat is an example of a channel. [Channels](#) are the location where your assistant interacts with your users - for example, over the phone, on a website, or in Slack. At least one channel is required for every assistant.

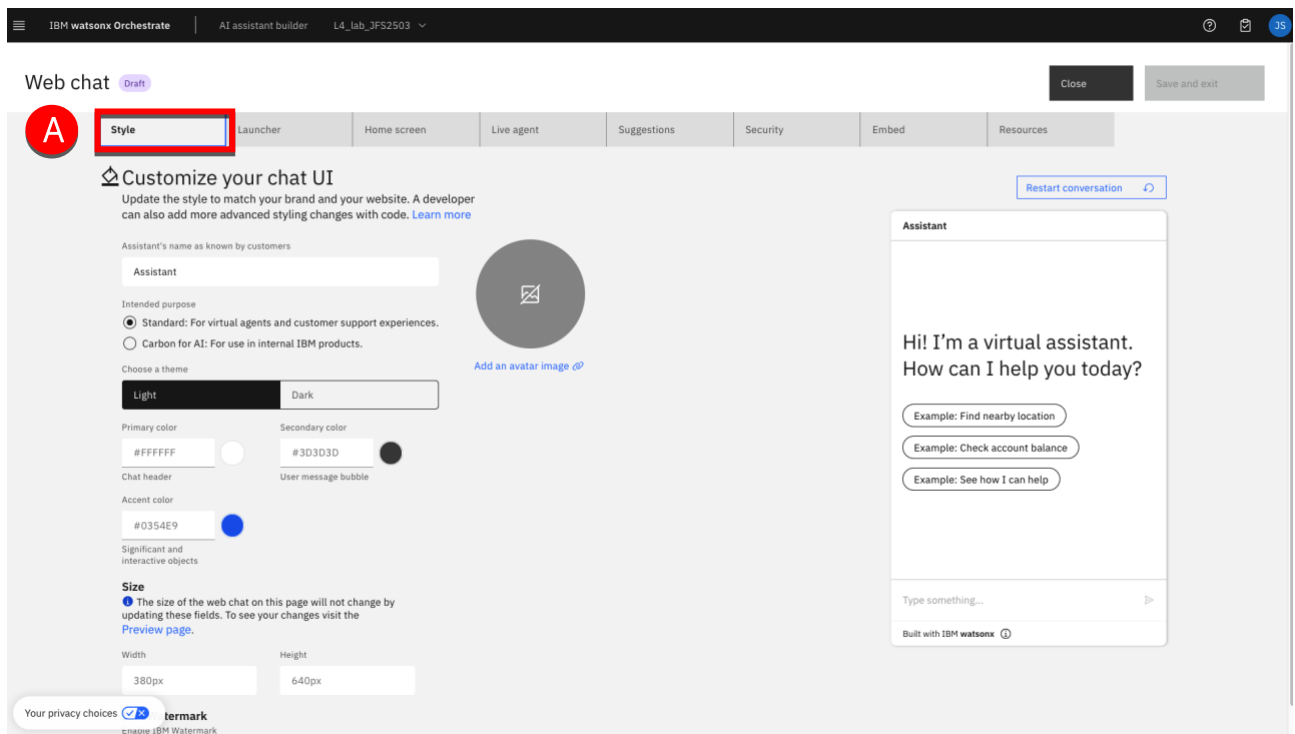
Optionally, you can interact with the assistant in the right-hand corner.

You might notice that the starting screen for the assistant looks different than the Greet Customer action you configured a tested earlier. This is expected behavior – greetings may vary depending on the channel being used. You can verify that this is the assistant you were editing before by typing 'Open an account please', for example, you will trigger the Open an Account action you recently created and edited.

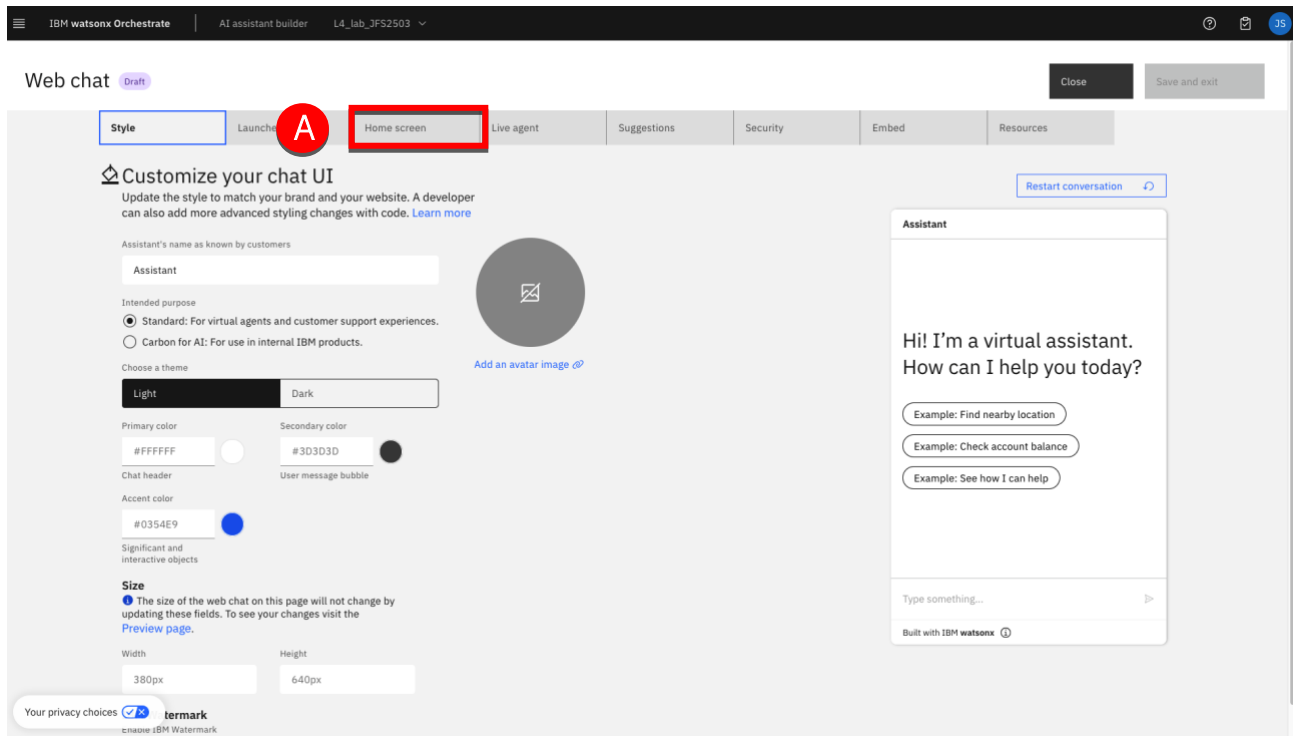
2. Click **customize web chat** (A).



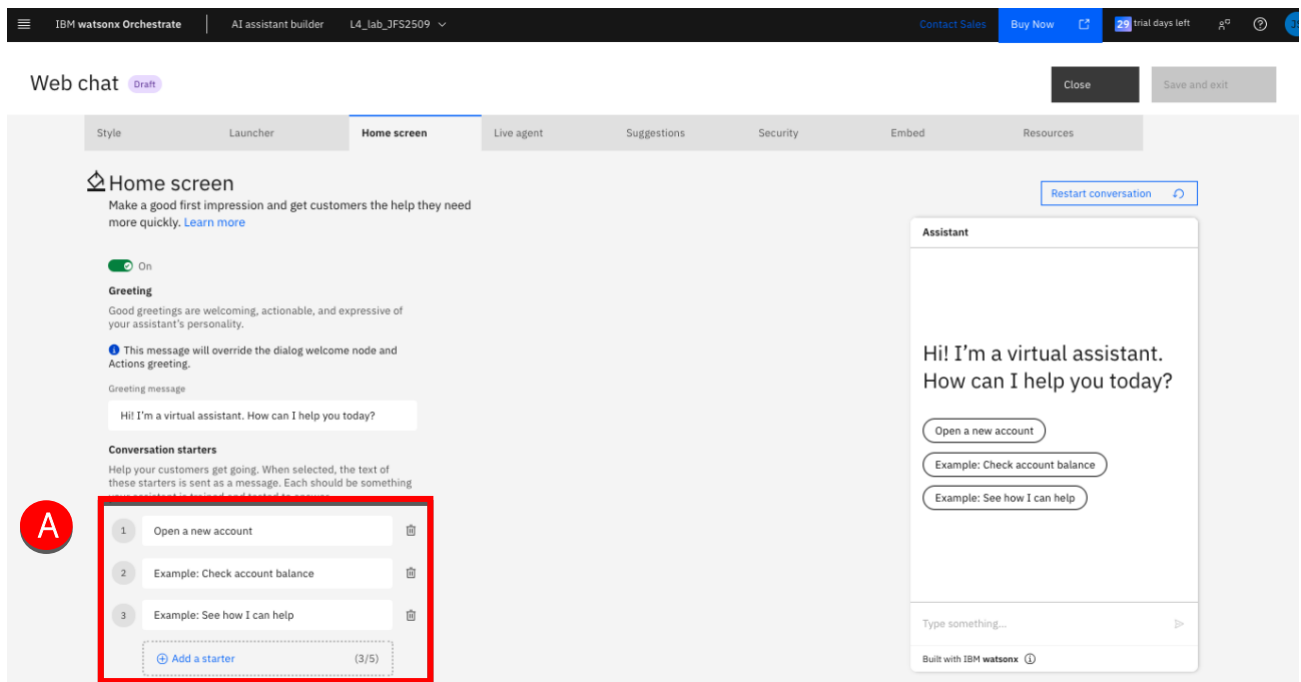
3. Observe the **Style tab** (A) that is now open. Optionally, you may alter the theme, primary color, secondary color, and accent color to change the appearance of the Assistant.



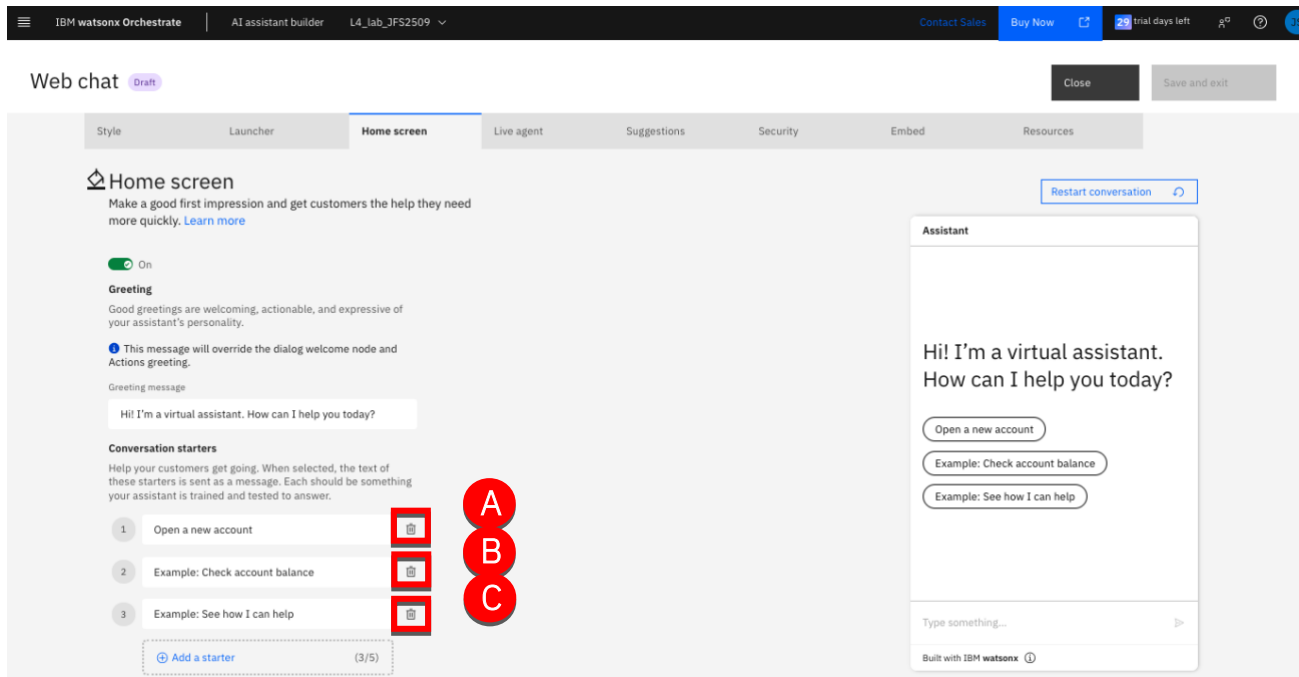
4. Click the Home Screen tab (A).



5. Note the 3 existing conversation starters (A). 3 conversation starters are automatically generated based on your use case.

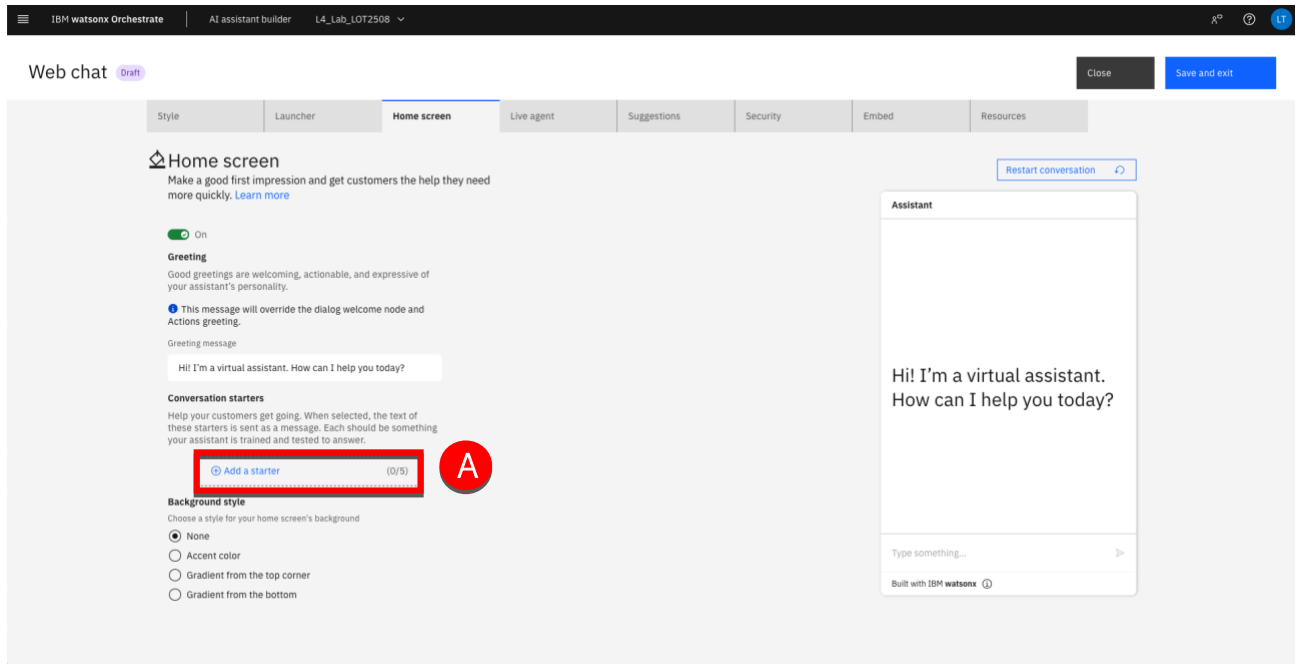


6. Delete the 3 existing conversation starters by clicking the Trash Icon next to them (A) (B) (C).

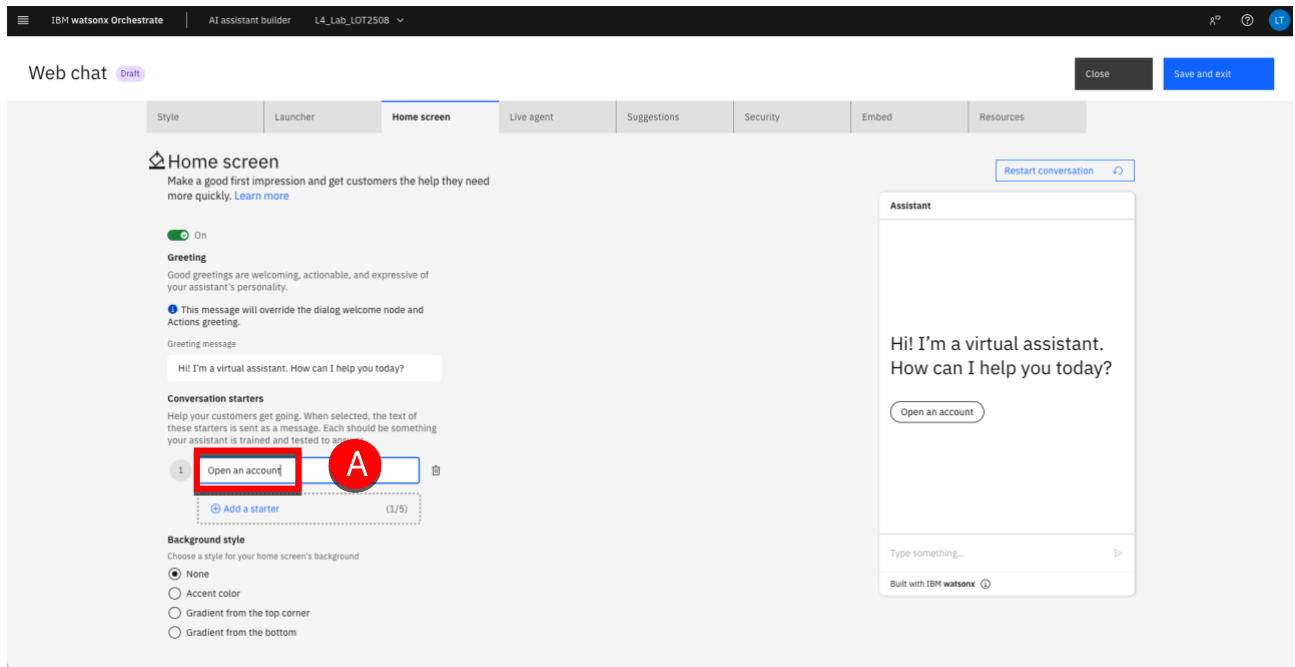


Typically, the generated conversation starters are appropriate for a banking scenario. However, you are removing the generated conversation starters to ensure consistency with this lab guide.

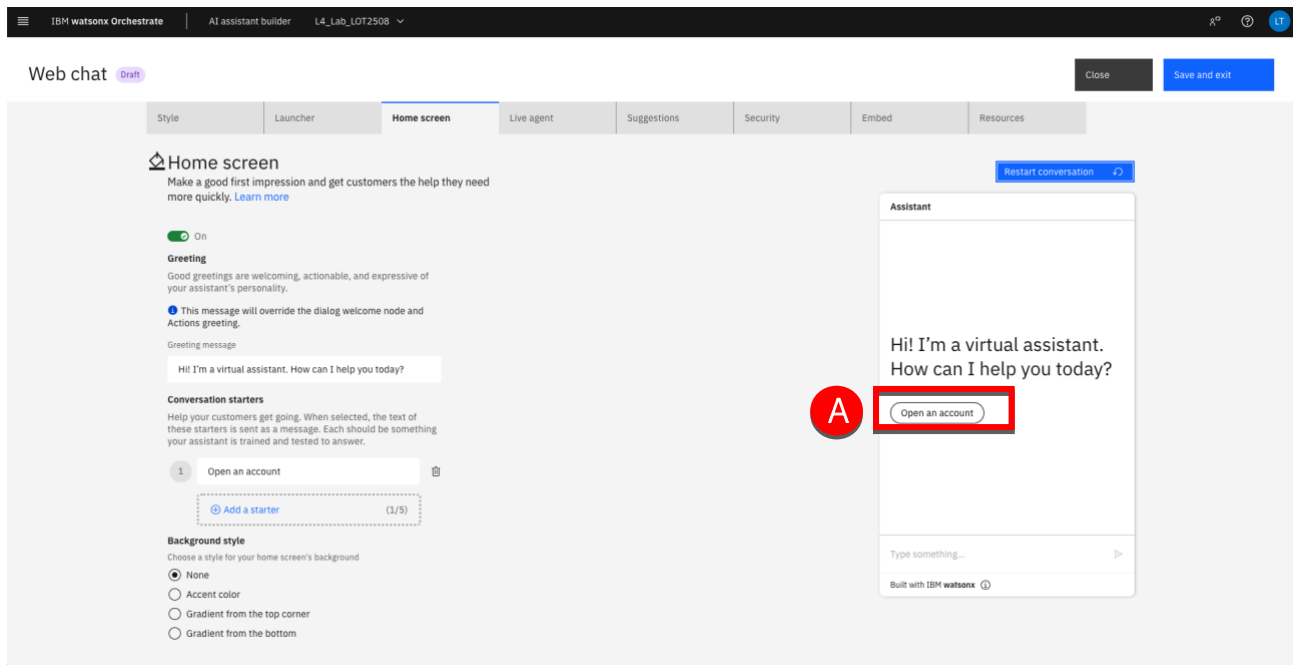
7. Click **Add a starter** (A).



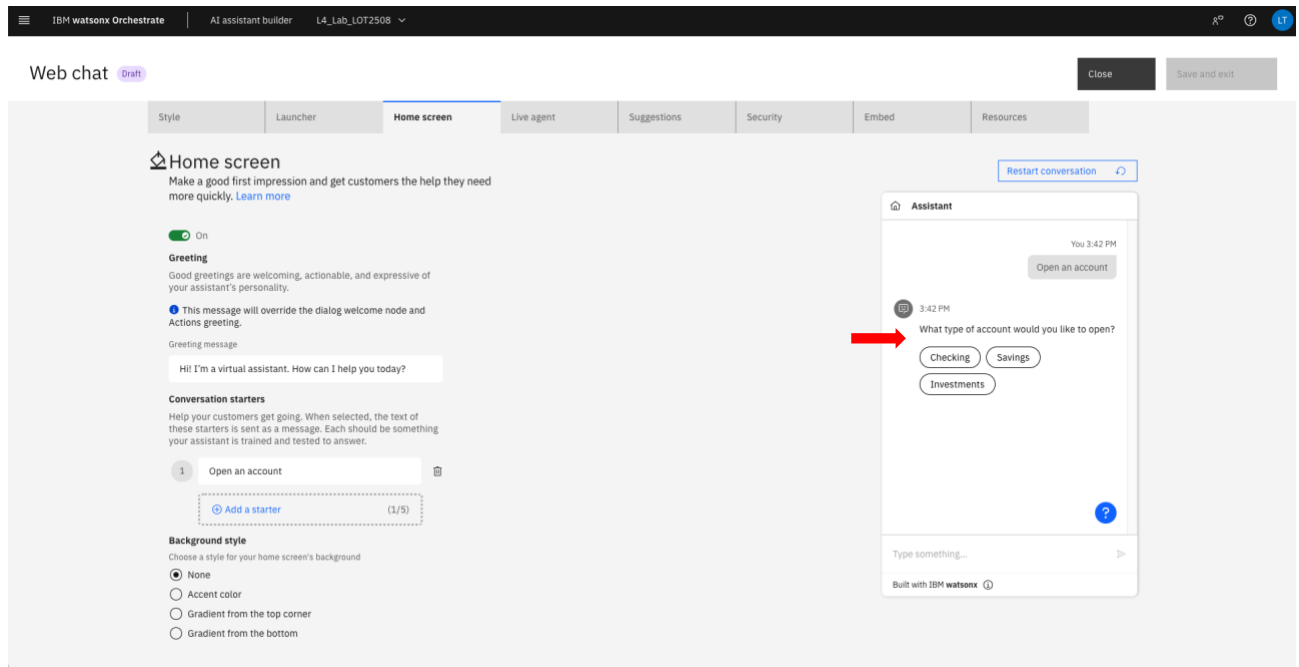
8. Delete the prepopulated text in the starter, then type Open an account (A).



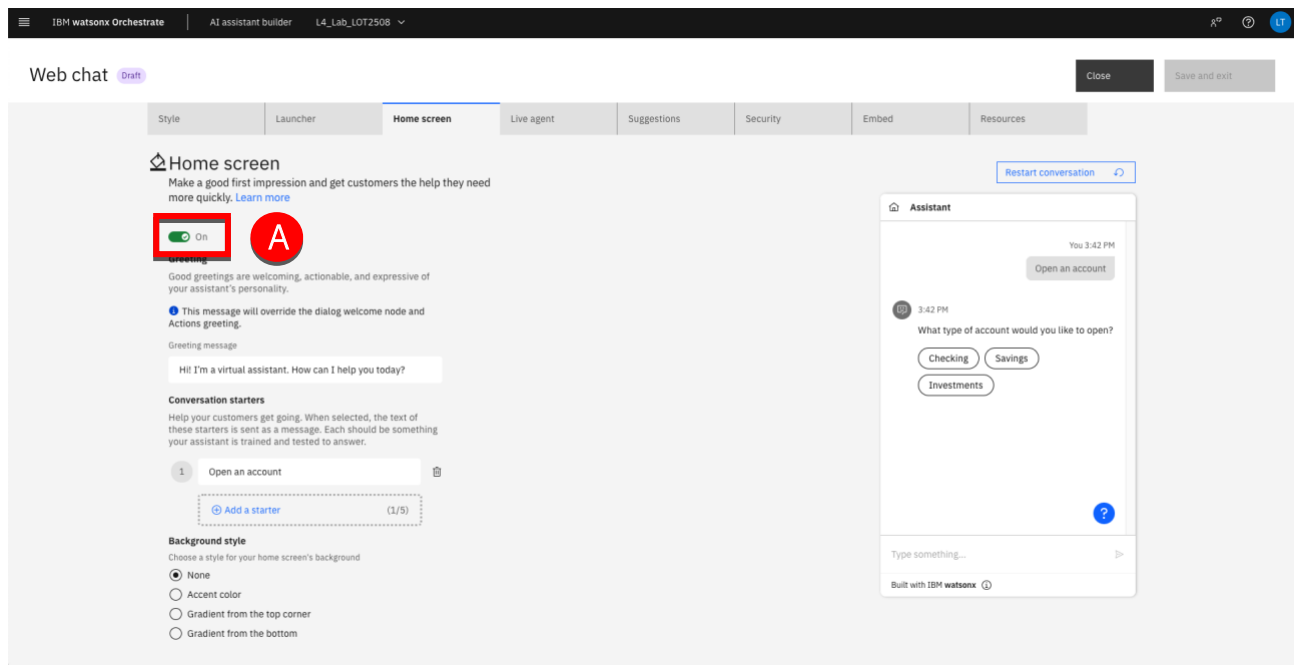
9. In the Assistant window on the left, click Open an account (A).



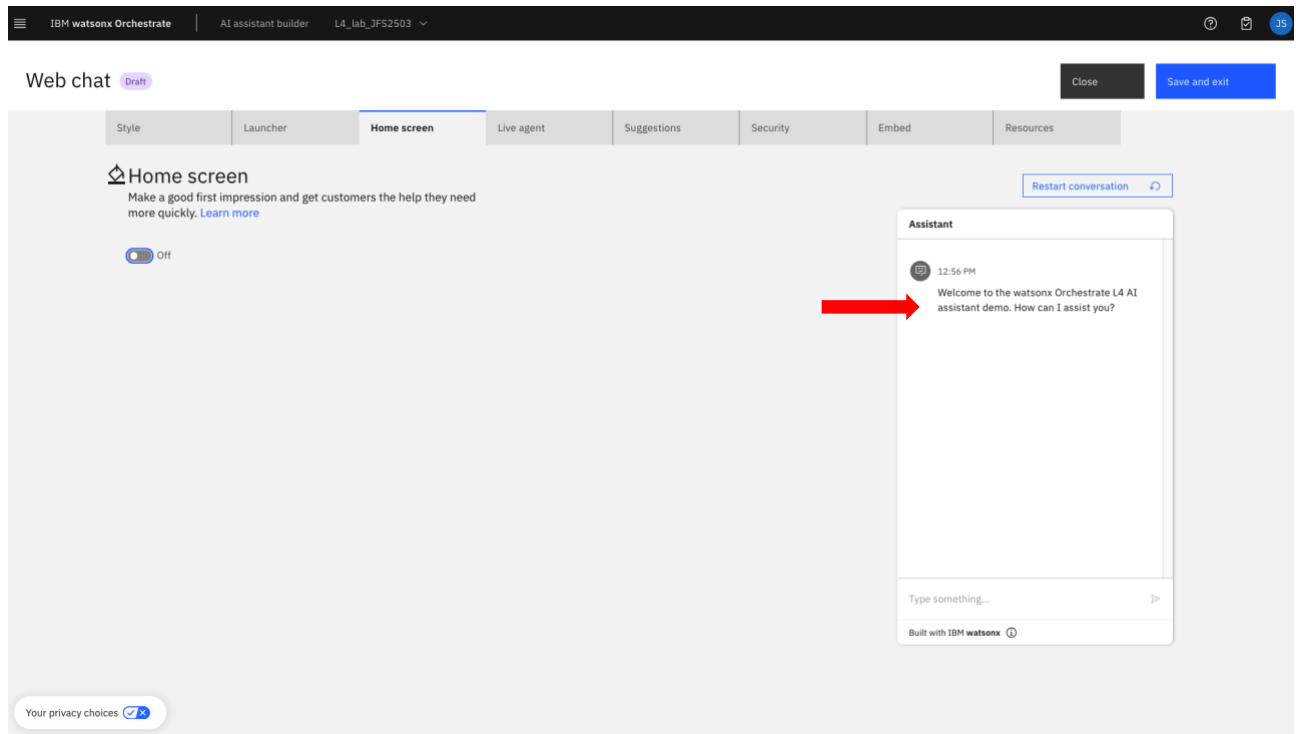
10. Observe that the assistant recognized the user input and started the Open an account action. Optionally, you can continue progressing through the action.



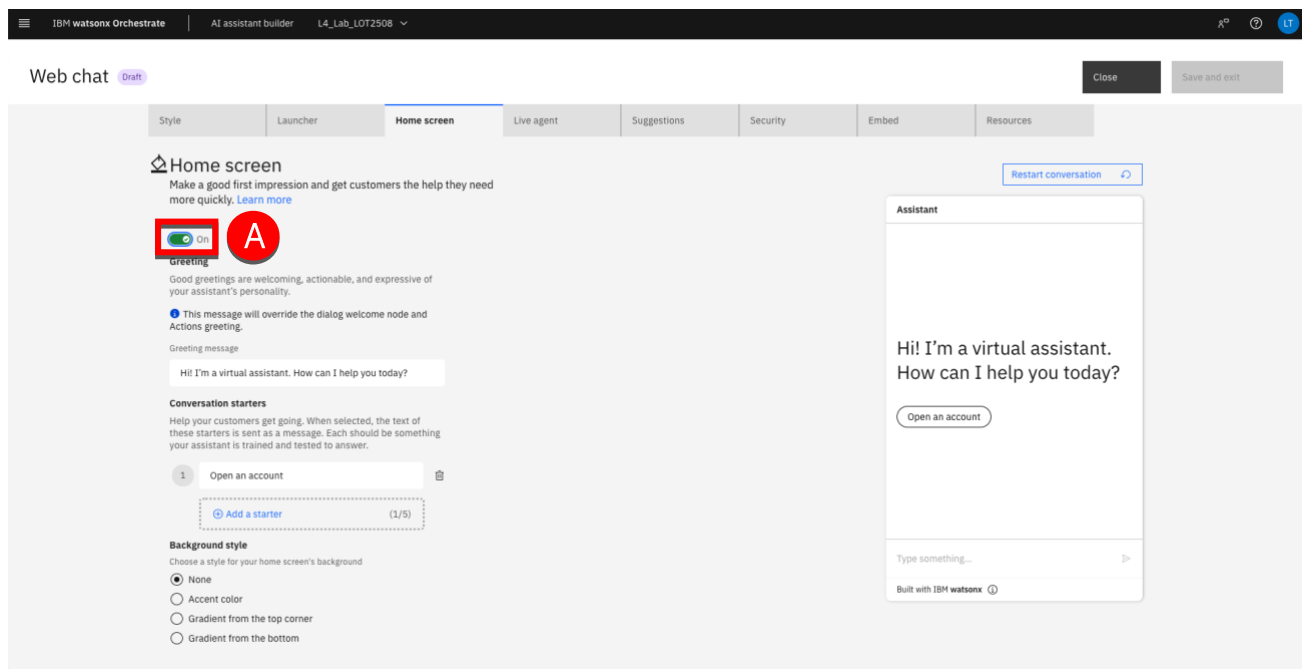
11. Disable the Home Screen setting switching the radio button to Off (A).



12. Observe that the conversation starters are no longer listed. Instead, the Greet Customer action is displayed.



13. Re-enable the Home screen (A).



14. Select the **Embed** tab (A).

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. At the top, the navigation bar includes 'IBM watsonx Orchestrate', 'AI assistant builder', and a dropdown menu showing 'L4\_lab\_JFS\_2503'. Below this, the 'Web chat' section is in 'Draft' mode, with buttons for 'Close' and 'Save and exit'. The main configuration area has several tabs: 'Style', 'Launcher', 'Home screen' (selected), 'Live agent', 'Suggestions', 'Security', 'Embed' (highlighted with a red box and labeled 'A'), and 'Resources'. The 'Home screen' tab shows settings for the assistant's initial display, including a 'Greeting' message and 'Conversation starters'. The 'Embed' tab shows a preview of the assistant's chat interface, displaying a greeting message and two buttons: 'Check account balance' and 'Open an account'.

15. Observe the HTML code in the Embed tab. Optionally, click the **Learn more** button (A) to see more details about configuring the web chat to run in a custom website. Embedding the assistant is demonstrated in the ‘Integrate your AI Assistant in your web interface’ lab associated with the watsonx Orchestrate L4 course.

IBM watsonx Orchestrate | AI assistant builder | L4\_lab\_3FS2503

Web chat Draft Close Save and exit

Style | Launcher | Home screen | Live agent | Suggestions | Security | **Embed** | Resources

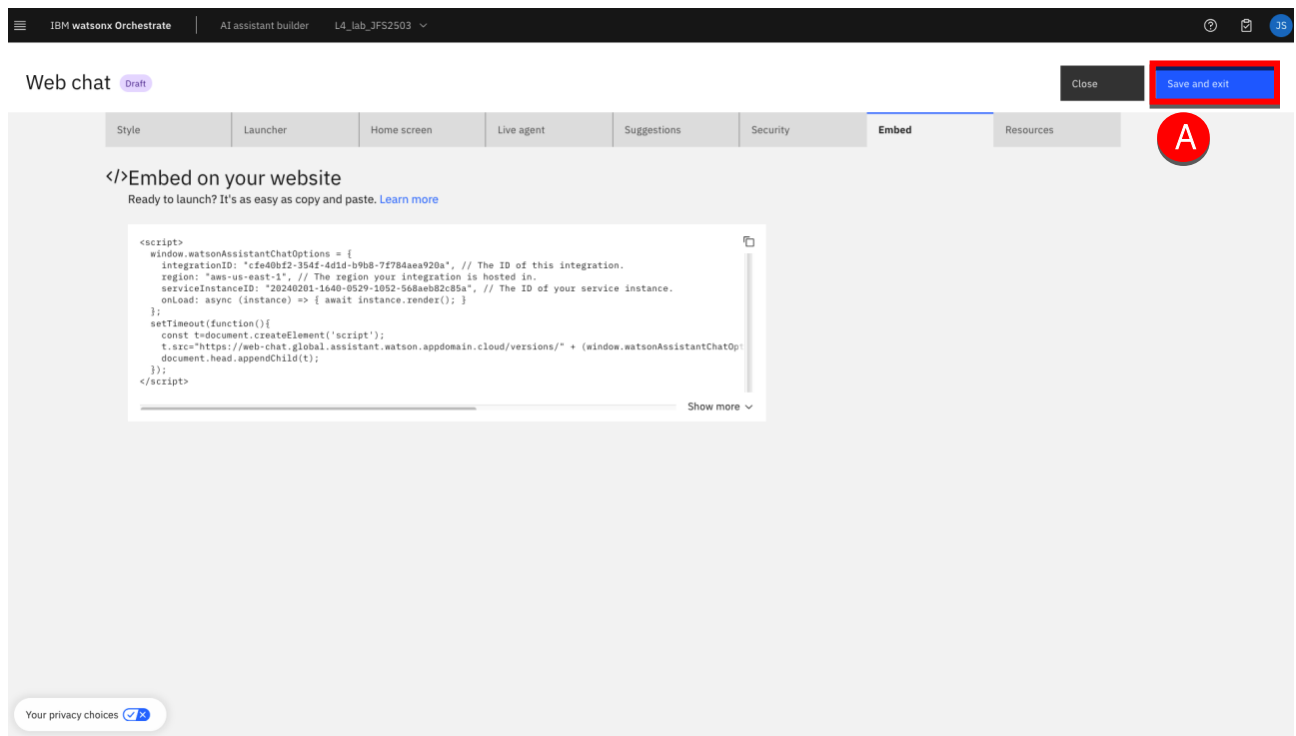
</>Embed on your website  
Ready to launch? It's as easy as copy and paste. [Learn more](#) **A**

```
<script>
window.watsonAssistantChatOptions = {
  integrationID: "cfe40b22-354f-4d1d-b9b8-7f784aea920a", // The ID of this integration.
  region: "aws-us-east-1", // The region your integration is hosted in.
  serviceInstanceID: "20240201-1640-0529-1052-568aeb82c85a", // The ID of your service instance.
  onLoad: async (instance) => { await instance.render(); }
};
setTimeout(function(){
  const t=document.createElement('script');
  t.src="https://web-chat.global.assistant.watson.appdomain.cloud/versions/" + (window.watsonAssistantChatOptions.integrationID) + ".js";
  document.head.appendChild(t);
});
</script>
```

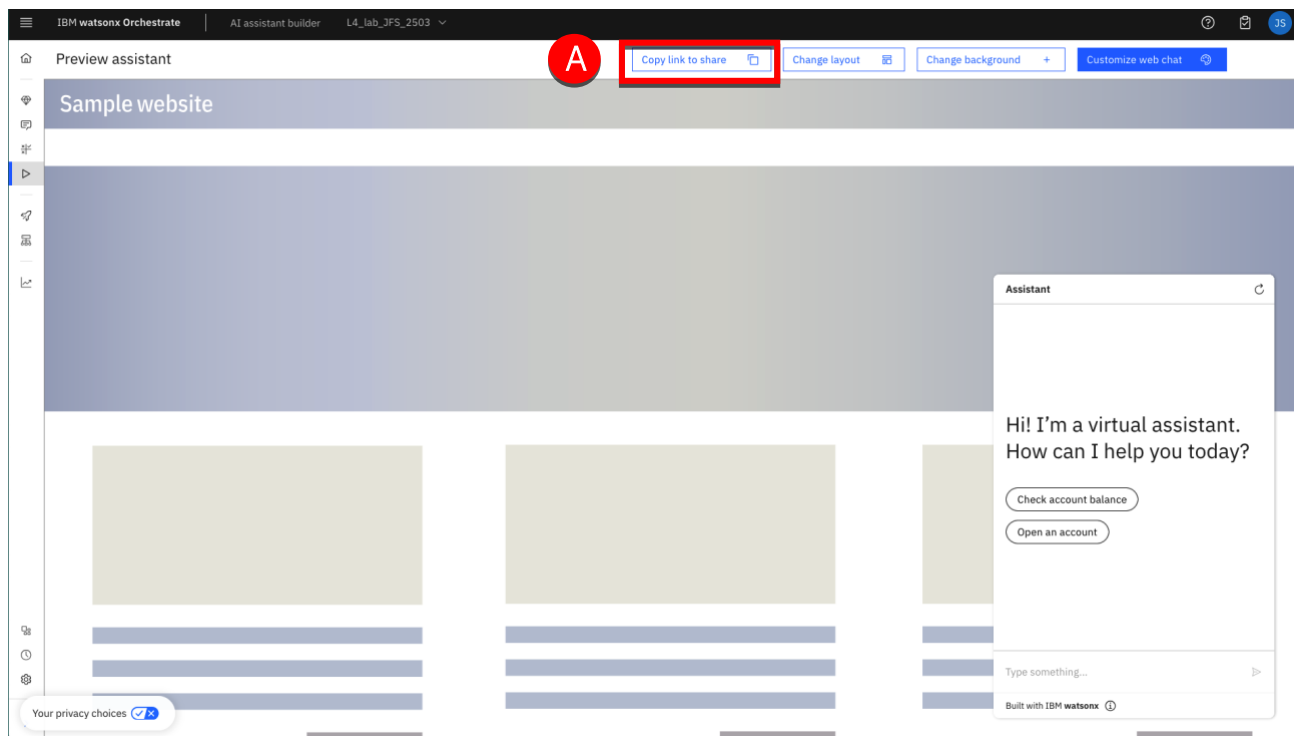
[Show more](#)

Your privacy choices

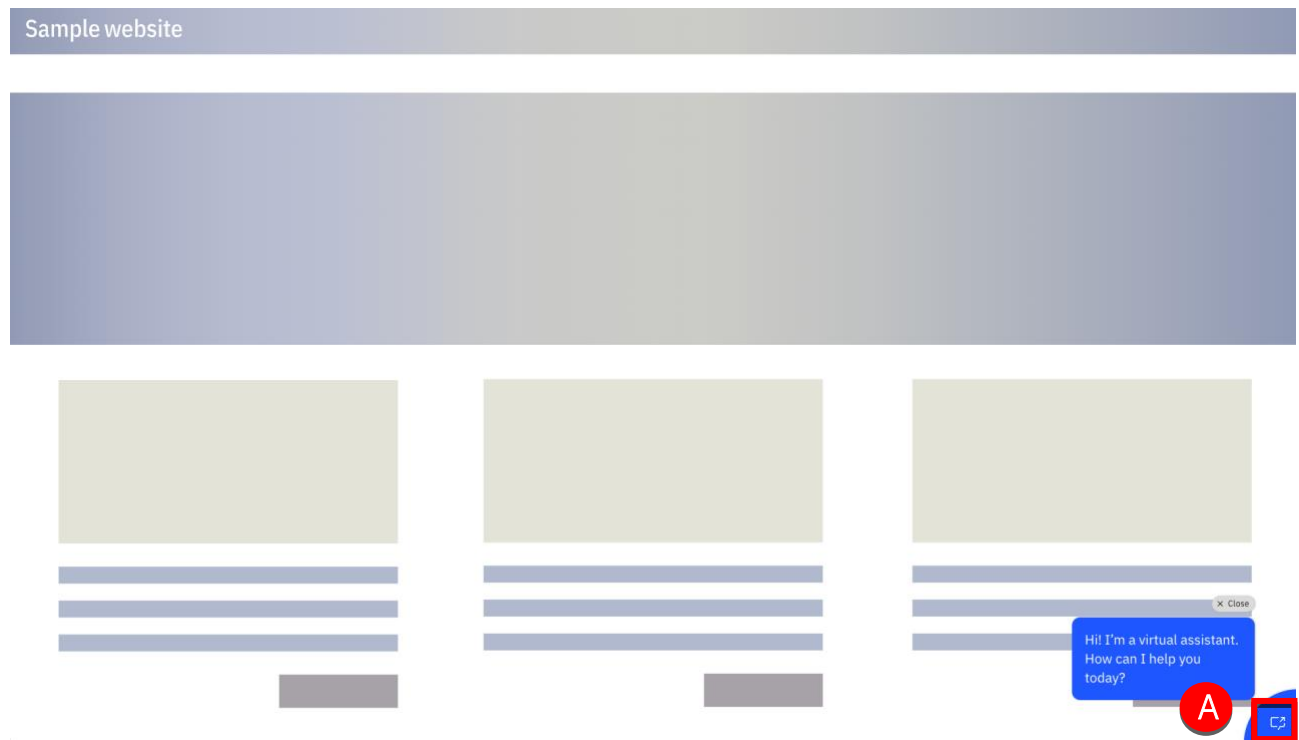
16. Click **Save and exit** (A) to return to the preview assistant window.



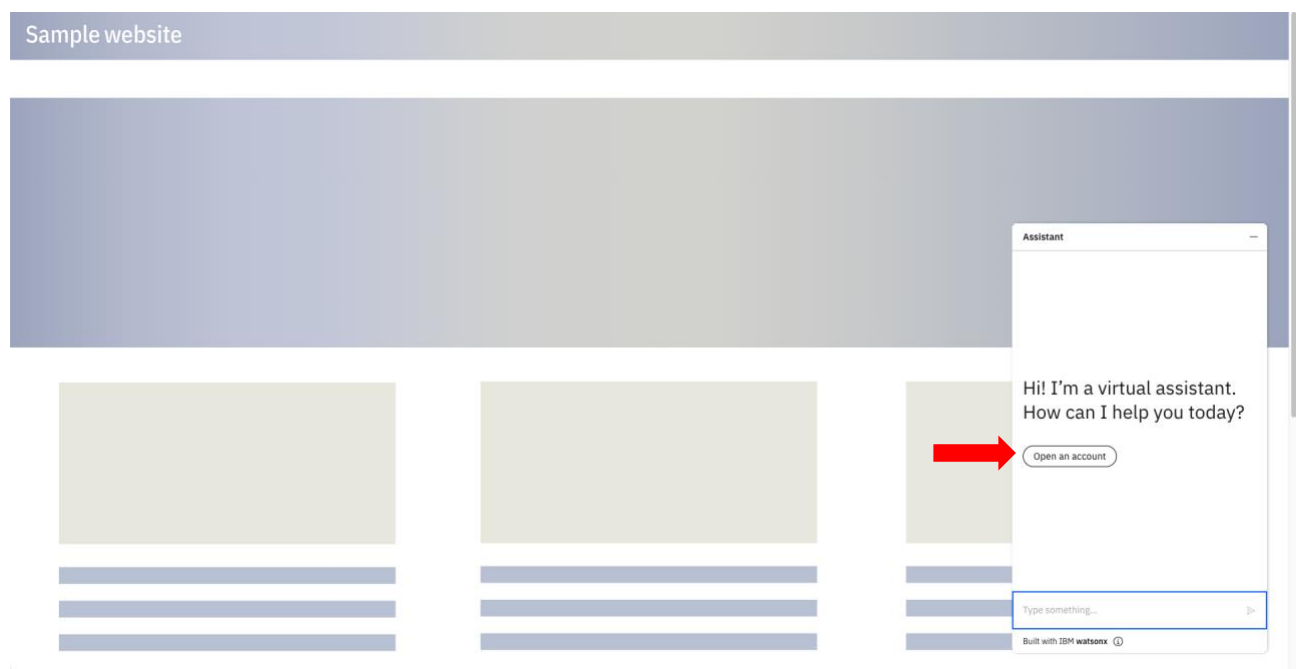
17. Click **Copy Link to share** (A).



18. Open a new window in your web browser and paste the copied link. Then, if the Webchat is not already open, click the virtual assistant in the bottom right-hand corner (A).



19. Observe the web chat of your assistant. Note that it has the updated conversation starters.



Additionally, note that you can interact with and preview your assistant outside of the watsonx Orchestrate AI assistant editor.

## Environments and publishing the assistant

Now, you will publish a first version of the assistant. So far, you have been building an assistant in a Draft environment.

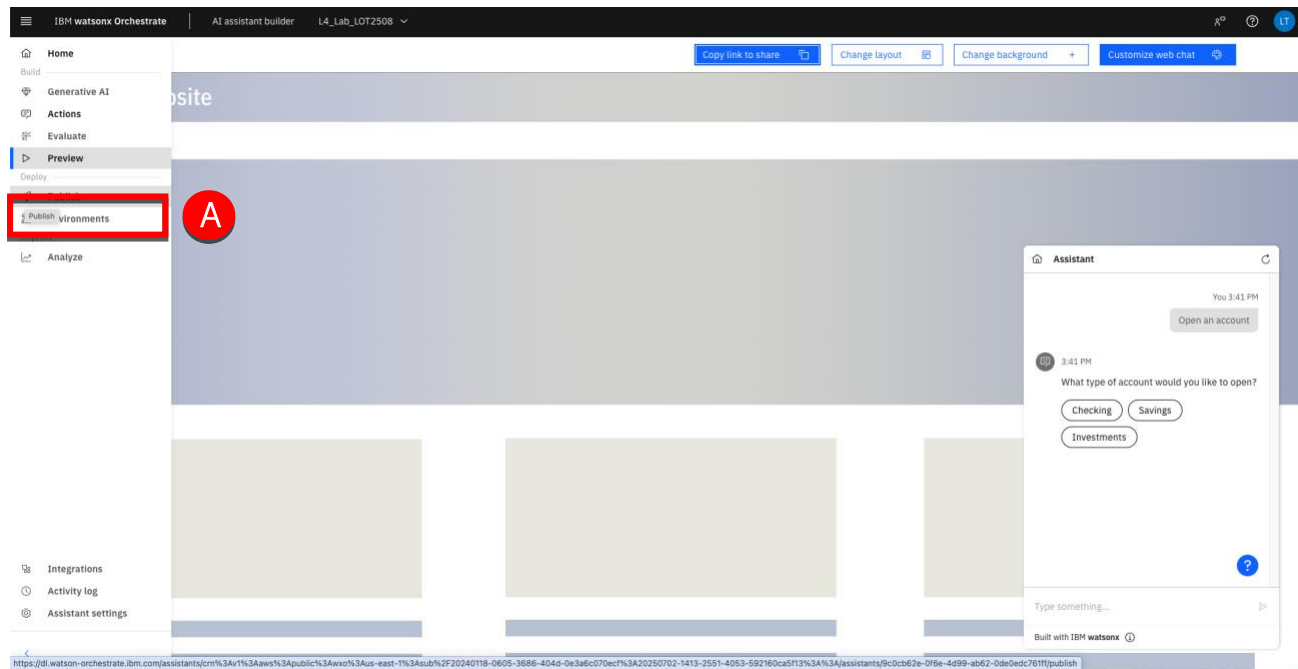
[Environments](#) are ways to isolate your work with an assistant. Each new assistant comes with two environments: the draft environment and the live environment. Your customers interact with assistants on the live environment and cannot interact with assistants on the draft environment. You can build and iterate on your content separately from what your customers see. You do not want customers to stumble upon an incomplete action that leads them to a dead end.

Each environment contains its own content, channels, and extensions. Environments also have their own IDs, URLs, and service credentials that can be referenced by external services.

### Publish in Draft environment

To start, you will publish your current assistant to a draft environment.

1. Return to the Preview assistant tab in your browser, then expand the lefthand pane and click **Publish (A)**.



[Publishing](#) is a way to maintain a healthy lifecycle management process. You can create incremental versions of your content over time, making it easier to manage deployment of changes and roll back (revert) to prior versions if necessary.

2. Review the **Unpublished content**, then click **Publish (A)**.

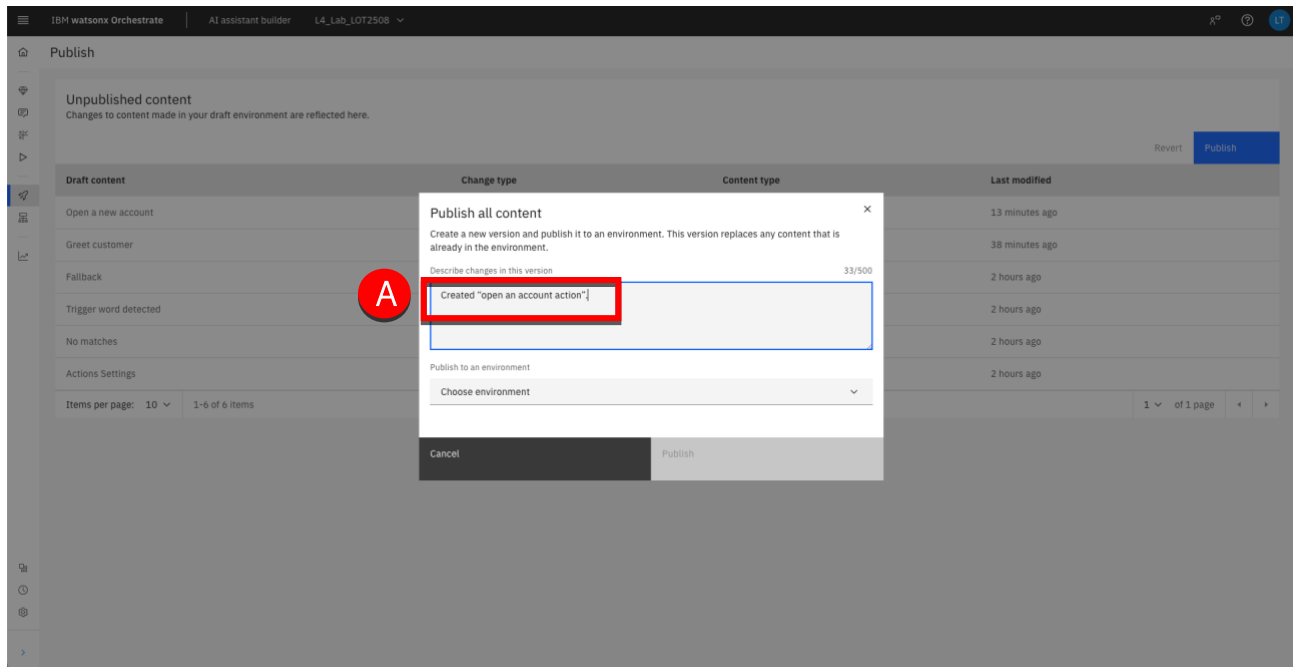
The screenshot shows the 'Publish' page in the IBM watsonx Orchestrator interface. The page title is 'Publish'. Below the title, it says 'Unpublished content' and 'Changes to content made in your draft environment are reflected here.' There is a red circle with a white 'A' next to a 'Publish' button. A red arrow points to the 'No matches' row in the table. The table has four columns: 'Draft content', 'Change type', 'Content type', and 'Last modified'. The table lists several draft items, including 'Open a new account', 'Greet customer', 'Fallback', 'Trigger word detected', 'No matches', and 'Actions Settings'. The 'No matches' row is highlighted with a red arrow. The 'Actions Settings' row is also visible. At the bottom of the table, there is a pagination bar showing 'Items per page: 10' and '1-6 of 6 items'.

| Draft content         | Change type | Content type | Last modified  |
|-----------------------|-------------|--------------|----------------|
| Open a new account    | Updated     | Actions      | 12 minutes ago |
| Greet customer        | Updated     | Actions      | 37 minutes ago |
| Fallback              | Updated     | Actions      | 2 hours ago    |
| Trigger word detected | Updated     | Actions      | 2 hours ago    |
| No matches            | Updated     | Actions      | 2 hours ago    |
| Actions Settings      | Updated     | Settings     | 2 hours ago    |

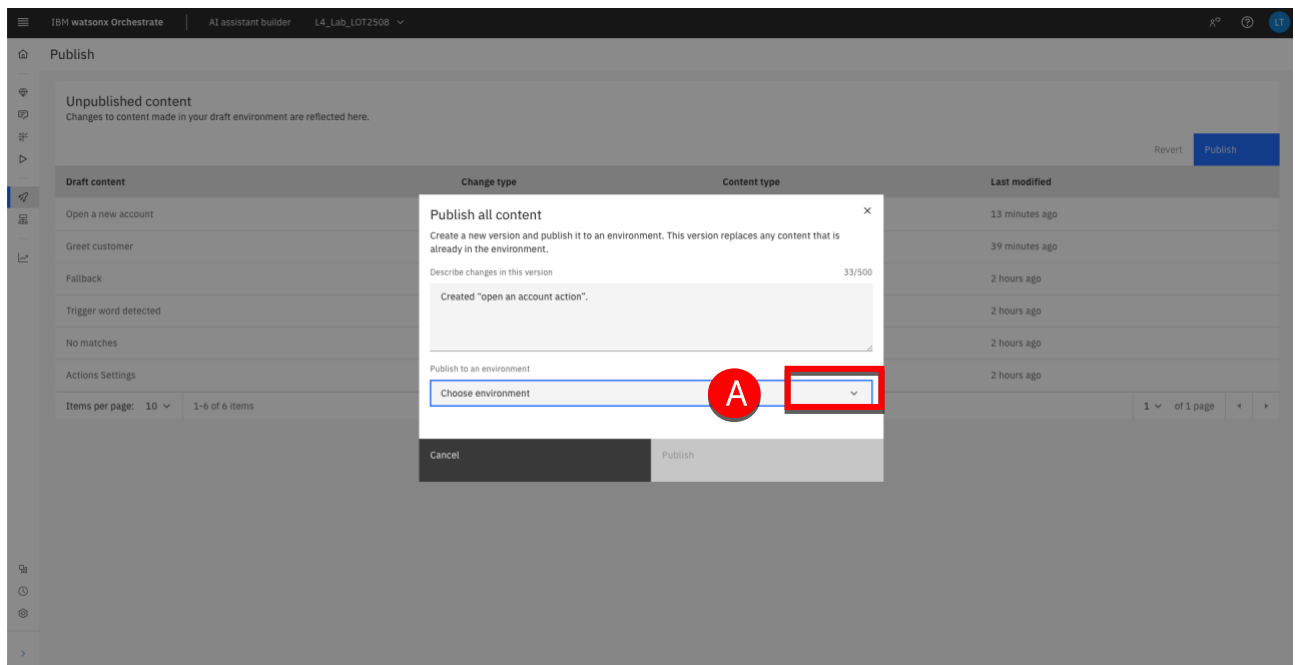
When you publish your content, AI assistant builder creates a snapshot of the draft content, resulting in a version. This version contains all the content from actions, including settings and variables.

**Note:** Published Versions do not contain integration configurations or environment settings. Integration configurations and environment settings must be configured independently for each draft and live environment.

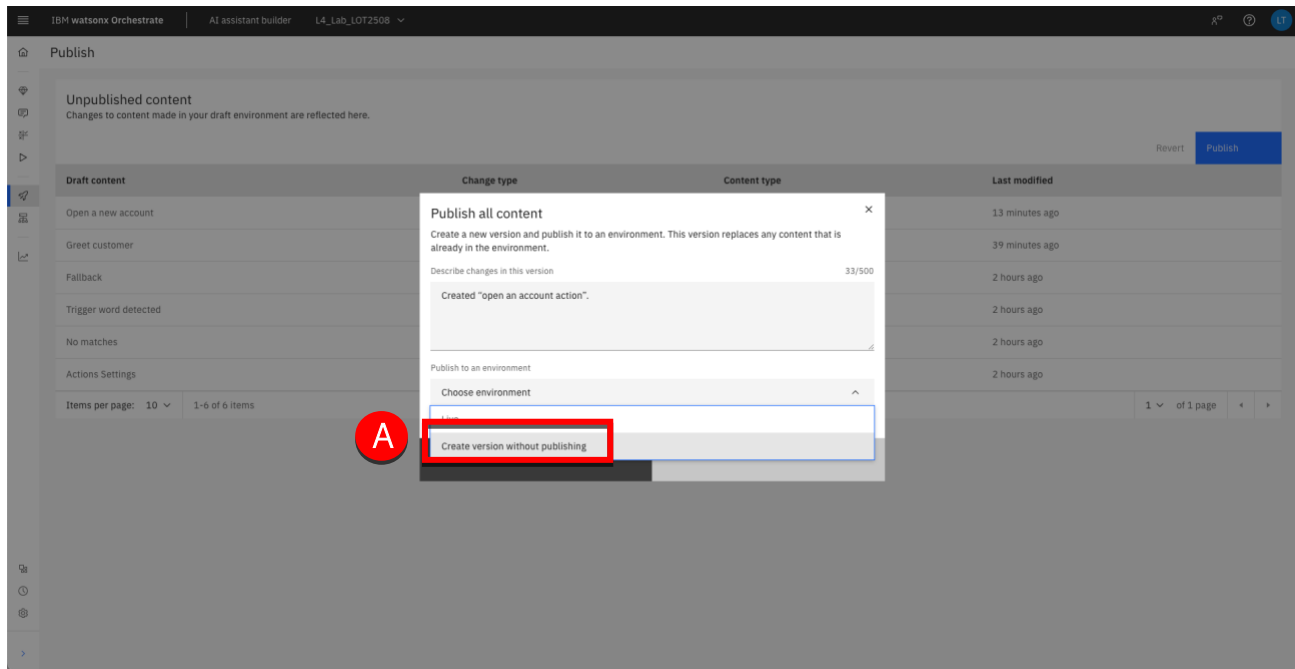
3. In the publish window, type Created “open an account” action. in the **Describe changes in this version** pane (A).



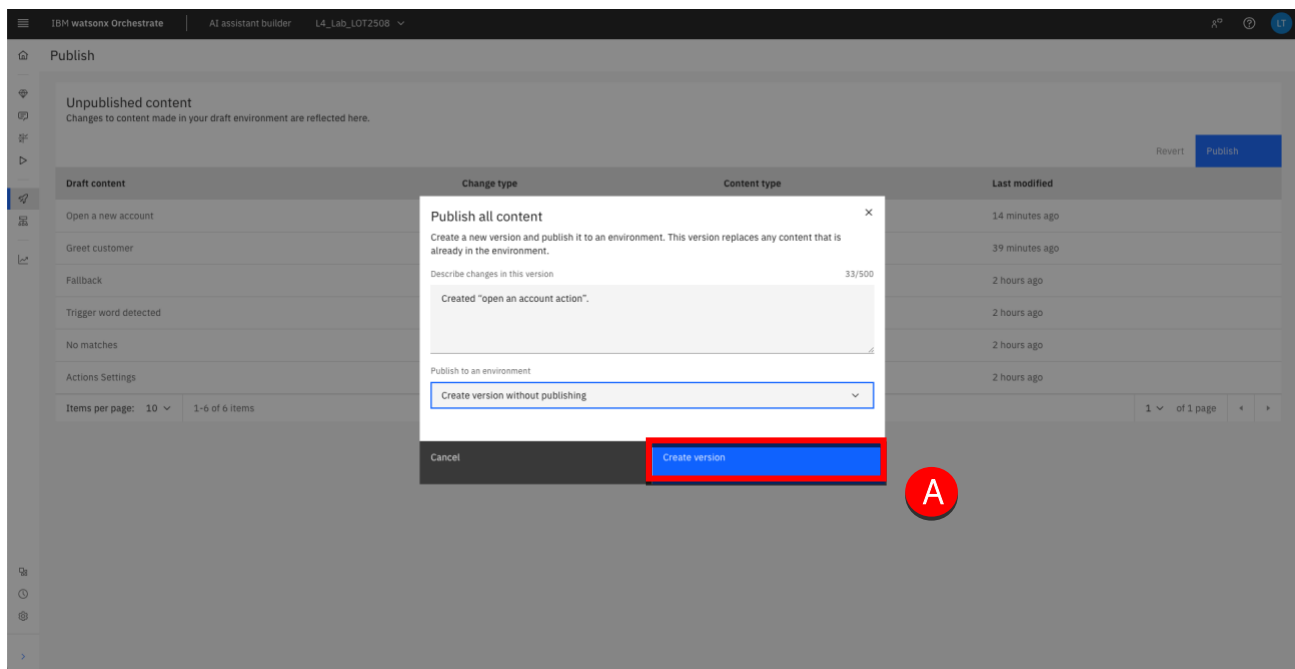
4. Open the options in the **Publish to an environment** drop down window (A).



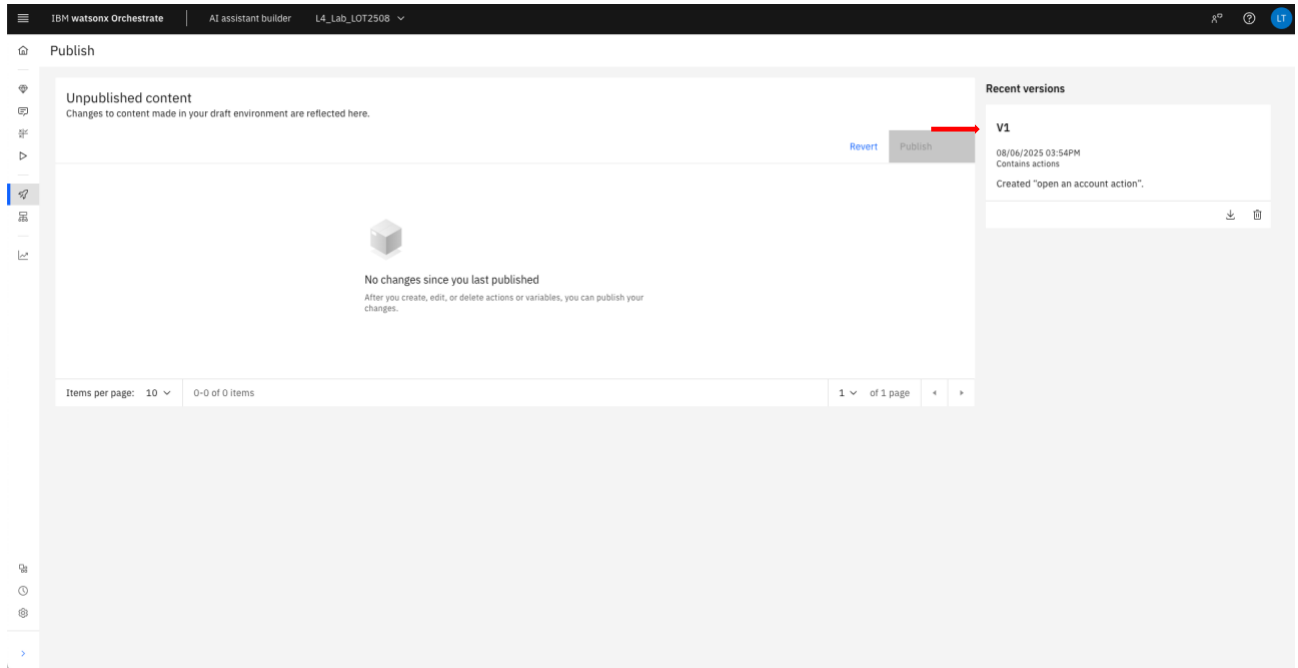
5. Select **Create a version without publishing** (A).



6. Click **Create version** (A).



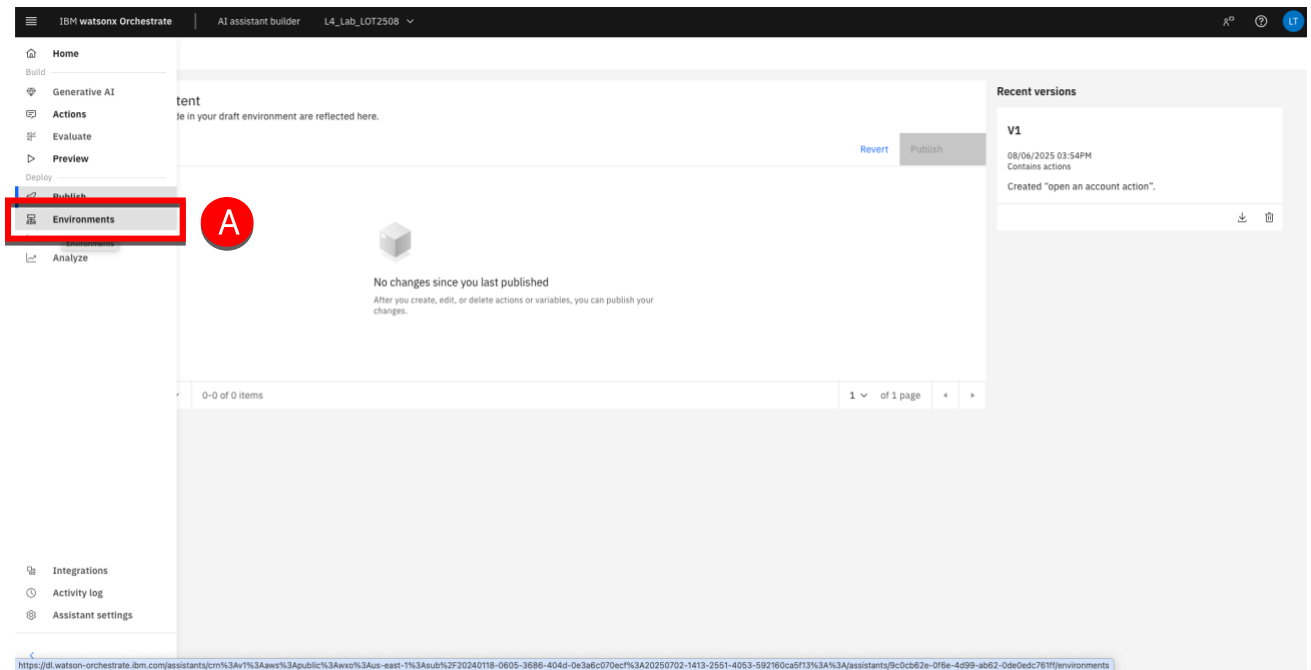
Let the AI assistant finish publishing your assistant in the Draft environment. Once published, observe that V1 has been created.



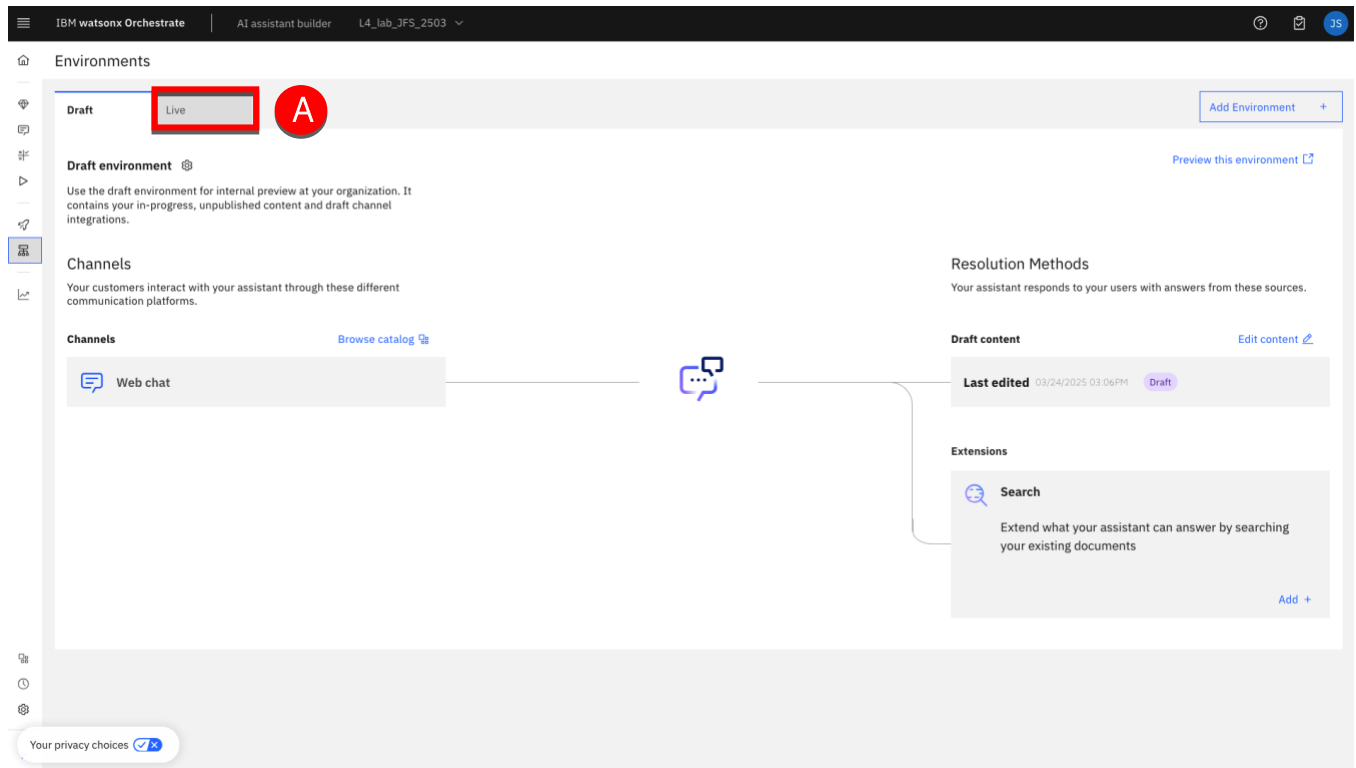
## Publish in the Live environment

Now, you will publish your assistant in a Live environment.

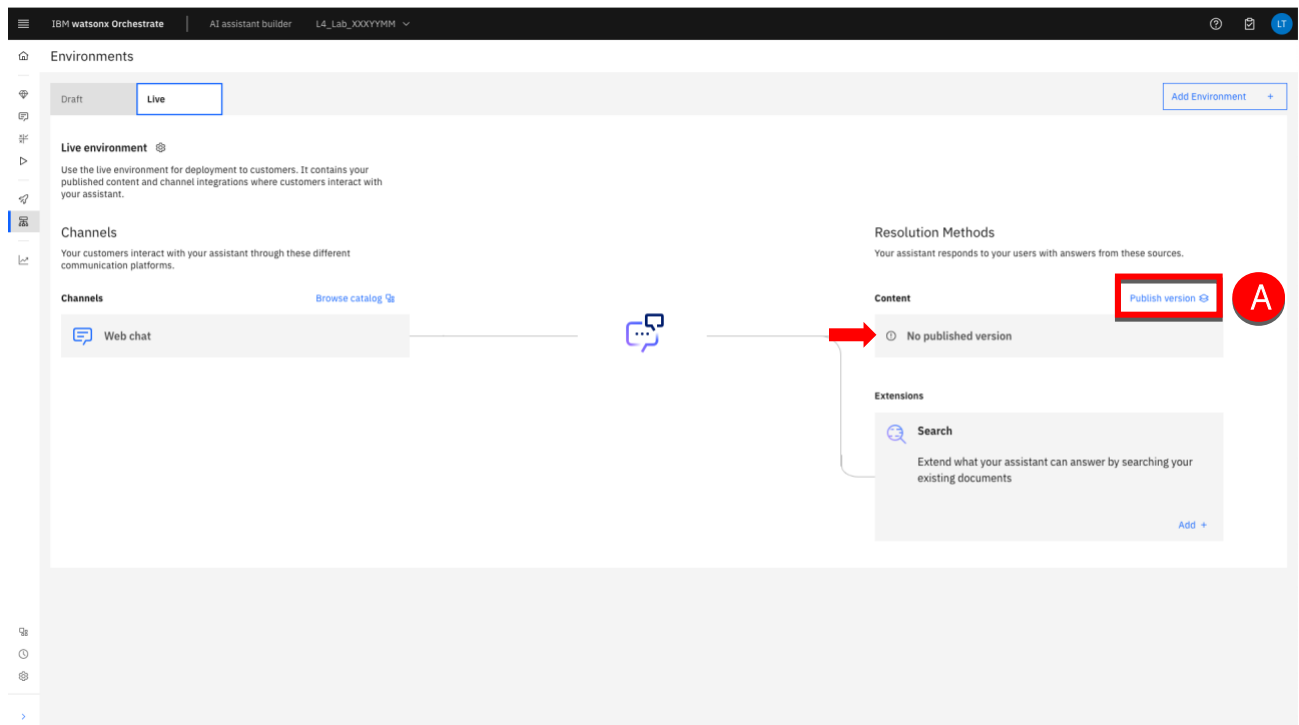
1. In the left pane, click **Environments** (A).



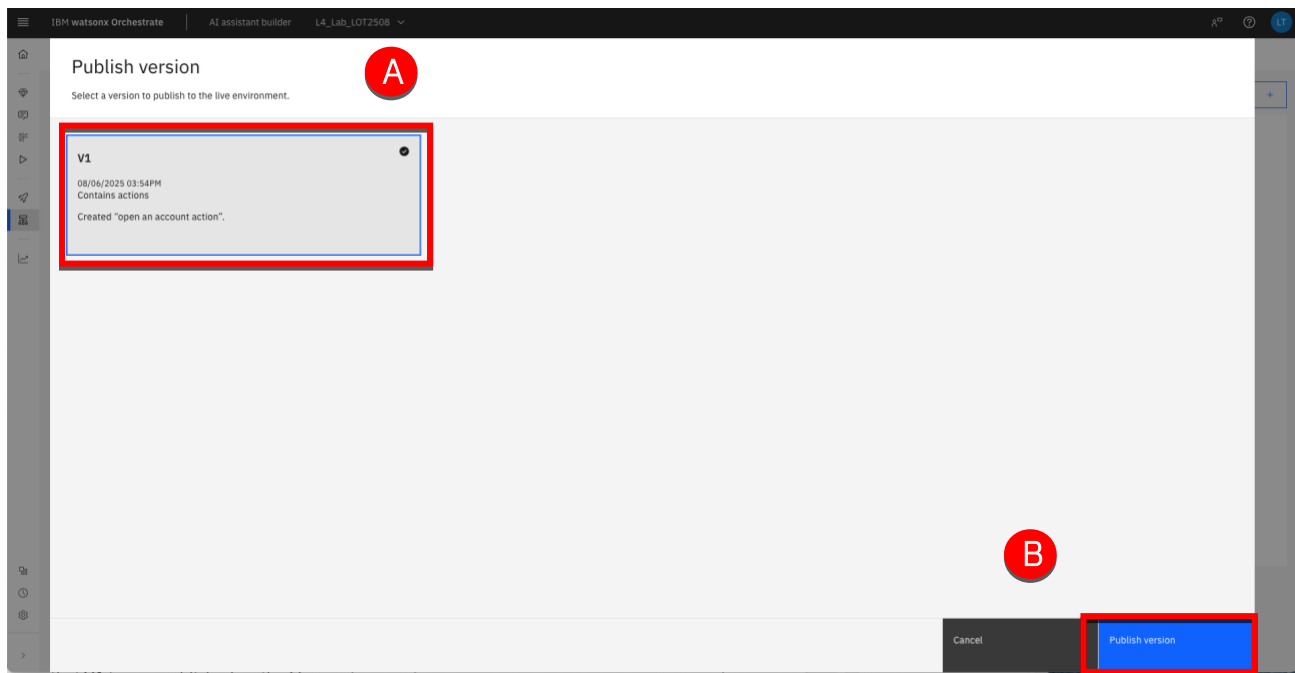
2. In the Environments page, select the **Live** tab (A).



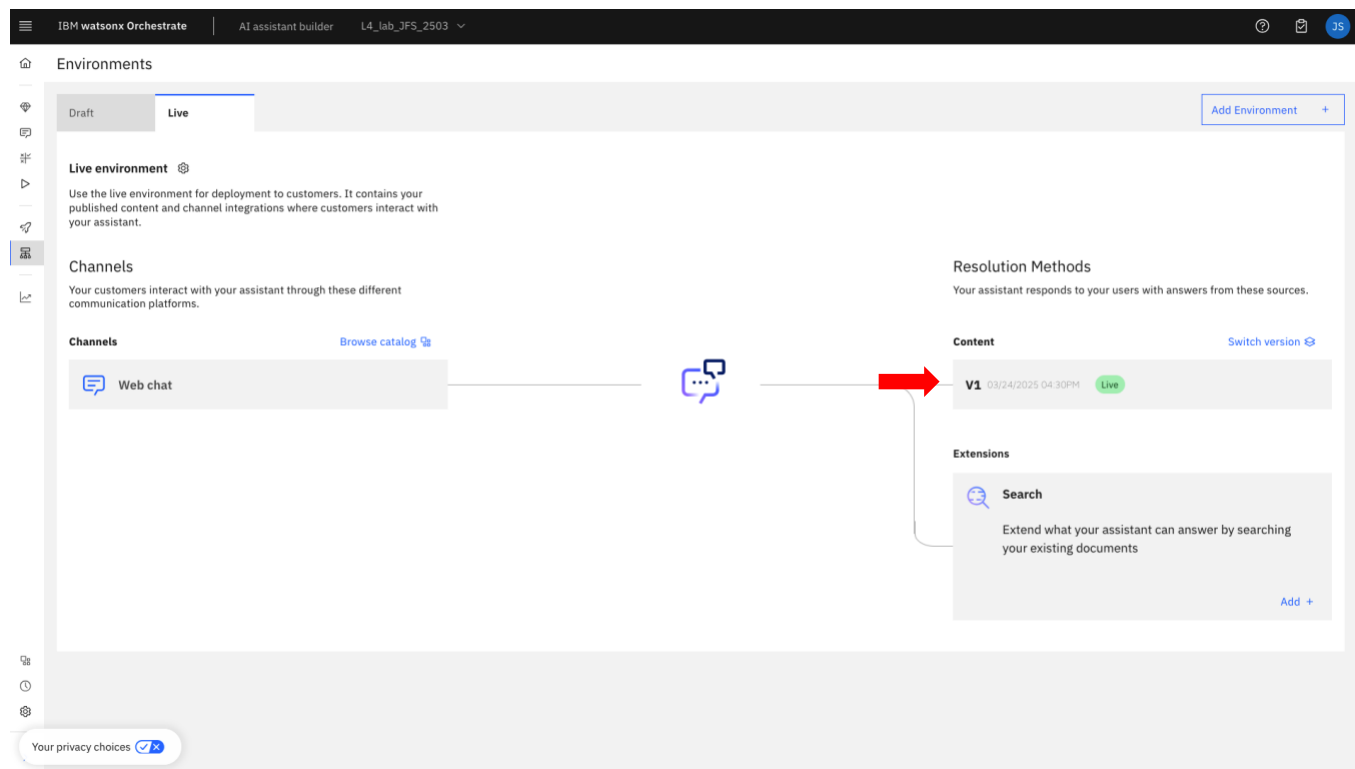
3. Observe that no versions have been published in the Live environment. Click **Publish version** (A).



4. In the publish version window, Select **V1** (A) then click **Publish version** (B).



5. Observe that **V1** is now published on the **Live** environment.



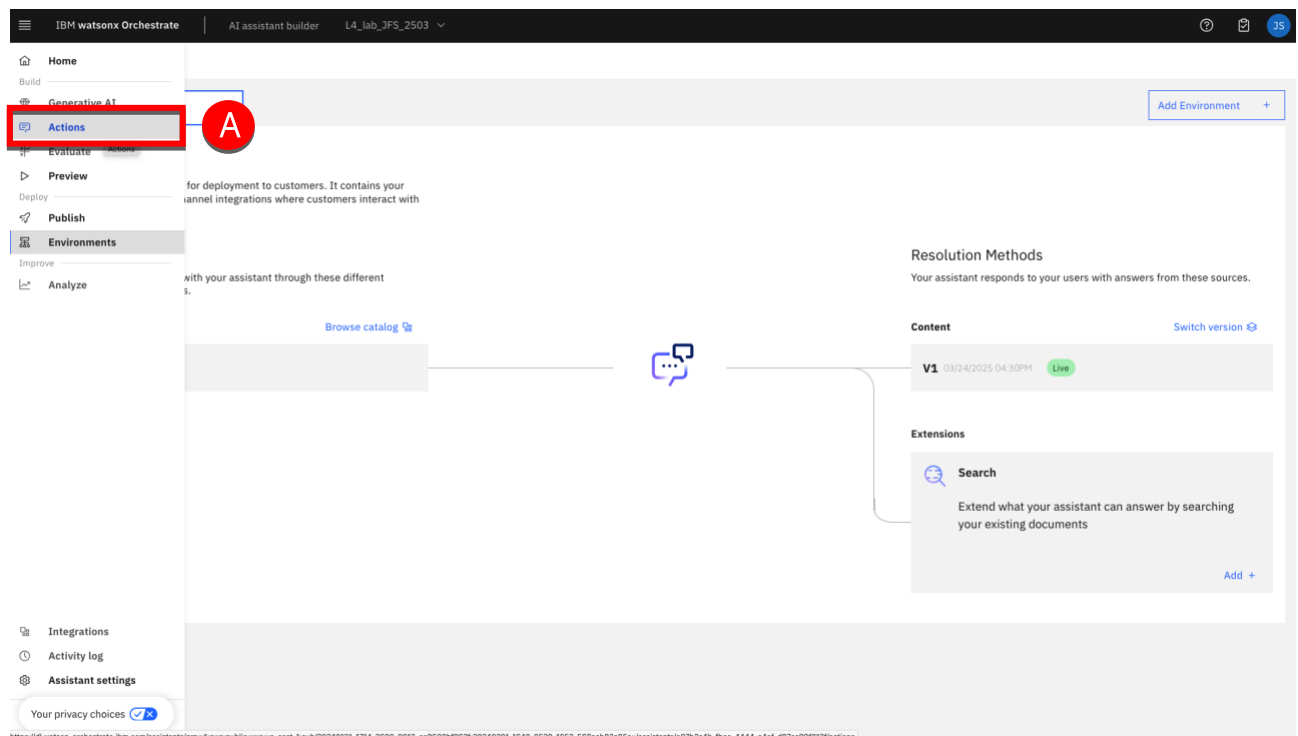
## Iterate on your assistant

Your assistant is now published in a live environment. Even though your assistant is published and Live, you want to continue to build and iterate.

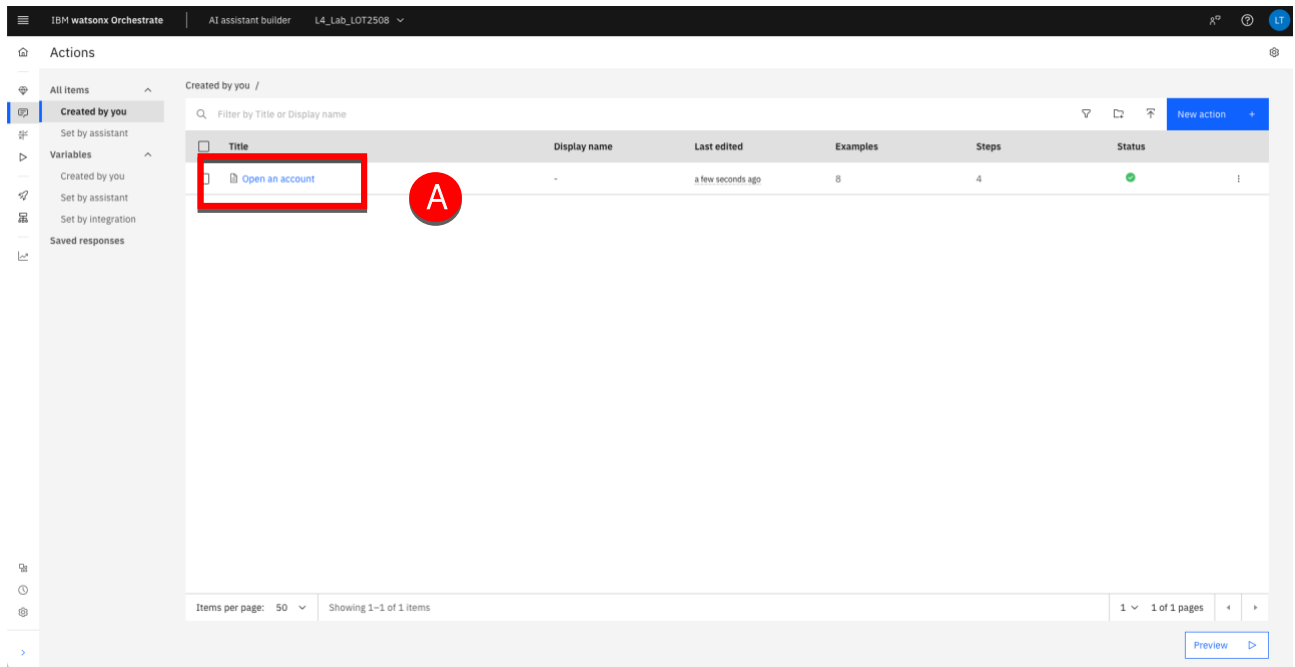
Specifically, you want to update the **Open an account** action to ask for a properly formatted investor number – not just any number.

For this scenario, a properly formatted investor number has 3 digits, followed by dash, then 6 digits. An example is 123-456789

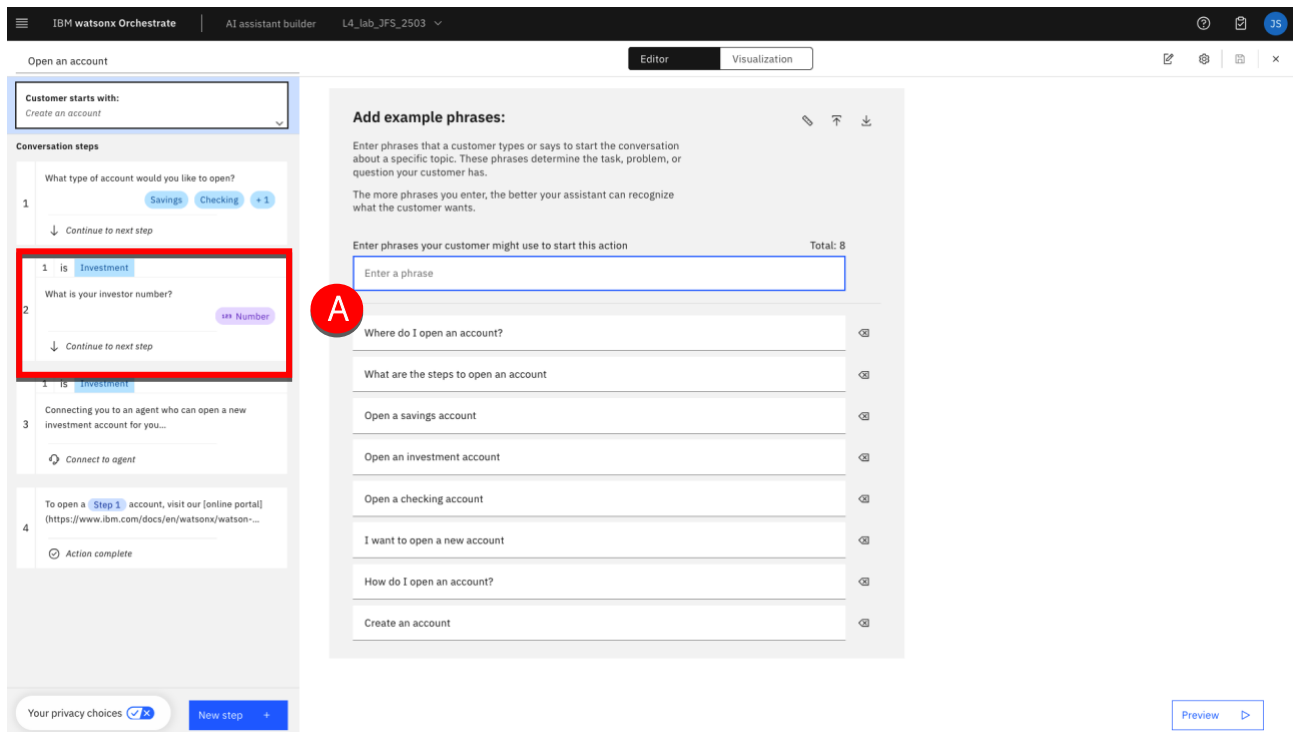
1. In the left pane, go to the **Actions** page (A).



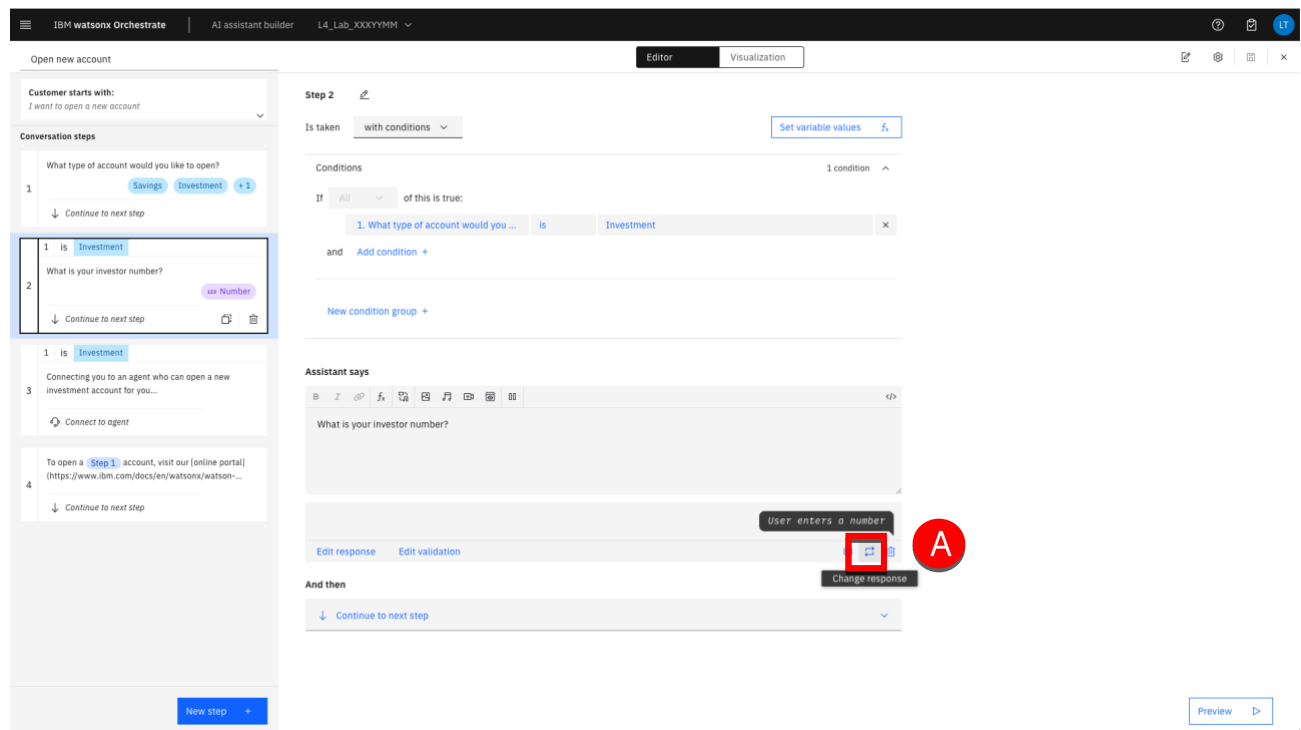
2. Click Open an account (A).



3. Select Step 2 (A).

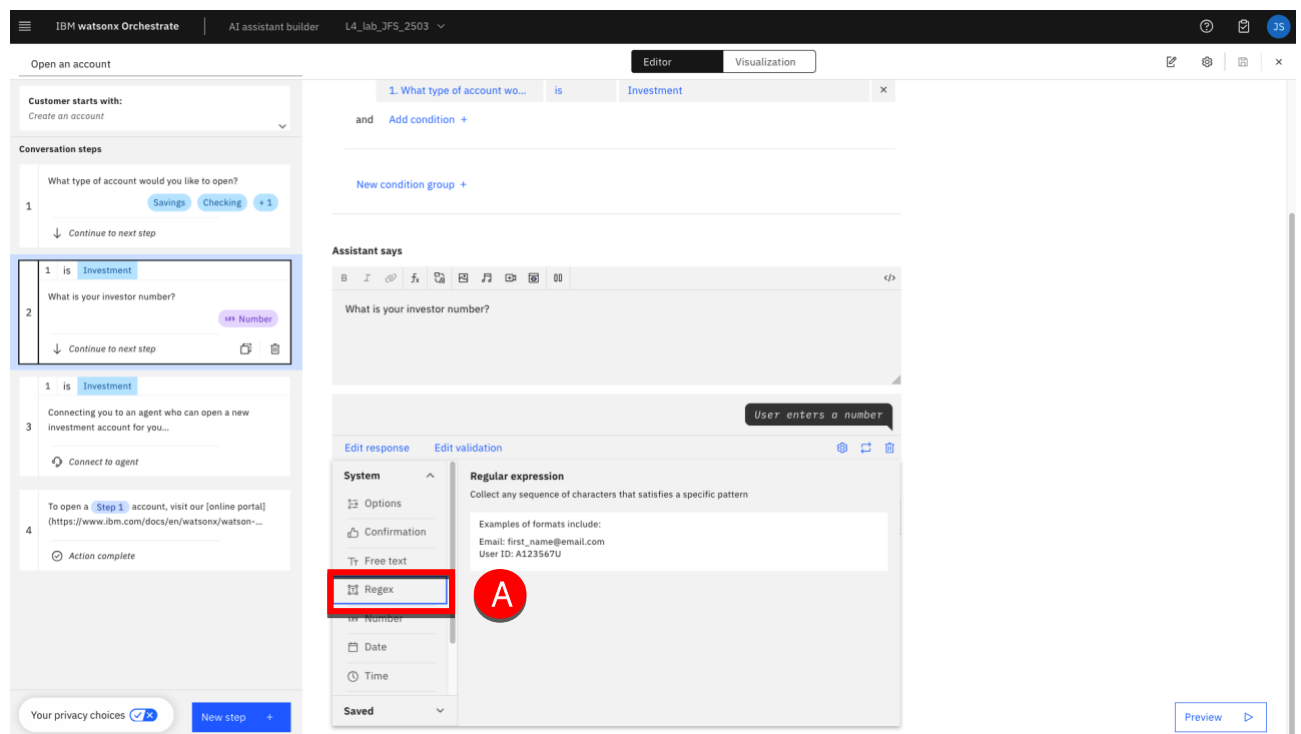


4. In the define customer response section, click the **Change response** icon (A).



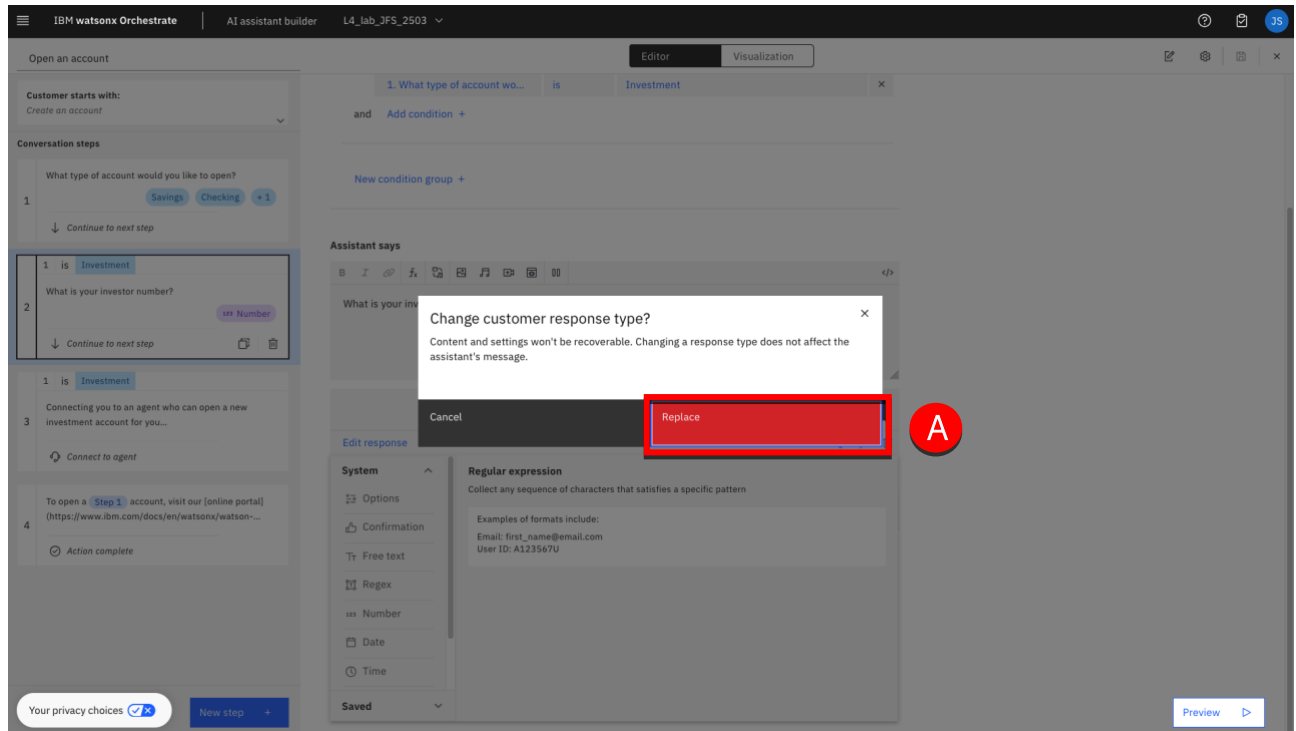
Let's now define the regular expression.

5. Select **Regex** (A).

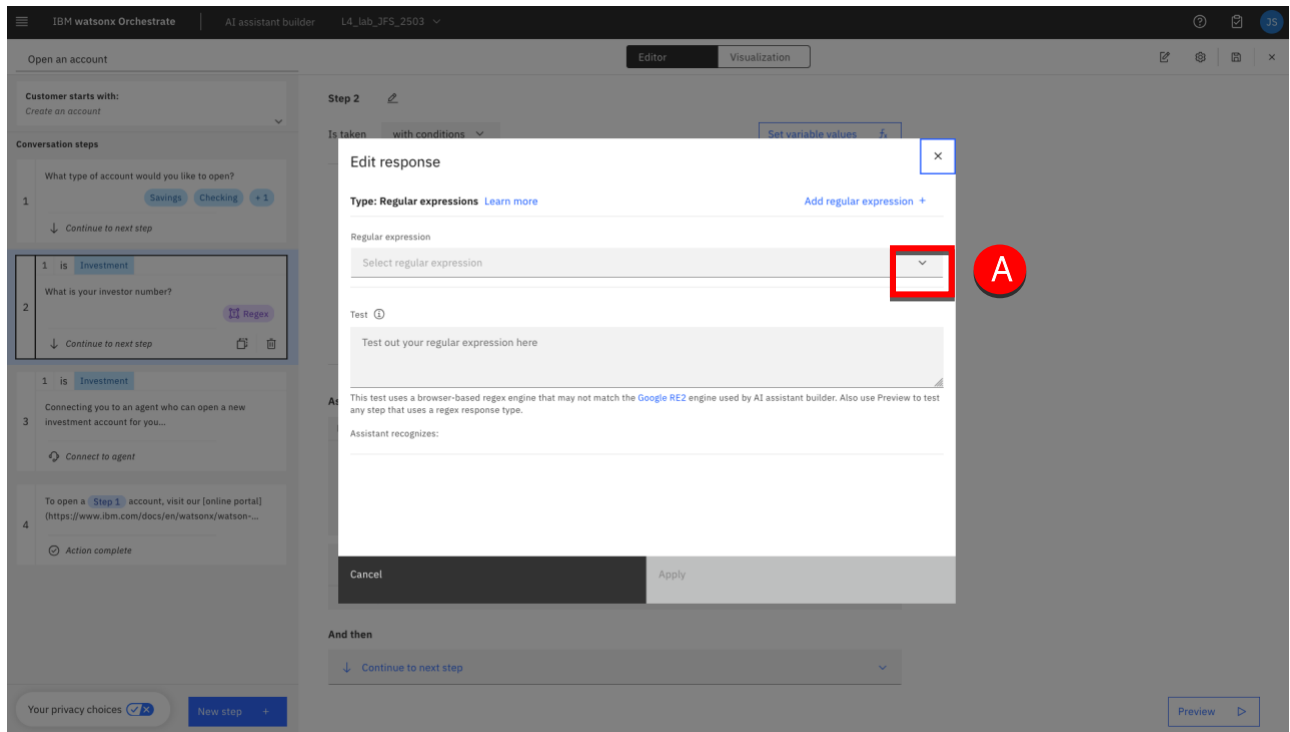


Regular expressions (Regex) are a sequence of characters that specifies a specific match pattern in text. You can think of regular expressions as a powerful and flexible “Find” / “Find and replace” method for text data. Discussions on the syntax and advanced applications of regular expressions is beyond the scope of this course, but there are many resources available online to learn about regular expressions.

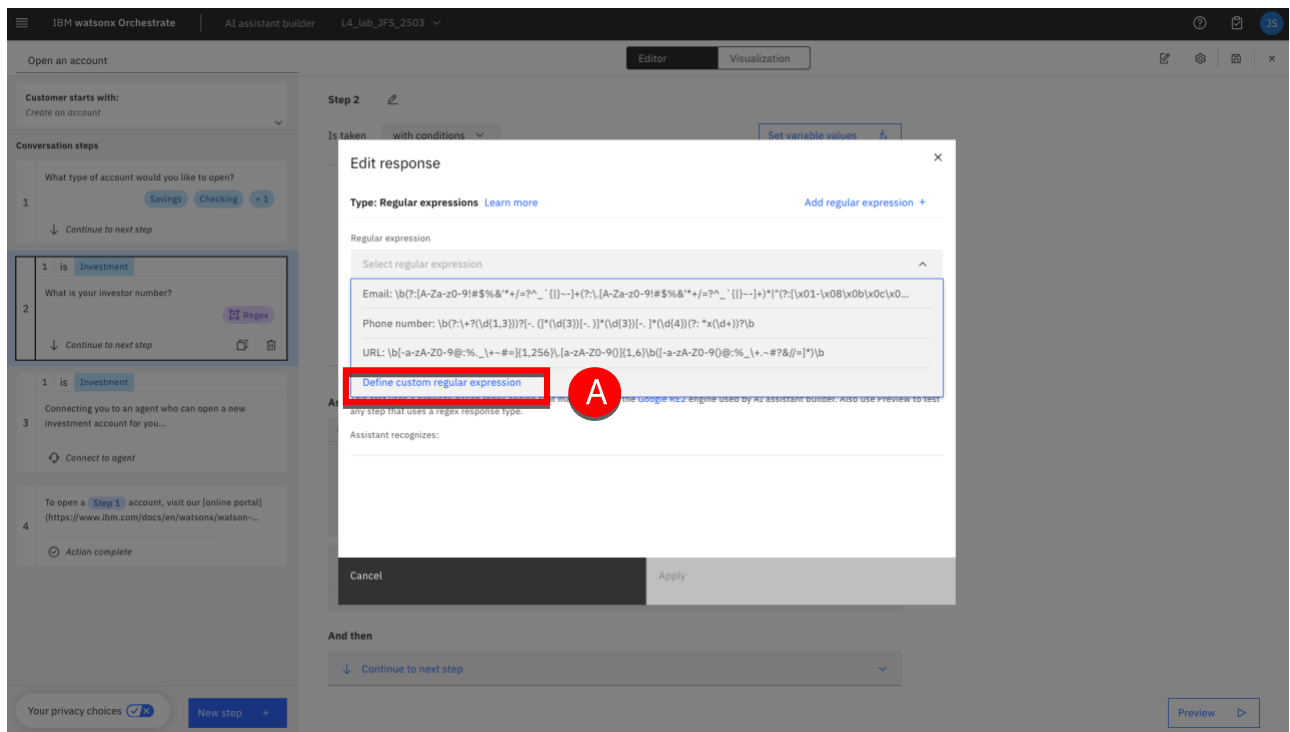
6. Click **Replace** (A).



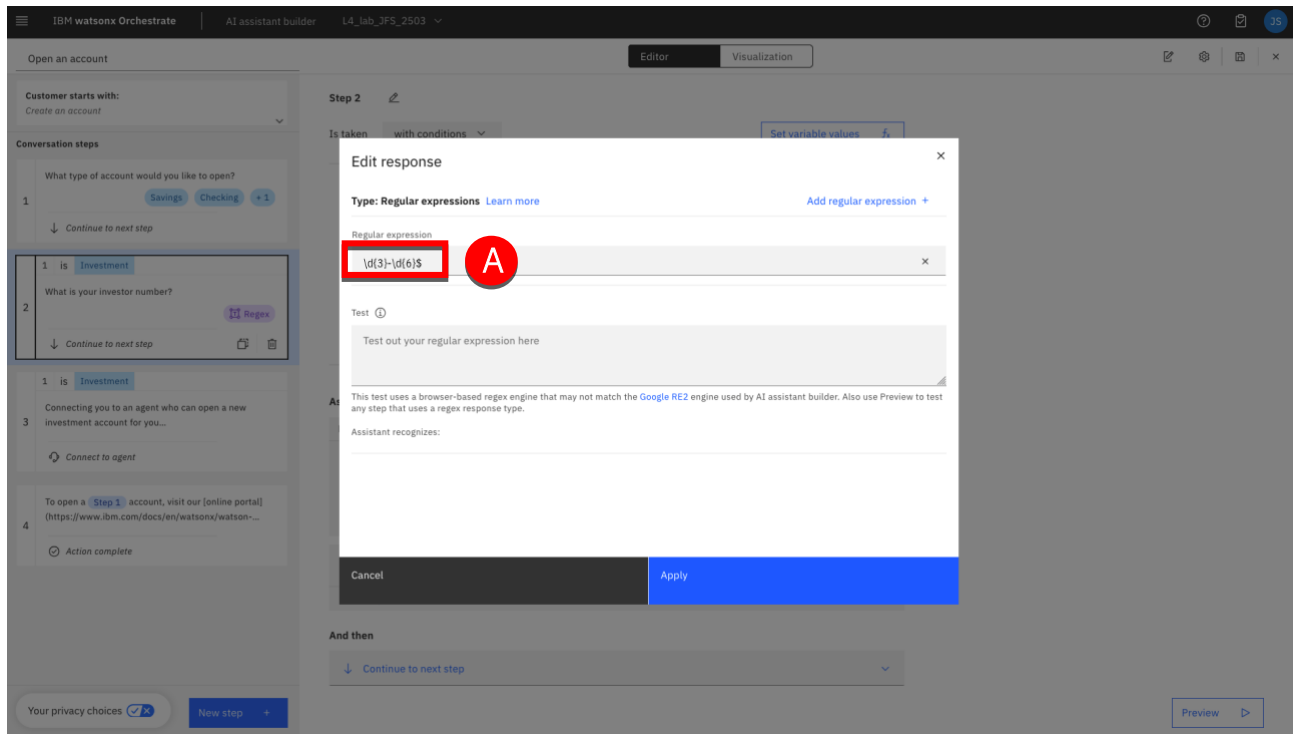
7. Expand the **Regular expression** dropdown menu (A).



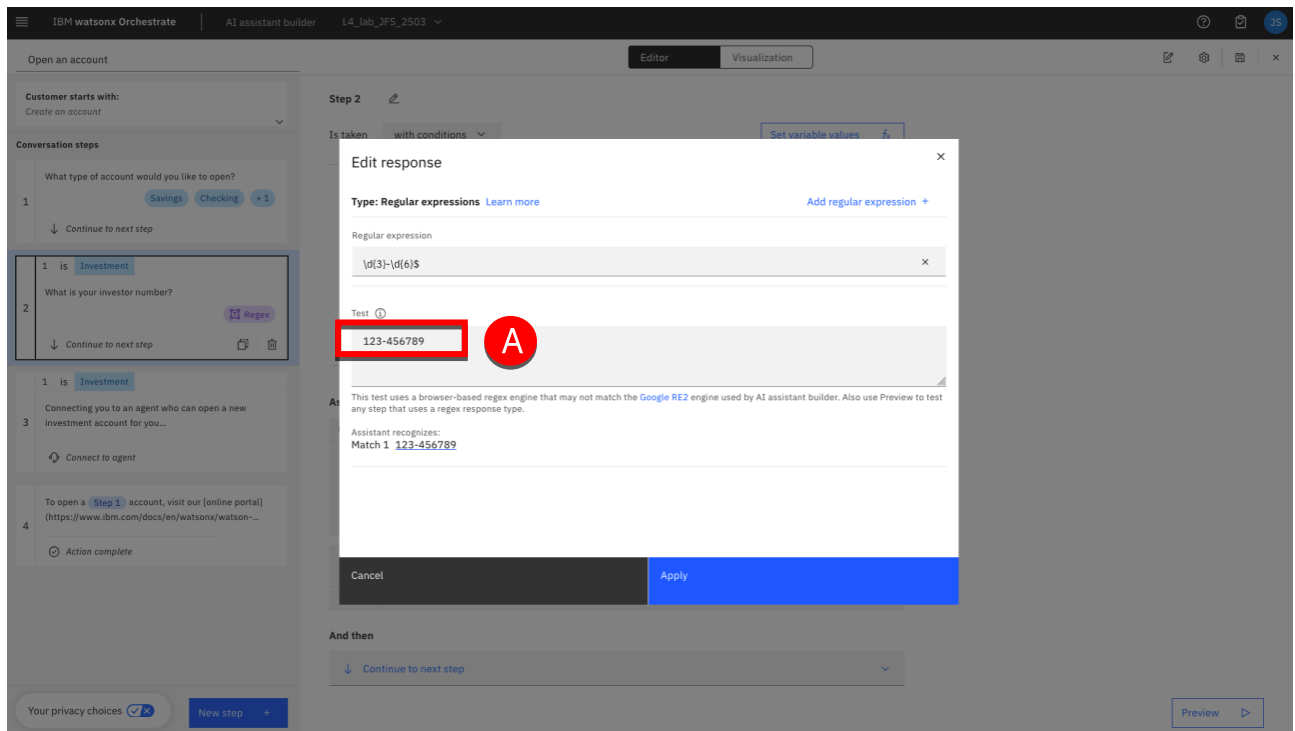
8. Select **Define custom regular expression** (A).



9. In the **Regular expression** field, type `\d{3}-\d{6}$` (A).



10. Next, you test your regular expression by typing 123 -456789 in the **Test** text field (A).

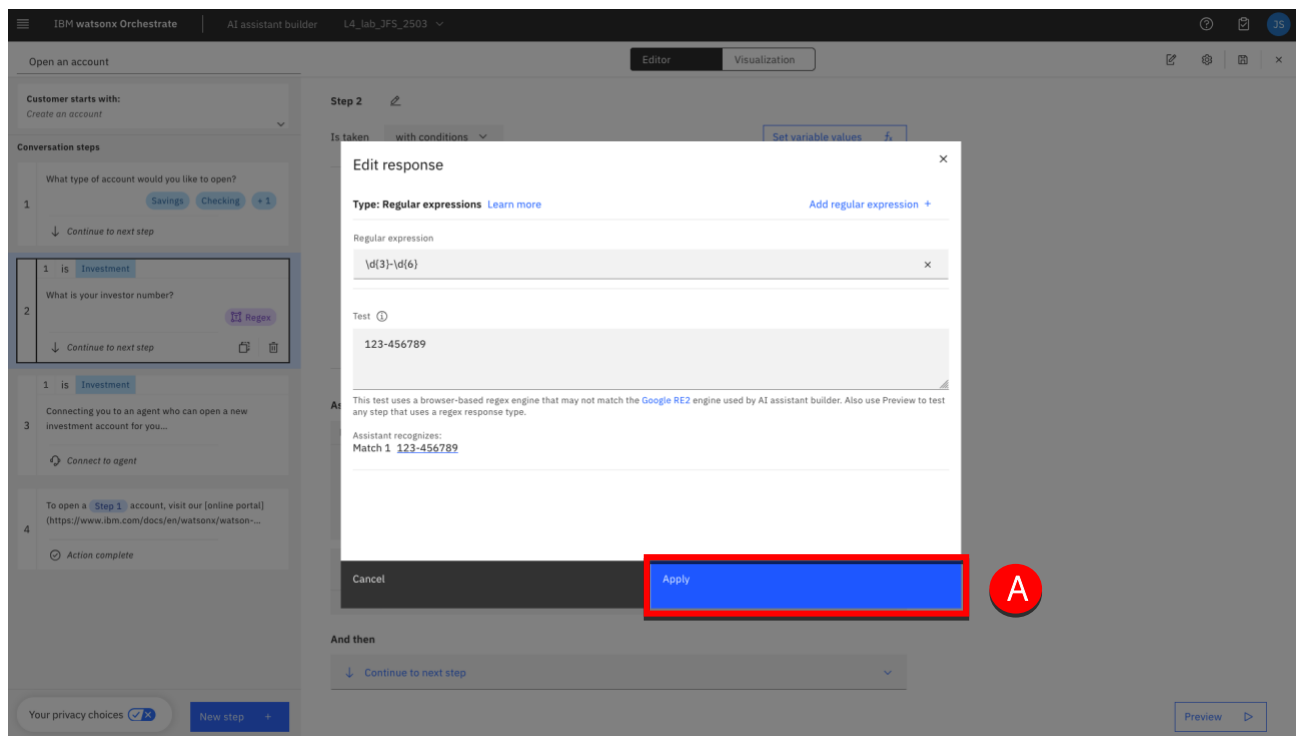


Note that the assistant matches the string once it follows the specified format.

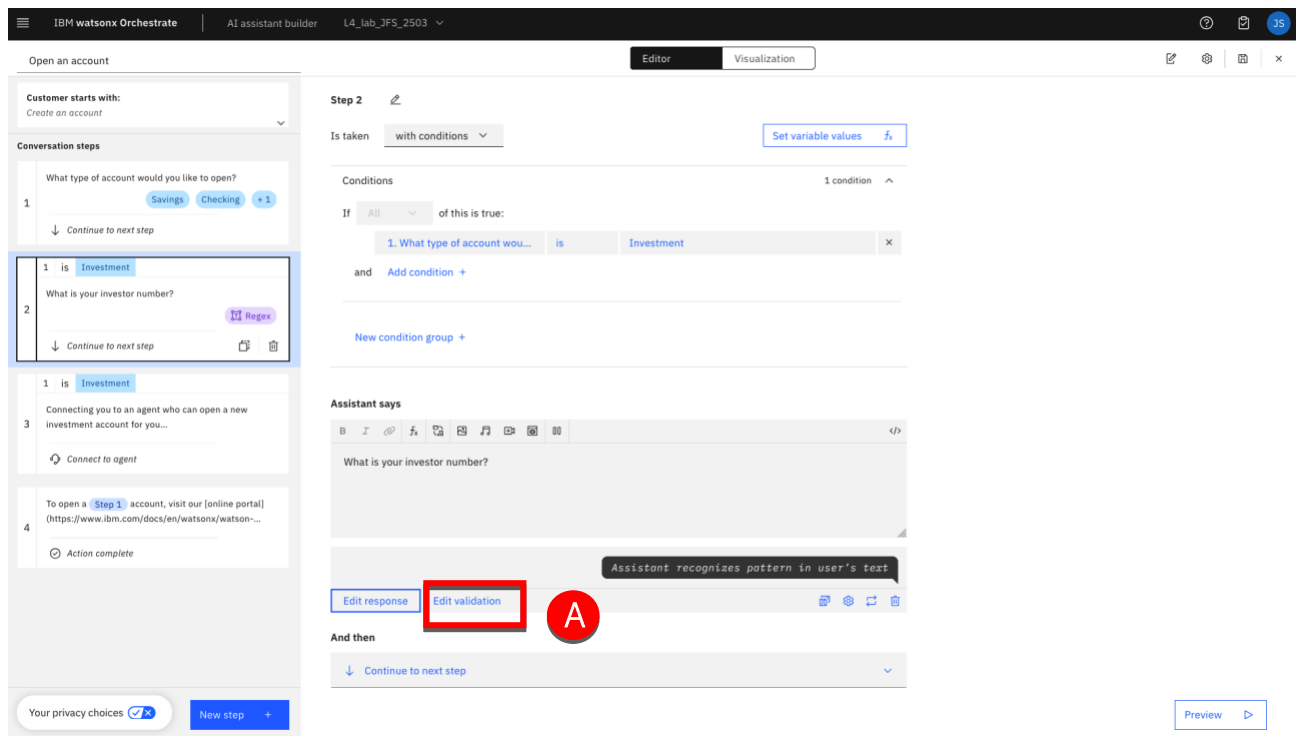
The regular expression `\d{3}-\d{6}$` matches a numeric pattern with the following structure:

- `\d{3}` : Exactly three digits (e.g., 123)
- `-`: A hyphen separating the two groups
- `\d{6}` : Exactly six digits (e.g., 456789)
- `$`: End of the chain.

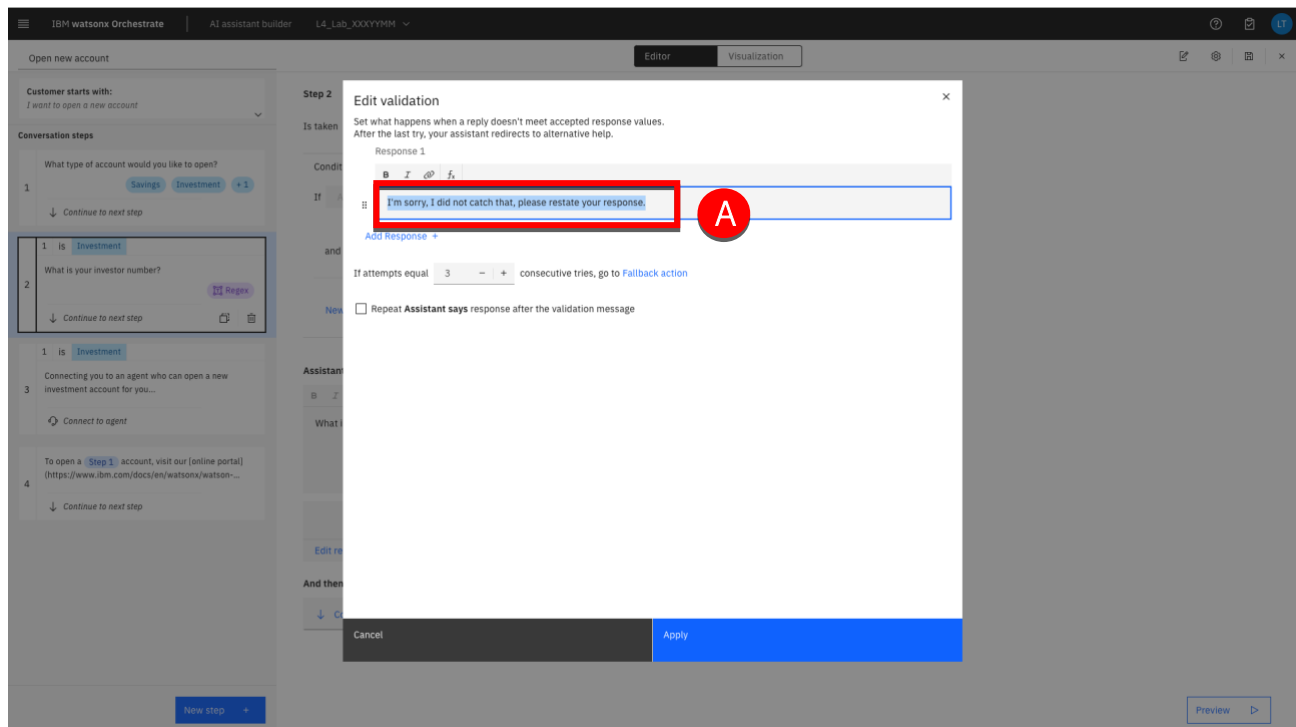
11. Click **Apply** (A).



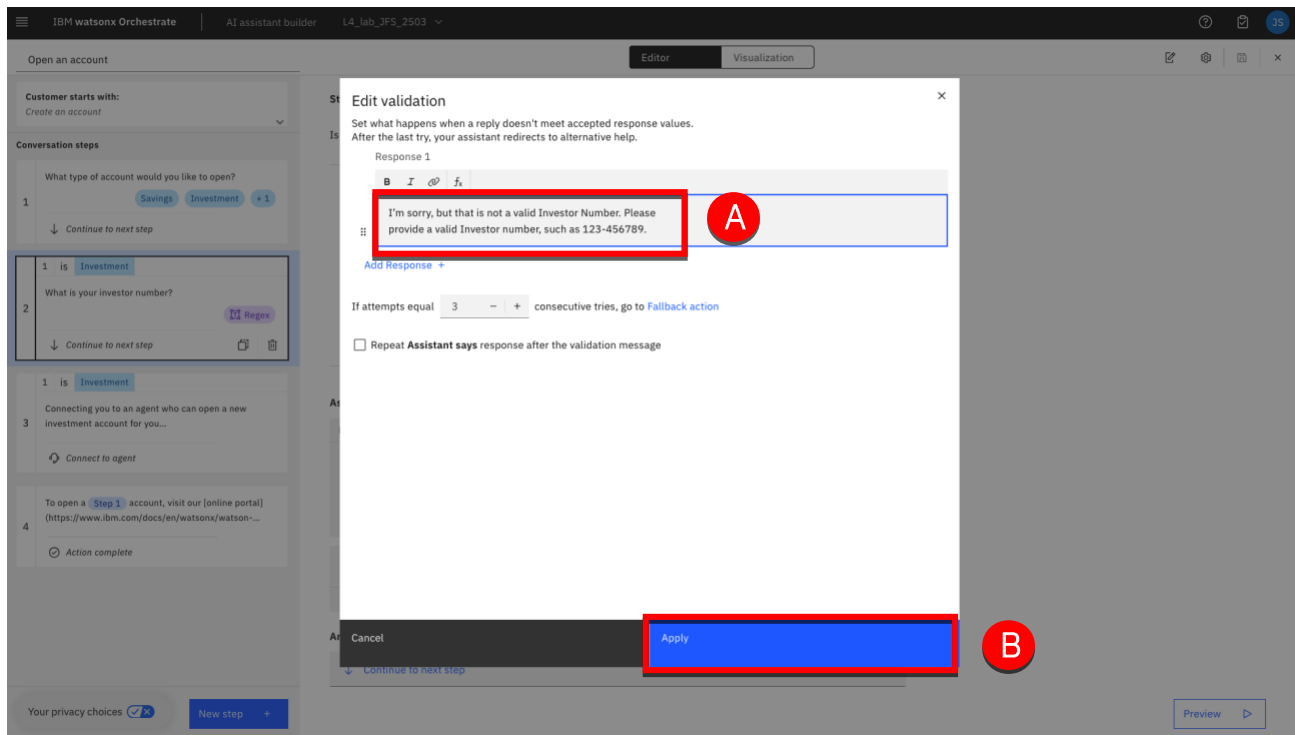
12. In the define customer response section, click **Edit validation** (A).



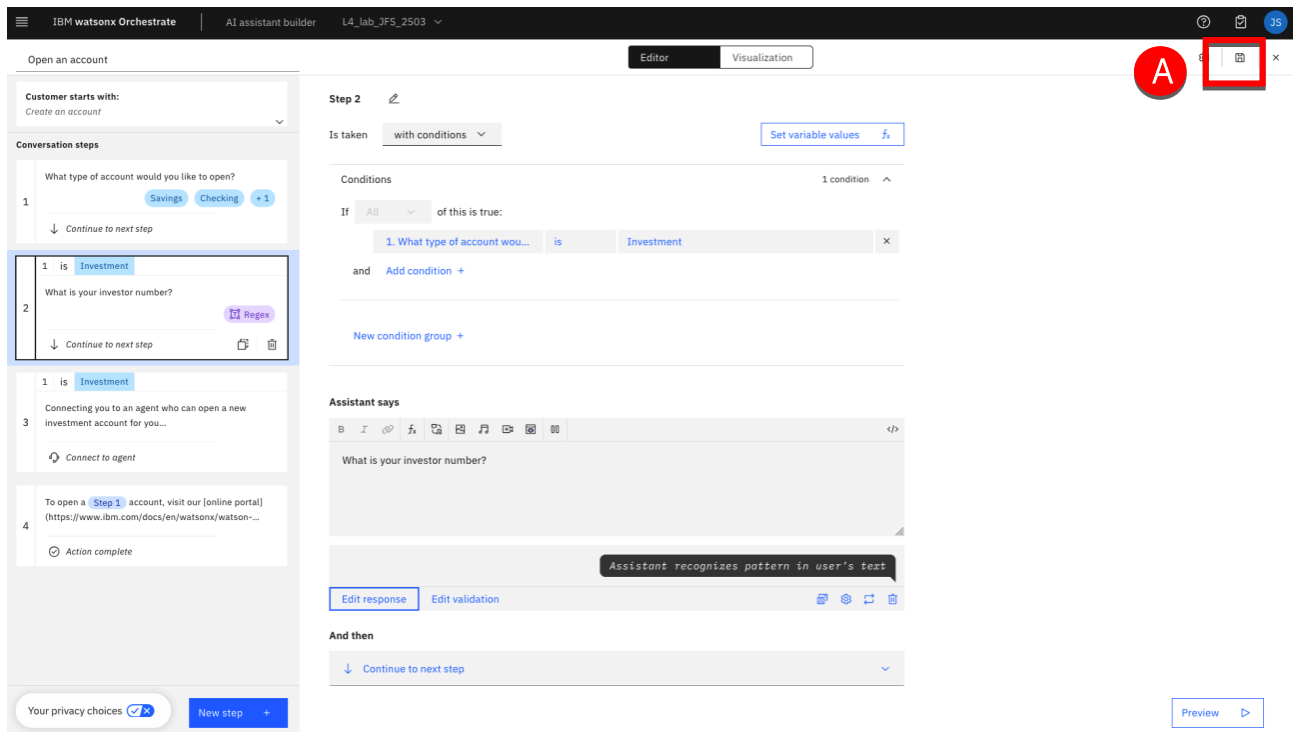
13. Observe the Edit validation window. Select the text in **Response 1** (A).



14. Delete the text in **Response 1**, then type I'm sorry, but that is not a valid Investor Number. Please provide a valid Investor number, such as 123-456789. (A) then click **Apply** (B).



15. To ensure these changes are saved, click the save icon (A).



16. Click **Preview** (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface. The left sidebar displays the 'Conversation steps' for 'Open an account'. Step 2 is highlighted with a blue box. The main area shows the configuration for Step 2, including conditions and the assistant's response. A red circle with the letter 'A' is placed over the 'Preview' button in the bottom right corner.

17. Type **Open account please** (A).

The screenshot shows the IBM watsonx Orchestrate AI assistant builder interface with the 'Preview' window open. The 'Preview' window shows a chat conversation where the user has typed 'Open account please'. A red circle with the letter 'A' is placed over the input field in the 'Preview' window.

18. Click Investment (A).

The screenshot displays the IBM watsonx Orchestrate AI assistant builder interface. The top navigation bar includes the IBM logo, the text "IBM watsonx Orchestrate", "AI assistant builder", and the identifier "L4\_lab\_3FS\_2503". The main workspace is divided into three panels: "Conversation steps", "Conditions", and "Assistant says".

**Conversation steps:** This panel on the left shows a sequence of steps for the "Open an account" flow. Step 1 asks "What type of account would you like to open?" with buttons for "Savings", "Investment", and a "+ 1" button. Step 2 asks "What is your investor number?" with a "Regex" button. Step 3 is a "Connect to agent" step. Step 4 provides a link to the online portal. A red box labeled 'A' highlights the "Investment" button in Step 1.

**Conditions:** The middle panel shows a condition for Step 2: "If All of this is true: 1. What type of account wou... is Investment".

**Assistant says:** The bottom panel shows the assistant's response, "What is your investor number?".

**Preview:** On the right, a "Preview" window shows a chat interface. It includes a greeting "Welcome to the watsonx Orchestrate L4 AI assistant demo. How can I assist you?", a user input "Open account please", and a list of suggested actions: "Checking", "Savings", and "Investment". The "Investment" button is highlighted with a red box, corresponding to the label 'A'.

19. Type 123-456 (A) (Not a valid Investor number).

The screenshot displays the IBM Watsonx Orchestrate AI Assistant Builder interface. The top navigation bar includes the IBM logo, the text 'IBM watsonx Orchestrate', and the current project name 'AI assistant builder L4\_lab\_JFS\_2503'. The interface is split into three main panels. The left panel, titled 'Open an account', shows a 'Conversation steps' section with a list of four steps. Step 2 is highlighted with a blue box and contains the question 'What is your investor number?' with a 'Regex' validation icon. The center panel, titled 'Step 2', shows the 'Conditions' section with a condition group containing one condition: '1. What type of account would you like to open? is Investment'. The right panel, titled 'Preview', shows a chat window with a user input '123-456' highlighted in a red box. A red circle with the letter 'A' is placed next to the input field.

Note: If you type an invalid investor code that is short, such as 123, this code will not be accepted and may trigger an unintentional topic change – known as a [Digression](#). The Assistant II lab will show you how to disable undesired digressions and demonstrate how to troubleshoot issues related to this behavior.

If you encounter this problem, restart the conversation in the **Preview** window and try again, entering a valid Investor number.

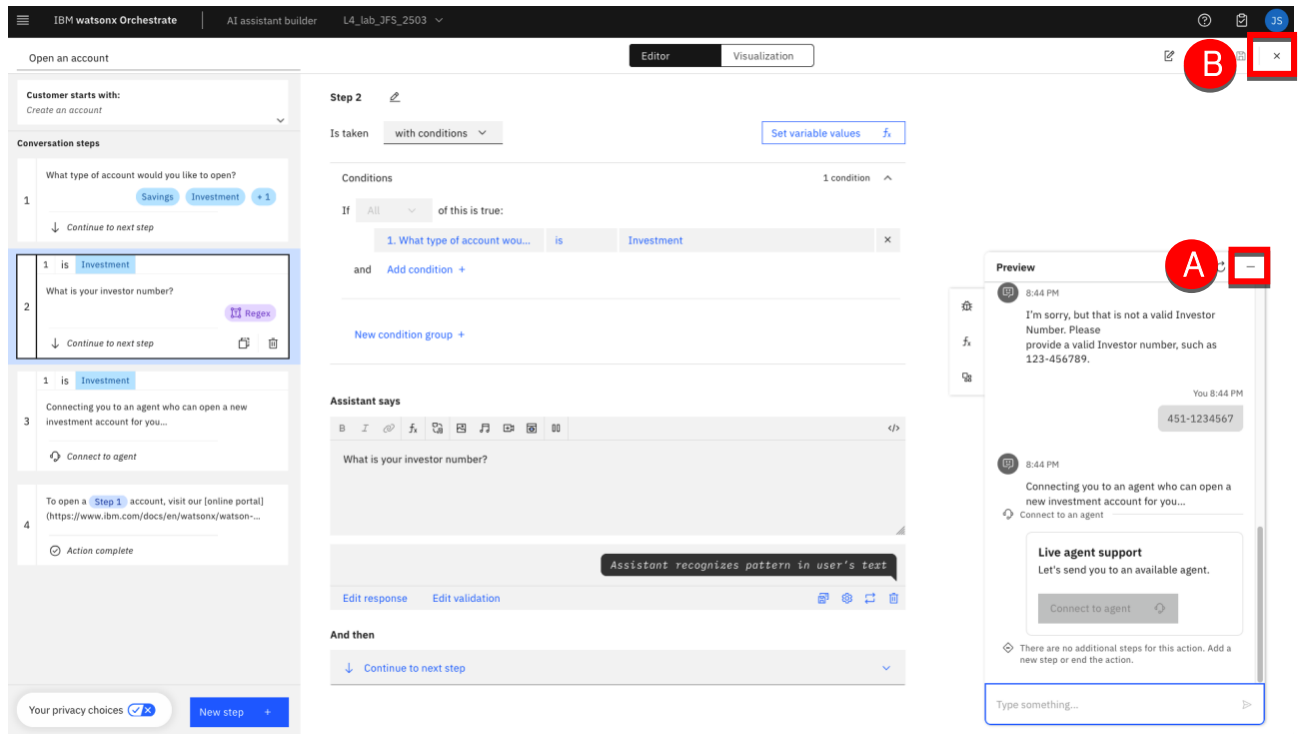
20. Type 451-1234567 (A).

The screenshot displays the IBM watsonx Orchestrate interface for an AI assistant builder. The main workspace is titled "Step 2" and shows a configuration for a step that is "Is taken with conditions". The conditions are set to "If All of this is true: 1. What type of account would you like to open... is Investment". The "Assistant says" section contains a text prompt: "What is your investor number?". Below this, there is a "Preview" window showing a chat conversation. The chat history includes a user message "What type of account would you like to open?" with buttons for "Checking", "Savings", and "Investment". The assistant responds with "Investment". The user then asks "What is your investor number?". The assistant responds with "I'm sorry, but that is not a valid Investor Number. Please provide a valid Investor number, such as 123-456789." The input field at the bottom of the preview shows the number "451-1234567" entered by the user, which is highlighted with a red box and a red letter 'A'.

21. Verify that after you enter a valid Investor number, the assistant connects you with an agent.

This screenshot shows the same IBM watsonx Orchestrate interface as the previous one, but with a different configuration for Step 2. The conditions remain the same, but the "Assistant says" section now contains a text prompt: "Connecting you to an agent who can open a new investment account for you...". Below this, the "Preview" window shows the chat conversation. The user enters the number "451-1234567". The assistant responds with "I'm sorry, but that is not a valid Investor Number. Please provide a valid Investor number, such as 123-456789." The user then enters the number "451-1234567". The assistant responds with "Connecting you to an agent who can open a new investment account for you...". A red arrow points to the "Connect to an agent" action in the chat preview. Below the chat preview, there is a "Live agent support" section with a "Connect to agent" button. The input field at the bottom of the preview shows the text "Type something...".

22. Close the **Preview** (A) then click the **x** button (B) to exit out of the action editor.

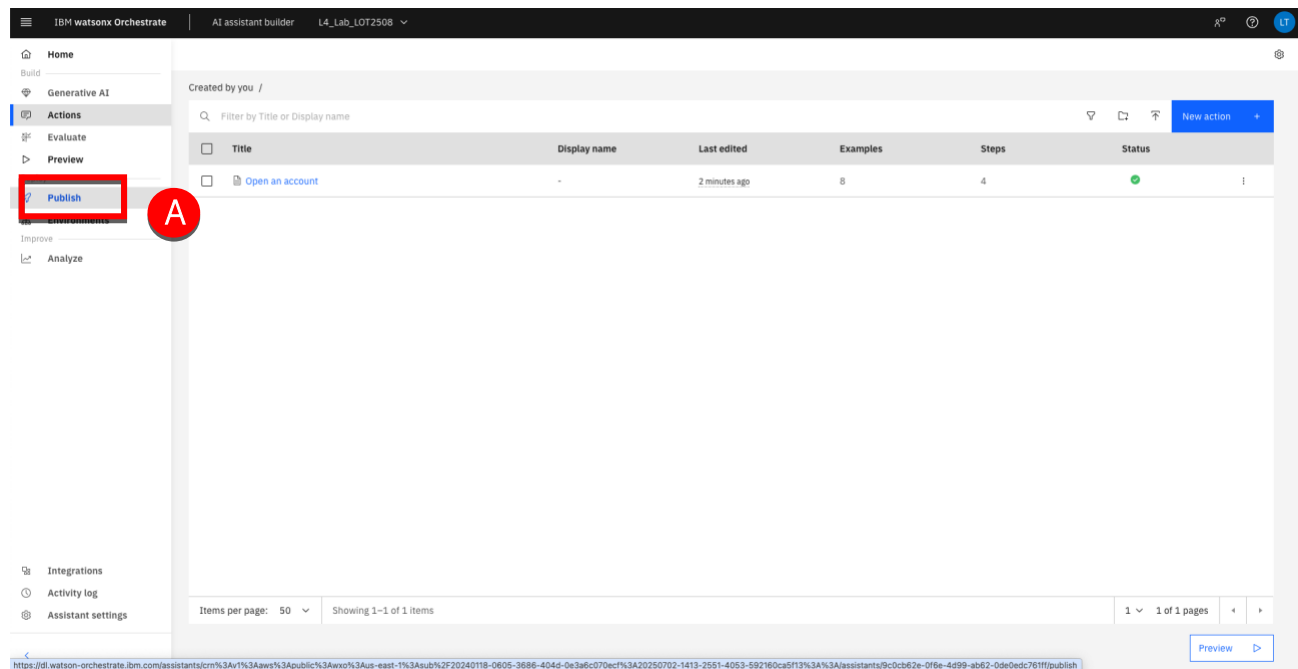


You have updated an action in your draft environment.

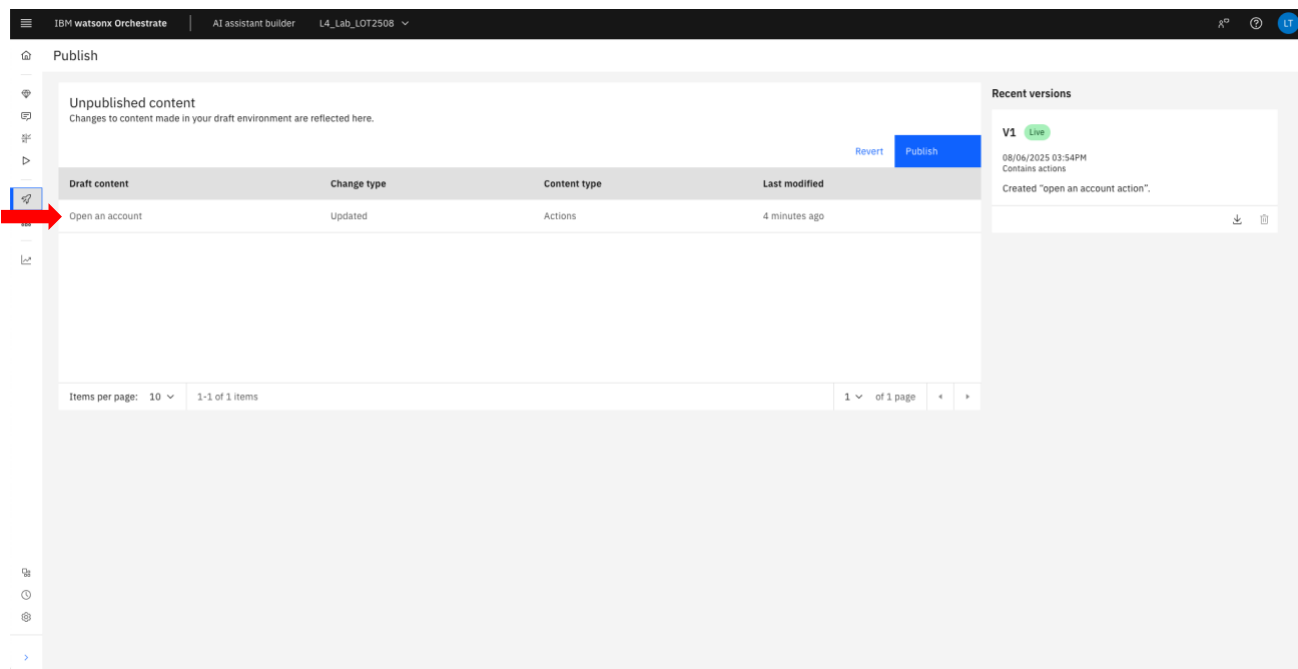
## Republish your assistant and update the environment

You have made an update to your draft environment. Now you need to republish your assistant in the draft environment, then push the changes to a live environment.

1. In the left pane, click **Publish** (A).

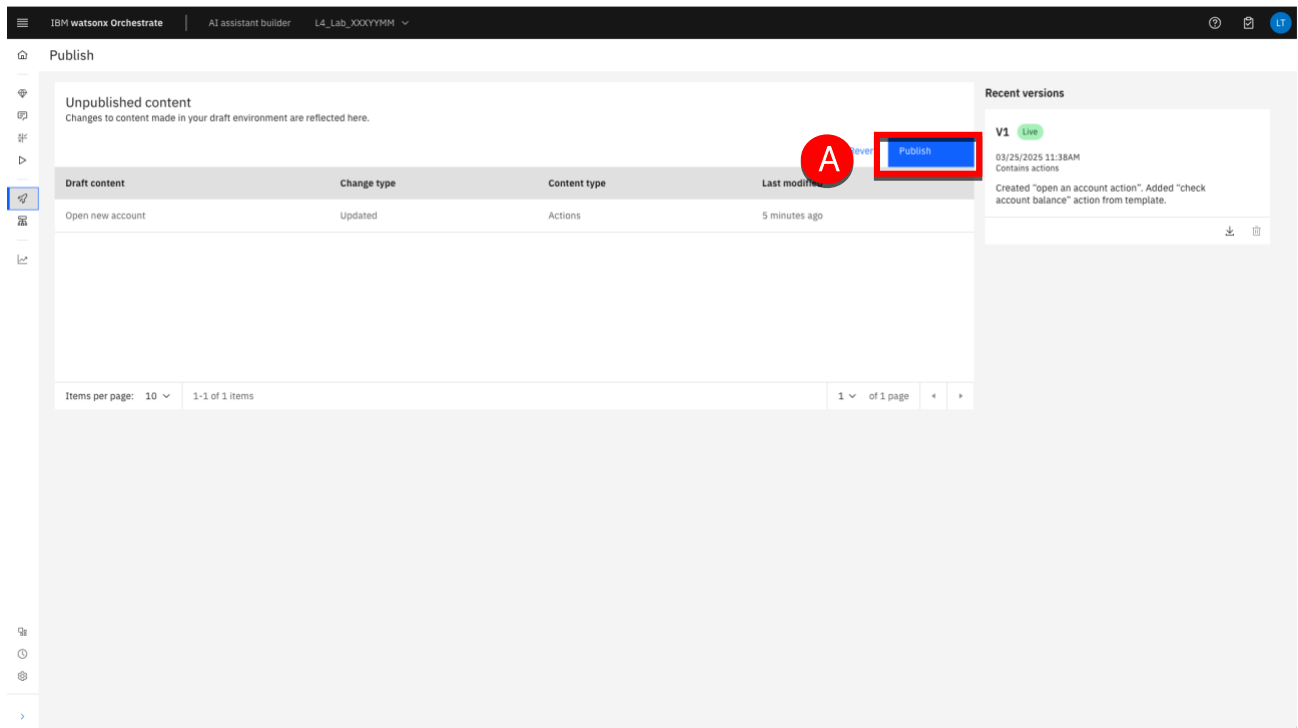


2. Observe the unpublished content listed.

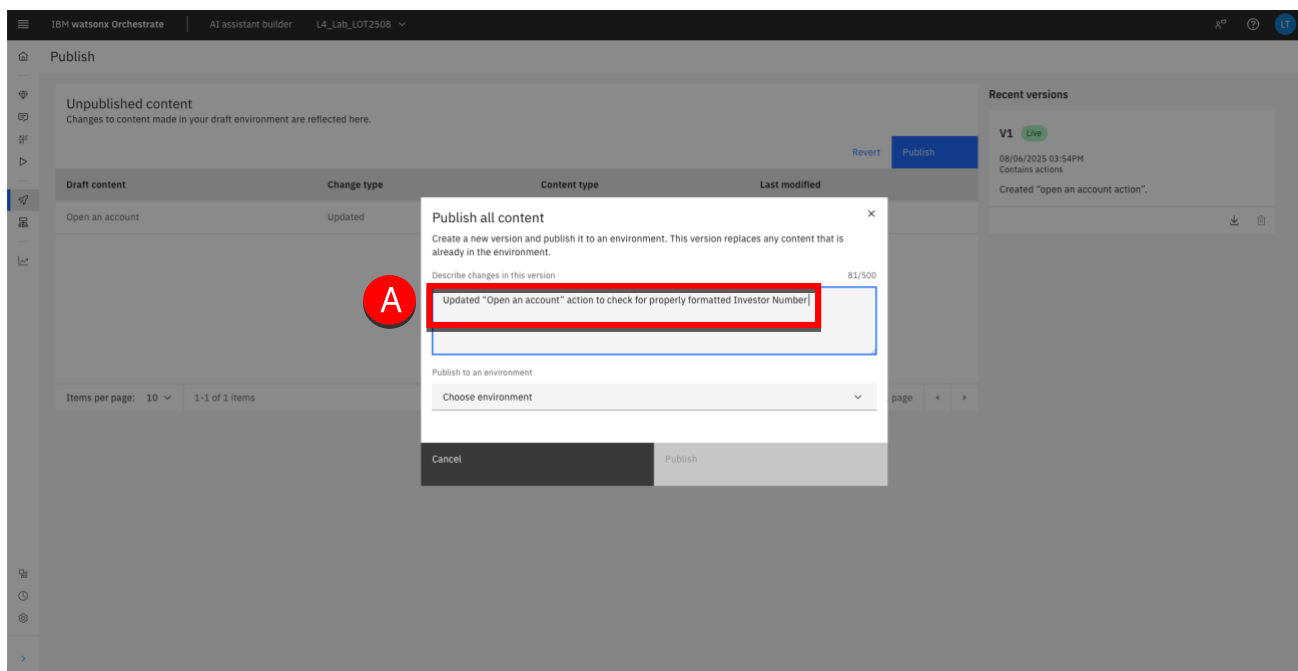


You will see that your Open an account action has been edited. This is because you updated the step that collects the Investor Number from the user.

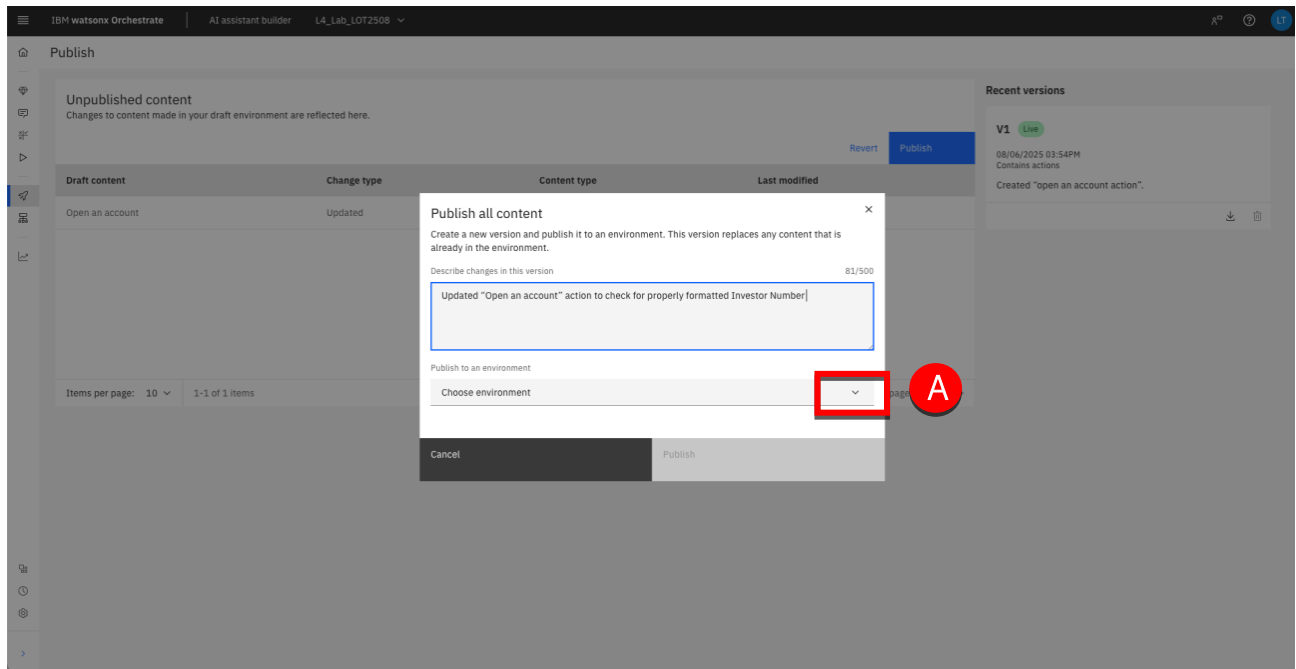
3. Click **Publish** (A).



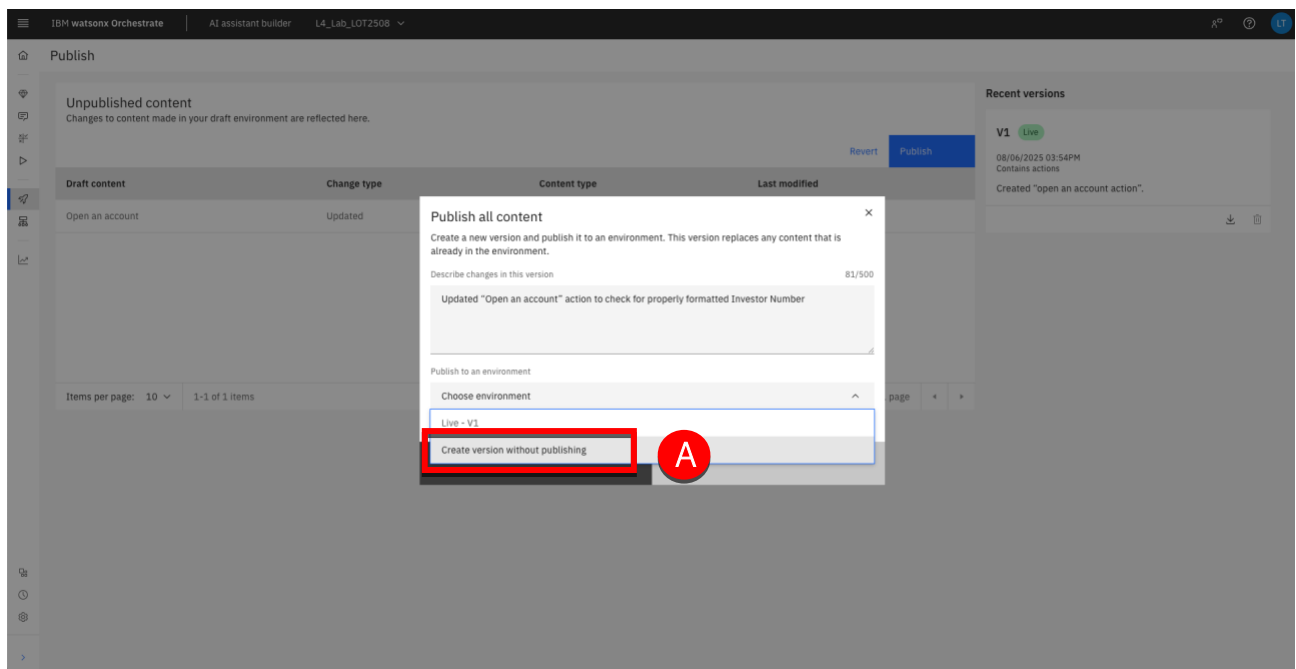
4. In the **Describe changes in this version** field, type Updated “Open an account” action to check for properly formatted Investor Number (A).



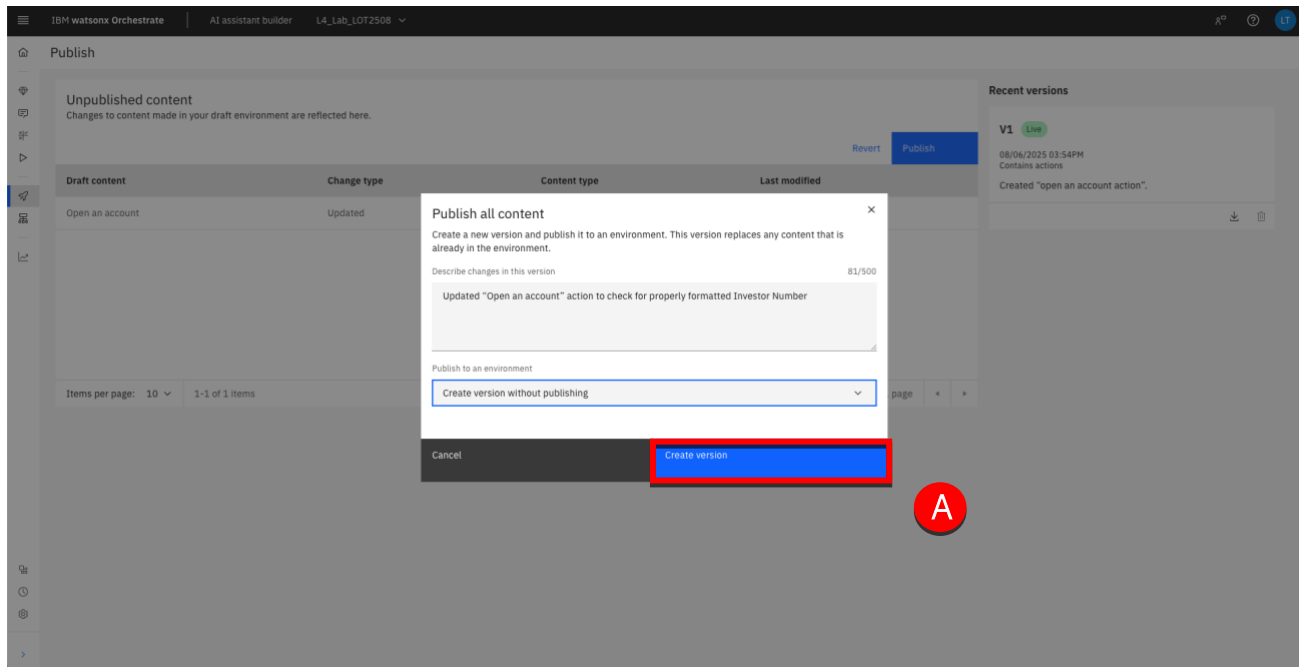
5. Expand the **Publish to an environment** dropdown menu (A).



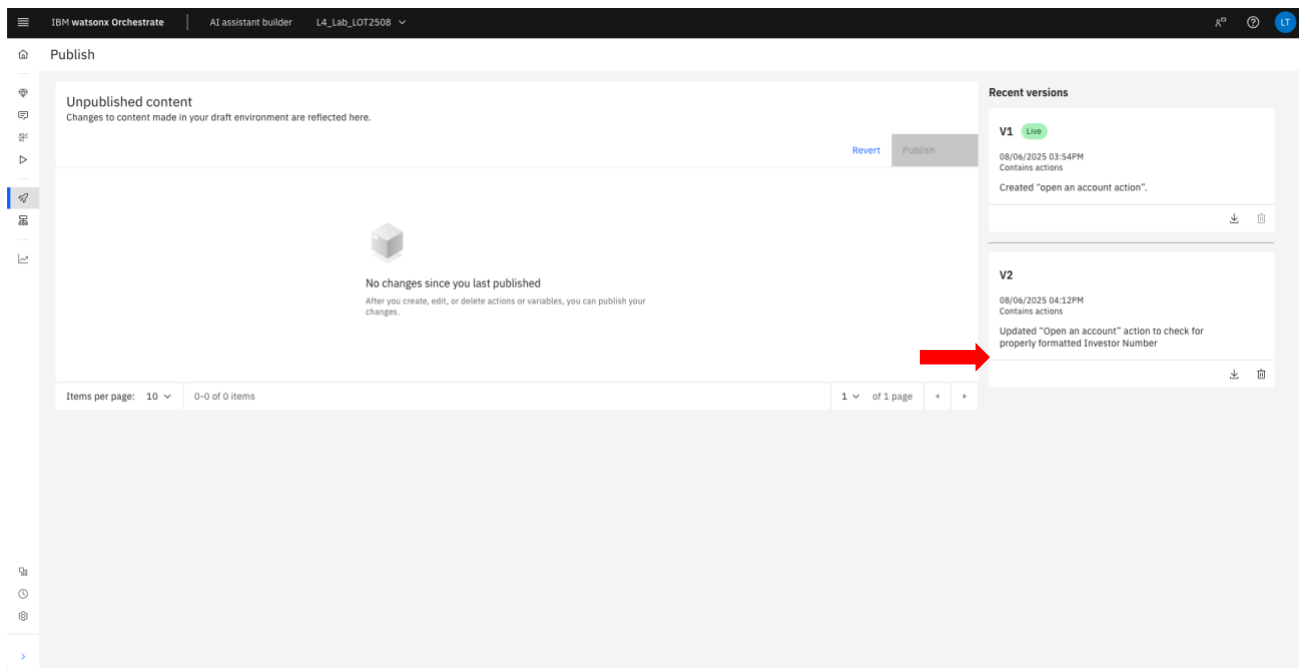
6. Select **Create Version without publishing** (A).



7. Click **Create version** (A).

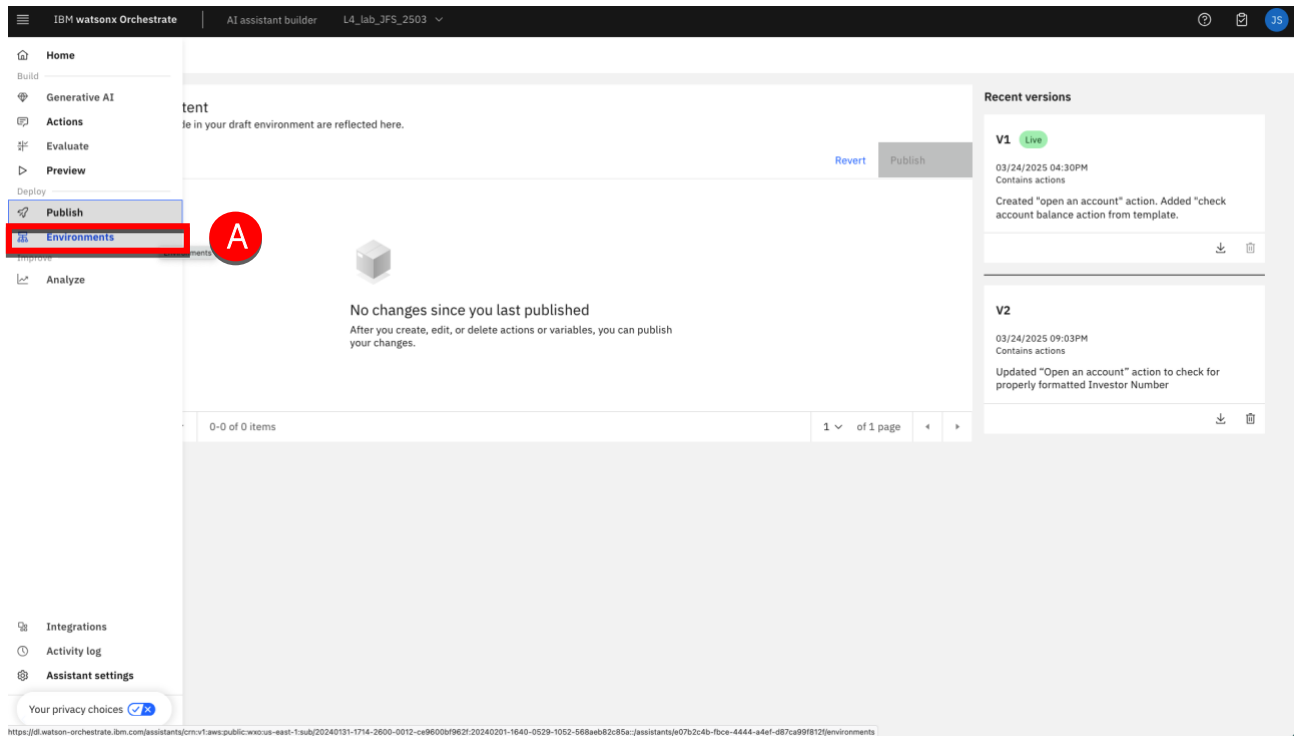


Observe the **Publish** page, specifically the V2 version that was recently created but not published in the Live environment.

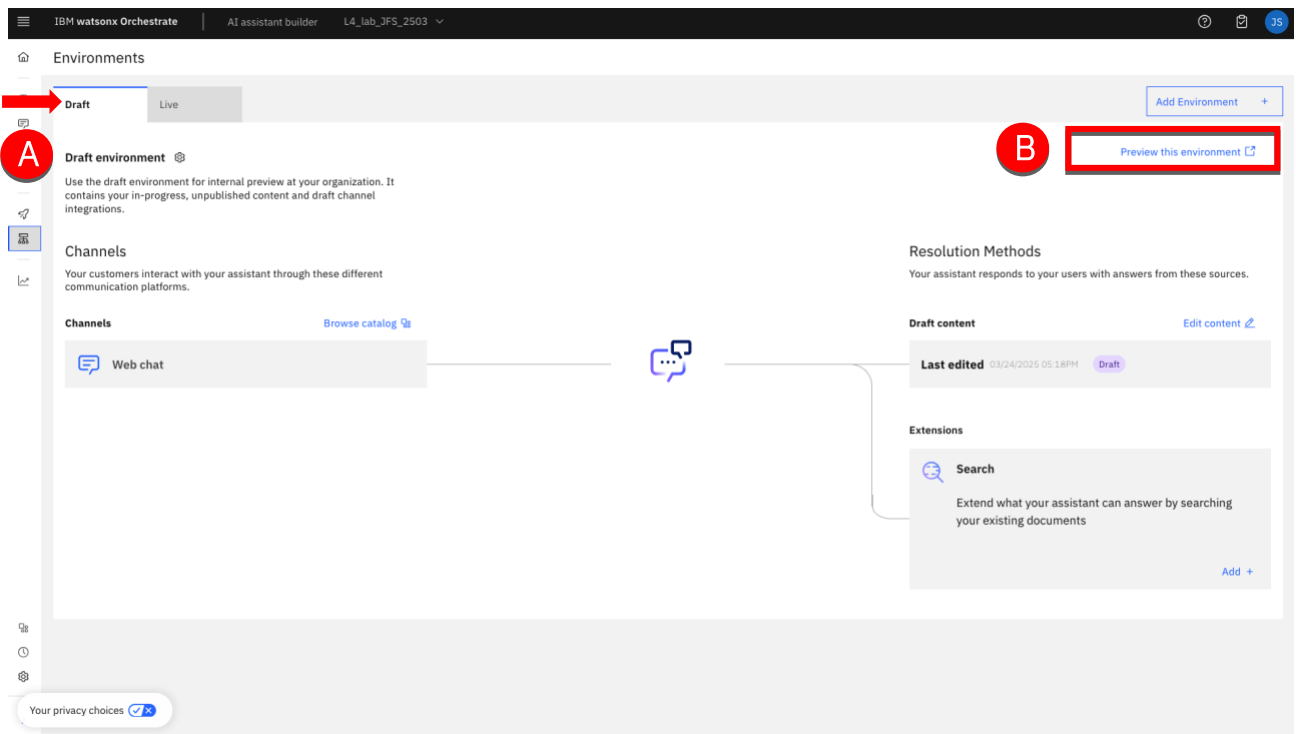


Note that V2 version was created, but V1 is still the Live environment

8. Go to the lefthand pane, then click **Environments** (A).

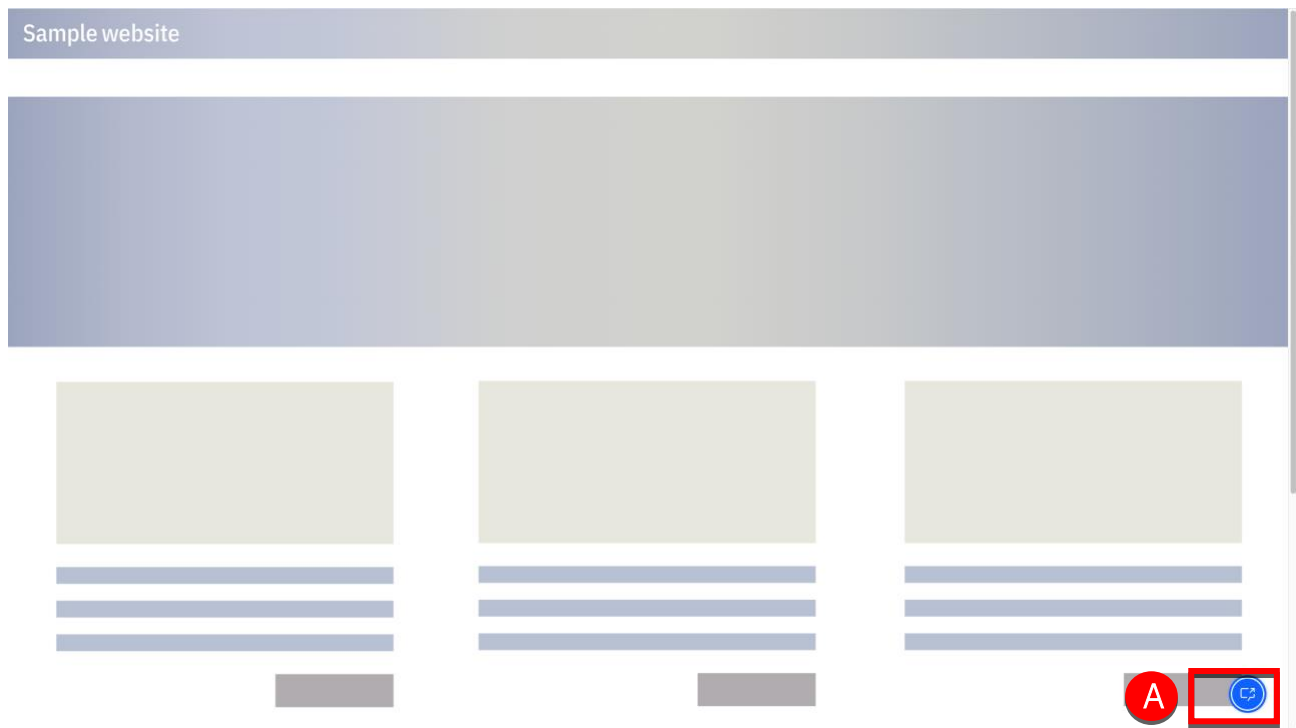


9. In the Environments page, verify you are viewing the Draft environment (A) then click **Preview this environment** (B).



This opens a new tab in your browser.

10. If it is not already open, open the virtual assistant (A).



11. Repeat the steps 17 – 21 in the [Iterate on your Assistant](#) section of this lab to test the Assistant's responses.

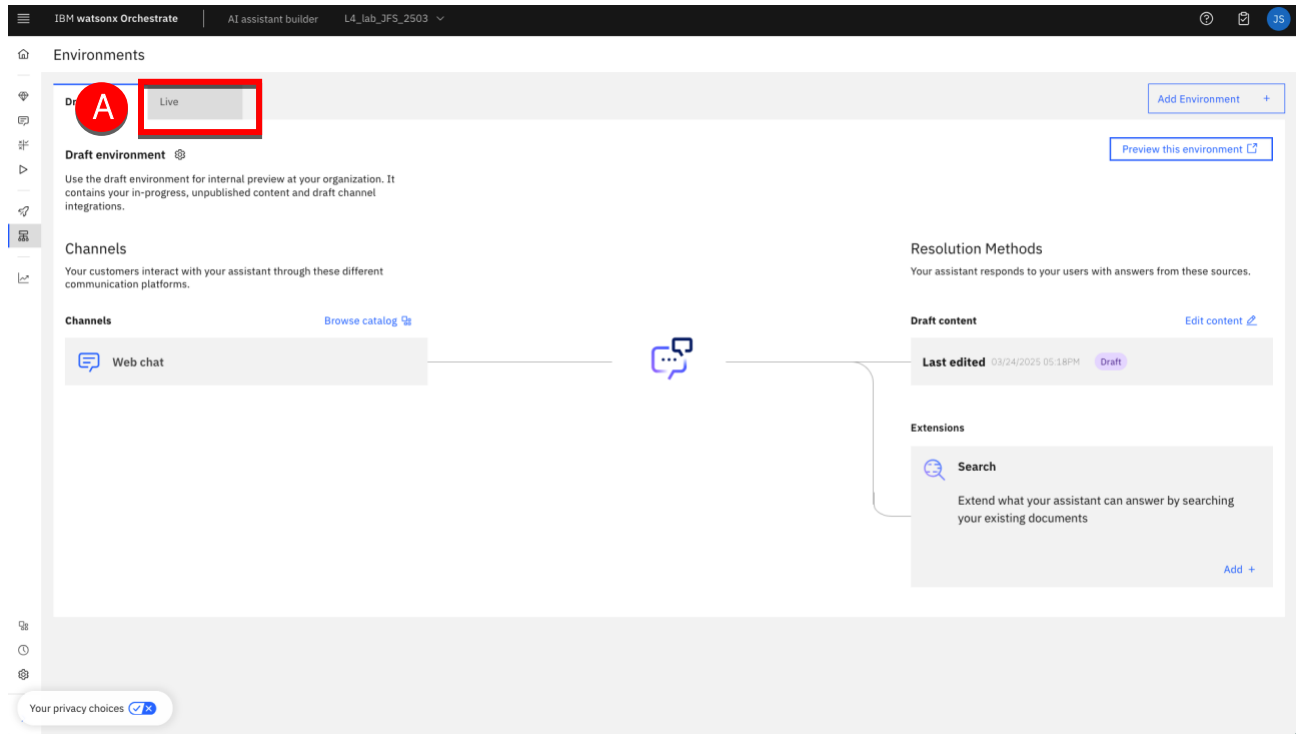


Notice that the changes in the Investor number have been demonstrated in the draft environment.

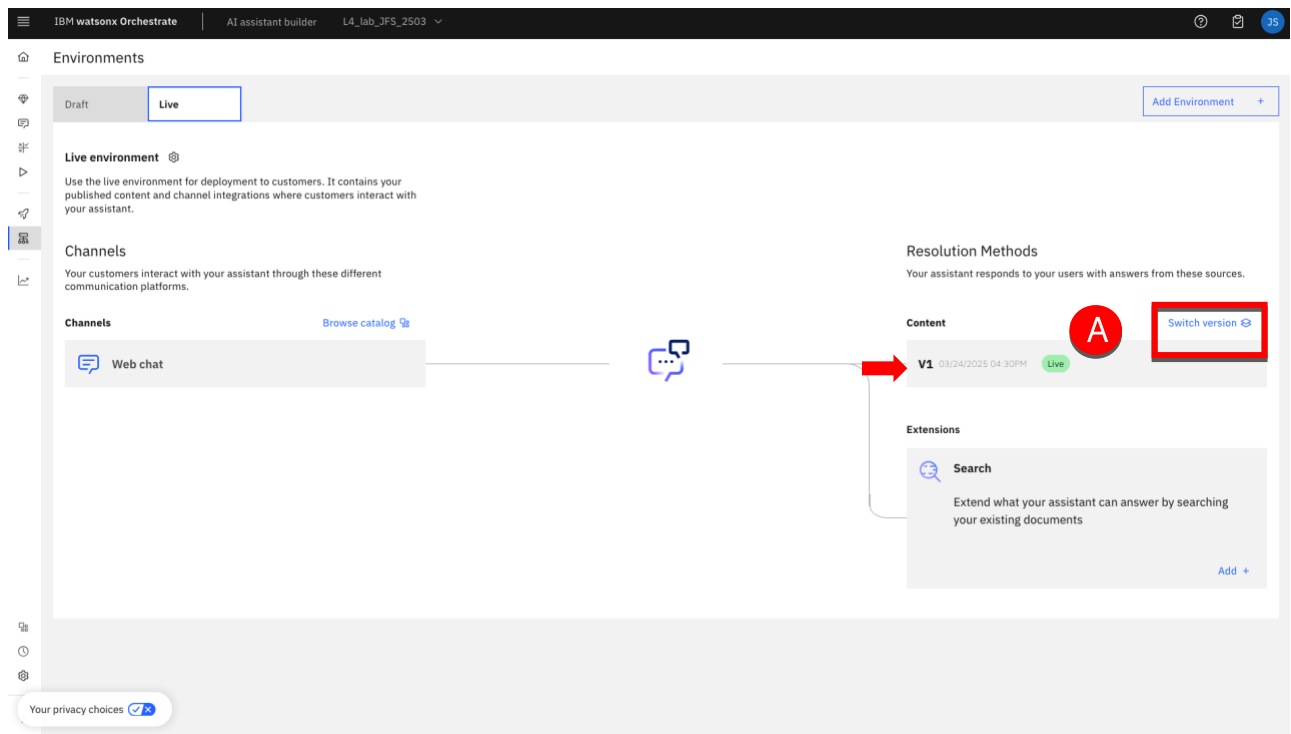
You can share these changes with internal testers.

Assuming that you and your enterprise is happy with the changes, you push these changes to the Live environment.

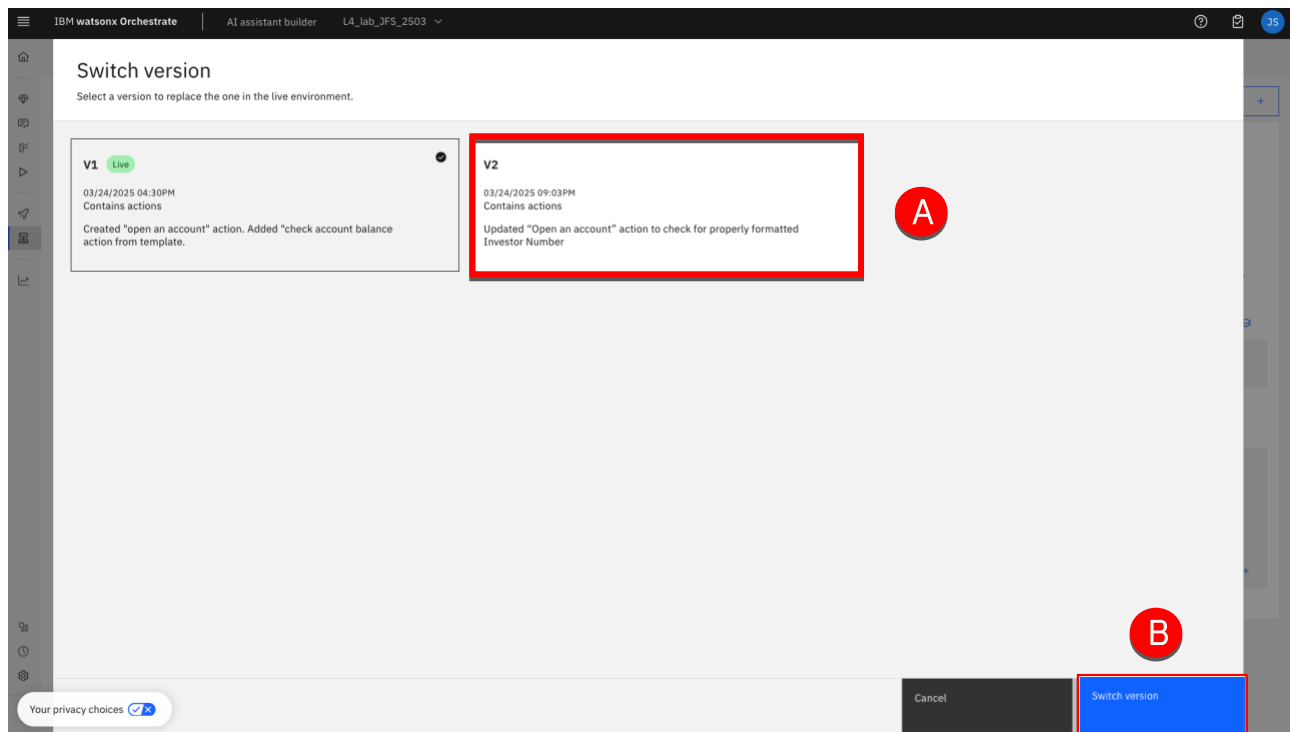
12. Return to the **Environments** tab in your browser then select the **Live** tab (A).



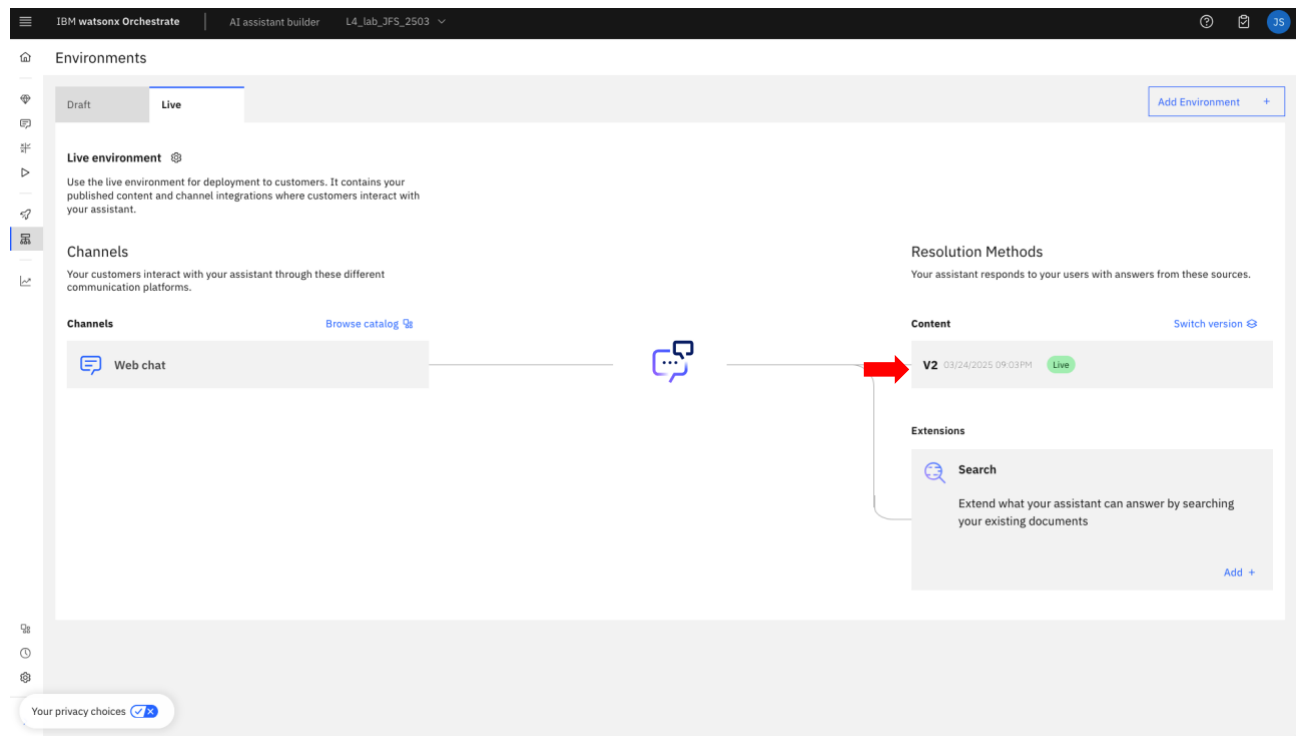
13. Notice that the Live environment still using the **V1** content you published. Click **Switch version** (A).



14. Select **V2** (A), then click **Switch version** (B).



15. Observe that the Live environment is now using the **V2** content.



You have completed the lab exercises.

## Summary

To review, you:

- Learned the action fundamentals and how to create an action.
- Instantiated business logic in the action using Action variables, conditions, and defining customer responses.
- Configured the Greet customer action.
- Configured the Preview settings, then tested the action using a preview link.
- Saved the initial version of your assistant, then published it to the Live environment.
- Iterated on your assistant, updating an action.
- Saved the iterated version of your assistant, then published the assistant to the Live environment.

## Next steps

The Assistant II (Integrate your AI Assistant in your web interface) lab will continue working where this lab ends.

If you plan on immediately continuing to the Assistant II lab, you may open the lab guide for the Assistant II lab and begin completing the Assistant II exercises.

If you are not immediately continuing to the Assistant II lab, follow the instructions in the [Export an assistant](#) section of this lab guide to save your assistant's files to your local computer. This allows you to continue working with the same assistant content later.

Next, follow these instructions in the [Delete your assistant](#) section of this lab guide. This will ensure that the watsonx Orchestrate tenant is not hosting unnecessary assistants.

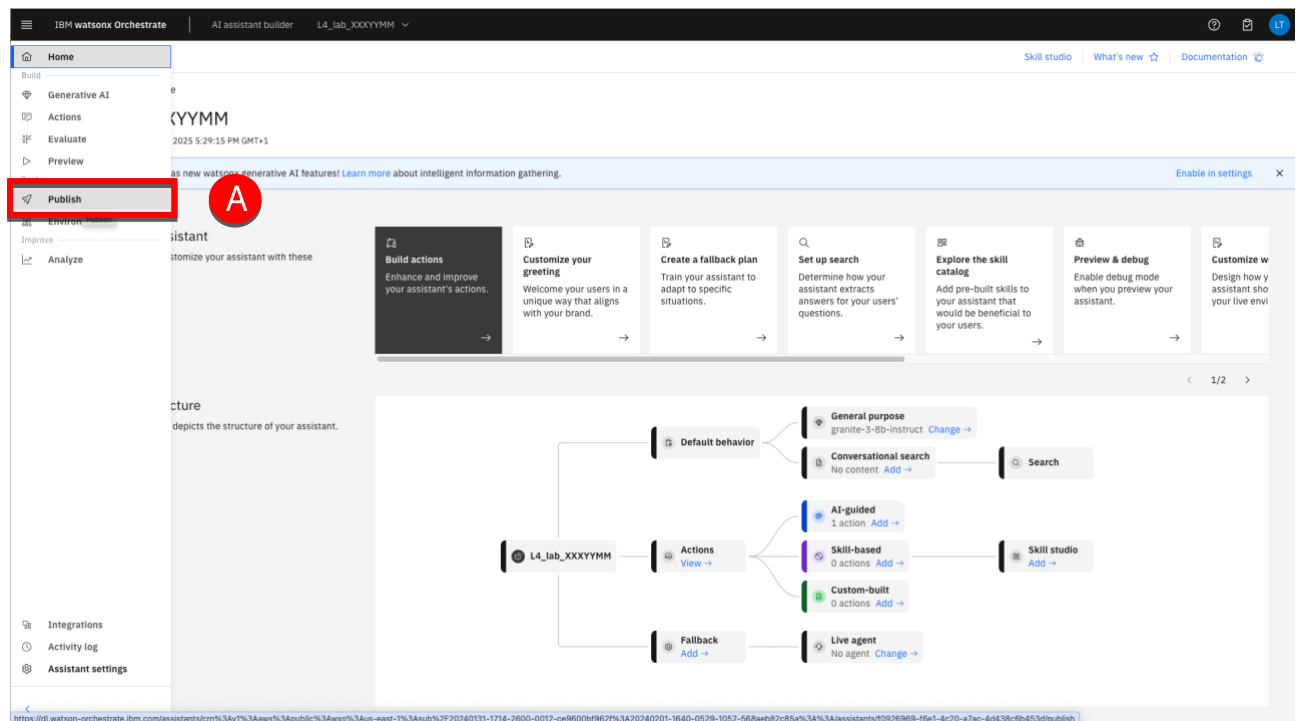
## Annex - Export / re-import/ delete your assistant archive

This section describes how to export your work in an archive that can be re-imported later, should you need to resume your work.

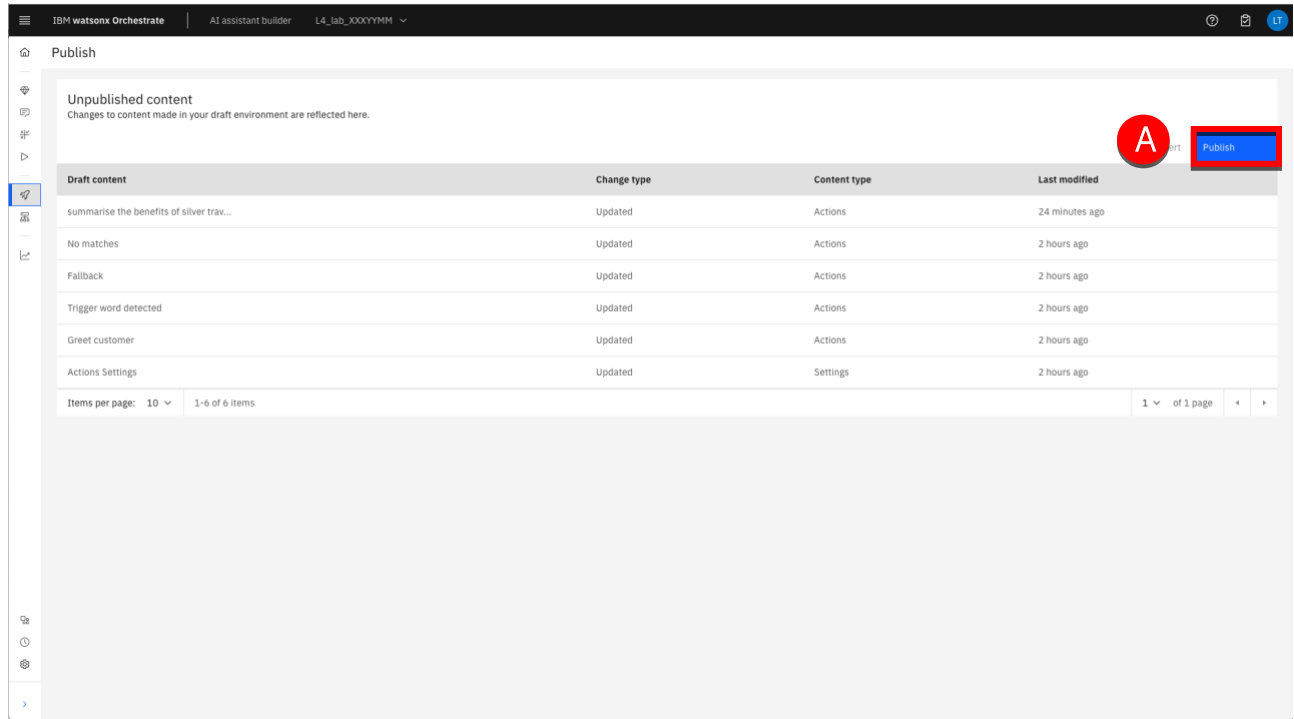
## Export an assistant

## Create a version

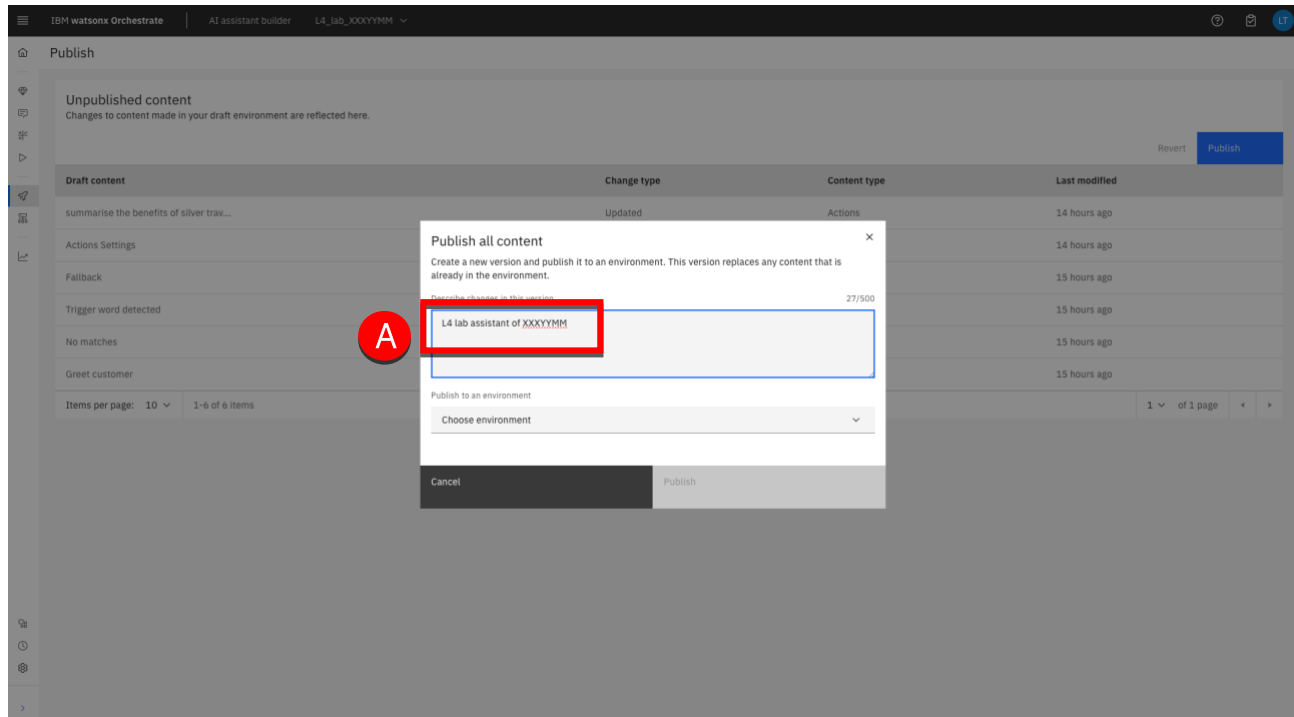
1. Go to the assistant home page.
2. Click **Publish** (A).



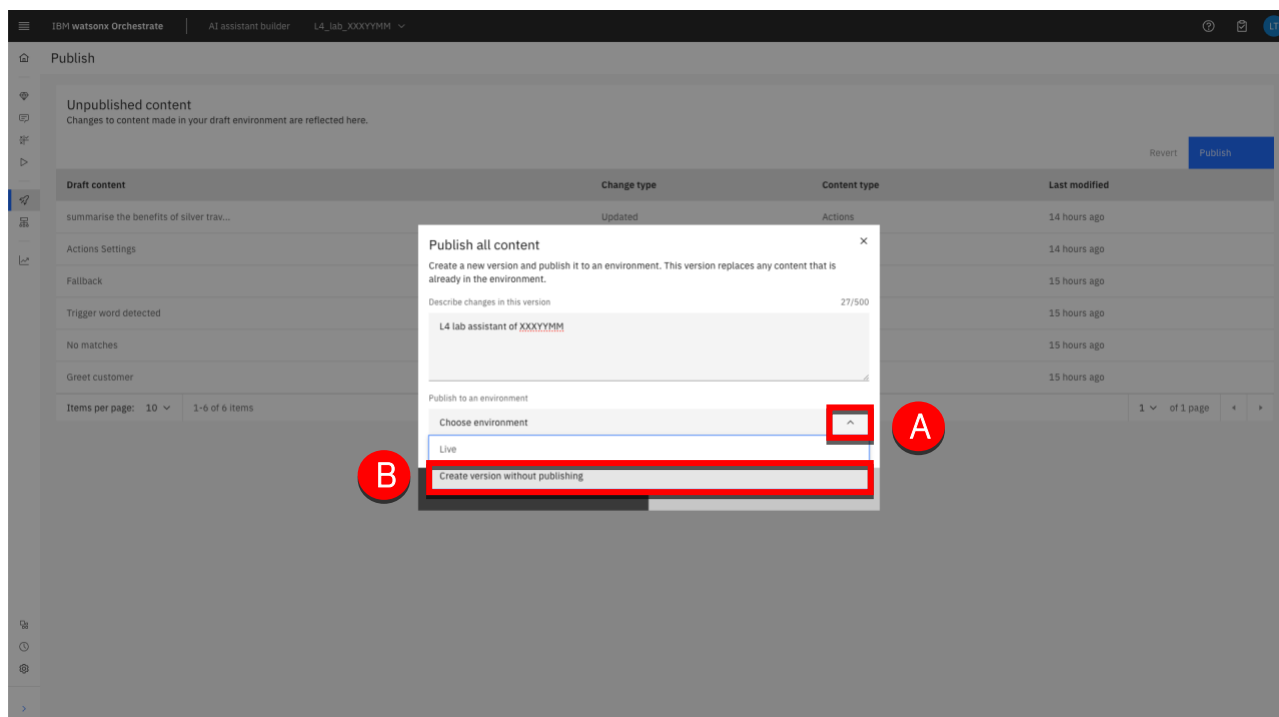
3. Click **Publish** (A). If Publish is greyed out, you may skip ahead to the [Create the archive](#) section.



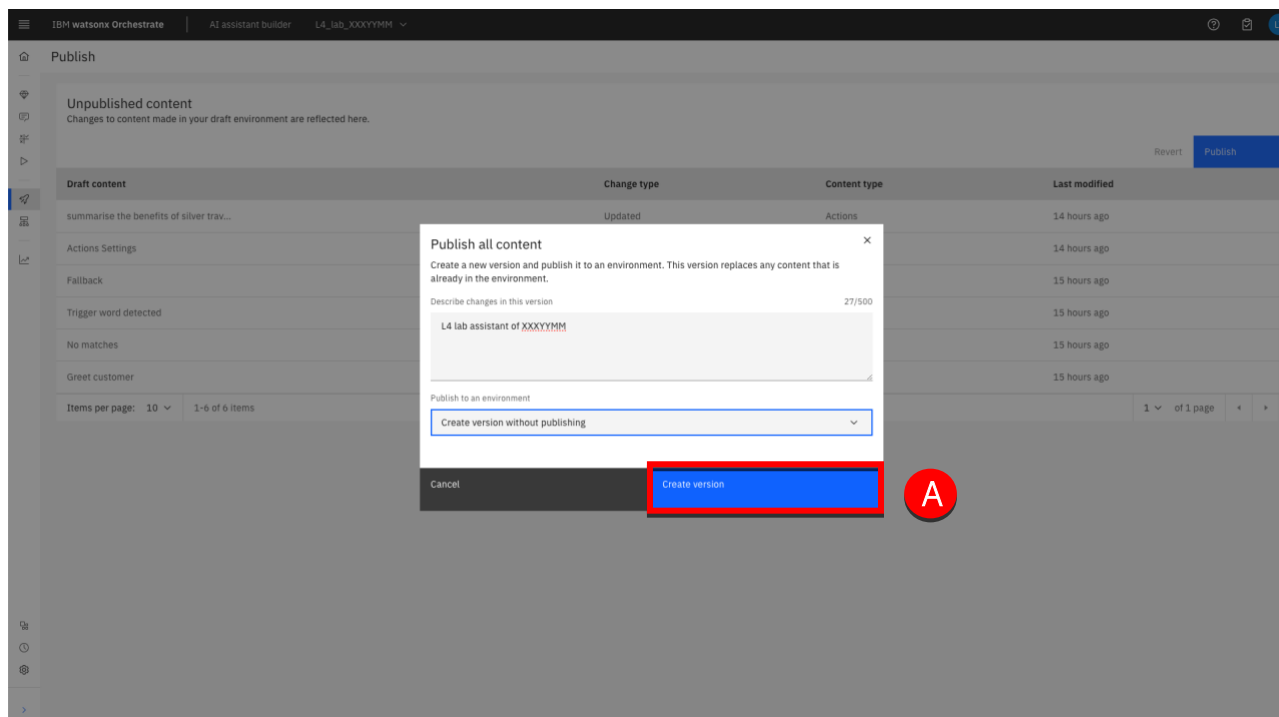
4. Enter **L4 lab assistant of XXXYYMM** (A) (XXXYYMM being you proper identification code).



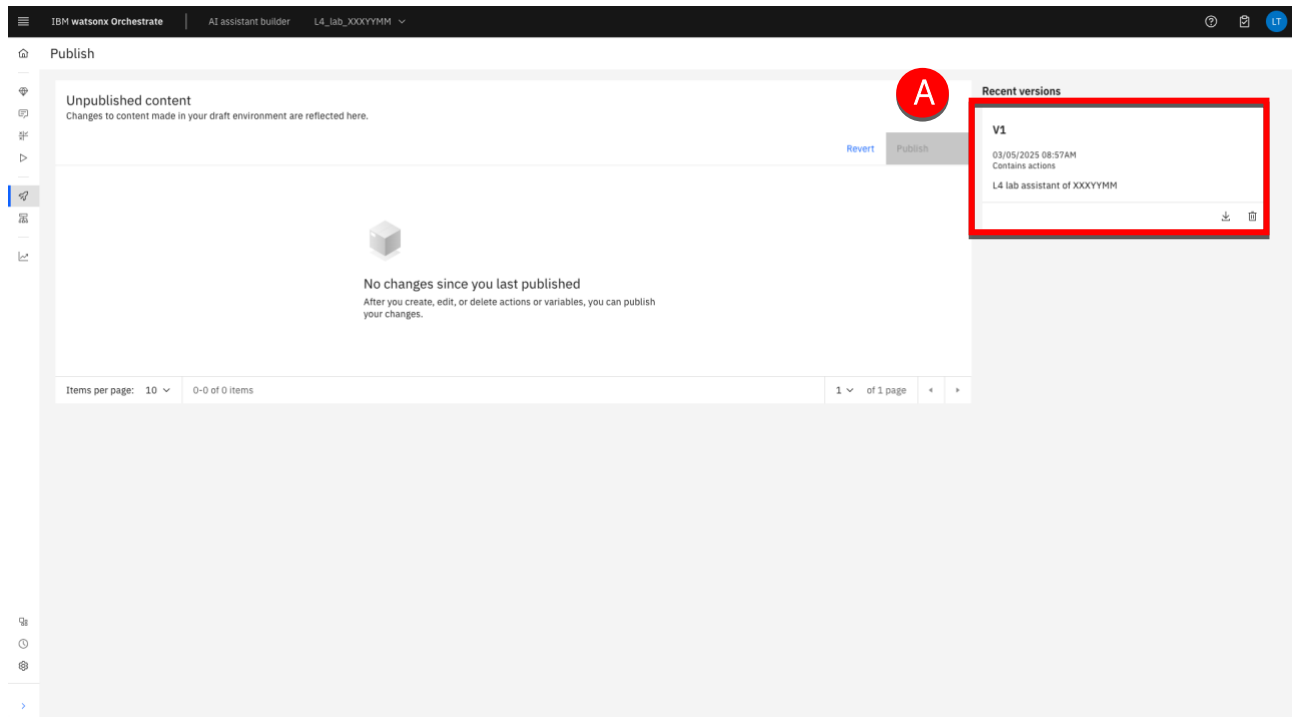
5. Expand the **Publish to an environment** menu (A) then select **Create version without publishing** (B).



6. Click **Create version** (A).

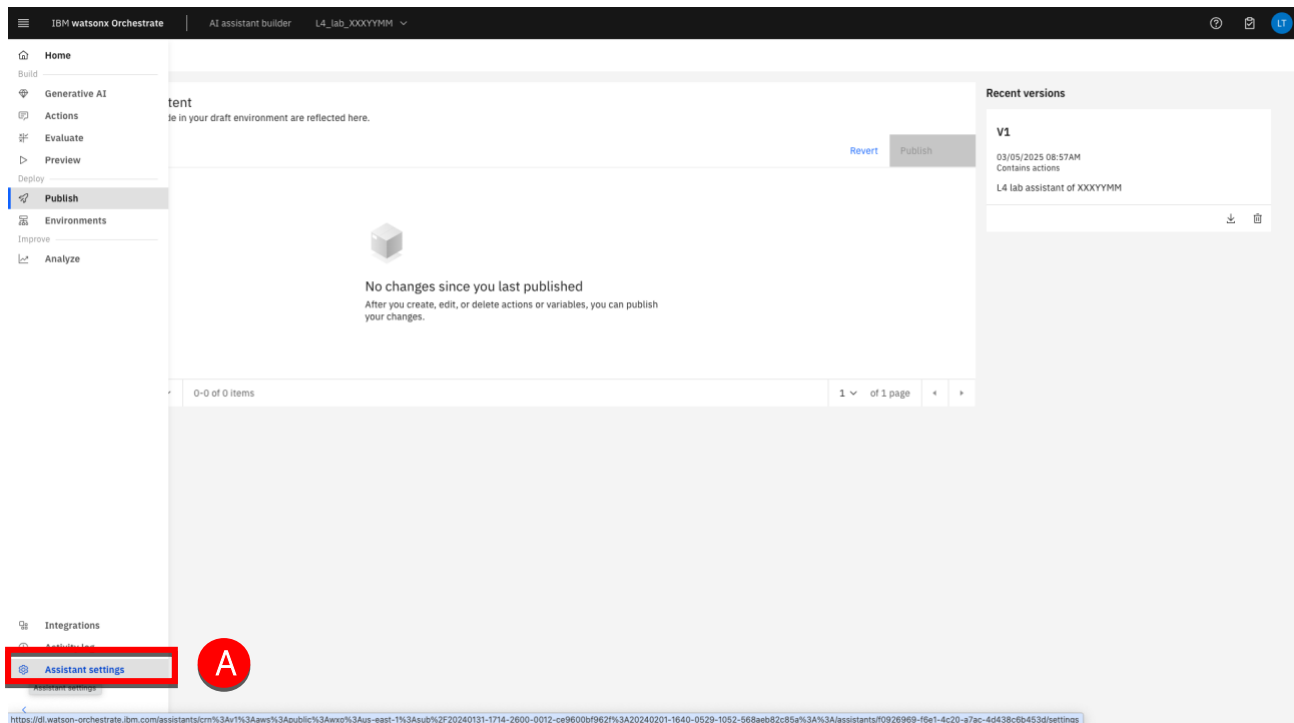


7. Wait for your version to be created (A).

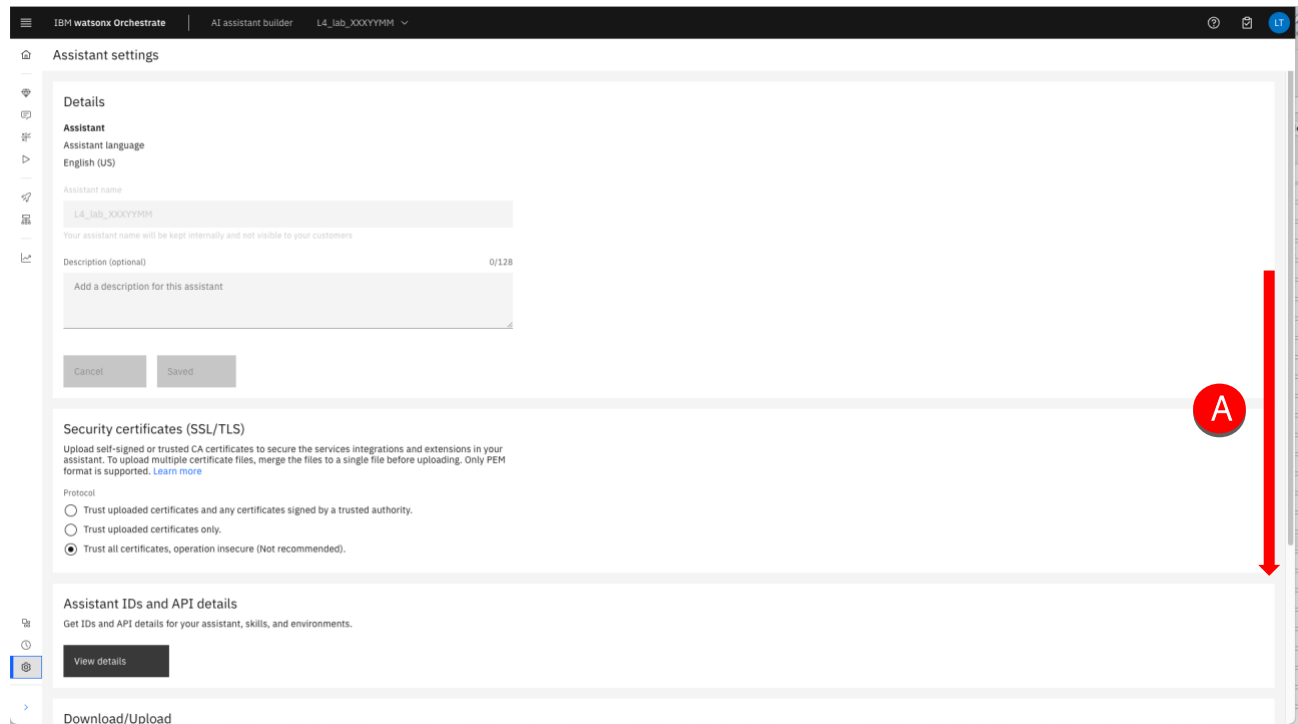


## Create the archive

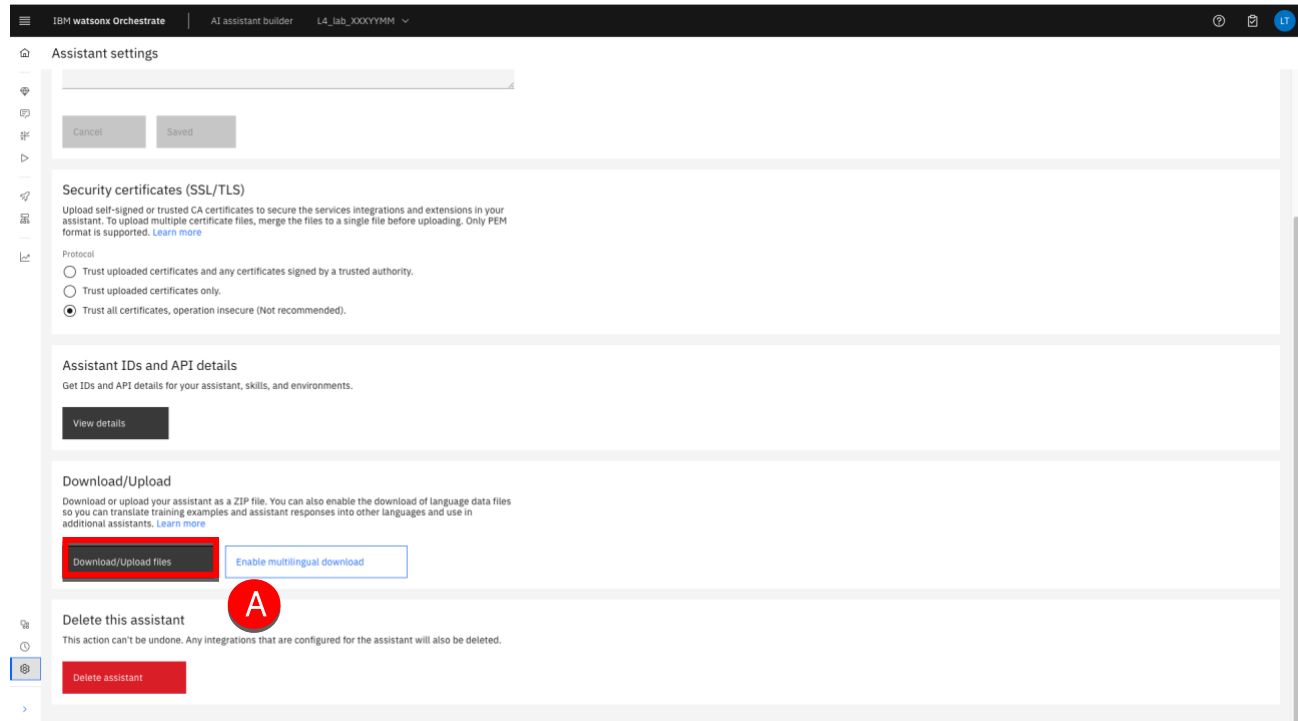
1. Click Assistant settings (A).



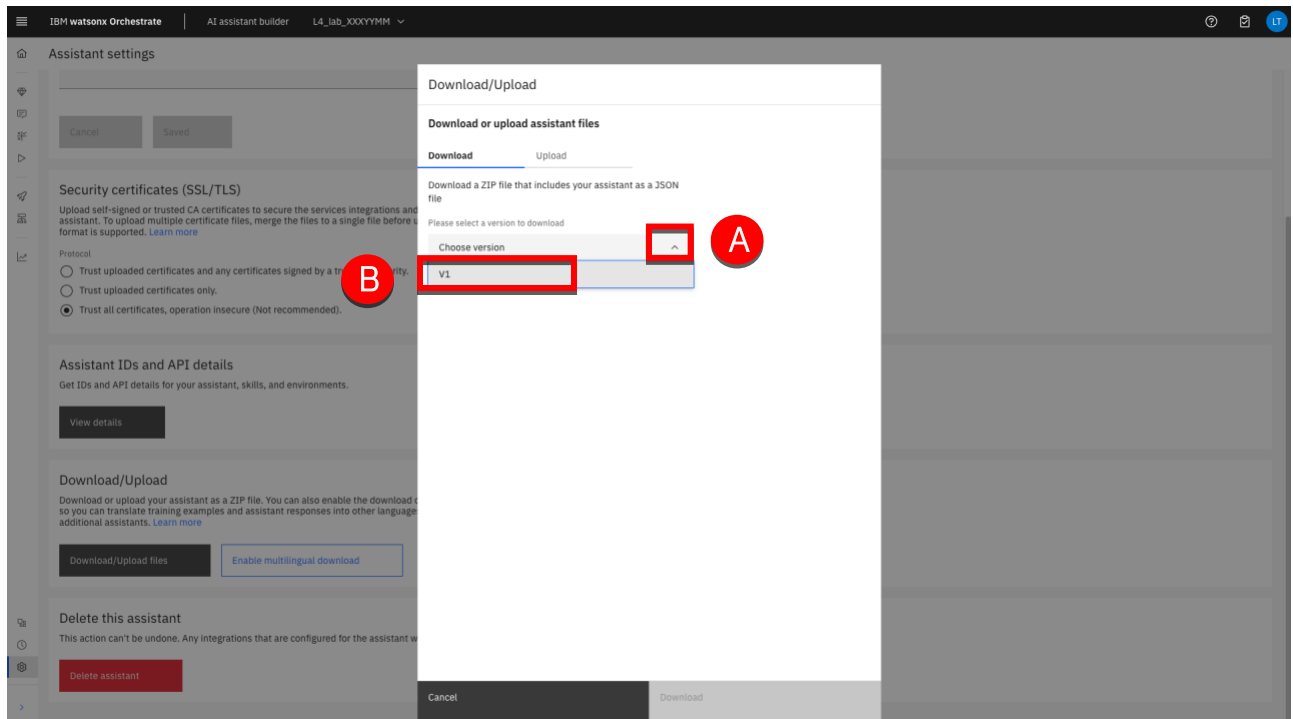
2. Scroll down (A).



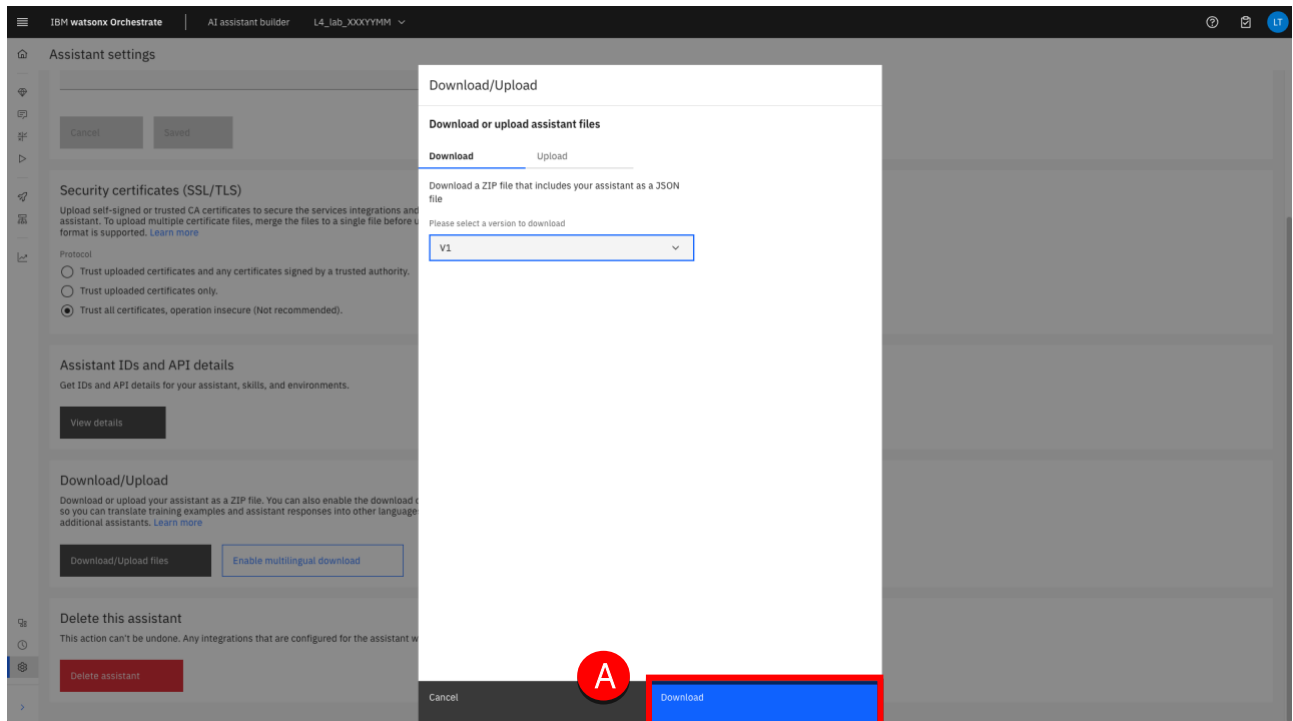
3. Click Download/Upload files (A).



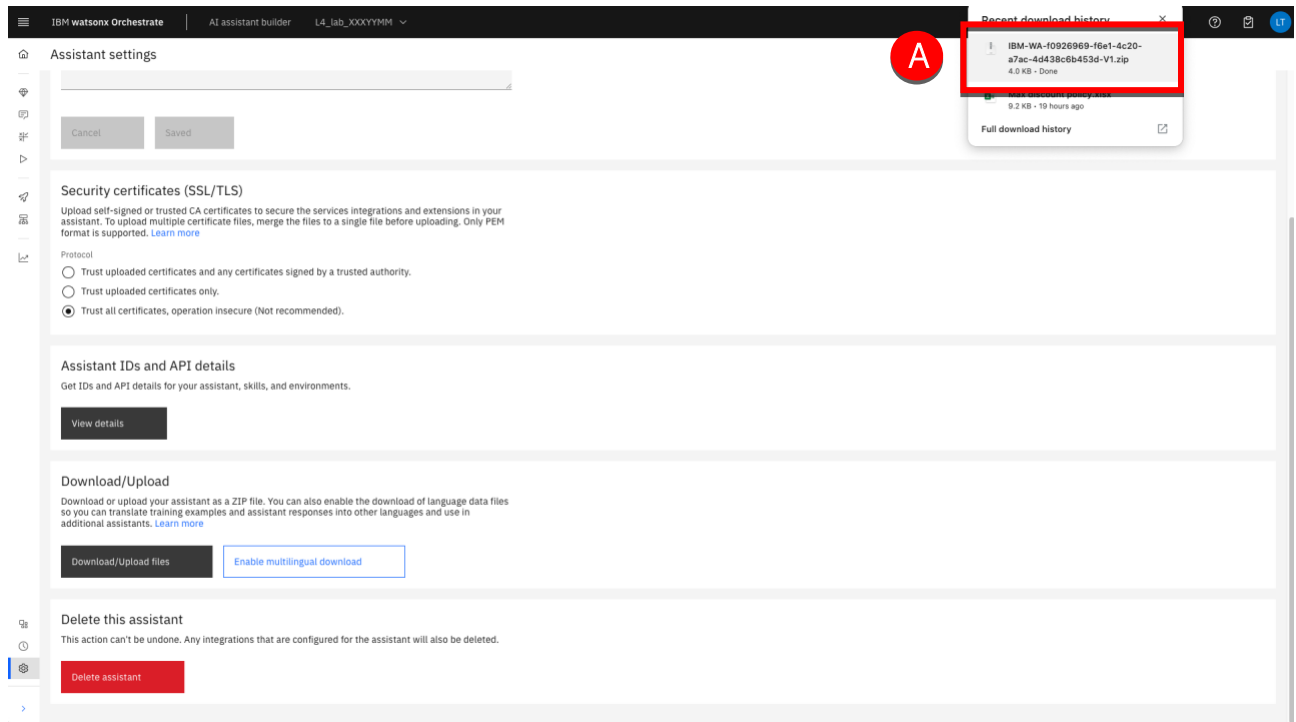
4. Expand the **Choose version** menu (A) then select your last published version (B).



5. Click **Download** (A).



6. Get your archive file from your download folder (A) (May change depending on your browser).

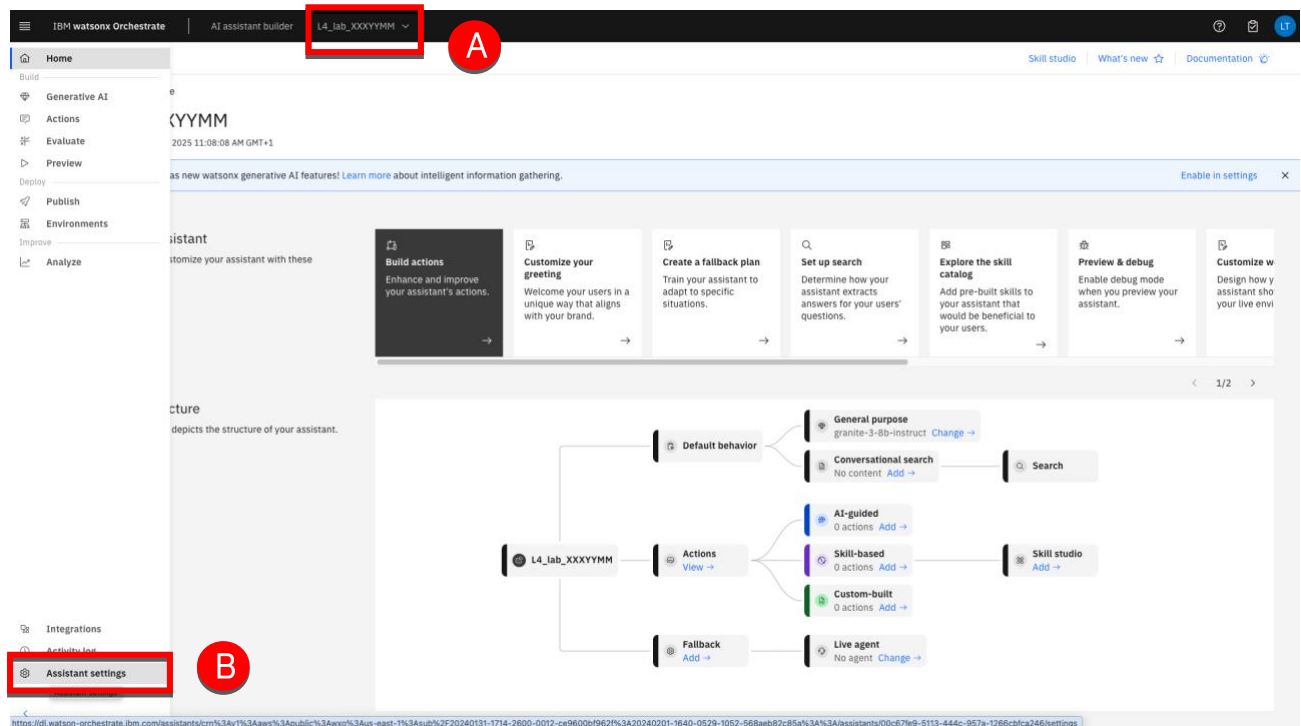


**Note:** the name of the generated archive file will look like IBM-WA-f0926969-f6e1-4c20-a7ac-4d438c6b453d-V1.zip.

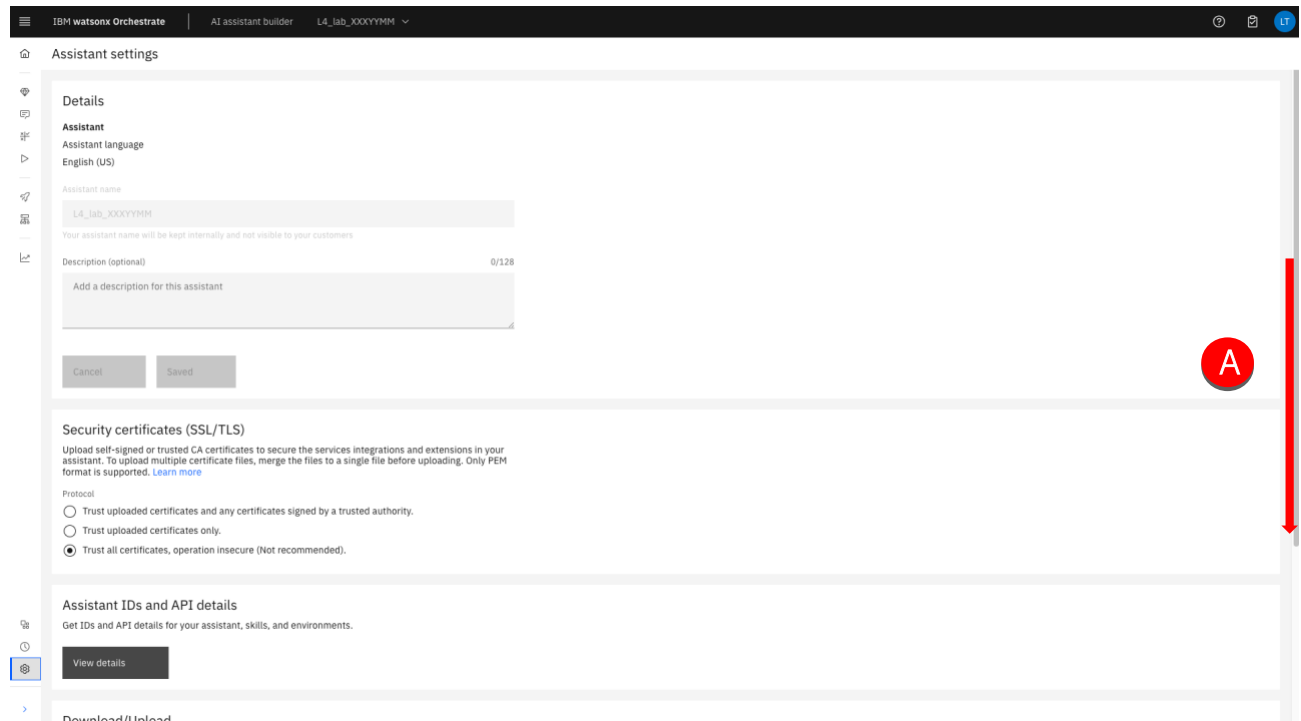
## Import an assistant

When you import actions from an archive, these actions are added to the current Assistant in AI assistant builder. It is important to create a new empty assistant with a proper name before importing any archive file.

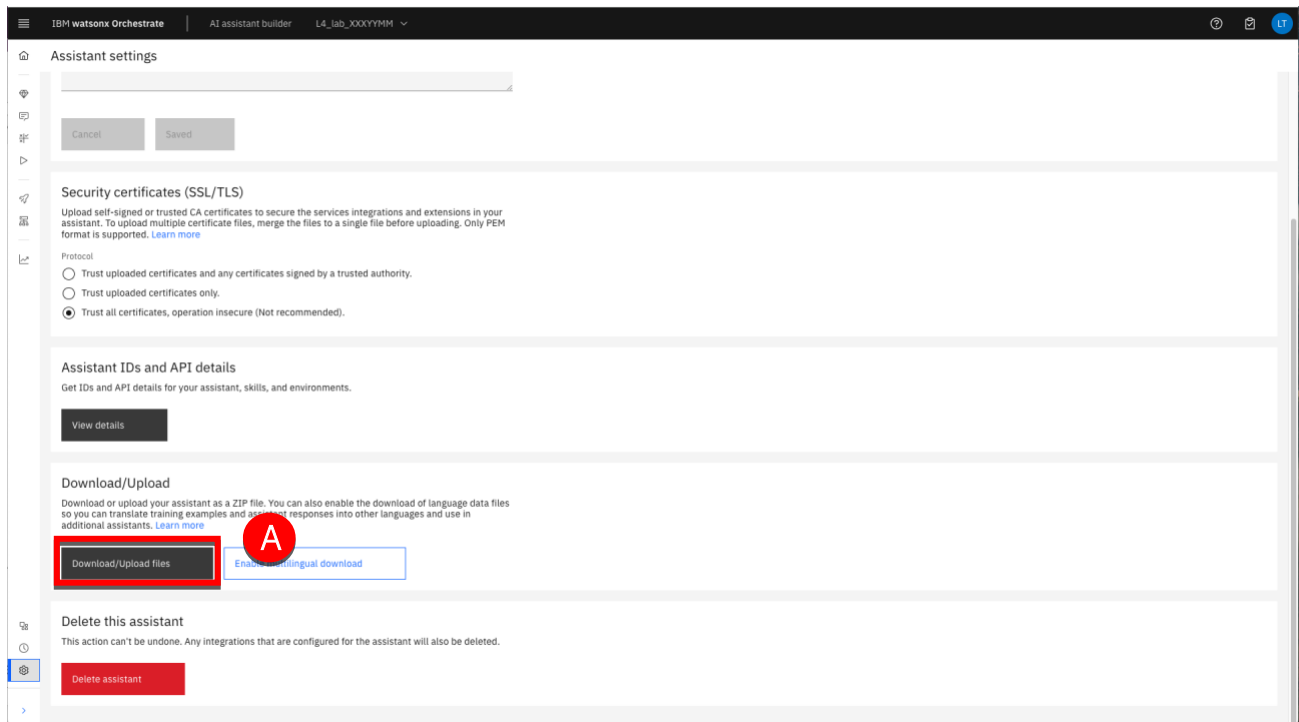
1. Create a new assistant following the [Create your L4 assistant instructions](#) in this guide.
2. Check that you are on your new assistant (A), click Assistant settings (B).



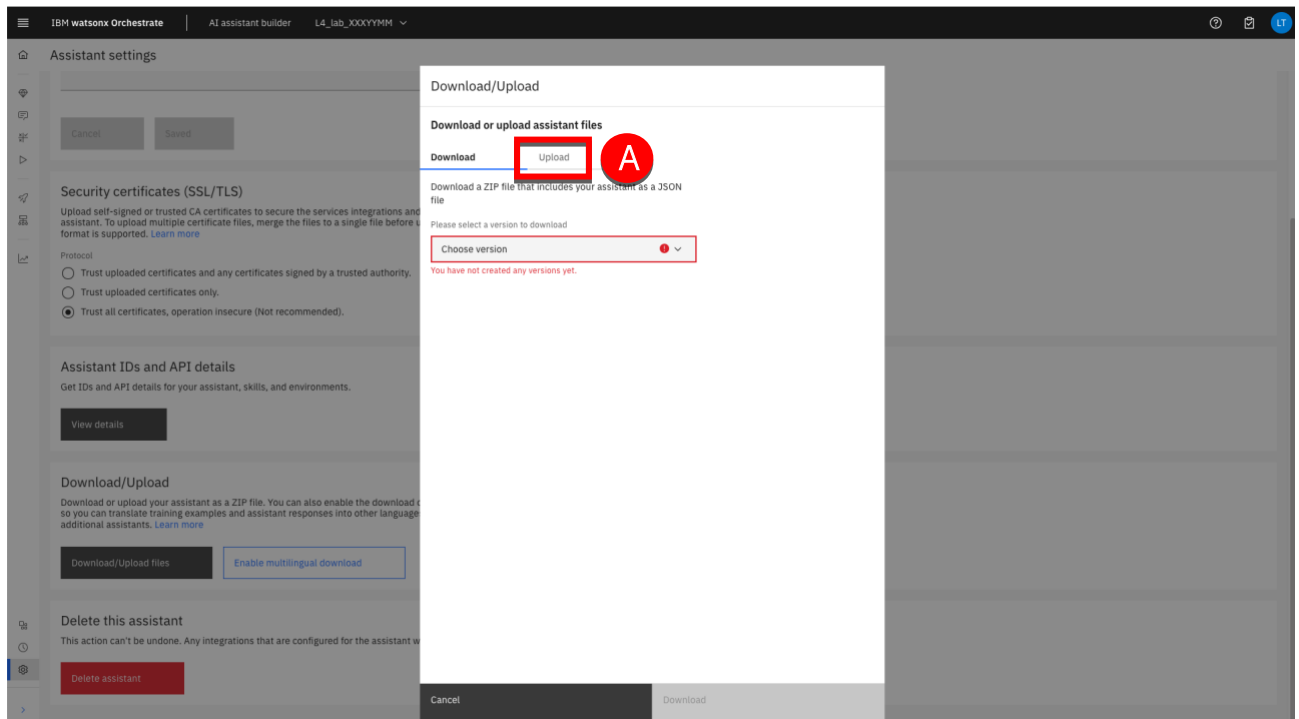
3. Scroll down (A).



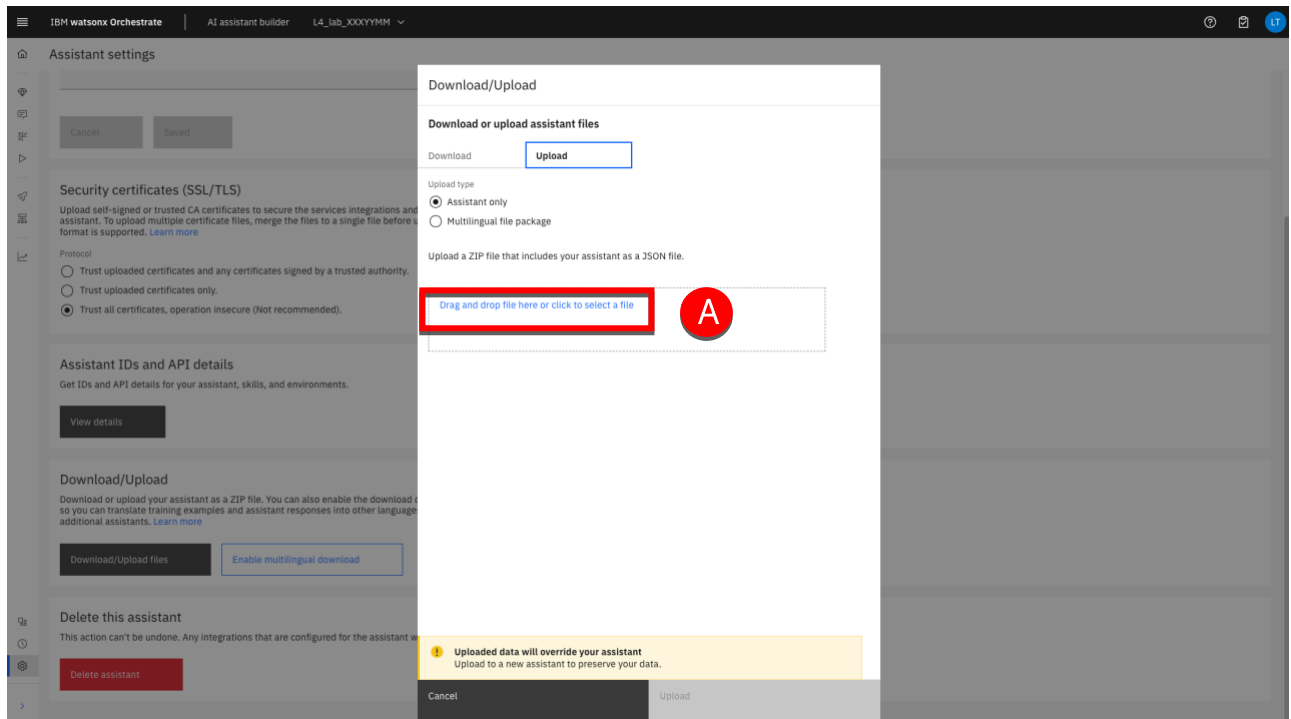
4. Click **Download/Upload files** (A).



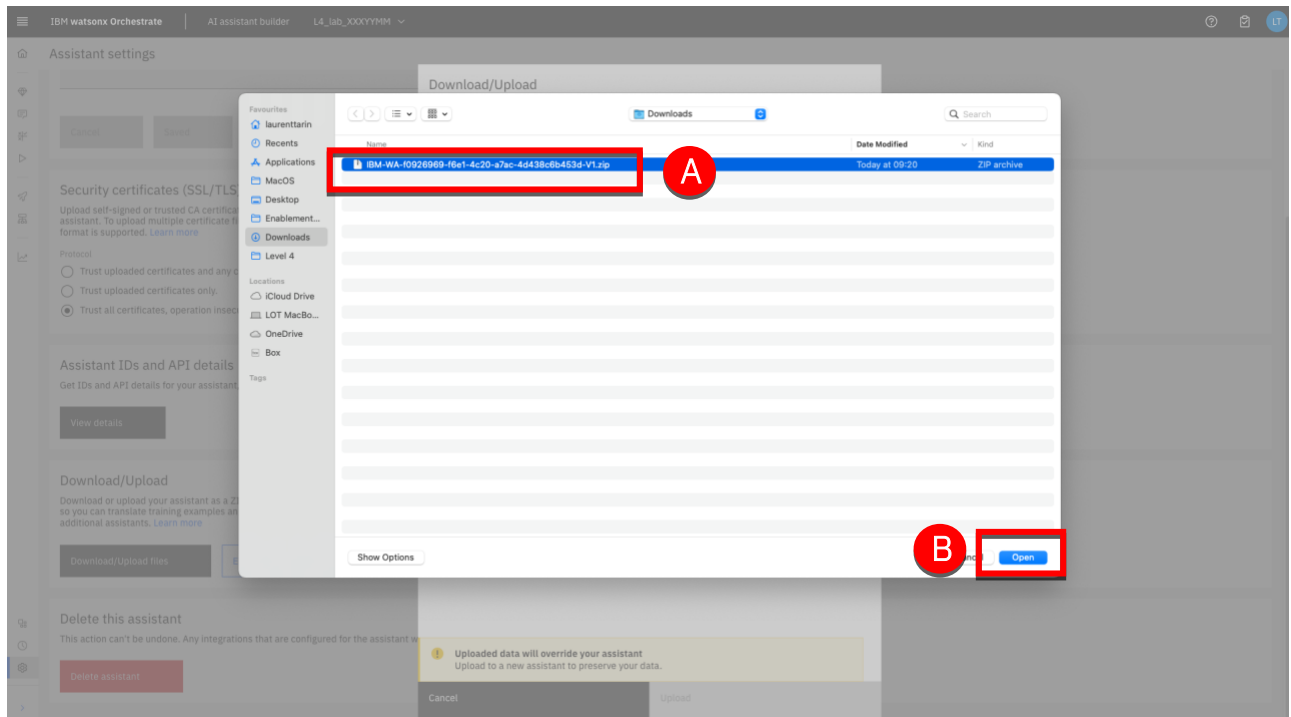
5. Select the **Upload** tab (A).



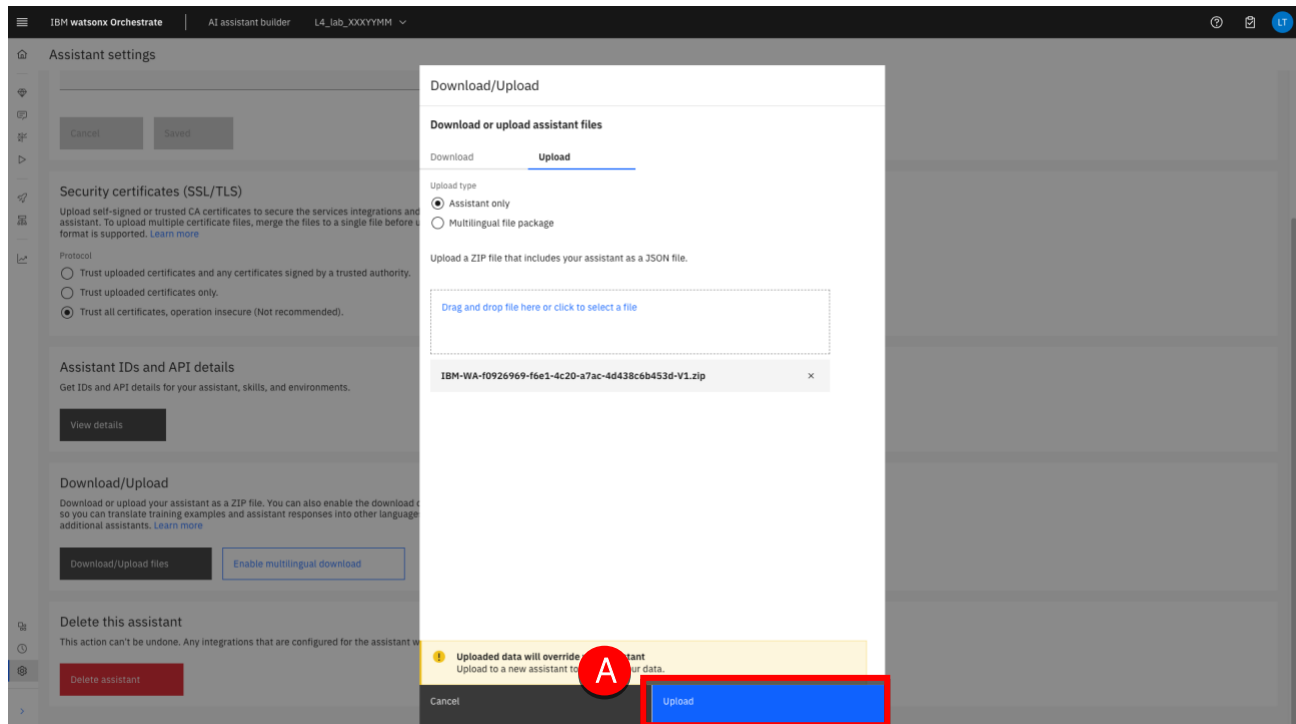
6. Click the **Drag and drop file here or click to select a file** link (A).



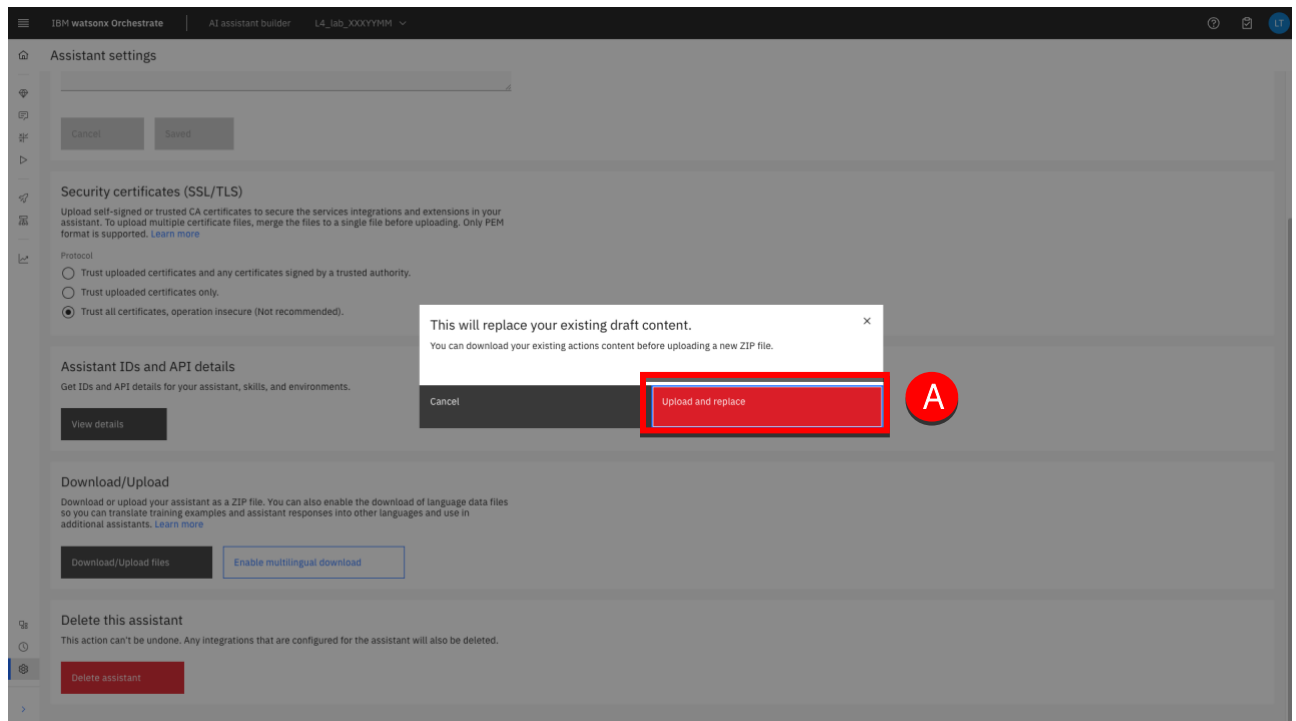
7. Select your archive file (A) then click **Open** (B).



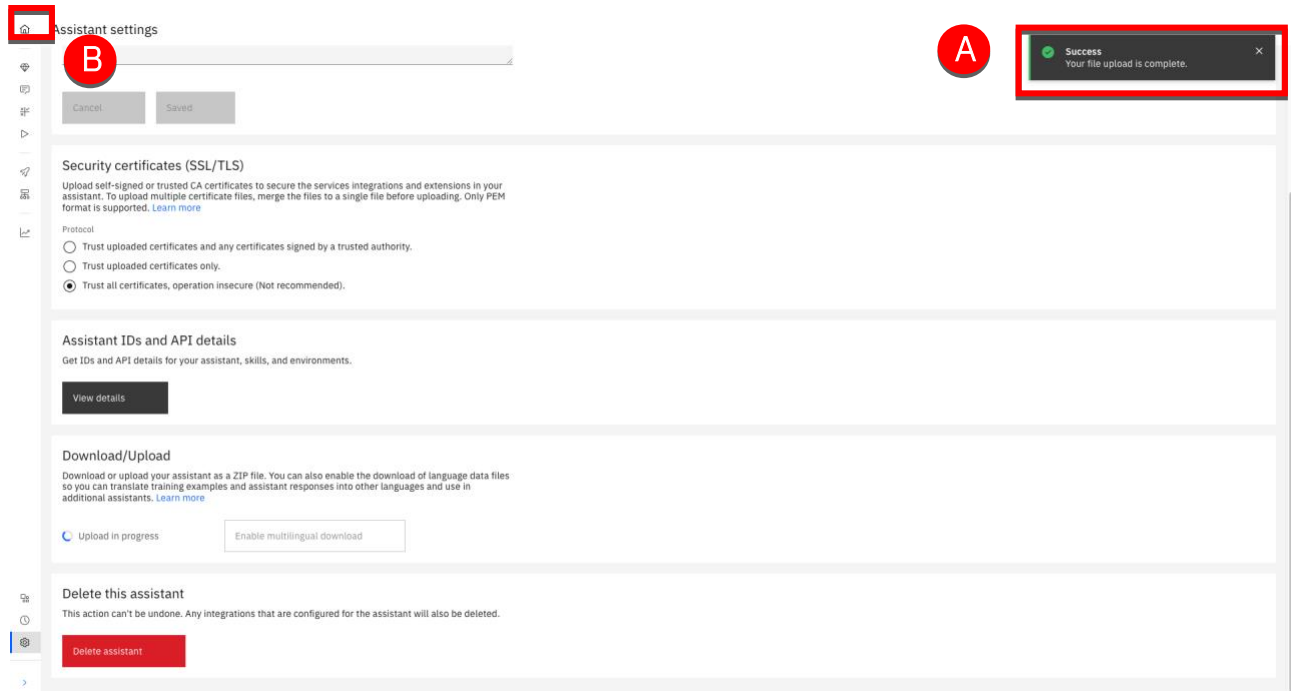
8. Click **Upload** (A).



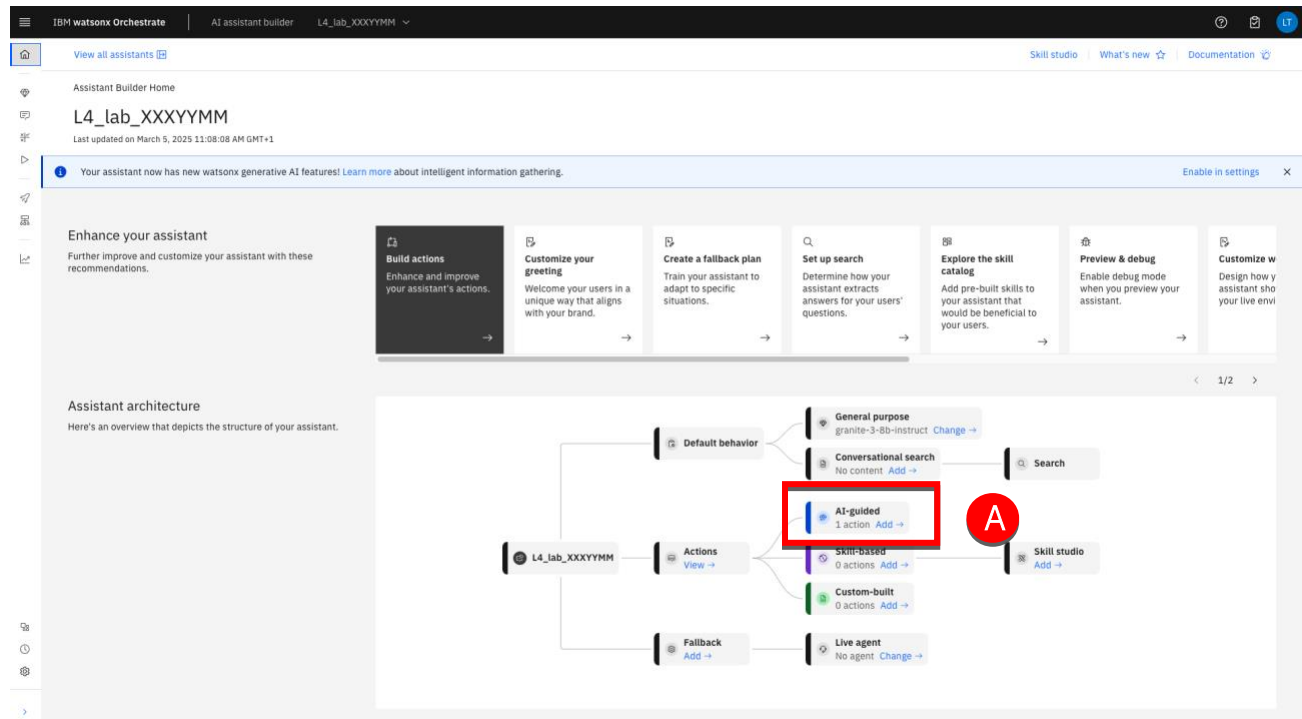
9. Click **Upload and replace** (A).



10. Wait for the **Success** notification that will be shortly displayed on the top right of your screen (A). Click the home icon (B).



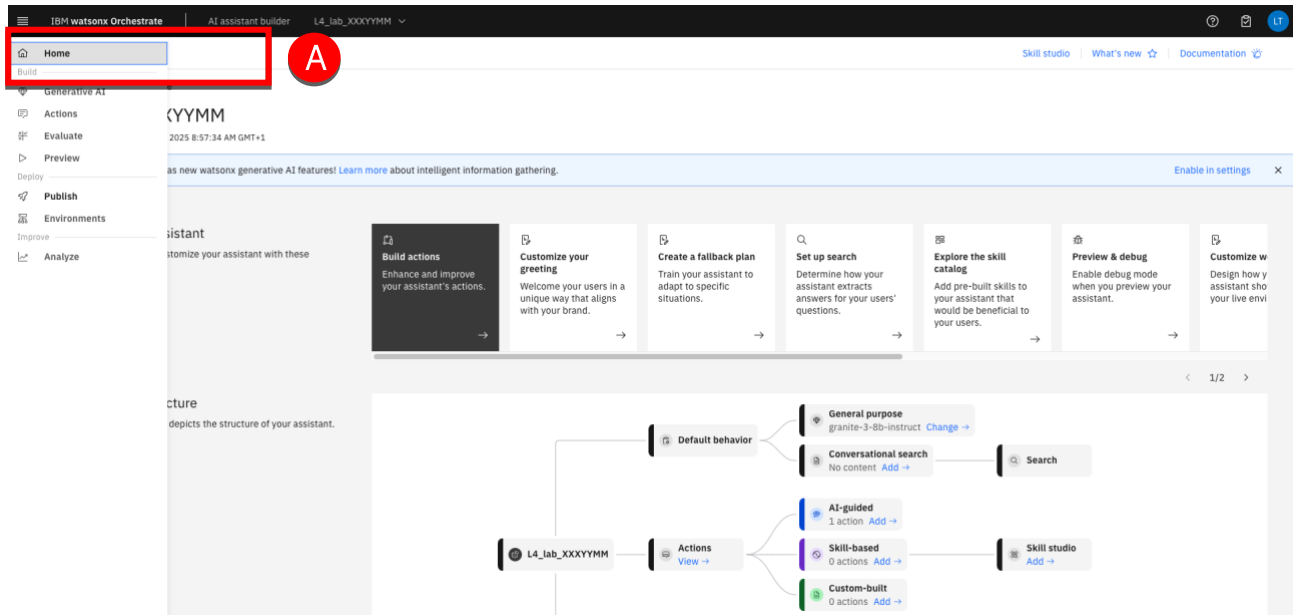
11. Check your previous actions are there (A) (AI-guided actions in the example below).



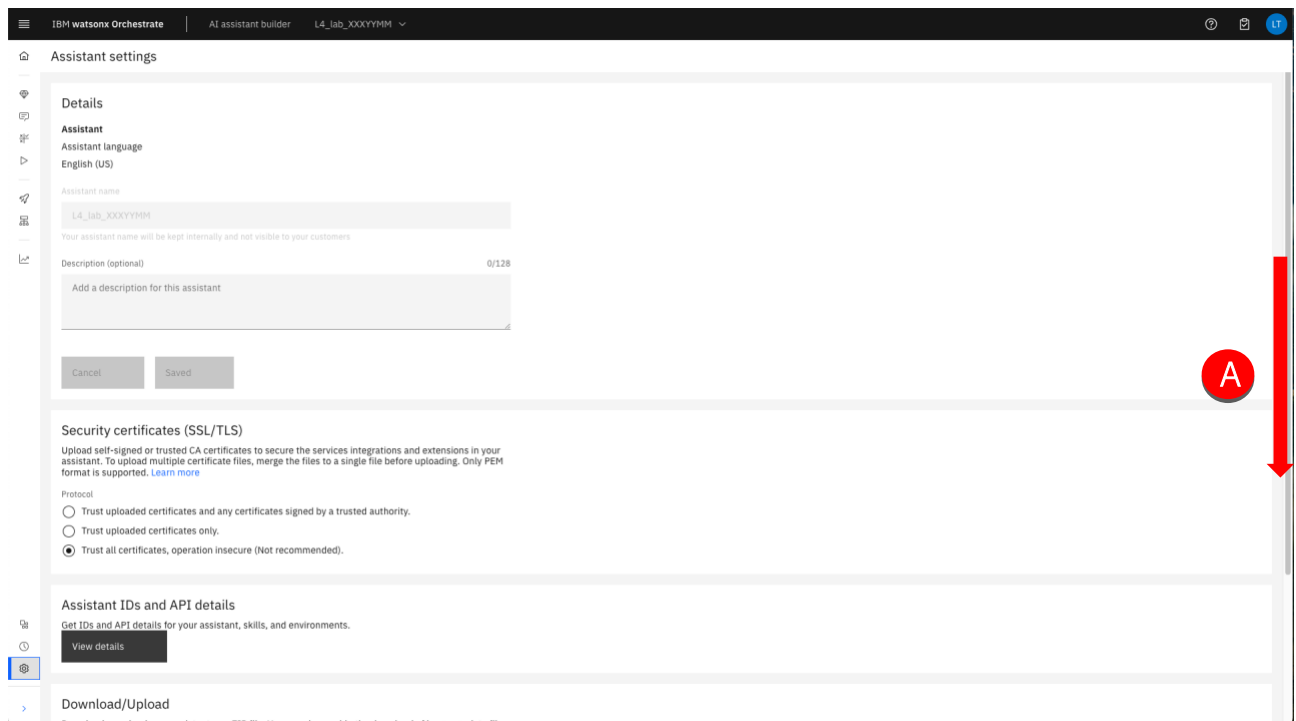
## Delete your assistant

Assistant must be deleted when you have performed all the hands-on labs related to assistants or when you have exported your archive before taking a break in the learning plan. Follow these instructions to delete your assistant.

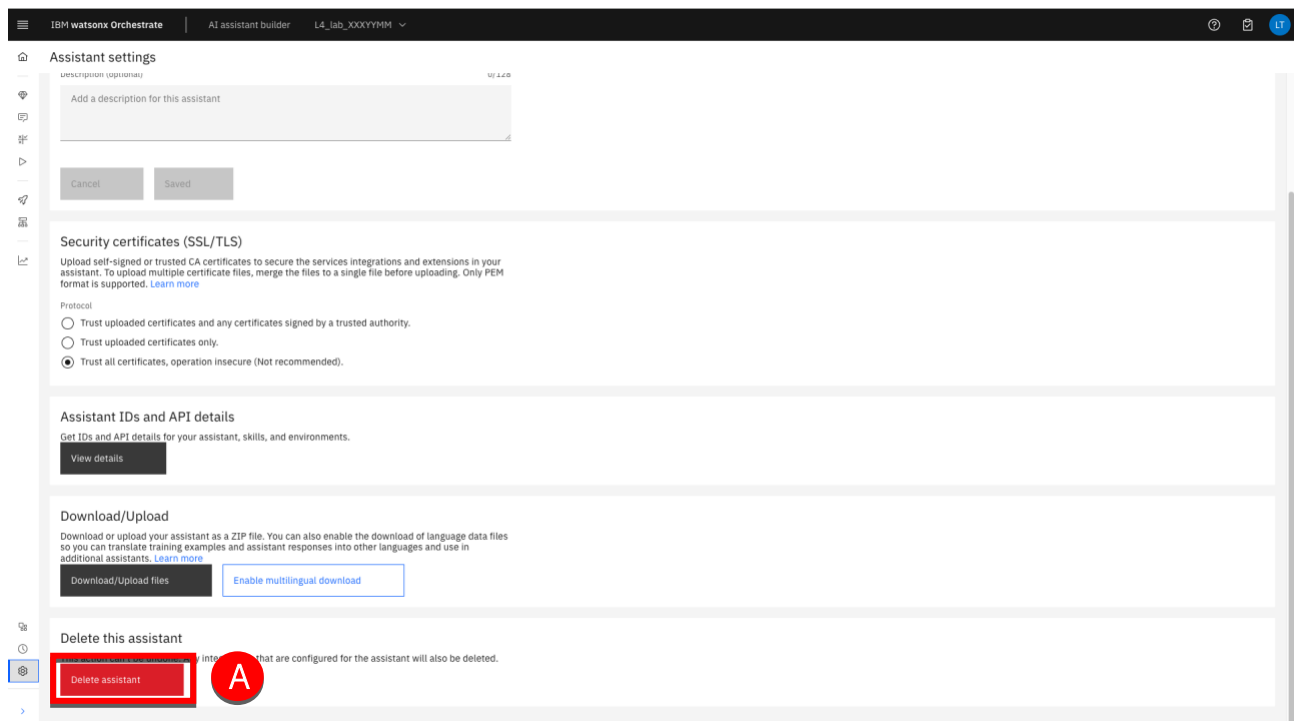
1. Return to the AI assistant builder home page.
2. Select the assistant to delete (A). Click **Assistant settings** (B).



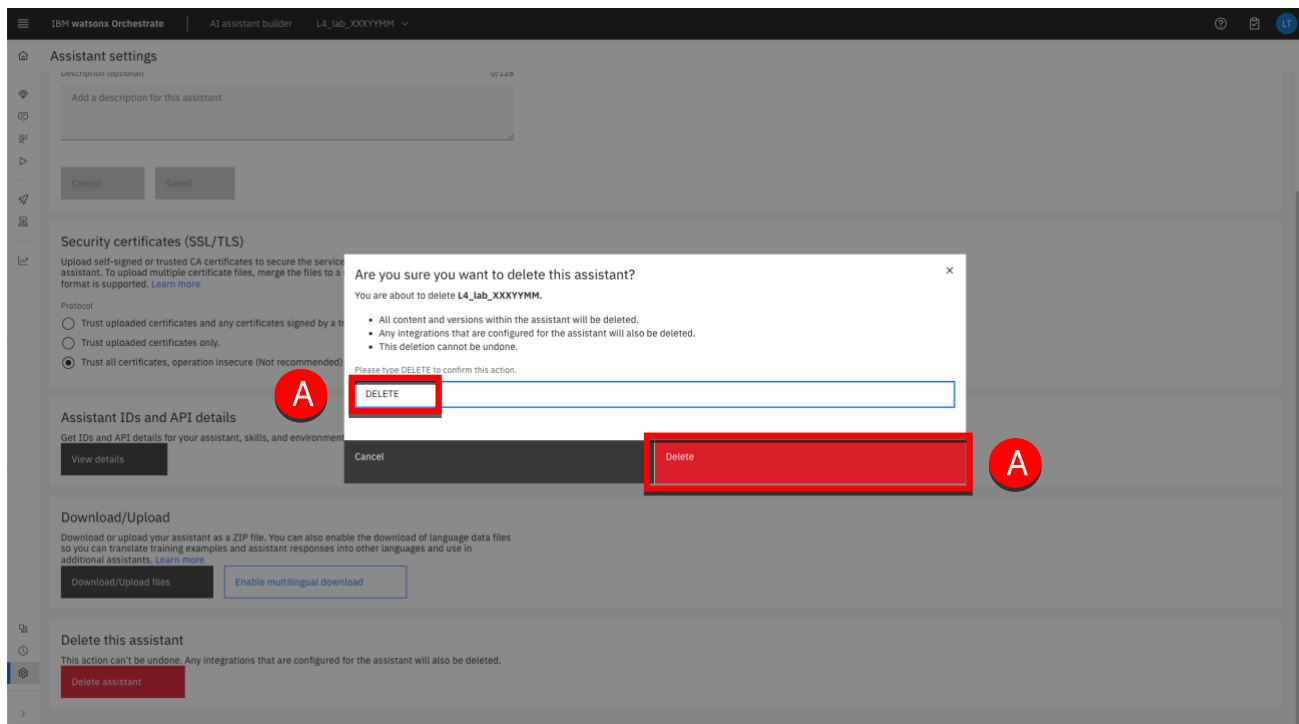
### 3. Scroll down (A).



### 4. Click Delete (A).



5. Enter DELETE (A) then click **Delete** (B).



Note: Once your assistant deleted, AI assistant builder will automatically open another assistant.

